



Europe under water: The macroeconomic cost of flooding and the economic case for adaptation

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Executive Summary



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Flood losses are rising rapidly as economic activity continues to concentrate in high-risk areas. Floods are Europe's most frequent and costliest natural hazard, with total economic losses climbing steadily, to EUR226bn in the first quarter of this century. While flood frequency in Europe has remained broadly stable at 46 per year since 2000, their cost has risen sharply. Losses from floods climbed 17.8% from EUR63.1bn between 2000-2009 to EUR 74.3bn from 2010-2019. From 2020-2025 alone, the cost reached EUR88.6bn, a 40% increase from the first decade of the century. Insurance covers only a fraction of the damage: The July 2021 floods alone caused a record EUR38bn in damage, of which only EUR9bn was insured, leaving households, businesses and governments to bear the remainder. Looking ahead, annual river-flood losses across the EU and UK could rise more than six-fold, from EUR7.8bn today to nearly EUR50bn by 2100 in a 3°C warming scenario, driven by more intense rainfall and continued development in exposed areas.

Floods are local events in their physical origin, but their economic consequences ripple through the entire economy. To quantify these economic effects, we simulate a one-off flood in 2027, calibrated to each country's average historical maximum flood depth over 2015-2024, and compare the economic trajectory with a no-flood baseline through 2030. The analysis links observed flood depth to real gross fixed capital formation and real household net disposable income, tracing how flood damage spreads through investment, consumption and public finances. Investment is hit hardest: cumulative gross fixed capital formation losses between 2027 and 2030 range from 10.5% in Norway to 14.6% in the Netherlands, with Germany at 12% recording the largest absolute loss at around EUR84bn. Real household net disposable income meanwhile falls by 3.9-5.4% in 2027-30 as reconstruction costs and labour-market effects accumulate, in turn reducing households' capacity to absorb uninsured losses and rising insurance costs.

Floods create a stagflationary shock, raising prices, slowing growth and eroding fiscal space. Consumer prices rise as damaged infrastructure, disrupted logistics and lower local availability of goods and services create supply bottlenecks. The cumulative price-level impact remains contained, from +0.2% in the United Kingdom to +1.0% in Greece, but it still adds pressure on households already facing an income shock. Private consumption declines in all countries, with cumulative losses ranging from -0.3% in Norway to -0.9% in Czechia; in absolute terms, the United Kingdom records the largest loss at about EUR61bn, ahead of Germany and France. GDP losses then range from around -0.4% in Norway to -1.0% in Spain, with Germany and France losing approximately EUR108bn and EUR79bn respectively. The impact on public finances is also visible, with cumulative 2027-2030 government deficits widening by 1.3pp of GDP on average, from -0.3pp in Norway to -2.4pp in Spain.

The economic case for flood prevention is overwhelming: well-targeted adaptation pays for itself many times over. Adapting to flooding requires more public capital than any other climate hazard, accounting for around 65% of all predominantly public measures in our adaptation taxonomy. Yet target adaptation investments in natural flood retention, resilient infrastructure and risk-sensitive land-use planning can largely offset future losses. Concretely, flood adaptation means keeping people and assets away from high-risk floodplains where possible, restoring natural retention areas, upgrading dikes, drainage, sewers and stormwater systems, and flood-proofing buildings and critical infrastructure. These measures should be combined according to the type of risk: basin-level retention and land-use planning for fluvial floods, local drainage and blue-green urban infrastructure for pluvial floods, supported by early-warning systems, emergency planning and insurance incentives. Flood resilience investments yield roughly four times their cost in avoided damages. Land-use planning offers similarly high returns: prohibiting development in floodplains and integrating green, water-retaining infrastructure into cities offer similarly high returns. This shows that more than climate policy, adaptation is preventive fiscal policy.

Europe's challenge is to deliver effective flood adaptation solutions at the speed and scale required. The measures with the highest returns, keeping development out of floodplains and restoring natural flood retention, are also the most politically difficult to implement, while major flood protection infrastructure requires lengthy planning and construction. Germany illustrates this implementation gap: despite the EUR38bn flood disaster in 2021, only around EUR500m of the EUR6–7bn planned under the National Flood Protection Programme launched in 2013 has been spent. The main barriers are institutional fragmentation and lengthy approval procedures rather than limited fiscal capacity.

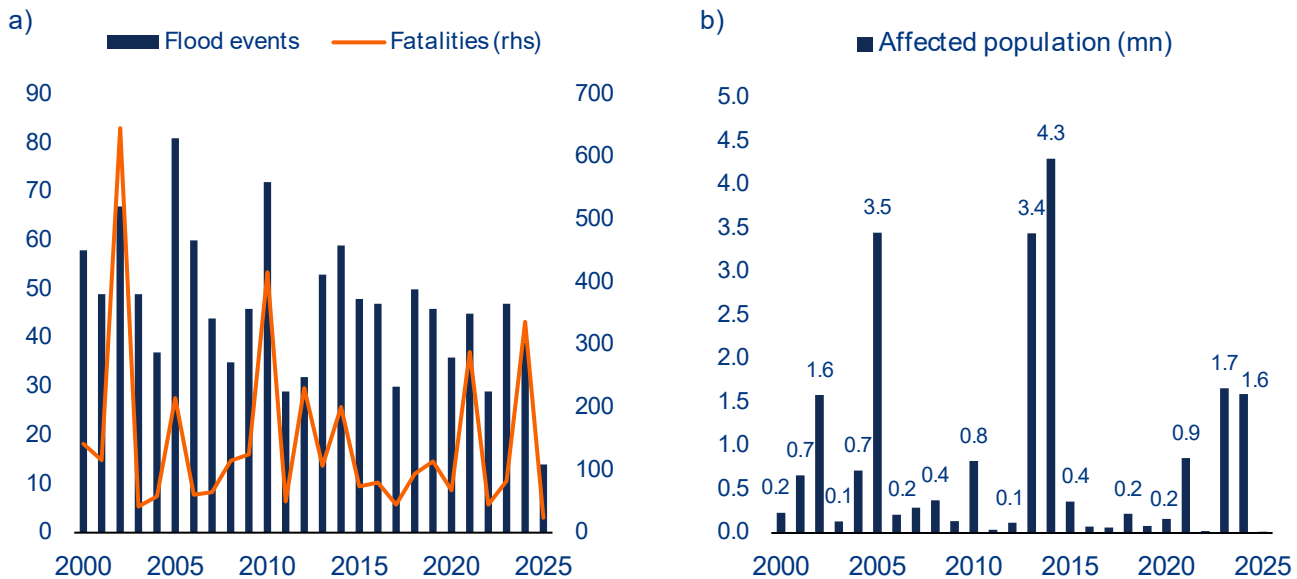
Long-term flood resilience requires integrating prevention, adaptation and insurance coverage into a single risk-management strategy. Structural flood defences, resilient building standards, property-level adaptation and greater public awareness must be complemented by sustainable public-private insurance partnerships. Experience from Spain's Consorcio, France's CCR and reforms under discussion in Germany and Ireland shows that risk transfer remains sustainable only when accompanied by meaningful risk reduction. Accelerating proven adaptation measures would strengthen public finances, reduce uninsured losses and safeguard the long-term insurability of flood risk.



The scale and cost of flooding in Europe

Europe has recorded an average of 46 flood events per year since 2000 - with close to two events per week in the most active years - but the number of floods has not shown a clear upward trend. Figure 1 plots annual flood events alongside fatalities between 2000 and 2025. Annual flood frequency fluctuates widely, from fewer than 30 events in quieter years to more than 80 in the most active, yet recent years are no more flood-prone than the early 2000s. What has changed is the severity of the worst events. Fatalities and affected populations are concentrated in a handful of catastrophe years, notably the Central European floods of 2002, the devastating floods in western Germany

and Belgium in July 2021, and the three major flood events of 2024 - the October storm in eastern Spain and the June and September floods across Central Europe. Between these episodes, flood-related deaths remain comparatively low, giving the series a pattern of intermittent but highly destructive shocks rather than a steady upward trend. The affected population follows a similar pattern, with additional peaks in 2005, 2013, 2014 and 2023, reflecting years in which floods displaced or affected large numbers of people despite relatively low mortality.

Figure 1: Flood events and associated fatalities (a), and affected population in million (b) in Europe, 2000 to 2025

Sources: ClimRad¹, EarthianAI, Allianz Research

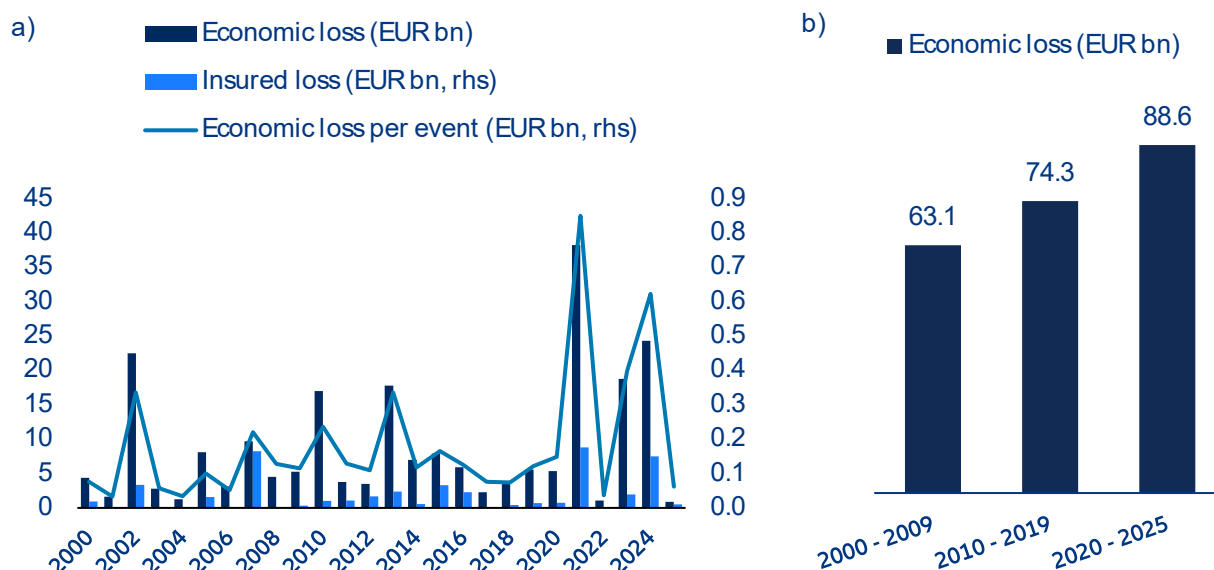
Economic losses have risen sharply as more people and assets are exposed to flood risk. The July 2021 floods remain Europe's costliest flood disaster on record, causing around EUR38bn in economic losses, followed by the Europe-wide floods of 2024 (EUR24bn) and the Central European floods of 2002 (EUR22bn; Figure 2a, 2025 prices). As with fatalities, flood losses are concentrated in a handful of catastrophe years rather than accumulating steadily over time. There is also no sign of moderation: the two most expensive years in the record - 2021 and 2024 - occurred within the past five years. The economic cost per event has risen sharply over this period. Between 2000 and 2019, a flood event caused on average around EUR0.1bn in economic losses; between 2020 and 2025 that figure climbed to approximately EUR0.4bn, with the 2021 floods alone averaging EUR0.8bn per recorded event. The rising economic cost of floods is clear when we look at the trend since 2000. Total economic losses grew 17.8% between the 2000–2009 and 2010–2019 decades, rising from EUR 63.1bn to EUR 74.3bn. The 2020–2025 half-decade alone has already reached EUR 88.6bn in losses, driven largely by the disastrous events of 2021, 2023, and 2024 – an increase of 19.1% over 2010–2019 and 40% over the first decade of the century (Figure 2b). This increase reflects not only more damaging weather but, above all, growing exposure. As urbanisation and economic activity continue to concentrate in flood-

prone areas, similar flood events destroy substantially more capital than they did two decades ago. Because exposure is shaped by land-use decisions, building standards and flood protection, a large share of these future losses remains within policy control.

Most flood losses remain uninsured, leaving households, businesses and governments to absorb the economic cost. In 2021, insured losses of around EUR9bn covered less than one-quarter of total damage; in 2002 the insured share was around 15%, and in 2010 below 10%. These swings reflect which countries flood in a given year, not a steady trend, because flood cover differs sharply across Europe. It is bundled into standard home insurance in some countries, such as the UK, but remains optional in others, including Germany, so a year dominated by losses in low-coverage markets pulls the European insured share down. Moreover, in several European markets, protection and modelling have historically been stronger for river flooding than for localised heavy-rain and surface-water flooding. Where losses are uninsured, they fall directly on households and businesses or are ultimately borne by the public sector through emergency relief and reconstruction. This protection gap amplifies the macroeconomic consequences of flooding, weakening household balance sheets, constraining investment and increasing pressure on public finances.

¹ ClimRad is a climate data tool developed by Allianz Research through a collaboration with EarthianAI

Figure 2: (a) Economic and insured flood losses in Europe, 2000 to 2025 (EUR billion); (b) Evolution of economic losses in Europe per decade



Sources: ClimRad, EarthianAI, Allianz Research

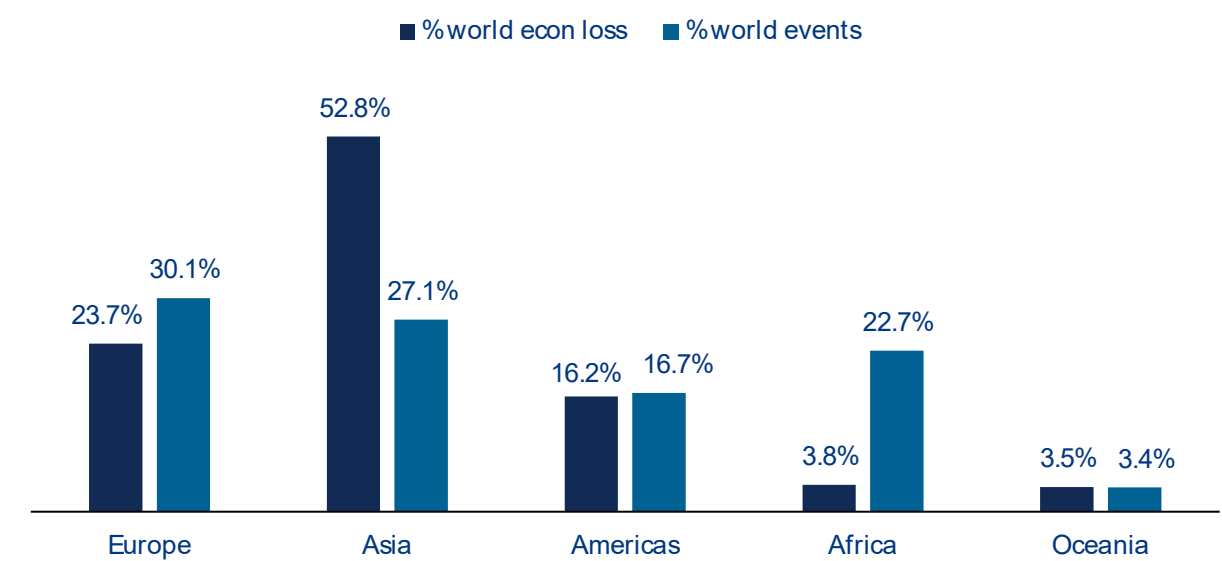
Floods impose a disproportionately large economic burden on Europe. Although Europe accounts for around 30% of recorded global flood events, it represents just under one-quarter of global flood losses (Figure 3). By contrast, Asia generates more than half of global flood losses from only around one-quarter of recorded events, reflecting the concentration of people and economic assets in highly exposed flood-prone regions.²

Within Europe, floods are by far the most costly natural hazard. Between 2000 and 2025, they accounted for around 55% of all natural-catastrophe losses while representing roughly 45% of recorded events (Figure 4). No other hazard comes close. Storms, both severe convective storms and extratropical cyclones, rank a distant second, while earthquakes account for around one-tenth of losses despite occurring only rarely, illustrating their low-frequency, high-severity nature. Floods combine both high frequency and substantial damage, making them Europe's largest source of cumulative natural-catastrophe losses.

Flood frequency alone is a poor guide to economic risk. Italy, Spain and France recorded the largest number of flood events between 2000 and 2025, but countries with fewer floods often experienced much larger impacts on their populations (Figure 5a-b). Bulgaria, Bosnia and Herzegovina and Czechia stand out as having particularly high numbers of people affected relative to the frequency of flooding, reflecting repeated exposure to major flood events. Ranking countries by economic losses reshuffles the picture again: Germany records by far the largest cumulative losses (around USD78bn), followed by Italy (USD42bn) and Spain (USD25bn) (Figure 5c). The ranking reflects not only flood frequency but also the concentration of economic assets in exposed areas and the occurrence of exceptionally destructive events such as the 2021 floods.

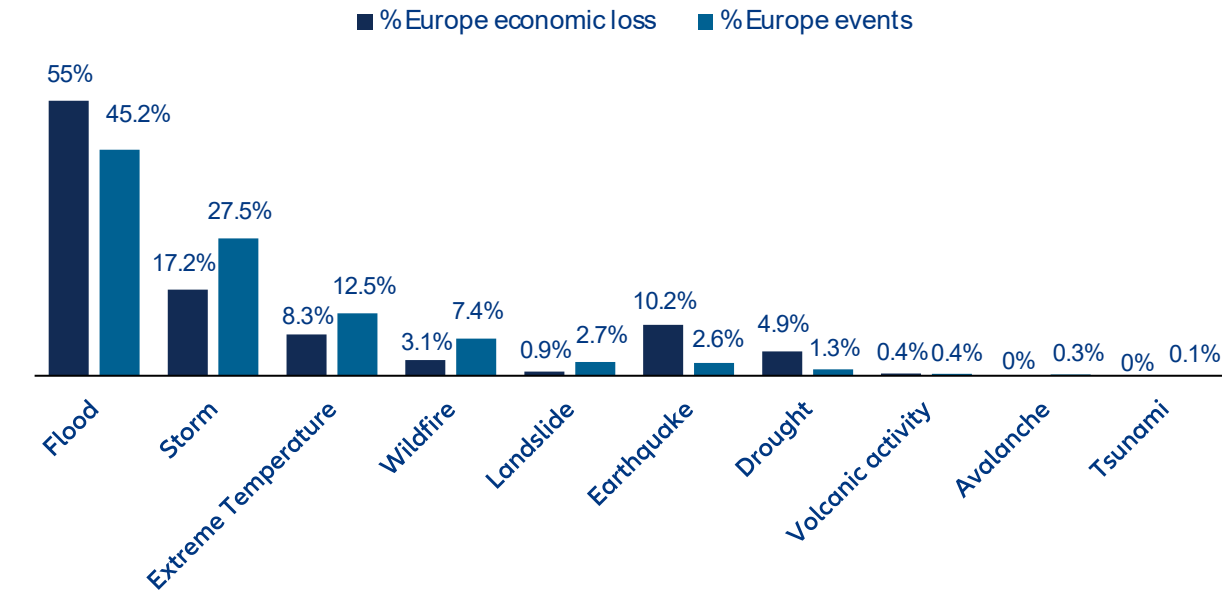
² Because flood events are captured through national reporting and emergency-mapping activations, coverage is not fully uniform across countries; this may affect individual figures but not the overall conclusions.

Figure 3: Regional shares of global flood economic losses and of recorded flood events



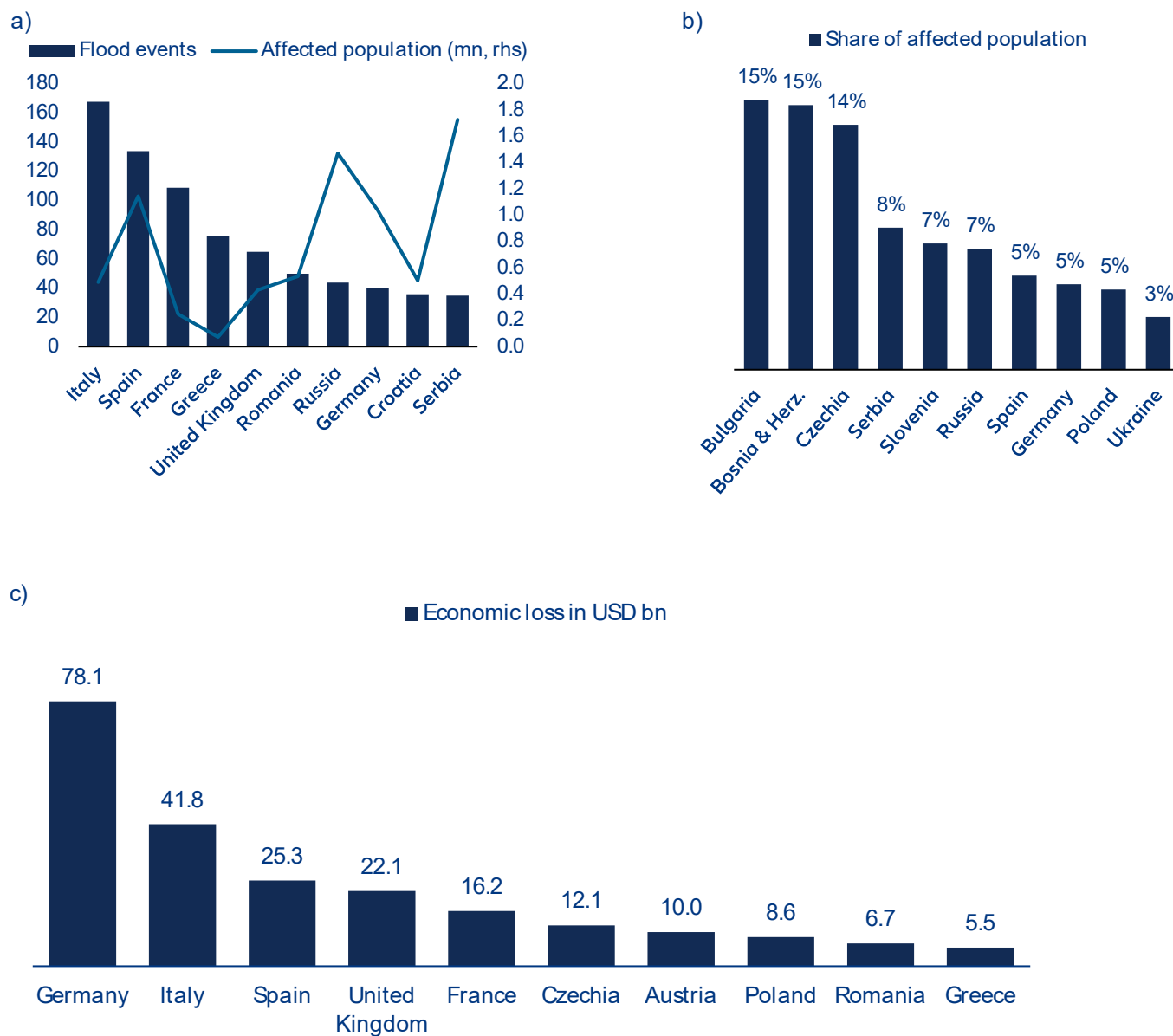
Sources: ClimRad, EarthianAI, Allianz Research

Figure 4: Shares of European natural-catastrophe losses and events, by hazard type



Sources: ClimRad, EarthianAI, Allianz Research

Figure 5: (a) Top 10 European countries ranked by number of flood events and by affected population from 2000 to 2025 (b) Top 10 European countries ranked by affected population from 2000 to 2025 (c) Top 10 countries ranked by total economic losses from 2000 to 2025



Sources: ClimRad, EarthianAI, Allianz Research



The macroeconomic impact of flood events in Europe

Floods become macroeconomic shocks that spread through the whole economy reducing investment, household incomes and fiscal space. Floods affect the macroeconomy through three channels. For households, the shock reduces real disposable income through repair costs, temporary displacement, lost working time and higher prices for essential goods and services. For firms, it reduces productive capacity through damage to buildings, machinery and inventories, while also raising short-term costs through emergency repairs, business interruption and supply-chain disruptions. These effects weaken investment. For the public sector, floods increase emergency and reconstruction spending while reducing tax intake as activity slows. We capture these channels in two steps. First, we estimate how observed flood depth³ affects real gross capital formation and real net disposable household income. Second, we feed these estimated shocks into a macro-econometric model to assess the wider macroeconomic effects. The exercise is calibrated as a one-off flood shock in 2027, based on the recent historical average of maximum flood depth over 2015 to 2024, and all results are measured relative to a no-flood baseline. This analysis is based on recently observed flood events; however, more extreme floods could have even greater economic impacts across European countries than indicated here. For example, more than 18% of the populations of Austria, Hungary,

Netherlands; Liechtenstein and Slovakia live in floodplains.⁴ The numbers should therefore be read as a standardized stress test rather than a macroeconomic forecast under a tail risk event.

Investment is hit the hardest. A flood destroys part of the existing capital stock, but it also changes the conditions under which firms decide whether to commit to new capital. Since the dependent variable is real gross fixed capital formation for the total economy, the estimate already includes any reconstruction investment recorded as fixed capital formation. The reported loss is therefore a net shortfall relative to the no-flood baseline, after rebuilding spending has partly offset the initial capital damage. (See Appendix A1 for details on our empirical methodology). Estimation results show that the cumulative loss in capital formation — over the period 2027 – 2030 — is large across all countries, ranging from around -10.5% in Norway to -14.6% in the Netherlands from 2027 to 2029 (Figure 6). The Netherlands records the largest relative loss, followed by Croatia, France and Greece at about -14%, while Spain and Ireland are close to -13.4%. Germany is less affected in percentage terms, at 11.9%, but records the largest absolute loss at about EUR83.6bn. France follows with EUR78.4bn, the United Kingdom with EUR66.5bn, Italy with EUR51.3bn and Spain with EUR36.6bn.

³ See Appendix A3 on how we calculate flood depth in Europe from 2015 to 2024

⁴ [Population characteristics in potential flood-prone areas | Maps and charts | European Climate and Health Observatory Climate-ADAPT](#)

Figure 6: Cumulative capital formation loss (2027–2029) from a one-off 2027 flood shock, calibrated to the 2015–2024 average maximum depth, relative to a no-flood baseline



Source: Allianz Research

Floods leave a lasting mark on household incomes.

Cumulative real net disposable income losses for the period 2027 – 2030 range from -3.9% in Slovakia to -5.4% in Croatia (Figure 7). Sweden, Italy and Germany sit just above -4%, while Belgium is at -4.3% and Finland at -4.5%. The largest losses are recorded in Spain (-4.7%), France (-4.9%), Greece (-5.0%), Ireland (-5.1%), the Netherlands (-5.3%) and Croatia (-5.4%). Household income however does not collapse in the year of the flood rather the impact is lagged, which is consistent with the way disaster losses are absorbed in annual data, as reconstruction costs, business interruption and labor-market effects accumulate over the following years. Here the macro analysis begins to connect with insurance affordability. Under risk-based pricing, the same homeowners in floodplains often face higher than average flood premiums and more frequent losses. A flood therefore creates a double pressure: it weakens the income base from which protection is paid, while insurers might re-assess the risk, which could lead to higher prices.

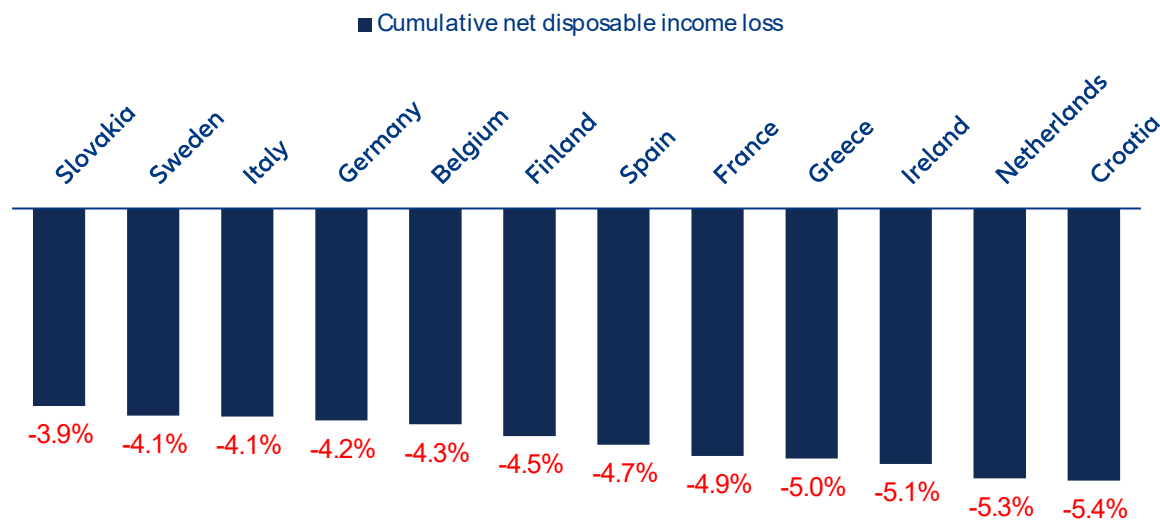
Flood shocks lead to significant GDP losses, with the strongest relative impacts in Spain, Croatia, Greece and the United Kingdom.

The largest cumulative GDP loss over 2027 to 2030 are in Spain (-1.0%), Croatia (-0.8%), Greece (-0.8%), the United Kingdom (-0.8%), France (-0.7%) and Germany (-0.7%), mainly reflecting the severity of flood events observed over the last decade. In monetary terms, Germany loses around EUR108bn relative to the baseline, followed by the United Kingdom (EUR106.3bn), France (EUR79.2bn), Spain (EUR59.2bn) and Italy (EUR56.5bn).

The key insight is that a flood is a local physical shock, but its macroeconomic effects spread beyond the flooded area.

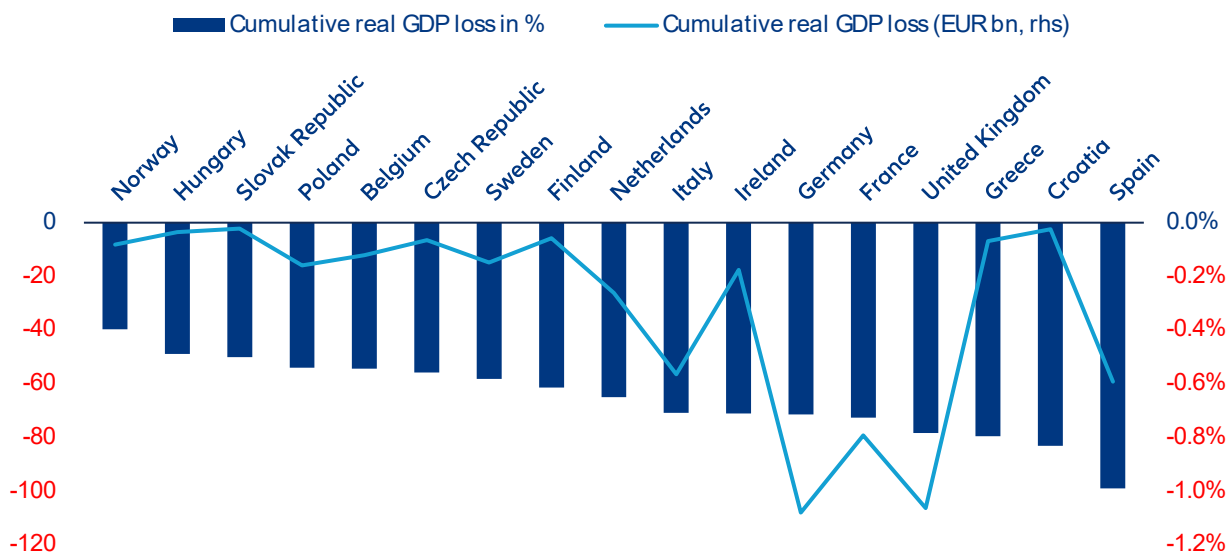
A flood lowers GDP not only through the immediate destruction of assets, but also through delayed investment, weaker household income, lower consumption and higher reconstruction needs. Reconstruction spending offsets part of the loss, but it does not erase it. The model result is consistent with the broader disaster literature: the immediate destruction is only the first-round effect; the more persistent cost comes from disrupted accumulation and weaker income and demand over the following years.

Figure 7: Cumulative net disposable income loss, constant 2013 price, (2027–2029) from a one-off 2027 flood shock, calibrated to the 2015–2024 average maximum depth, relative to a no-flood baseline



Sources: Allianz Research

Figure 8: Cumulative GDP loss (2027–2030) from a one-off 2027 flood shock, calibrated to the 2015–2024 average maximum depth, relative to a no-flood baseline

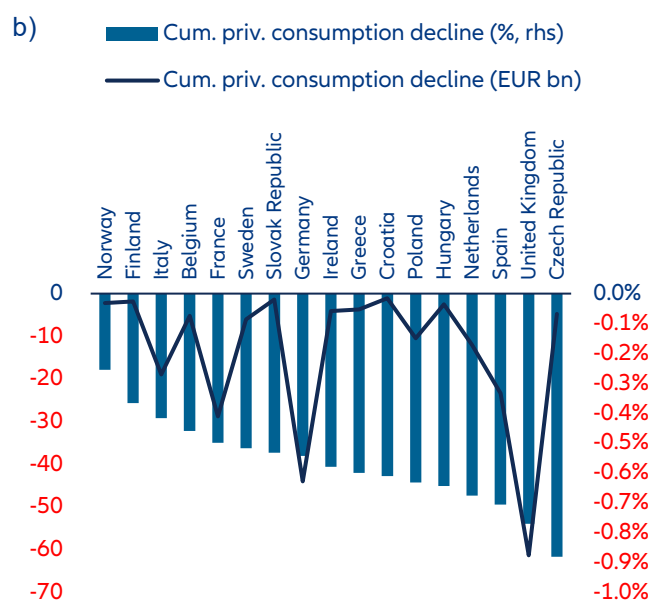
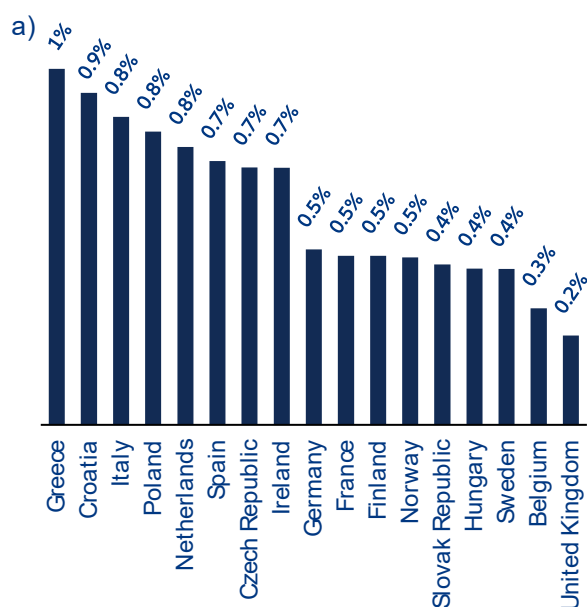


Sources: Allianz Research

Floods create a stagflationary shock by raising prices even as economic activity weakens. Damaged infrastructure, disrupted logistics and lower local supply temporarily increase prices across countries (Figure 9a). The cumulative (2027 – 2030) price-level impact is highest in Greece, at around +1.0%, followed by Croatia (+0.9%), Italy (+0.8%), Poland (+0.8%), the Netherlands (+0.8%) and Spain (+0.7%). At the lower end, the impact is more contained in Belgium (+0.3%) and the United Kingdom (+0.2%). While the inflation effect is moderate, it compounds the loss of household purchasing power.

Floods reduce household consumption by eroding real incomes and purchasing power (Figure 9b). Private consumption falls in every country, with cumulative losses ranging from around -0.3% in Norway to -0.9% in Czechia. The largest relative declines are recorded in Czechia (-0.9%), the United Kingdom (-0.8%), Spain (-0.7%), the Netherlands (-0.7%), Hungary (-0.6%) and Poland (-0.6%). In absolute terms, the largest losses are in the United Kingdom with about EUR61.3bn in private consumption relative to baseline, Germany EUR44bn, France EUR28.8bn, Spain EUR23.3bn and Italy EUR18.9bn.

Figure 9: Cumulative inflation rise (a) and private consumption decline (b) (2027–2030) from a one-off 2027 flood shock, calibrated to the 2015–2024 average maximum depth, relative to a no-flood baseline



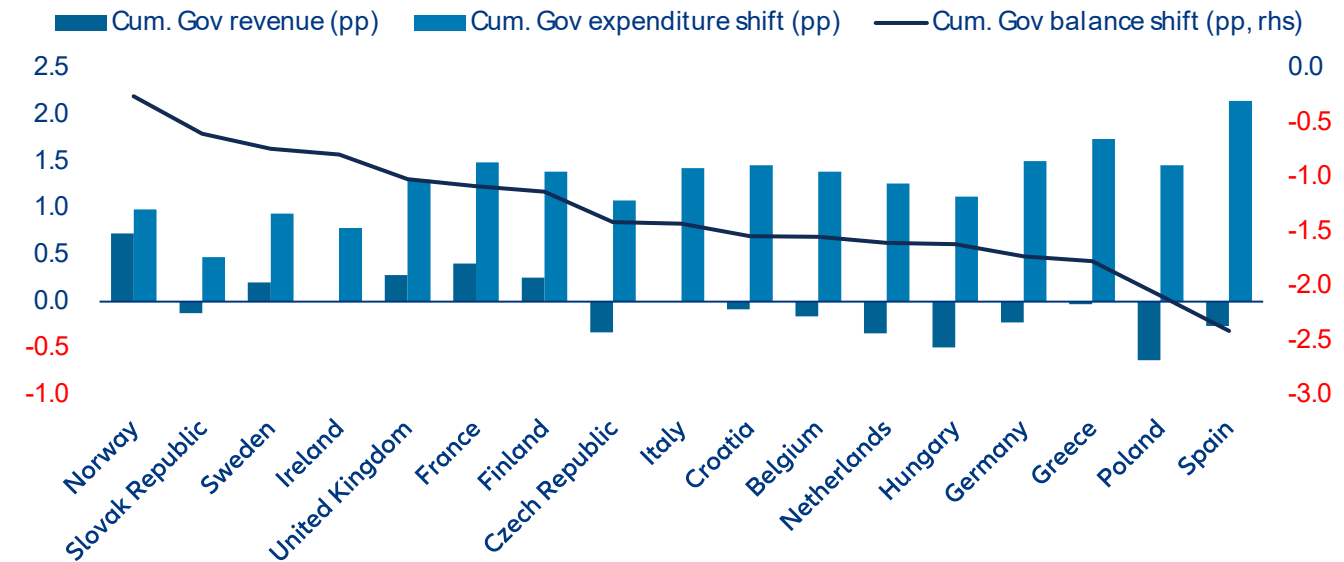
Sources: Allianz Research

Floods place sustained pressure on public finances, leaving governments with less fiscal space (Figure 10).

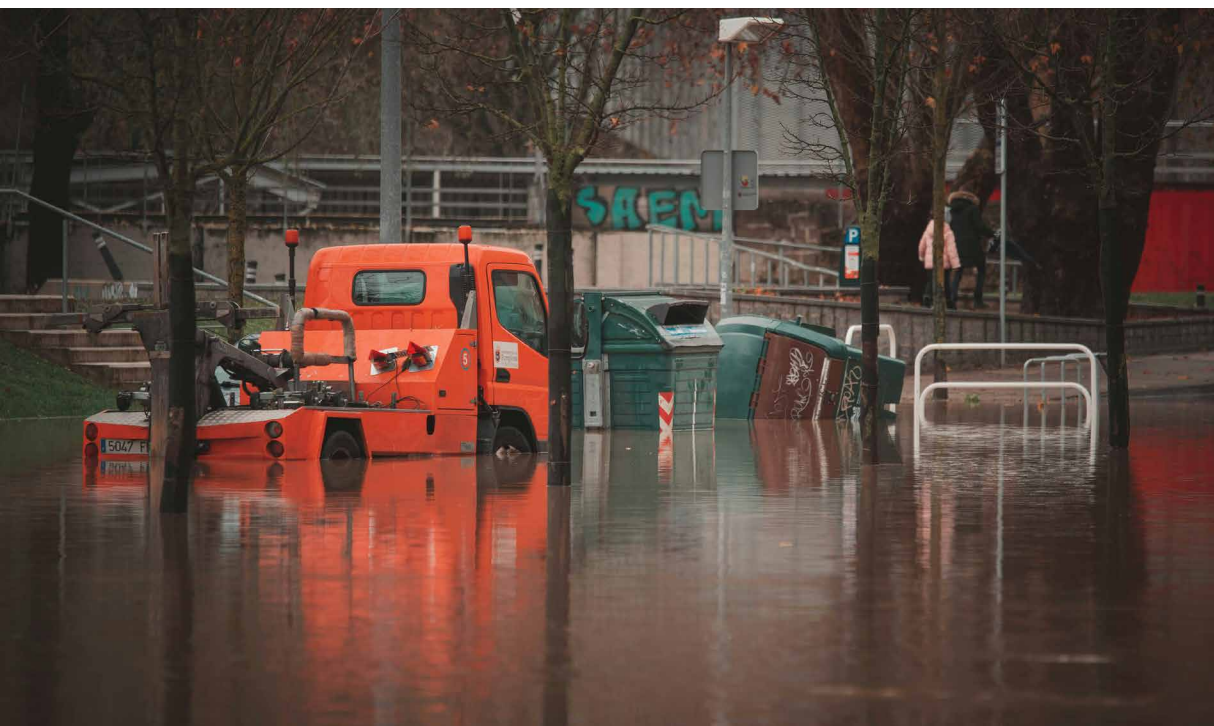
The government balance deteriorates in every country in the stress test, from a relatively contained -0.3 percentage points in Norway to -2.4pp at the upper end. The largest balance shifts are recorded in Spain (-2.4pp), Poland (-2.1pp), Greece (-1.8pp), Germany (-1.7pp), Hungary (-1.6pp) and the Netherlands (-1.6pp). The deterioration is mostly expenditure-driven, reflecting emergency response, reconstruction support and the public share of recovery costs. Spain records the largest expenditure shift, at about +2.2pp, followed by Greece (+2.2pp), Germany (+1.7pp), France (+1.5pp), Poland (+1.7pp), Germany (+1.5pp), France (+1.5pp), Poland

(+1.5pp) and Croatia (+1.5pp). The revenue channel is more mixed. It is positive in some countries, including Norway, France, the United Kingdom, Finland and Sweden, but negative in others, especially Poland (-0.6pp), Hungary (-0.5pp), the Netherlands (-0.3pp), Czechia (-0.3pp), Spain (-0.3pp) and Germany (-0.2pp). This mixed pattern is plausible: reconstruction and price effects can support some nominal tax bases, while weaker real activity and consumption reduce others. What is common across all countries is the pressure on the fiscal balance. Flood risk therefore creates a public finance problem in addition to a household income and capital accumulation problem.

Figure 10: Cumulative government balance (2027–2030) from a one-off 2027 flood shock, calibrated to the 2015–2024 average maximum depth, relative to a no-flood baseline



Sources: Allianz Research



Adaptation to flood-proof our economies

Floods are already Europe's costliest natural hazard, relying most heavily on public capital. Flood adaptation requires the most public capital of all climate hazards: of the 61 adaptation measures in our taxonomy, flood protection accounts for 65% of all primarily public measures, reflecting the dominance of large-scale infrastructure, retention and spatial planning in managing flood risk.⁵ The largest losses, the widest protection gap and the heaviest reliance on public investment together make closing the flood-adaptation gap not only an insurance, but a fiscal priority.

Concretely, flood adaptation is a layered risk-reduction strategy rather than a single type of investment. It starts with spatial planning: avoiding new development in floodplains, restoring natural retention areas and giving rivers more room to absorb peak flows. It then requires blue-green and grey infrastructure, including retention basins, dikes, levees, flood walls, sustainable drainage systems, porous pavements, green roofs, stormwater management, wastewater and sewage upgrades and pumping equipment. In already exposed areas, adaptation must move down to the asset level through flood-adapted construction, elevated foundations, protected utilities, backflow prevention, water-resistant

materials and the relocation of vulnerable equipment above expected flood levels. Finally, these physical measures need to be complemented by risk mapping, early-warning systems, evacuation planning and insurance incentives that reward actual risk reduction. The point is not to choose between nature-based solutions, infrastructure and property-level protection, but to combine them according to the type of flood risk: fluvial flooding requires river-basin management and retention, while pluvial flooding depends more on local drainage, urban design and building-level resilience.

One of the most positive developments since 2021 is the rapid improvement in flood risk information. Traditionally, flood management focused on river floodplains, mapping areas at risk from overflowing rivers and identifying exposed assets. While these remain important, the 2021 European flood event demonstrated that destructive floods are often compound events, combining fluvial flooding, when a river bursts its banks, with pluvial flooding, flash floods caused by intense rainfall and surface water floods from overwhelmed drainage systems. These flood types differ fundamentally. Fluvial flooding typically allows somewhat longer warning times and can often

⁵ From invisible to investible: An investment taxonomy for climate adaptation | Allianz

be managed through dikes, retention areas, and coordinated water management. The German Insurance Association (GDV) reports that less than 8% of addresses in Germany lie in the river floodplains⁶. Pluvial flooding, by contrast, can occur outside mapped floodplains, develops rapidly, and depends heavily on local factors such as urban drainage, topography, and land use.

Fragmented flood management leads to structural challenges for governance, but high-resolution flood mapping can help. In Germany, for example, river flood management is largely organized at the state and basin level, while pluvial flooding is primarily a municipal responsibility. This creates a structural challenge: municipalities play a critical role in managing risk but often lack the resources, data, or capacity to evaluate and implement effective measures at scale. Tools such as the surface water flood maps in North Rhine-Westphalia highlight areas at risk from pluvial flooding, including estimates of potential water depth and velocity⁷. These maps provide an essential bridge between regional-level hazard assessment and local decision-making. In practice, this makes flood risk operational rather than abstract: municipalities can use these maps to prioritise sewer and drainage upgrades, identify locations where temporary pumping may be needed and prepare emergency access routes. They also improve public risk communication by showing which homes, schools, care facilities and critical services are most exposed, allowing evacuation planning and adaptation spending to be targeted more precisely.

Artificial intelligence is accelerating the ability to evaluate adaptation options. AI-supported approaches allow rapid scenario testing, enabling stakeholders to simulate the impact of measures such as retention basins, drainage upgrades, or structural protections before committing to costly engineering studies. While they do not replace detailed local studies, they enable municipalities, planners, and citizens to better understand and discuss risk. The value of maps and models extends beyond identifying risk, to shaping behavior, such as guiding land use, informing building standards and supporting emergency planning.

Germany after the 2021 floods as a case study for European analysis: stronger frameworks, but

persistent implementation challenges. The 2021 floods made the implementation gap especially visible, and the five years since then provide a useful test of how far policy frameworks, risk mapping and local adaptation planning have actually moved from recognition to execution.

Germany has taken some important steps since the 2021 floods. One immediate response was the strengthening of early warning systems, including the introduction of alerts via Cell Broadcast (DE-Alert) to rapidly reach citizens in a potentially affected area via mobile phone during emergencies. Germany also adopted the Federal Climate Adaptation Act⁸, which entered into force in 2024. This creates a binding framework for adaptation policy across federal, state, and municipal levels. It requires systematic risk analysis, coordinated planning, and measurable targets for adaptation progress. Concrete implementation is still at an early stage, as the federal framework only entered into force in mid-2024. But some measures are already visible at local level, such as improved flood risk communication, even if large-scale infrastructure projects remain slow to deliver. The updated German Climate Adaptation Strategy⁹ complements this by establishing measurable targets for climate adaptation, including goals for infrastructure, land use, water management, spatial planning, business and civil protection.

There have also been improvements in flood mapping and risk communication, but delivery at scale remains slow. In response to the 2021 flood along the Ahr River, the German state of Rhineland-Palatinate revised the flood hazard maps a few months after the event, which incorporated improved hydrological understanding to better represent extreme runoff dynamics and flood extents¹⁰. A full revision of all floodplains in Rhineland-Palatinate was completed in February 2026¹¹.

Despite progress, delivery at scale remains slow. Germany's federal system spreads planning and disaster-response responsibilities across multiple levels of government and aid organisations, creating coordination challenges, and large-scale projects such as retention basins face long planning and approval timelines. The National Flood Protection Programme

⁶ „ZÜRS Geo“ - Zonierungssystem für Überschwemmungsrisiko und Einschätzung von Umweltrisiken

⁷ LANUK Klima

⁸ BMUKN: The Federal Climate Adaptation Act

⁹ BMUKN: German Strategy for Adaptation to Climate Change

¹⁰ Überschwemmungsgebiet . Struktur- und Genehmigungsdirektion Nord

¹¹ Hochwassergefahren- und -risikokarten . Hochwasserrisikomanagement in Rheinland-Pfalz

(NHWSP) captures both the ambition and the lag: it bundles around 109 priority, supra-regional measures worth an estimated EUR6–7bn - co-financed 60% by the federal government and 40% by the Länder - targeting close to 1 billion m³ of new retention and roughly 40,000 ha of reconnected floodplain,¹² and federal modelling confirms these measures would lower the peak levels of future major floods by 10–50 cm over long river stretches¹³. Yet more than a decade after the 2013 floods, only a little over EUR500m had actually flowed through the dedicated „Preventive Flood Protection“ framework, with many NHWSP measures still in planning or early construction. The same lag appears locally: in the Ahr valley, communities and the district of Ahrweiler published their technical study of catchment-wide flood protection only in 2025 - four years after the disaster - with concrete measures still to be designed and built. This highlights a broader European challenge: adaptation frameworks have improved markedly, but delivery on the ground lags well behind.

The most effective flood adaptation measure — floodplain protection — is also the most politically challenging. Floodplains are natural water storage zones. When they are developed, flood risk intensifies. Giving rivers more room to flood and avoiding new development in floodplains are therefore key to risk reduction. However, this principle often conflicts with local economic and political realities. Municipalities depend on population and economic growth, which lead to development pressures—particularly in urban and waterfront areas. Additionally, floodplains are attractive to development because they are flat and require minimal grading. Another challenge is posed by rebuilding in affected locations after a major flood, either because it is permitted or required by authorities, or because homeowners seek to restore the communities they lost and remain emotionally attached to. Yet unless new construction incorporates resilience measures, these properties remain vulnerable to future floods.

Development in flood-prone areas requires strict building and planning requirements. Examples such as Hamburg’s HafenCity demonstrate that building in exposed areas can be compatible with risk management, if designed appropriately. This includes elevated structures, floodable public spaces, protected utilities, and clear evacuation and emergency strategies. The key is not simply whether to build, but how—and

under what conditions. If exposure cannot be avoided, it must at least be made resilient. In this way, development and nature-based flood solutions can be combined rather than treated as mutually exclusive.

Large-scale structural measures remain an important component of flood risk management, but should be used selectively. Levees, flood walls, and retention basins can significantly reduce the frequency and severity of moderate flood events. However, they also have limitations. Extreme floods can exceed design levels and overtop defenses. When this occurs, the impact can be even more severe, particularly if communities behind defenses have become less aware of the underlying risk or have accumulated additional values due to new construction. Structural protection is most appropriate where high exposures already exist, and relocation and individual property level retrofitting is not feasible or economically viable. In particular, it should not be used to justify new development as this would lead to increasing loss potential in catastrophic flood events. Importantly, it must be complemented by residual risk awareness, emergency planning, and building-level resilience.

While infrastructure plays a crucial role, properties need to be adapted at an individual building level. Especially in areas facing pluvial flooding and residual risk, flood-adapted building should become standard for all new construction. Already in 1995 the LAWA (Bund/Länder-Arbeitsgemeinschaft Wasser, Germany Working Group on Water Issues) concluded that adapted construction and individual prevention at the property level is the most cost effective and sustainable risk reduction method. Resilient renovation includes:

- Raising critical systems (electricity, heating)
- Choosing more resilient heating systems (e.g. replacing oil systems that cause contamination and avoiding wood pellets that expand when wet and can damage structures)
- Sealing entry points
- Installing backflow prevention
- Using water-resistant materials for walls, insulation and floors.
- Designing spaces for easier recovery

Tools such as the Hochwasserpass, a building specific flood risk certificate, help translate technical risk information into actionable guidance for property owners. They increase awareness, support

¹² 10 Jahre Nationales Hochwasserschutzprogramm (NHWSP)

¹³ Untersuchungen zur Ermittlung der Wirkungen von präventiven Hochwasserschutzmaßnahmen im Rahmen des Nationalen Hochwasserschutzprogramms

preparedness, and empower homeowners to take preventive action. Adapted construction also includes:

- Avoiding living spaces in basements
- Elevated foundations that allow for waterflow below or at the design flood level
- Waterproof material for walls, insulation and floors (at least up to the design flood level)
- Low value occupancies in ground floor below the design flood level.

These measures do not fully eliminate risk, but they significantly reduce the damage and recovery time. This is particularly important given the long disruption associated with flood events.

The economic case for targeted flood adaptation is strong and could significantly reduce economic damage. Without action, river-flood losses in the EU and UK are projected to rise more than six-fold to nearly EUR48–50bn a year under 3°C of warming by 2100, with close to 500,000 people exposed each year.¹⁴ Coastal flooding compounds the picture: absent adaptation, EU+UK annual coastal-flood damage could climb from about EUR 1.4bn today to nearly EUR240bn by 2100, of which roughly 95% could be avoided by pairing moderate mitigation with raised defenses.¹⁵ Using flood-risk modelling and cost-benefit analysis, the European Commission Joint Research Council (JRC) finds that reducing flood peaks with retention (detention) areas is the most economically attractive strategy: it can cut projected 2100 river-flood losses from about EUR50bn to EUR 8bn a year – an ~84% fall in expected losses, roughly back to today's risk level – while returning about EUR4 for every EUR1 invested over 2020–2100.¹⁶ Strengthening dykes delivers a somewhat lower risk reduction (~70%) for comparable annual outlay, and building-level “damage-reduction” measures (the property-level retrofits discussed above) are found to be cost-efficient in most EU countries. These returns sit well within the wider evidence base: across all hazards, the Global Commission on Adaptation puts benefit-cost ratios for resilience investment at 2:1 to 10:1, and estimates that USD1.8trn invested globally across five adaptation priorities in 2020–2030 could generate USD7.1trn in net benefits.¹⁷

Adaptation is therefore preventive fiscal and

insurance policy. As insurance premiums reflect expected losses, public investment in flood defences and resilient construction is also an investment in insurance affordability and macro-financial stability. Despite these climate-induced trends, insurers have expanded flood coverage in many countries over the past 25 years. The protection gap for residential buildings in Germany shrank from 80% in 2003 to 43% in 2024, even as the building stock grew by three million structures. While the gap remains significant, it is not caused by a lack of insurance availability. Instead, it reflects limited awareness of flood risk among property owners and affordability challenges for those exposed to higher levels of risk.

A balanced approach to flood insurance: risk-based pricing and risk reduction. Coverage absorbs losses for individual homeowners after floods, but it also signals the presence of risk. Pricing insurance based on risk is therefore essential, as it provides incentives for adaptation and discourages development in areas highly prone to flood events. Risk-based pricing is an important element of fairness within the insurance pool, as it helps ensure that premiums more closely reflect the level of risk each policyholder contributes, thereby limiting cross-subsidization between lower- and higher-risk properties. However, without complementary measures, it can create affordability challenges for vulnerable households. Effective approaches combine:

- Risk-based pricing
- Ability for customers to choose limits and deductibles
- Support for risk reduction
- Incentives for resilient behavior
- Public-private partnerships with a focus on risk reduction

Public-private partnerships (PPPs) have long been used in Europe to address flood risk where private insurance markets alone cannot provide affordable and comprehensive coverage. Established systems such as Spain's Consorcio de Compensación de Seguros and France's Caisse Centrale de Réassurance (CCR) represent two mature models. Both operate on a risk-transfer mechanism in which risk is pooled nationally and backed by the state, ensuring broad coverage also in high-risk areas. In Spain, the Consorcio provides compensation for extraordinary risks—including floods—through a mandatory surcharge on insurance policies. France's

¹⁴ JRC Publications - Adapting to rising river flood risk in the EU under climate change

¹⁵ JRC Publications - Adapting to rising coastal flood risk in the EU under climate change

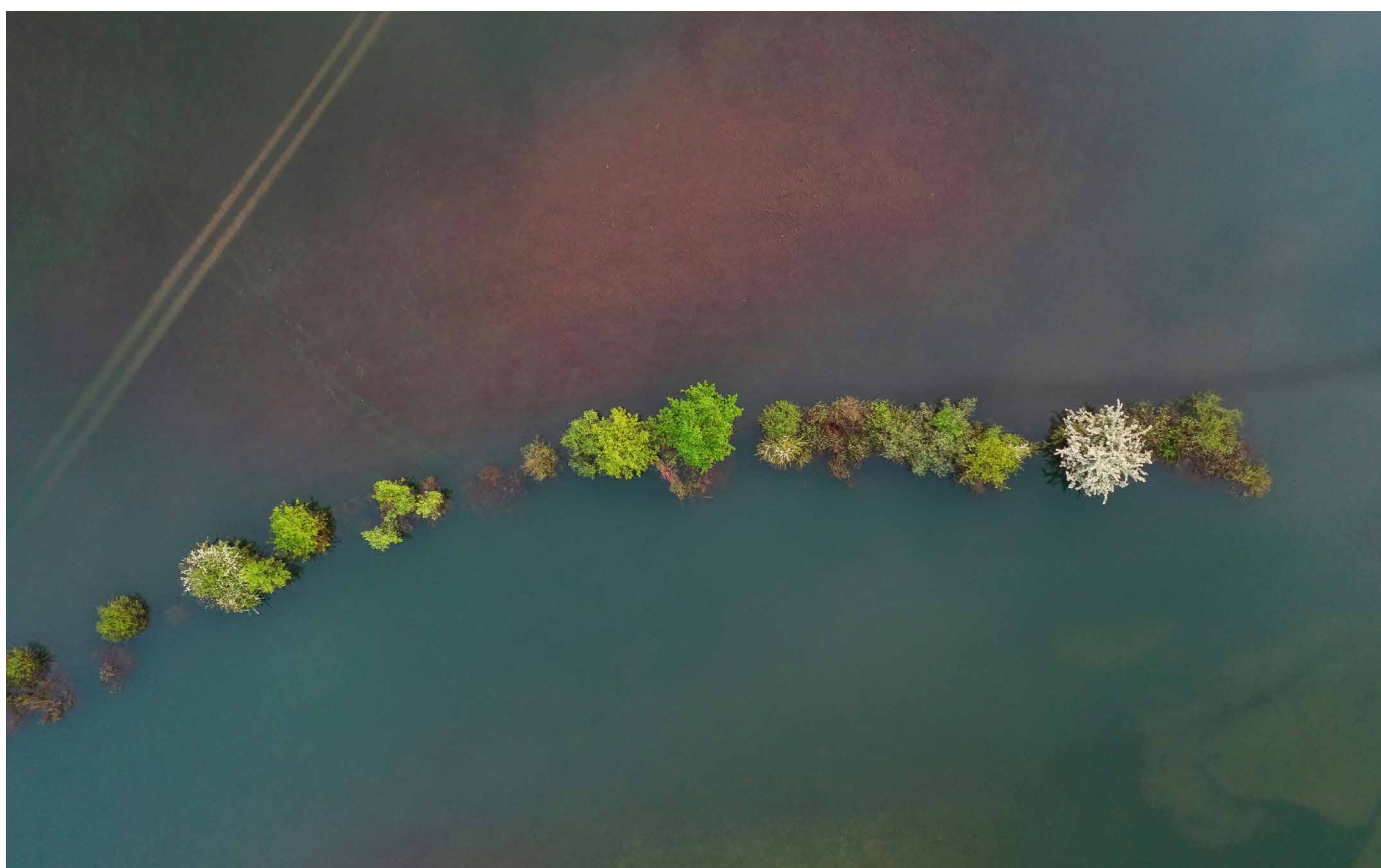
¹⁶ Cost-effective adaptation strategies to rising river flood risk in Europe | Nature Climate Change.

¹⁷ Adapt now: a global call for leadership on climate resilience - Global Center on Adaptation

CCR works through a Nat Cat scheme, which combines private insurers with a state backed reinsurer. The premiums increased significantly in 2025, after years of increasing risks and little focus on mitigation. Encouragingly, the Barnier Fund, which had been introduced dedicated to funding prevention, has recently returned to the political agenda. In the past it was chronically underfinanced and underused.

Ireland and Germany are exploring PPP approaches to resolve existing protection gaps and growing affordability pressure. In Ireland, a government-led working group is developing a long-term strategy to address access to flood insurance. The group has acknowledged that the problem “cannot be solved by insurance alone” and requires a public-private partnership approach. In Germany, the GDV is calling for a comprehensive natural hazard strategy that would

cover extreme risks by combining universal insurance coverage (with an opt-out solution), binding adaptation measures and state participation. For both countries, these exploratory frameworks highlight an important implication for resilience investment: PPPs are most effective when paired with risk reduction. If insurance pricing reflects underlying risk, then public funding and policy should prioritize measures such as floodplain restoration, resilient construction, and infrastructure upgrades that reduce expected losses over time. In this way, PPPs can move beyond simply redistributing risk toward actively lowering it, thereby supporting long-term affordability and reducing future economic losses.



Appendix

A1 – The impact of floods on capital formation

Floods bear on capital through two channels that pull a measured investment series in opposite directions.

They destroy or damage part of the existing physical stock (buildings, factories and equipment, transport and utility infrastructure, agricultural land and crops) and they raise the uncertainty under which firms commit to new projects. Both of these depress investment in the immediate aftermath. Post-flood reconstruction is recorded as gross investment, so a flood can leave a positive imprint on the same aggregate once rebuilding gets under way. The net effect on a flow measure of investment is therefore an empirical question, and the sign and timing are what the regressions below are designed to recover.

The wider literature frames why the answer matters. Hsiang and Jina (2014)¹⁸ find that the national-income losses from tropical cyclones are persistent and do not recover within two decades, pointing to a capital channel that compounds rather than heals. At the city scale, Kocornik-Mina et al. (2020)¹⁹ reach a more sanguine conclusion for floods specifically, documenting rapid recovery of local economic activity and little permanent relocation. The companion analysis in this paper places investment at the center of the heat transmission, in line with Casey et al. (2024)²⁰, who argue that standard climate-economy models understate damages to capital-intensive activity. The exercise here asks the parallel question for floods: whether European gross fixed capital formation carries a detectable, datable response to flood events.

The analysis uses a balanced country-year panel of 36 European countries²¹ over 2010–2024 (540 observations), with no missing values in the outcome. The dependent variable is the log of gross fixed capital formation for the total economy (the AMECO series labelled OIGT), measured in volume terms at 2020 constant prices. Because the series is real and expressed on a national-currency base, country fixed effects absorb level differences in currency base and price level without contaminating the estimates with inflation.

Flood exposure is constructed from Copernicus Emergency Management Service (CEMS) activations aggregated to the country-year level²². Two treatment forms are used. The first is a binary indicator equal to one when a country records any flood activation in a given year; this is the case in 47 of the 540 country-years, spread across 17 countries. The second, which is the severity measure carried into the headline specification, is the average across affected NUTS-2 regions of each region's maximum inundation depth in centimeters, entered as $\log(1 + \text{depth})$. This measure is a regional average of regional maxima rather than the single deepest point recorded nationally; among treated country-years it averages roughly 125 cm.

Specification

The baseline is a two-way fixed effects (TWFE) model with country and year fixed effects, estimated in levels of log OIGT (Eq. 1). Identification rests on the meteorological exogeneity of flood timing conditional on the fixed effects: year effects absorb common shocks (the pandemic, euro-area financial conditions), country effects absorb time-invariant exposure and structure, and the flood term picks up within-country, within-year deviations.

¹⁸ [The Causal Effect of Environmental Catastrophe on Long-Run Economic Growth: Evidence From 6,700 Cyclones | NBER](#)

¹⁹ [Flooded Cities - American Economic Association](#)

²⁰ [Understanding climate damages: Consumption versus investment - ScienceDirect](#)

²¹ Main results consider only 17 European countries, where flood events were recorded for the period 2015 – 2024.

²² We use flood depth data from ClimRad, a tool developed by Allianz Research and EarthianAI. CEMS Rapid Mapping is activated on request by authorised users (national civil protection authorities, the ERCC and others), so activations reflect both the severity of an event and the decision to request mapping. This means that coverage is not fully uniform across countries or constant over time, as activation practice and user uptake have evolved since the service became operational in 2012. We therefore treat the activation count as an indicator of significant, officially recognised flood events rather than a complete census of all flooding. We rely on the severity measure (inundation depth) rather than activation frequency alone for the headline specification.

$$\ln(\text{OIGT}_{it}) = \beta \cdot \text{Flood}_{it} + \alpha_i + \lambda_t + \varepsilon_{it} \quad (\text{Eq. 1})$$

Because flood damage propagates into investment with a delay the preferred specification is a distributed-lag extension that enters the flood term contemporaneously and at four annual lags (Eq. 2). A Wald test on the sum of the lag coefficients summarizing the cumulative multi-year response. Standard errors are clustered at the country level throughout (36 clusters in the full panel).

$$\ln(\text{OIGT}_{it}) = \sum_{k=0}^4 \beta_k \cdot \text{Flood}_{i,t-k} + \alpha_i + \lambda_t + \varepsilon_{it} \quad (\text{Eq. 2})$$

Here Flood denotes the log average maximum depth, $\log(1 + \text{depth})$. The contemporaneous-only model (Eq. 1) is reported for reference; the distributed-lag model (Eq. 2) is the one from which the headline estimate is drawn. Because the number of clusters is modest, the inference on the headline coefficient is cross-checked with a restricted wild cluster bootstrap (Rademacher weights, B = 1999).

Results

Table A1 reports the distributed-lag estimates for the depth specification, where the effect of a single flood is allowed to play out over the event year and the four subsequent years. The contemporaneous coefficient is small but significant at the 10% level. The estimated effect grows over the following two years and is largest at the second lag, where a one-log-point increase in average maximum inundation depth is associated with a reduction in capital formation of about 1.1% (-0.0108 , $p < 0.05$). This is the only lag individually significant at the 5% level and the coefficient the paper carries forward. By the fourth year after the event the sign turns positive, though far from significant, which is what one would expect if reconstruction investment re-enters the capital formation measure and partially offsets the earlier shortfall. The depth regressor is zero in non-flood years, so moving from no flood to an average-severity event ($\log(1 + 125) \approx 4.8$) implies a cumulative shortfall on the order of 5% of capital formation two years out.

Table A1. Distributed-lag TWFE estimates: log gross fixed capital formation (OIGT) regressed on flood-depth severity in the event year and the following four years (lags 0–4).

Log avg-max depth	
Flood depth (t)	-0.0077* (0.0042)
Flood depth (t-1)	-0.0088* (0.0047)
Flood depth (t-2)	-0.0108** (0.0045)
Flood depth (t-3)	-0.0067 (0.0050)
Flood depth (t-4)	+0.0042 (0.0095)
Sum of coefficients	-0.0298
Fixed effects	Country + Year
SE cluster (clusters)	Country (36)
Observations	396

Notes: Dependent variable is log OIGT (AMECO gross fixed capital formation, total economy, 2020 constant prices). The treatment is the log average maximum depth, $\log(1 + \text{depth})$, entered at t to t-4. The common-sample size (N = 396) reflects the loss of early years to the lag window. Standard errors clustered by country in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Source: Allianz Research

A2 –The impact of floods on household disposable income

Floods reach household income through three channels that operate at different speeds. The most immediate is direct asset destruction and damage which erodes both wealth and the income those assets generate. The second is labor-market disruption, as flooded workplaces cost days of work and, in the worst-hit areas, push some workers out of employment entirely. The third is business interruption, as damaged premises, markets and transport links suppress local activity and the wages and profits that flow from it. None of these is instantaneous in annual data: most damage is incurred and absorbed over the months following an event, so the income response is expected to build over one to two years rather than only in the year of the flood itself.

The analysis uses a balanced panel of 72 NUTS-1 regions across 15 European countries over 2013–2023 (792 region-year observations). The dependent variable is net disposable household income per inhabitant (ESA 2010, item B6N), drawn from the Eurostat regional accounts (nama_10r_2hhinc). Because B6N is published in current euros, the series is deflated to real 2013 euros using a cumulative price index built by chaining annual HICP headline inflation to a 2013 = 100 base.

Specification

The baseline model is a two-way fixed effects (TWFE) model with NUTS-1 region and year fixed effects (Eq. 1). Region effects absorb time-invariant structure (geography, industrial mix); year effects absorb common shocks, such as ECB policy and pandemic transfers. Conditional on these, flood timing is treated as quasi-random, since extreme-precipitation events are driven by meteorology rather than by income trajectories.

$$\ln(Y_{it}) = \beta \cdot \text{Flood}_{it} + \alpha_i + \lambda_t + \varepsilon_{it} \quad (\text{Eq. 3})$$

Because damage is absorbed over the months after an event, the retained specification is a distributed-lag version of Eq. 3 that enters log maximum depth at t , $t - 1$ and $t - 2$, tracing the dynamic profile rather than compressing it into a single contemporaneous coefficient. This is the same depth-in-lags form carried forward in the capital analysis, which keeps the severity measure and the dynamic structure consistent across outcomes. The identifying assumption is that, conditional on the fixed effects, the timing of floods is unrelated to prior income trajectories; a pre-trends check in growth rates supports this, with pre-flood coefficients indistinguishable from zero (Wald $p = 0.285$).

Results

The retained estimates are the distributed-lag depth specification, reported in Table A2. The contemporaneous effect is small and insignificant (-0.33% per log-point of maximum depth, $p = 0.111$), in line with the annual-data timing, then becomes significant at the first and second lags (-0.30% , $p = 0.031$; -0.37% , $p = 0.027$), summing to a three-year cumulative effect of about -1.0% per log-point of maximum depth. Because both sides of the regression are in logs, the coefficients are elasticities of real income with respect to flood depth: a flood twice as deep is associated with a cumulative real-income loss roughly 0.7% larger over three years ($-0.0100 \times \ln 2$).

Table A2. Retained specification : distributed-lag log maximum depth, real household income per inhabitant

Log max depth	Coefficient	p-value
Flood depth (t)	-0.0033	0.111
Flood depth (t-1)	-0.0030**	0.031
Flood depth (t-2)	-0.0037**	0.027

Notes: NUTS-1 + year TWFE; log maximum depth entered at t, t-1, t-2. Coefficients are elasticities of real income with respect to maximum depth. SE clustered at country (15 clusters). N = 648. * p < 0.10, ** p < 0.05, *** p < 0.01.

Source: Allianz Research

A3 – The construction and interpretation of flood-depth data

Flood depth gives the macro shock its physical scale. The variable is not a count of flood events and it is not a regional-average water level. It is constructed from satellite-mapped inundation footprints, linked to official event records, and then aggregated to the country-year or regional panel used in the regressions. This matters because a depth of 125 cm should be read as the characteristic depth of flooded footprints within an event-region, not as the depth of water across the whole country or region.

The underlying event dataset combines several layers of information (Table A3). The primary source is the JRC satellite-derived flood-depth map catalogue for Europe, which provides 20 m raster maps for 2015-2024 with pixel values expressed in centimetres. These maps are derived from systematic flood detection in the Sentinel-1 radar archive, with water depth reconstructed from the mapped flood extent and terrain elevation. Permanent and seasonal water bodies are flagged separately and excluded from the depth statistics. Copernicus EMS Rapid Mapping products are used to attach named event information, area of interest, flood extent and sensing-date metadata. EU-Hydro, geoBoundaries and Natural Earth are then used to attribute the footprint to the nearest river, the relevant administrative unit and the country.

The JRC maps are spatio-temporal clusters rather than named disaster events. Each relevant map was therefore matched to a Copernicus EMS Rapid Mapping activation by comparing the map date and the first-level administrative units intersected by the flood raster with activation records. Candidate matches were then checked visually in QGIS by overlaying the satellite flood footprint with the CEMS delineation products. This step avoids treating anonymous raster clusters as events before they are anchored to a documented flood episode.

For each matched map, the raster is processed into contiguous flooded footprints. Each footprint receives GIS attributes: planimetric area, country, administrative unit, nearest EU-Hydro river reach and zonal water-depth statistics. The matched CEMS vector product is then joined to the footprint, adding event-level information such as activation code, area of interest, flood type when available, sensing source and acquisition date. The final record therefore corresponds to a flood footprint within one event-region, rather than to an entire national event.

Table A3. Flood-depth dataset: source inputs and role in the construction

Input	Role in the construction
JRC satellite-derived flood-depth maps for Europe	Primary source of flood extents and water depths; 20 m rasters, with depth values expressed in centimetres.
Copernicus EMS Rapid Mapping vector products	Event delineations, areas of interest, maximum flood extent where available, sensing source and event metadata.
EU-Hydro river network database	Attribution of flooded footprints to the nearest river reach.
geoBoundaries and Natural Earth administrative boundaries	Country and first-level administrative-unit attribution, plus screening of flood locations.
QGIS manual verification	Visual overlay of satellite footprints and CEMS delineation products before the consolidated event table is retained.

Source: Allianz Research based on JRC, Copernicus EMS Rapid Mapping, EU-Hydro, geoBoundaries and Natural Earth

The depth statistics are deliberately conservative. First, each contiguous flood footprint is summarized by the median of its 20 m pixels. This reduces sensitivity to isolated extreme pixels near channels, in topographic lows or in areas affected by elevation-model artefacts. Second, the event-region median, mean and maximum are calculated across these footprint medians. The reported maximum is therefore the characteristic depth of the most severely inundated footprint, not the single deepest pixel in the event. This distinction is important because the deepest pixel can be much larger than the economically relevant flooded-area statistic.

The statistics describe flooded land only. A mean or maximum depth of 125 cm does not mean that the wider NUTS region, or the country, was under 1.25 metres of water. It means that, within the inundated footprints identified for that event-region, the representative footprint depth was around 1.25 metres. Areas outside the mapped footprint are dry by construction and do not enter the depth statistic.

The statistics are also not exposure-weighted. A footprint in a sparsely populated field and a footprint in a dense urban area receive the same weight when the event-region depth distribution is formed. Exposure is captured separately through the affected-population attribute and through the macroeconomic panel structure. The depth variable should therefore be interpreted as hazard intensity, not as damage, exposed capital or population-weighted risk.

A photograph showing a variety of hands of different skin tones stacked on top of each other, resting on a thick, textured tree branch. The background is a lush green forest with sunlight filtering through the leaves. The text 'Our team' is overlaid in the center, with 'Our' in white and 'team' in orange.

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Allianz Research encompasses Allianz Group Economic Research and the Economic Research department of Allianz Trade.

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- **Yellow card only: As the Middle East conflict de-escalates – despite temporary flare-ups – we expect only a mild slowdown in global growth in 2026 to +2.5%, followed by a rebound to +2.9% in 2027, broadly in line with our previous baseline outlook.**

AI is strongly propping up the global economy, offsetting the drag from the energy shock and the trade war. The energy shock is still working through balance sheets, with consumer purchasing power recovering only in Q4 2026 and firms' profitability still exposed. The US is set to grow at +2.1% in 2026 supported by energy exports, a savings rate at its lowest since 2008, AI investment (a third of growth) and fiscal support (7.3% deficit). The Eurozone (ex-Ireland) will grow by just +0.9% in 2026 but rebound to +1.2% in 2027, held back by higher energy dependence, minimal AI offset and subdued growth in Germany as bold reforms are needed on top of the fiscal stimulus. China will stay resilient at +4.7%, led by exports and high-tech manufacturing, but faces headwinds from weak domestic demand and US tariffs. Global trade of goods will avoid a recession, growing by +2.9% in volume terms in 2026 and +2.4% in 2027 as the US trade war reloads with Section 301 following the expiry of Section 122 from 24 July, raising the US effective tariff rate from 8% to 13%.

- **Has inflation been knocked out? The peak should already be behind us.** With oil flows normalizing and crude prices even falling back below pre-war levels, the energy shock should prove short-lived, limiting second-round effects on broader inflation and avoiding a repeat of the 2022 inflation surge. We still expect some volatility during US-Iran negotiations triggering temporary oil price spikes but overall we see them dropping to USD75/bbl and USD67/bbl by the end of 2026 and 2027. Gas prices are still higher compared to pre-war levels, but with EUR41/MWh expected at the end of 2026 and EUR32/MWh in 2027 they remain in the trading range of the past three years. This suggests a far more muted inflationary impact than in 2022, when prices surged by over 500% to above EUR300/MWh. Headline inflation should therefore reach central bank targets in 2027 in major economies. We expect the ECB to stay on hold at 2.25% (around neutral) after having already delivered one insurance hike in June, while the Fed will raise policy rates at least once this year to 4.0% amid persistent inflation, driven by strong AI-related investment demand, before a normalization back to 3.5% in H2 2027.

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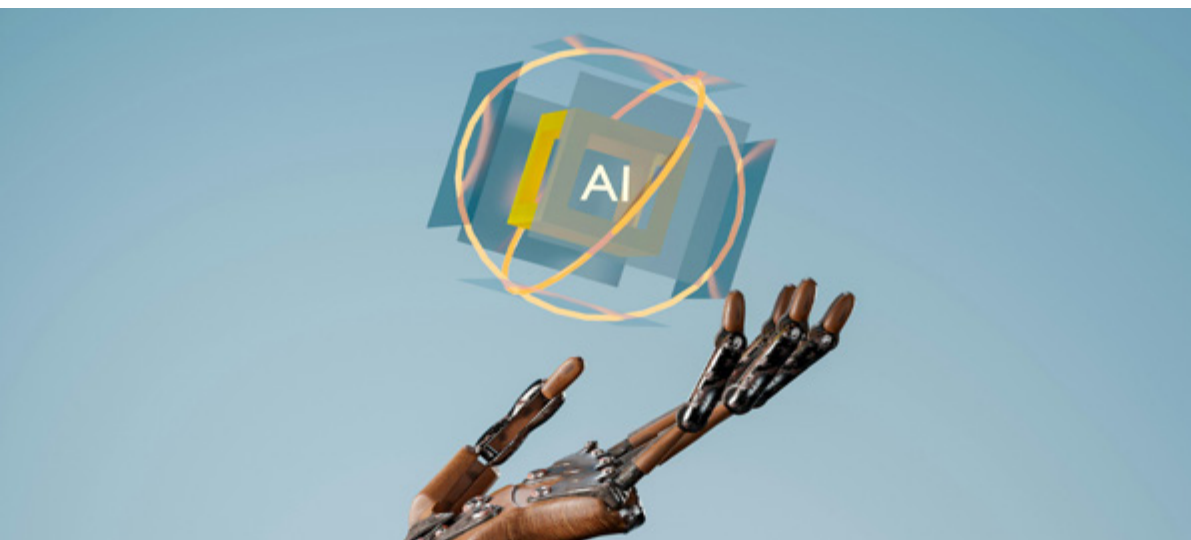
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• **The goalposts are shifting for corporates: Earnings held up in Q1 2026 but the delayed impact will hurt down the line.** Most European sectors posted positive EPS growth in Q1, while US earnings were buoyed by AI-driven tech. But a profitability squeeze might be building: Turnover growth was revised down for 12 out of 16 sectors, led by pharma (-2.0pps to 0%), utilities (-1.6pps), and motor vehicles (-1.3pps). Only electronics (+1.7pps to 11%), information & communication services (+0.7pps) and food & beverages (+0.4pp) bucked the trend. The auto sector remains most exposed via elevated net leverage meeting higher rates. Against this backdrop, we expect global insolvencies to increase by +4% in 2026 before plateauing in 2027.

• **Capital markets are playing the advantage, largely looking through near-term risks.** US rates should gradually decline to 4.35% by the end of 2026 and 4.1% in 2027 as inflation pressures and monetary tightening fade next year. German Bunds will hover around 3% amid ongoing supply pressures from quantitative tightening and a high fiscal deficit around 4% of GDP – above Italy and Spain. Equities remain supported by nominal growth and AI upside, although valuations leave little room for disappointment and relative value becomes more relevant. After strong year-to-date gains, further upside in 2026 looks limited with our full-year forecast of 13% total return for the S&P500 and 14% for the Eurostoxx, but in 2027 we see ongoing solid performance of 11% on both sides of the Atlantic. EM stocks are leading the charts (29% total return expected for 2026) driven by the strong weight of top South Korean and Taiwanese semiconductor companies in the index. Credit markets should continue to deliver attractive carry, even as ultra-tight spreads leave valuations stretched and skew risk to the downside. Rising issuance and falling interest coverage ratios are likely to drive a mild widening in spreads over the next two years – but elevated carry should still be enough to offset those valuation losses. Private markets remain a story of rising dispersion, with AI-related infrastructure and high-quality private credit outperforming more challenged segments.

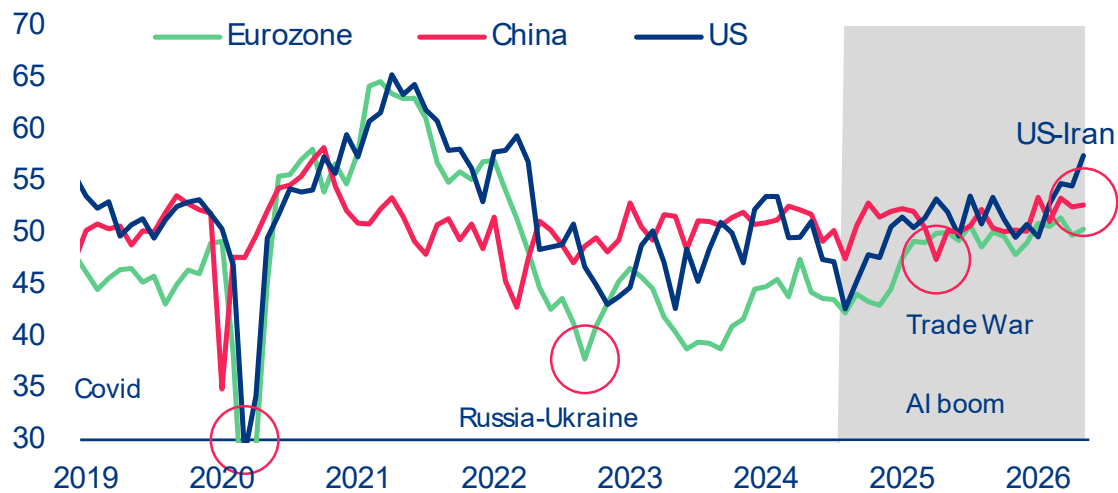
• **More shocks ahead? The number of risks ahead has certainly not decreased.** AI is doing the heavy lifting that geopolitics and fiscal policy cannot. But it is also concentrated, unevenly distributed and itself a source of downside risk if delivery disappoints as it would hit trade, investment and consumption. In addition, global policy uncertainty remains well above trend, with many risks ahead including a reloaded escalation of the trade war, US midterm elections, a subdued German fiscal push and polarization ahead of upcoming elections in key European countries. Climate risks remain omnipresent – the latest heatwave in Europe has passed, but risks of a severe El Nino year are still looming. Labor markets are still tight, but the medium-term outlook is more fragile. In several European economies, AI-driven layoffs could outweigh productivity gains, ultimately weighing on growth.



AI is propping up the world economy

With the de-escalation of the Middle East conflict, we expect only a temporary slowdown in global growth in 2026 before a rebound to +2.9% in 2027, broadly in line with our previous baseline outlook. The reopening of the Strait of Hormuz is unlikely to fully materialize before year-end, while the energy shock is still working its way through balance sheets, with consumer purchasing power only recovering in Q1 2027 as real wage growth returns to positive territory and firms' profitability remaining exposed to downside pressures. Energy prices are expected to gradually return to pre-war levels by end-2027, with Brent crude averaging USD83/bbl in 2026 before easing to USD70 in 2027, while European natural gas (Dutch TTF) averages EUR43/MWh in 2026 before falling to EUR34/MWh in 2027. However, past increases will continue to feed through supply chains, utility bills and rents over the coming quarters. Against this backdrop, global GDP growth is expected to slow to +2.5% in 2026, the weakest pace since the Covid-19 crisis, before rebounding to +2.9% in 2027, broadly in line with the 2023-2025 average. The slowdown (-0.5pp compared with 2025) remains managed despite the US-Iran war and the energy crisis, and is almost evenly split between advanced economies (-0.2pp) and emerging markets (-0.3pp). The AI boom, explaining around one-

third of US growth and driving close to two-thirds of Asian export growth, provides a genuine cushion to the global economy, but also increases concentration risks going forward. Overall, the US economy is expected to grow by +2.0-2.1% in 2026, broadly in line with potential growth. The Eurozone (excluding Ireland) is expected to grow by just +0.9% in 2026 before rebounding to +1.2% in 2027. China remains resilient, with GDP growth expected at +4.7%, driven by exports and high-tech manufacturing.

Figure 1: PMI new orders manufacturing index

Sources: LSEG Datastream, Allianz Research

Table 1: Real GDP growth forecasts, %

Growth (yearly %)	2023	2024	2025	2026f	2027f
Global	2.8	2.9	3.0	2.5	2.9
USA	2.9	2.8	2.1	2.1	2.0
Latin America	2.0	1.8	2.4	2.3	2.7
Brazil	3.3	3.0	2.6	2.0	2.1
UK	0.3	1.0	1.3	1.0	1.4
Eurozone	0.6	0.9	1.5	0.3	1.2
Eurozone excl. Ireland	0.7	0.9	1.1	0.9	1.2
Germany	-0.7	-0.5	0.4	0.7	1.2
France	1.9	1.4	0.9	0.5	1.1
Italy	1.0	0.6	0.7	0.6	0.7
Spain	2.5	3.5	2.8	2.2	1.7
Central and Eastern Europe	1.4	2.3	2.3	2.1	2.4
Poland	0.2	3.1	3.6	3.2	2.9
Russia	3.9	4.9	1.1	0.7	1.8
Türkiye	5.0	3.5	3.6	3.4	3.8
Asia-Pacific	4.3	3.9	4.4	4.3	4.0
China	5.4	5.0	5.0	4.7	4.4
Japan	0.7	-0.2	1.1	0.9	1.0
India	7.1	7.2	7.5	6.5	6.4
Middle East	2.1	2.4	3.2	-2.1	5.0
Saudi Arabia	0.5	2.6	4.5	2.7	5.3
Africa	2.7	3.5	4.2	4.0	4.0
South Africa	0.8	0.5	1.1	1.3	1.4

Sources: National, Allianz Research

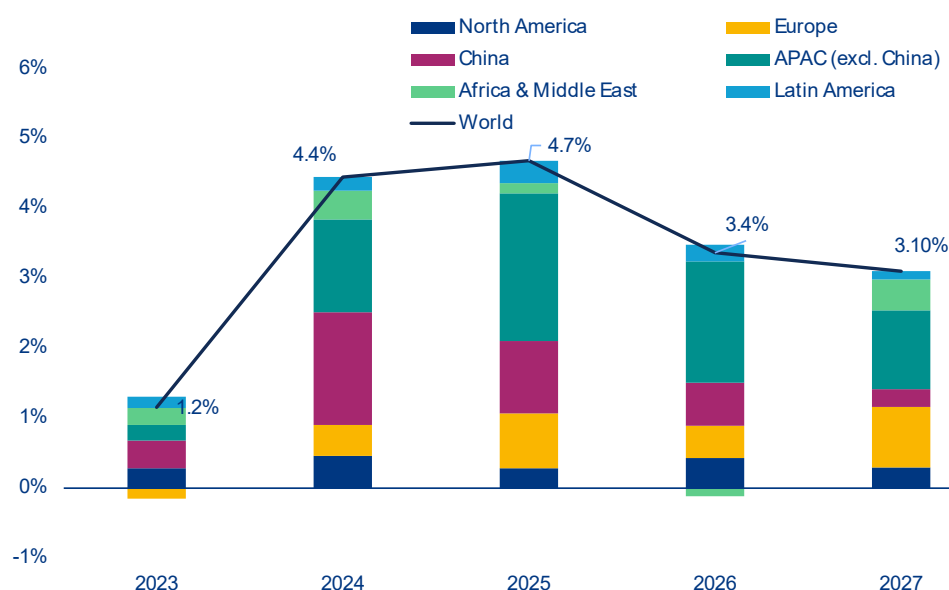
In 2026, 80% of global trade volume growth in goods and services is driven by Asia and the US (60% in 2027). Trade of goods in volume is projected to grow by +2.9% in 2026 and +2.4% in 2027. The closure of Strait of Hormuz from early March to mid-June 2026 contributed to a sharp contraction of global trade in Q2 2026 by blocking 4% of global trade. In parallel, the trade war initiated by the US in 2025 continues to cause losses to exporters. Starting 24 July, a new set-up built around Section 301 of the Department of Commerce will be established, bringing the US effective tariff rate back to 13%, what it was before the SCOTUS decision on the IEEPA tariffs (commonly known as “Liberation Day” tariffs). The pivot to Section 301 is expected to negatively affect Asian countries in particular, but exporters are adapting via rerouting or identifying new markets, reducing the impact.

A structural tailwind partly offsets these geopolitical headwinds: AI-linked supply chains. Surging demand for semiconductors, servers and data-center hardware is keeping both goods and services trade elevated through 2026, broadening the recovery beyond traditional manufacturing. This AI-driven investment cycle supports trade especially in North America and Asia, and cushions the cyclical losses from trade frictions. In 2027, this cycle of the AI capex investment is projected to slow after peaking in 2026 but it will continue to be the main driver of trade.

On the fiscal front, the energy shock hit at a moment when most countries were doing little to consolidate, given already weak growth and rising spending pressures – defence chief among them. Fiscal stances were broadly neutral at best, with governments balancing post-pandemic spending legacies against moderating growth outlooks and no major consolidation effort on the horizon. Learning from the 2022 energy crisis, most support measures implemented by both advanced and emerging economies were once again largely untargeted but remained contained, helped also by the more moderate scale of the energy-price spike relative to that earlier shock. As a result, we do not expect a significant impact on government budgets this year.

Shocks to be continued? The number of risks ahead has all but decreased. AI is doing the heavy lifting that geopolitics and fiscal policy cannot. But it is also concentrated, unevenly distributed and itself a source of downside risk if delivery disappoints as it would hit trade, investment and consumption. In addition, global policy uncertainty remains well above trend, with a reloaded trade war, US midterm elections, a subdued German fiscal push and polarization ahead of upcoming elections in key European countries all risks to watch. Climate risks remain omnipresent – the latest heat wave in Europe has passed but risks of a severe El Nino year are still looming.

Figure 2: Global trade of goods and services, volume and annual growth



Sources: Oxford Economics, Allianz Research

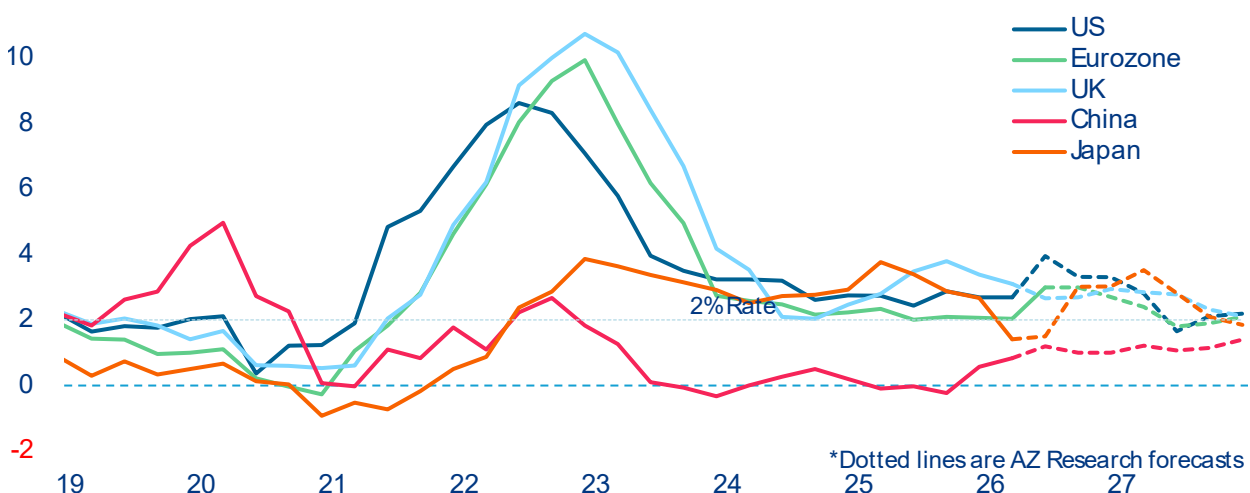


Peak inflation should be behind us

Inflation rates are past their peaks and should normalize towards central banks' targets by spring 2027, but core prices are expected to remain sticky in the US and the UK. The sharp decline in oil prices will pull down headline inflation rates across most markets, stretching well into 2027. In the US, after peaking at +4.0% y/y in Q2-26, we expect CPI inflation to decline and trough at +1.7% in Q2 2027, thanks to strong negative base effects of energy prices one year after the Iran shock. In

the Eurozone, we expect inflation to trough at +1.8% in Q2-2027, from a peak of +3.1% reached in Q2 2026. UK inflation should be a bit more steady as Ofgem utility price hikes in July and September should delay somewhat the pullback of energy inflation. We expect UK inflation to land at +1.9% by Q4 2027. Japanese inflation should reach +2% by Q3-2027. Against this backdrop, while lower energy inflation will pull down headline inflation rates decisively over the coming

Figure 3: Quarterly inflation, y/y, %



Sources: LSEG Datastream, Allianz Research

Table 2: Inflation forecasts, yearly, %

Inflation (yearly %)	2023	2024	2025	2026f	2027f
Global	6.1	4.7	3.8	5.1	3.3
USA	4.1	3.0	2.7	3.3	2.2
Latin America	14.8	16.6	7.6	4.0	3.3
Brazil	4.6	4.4	5.0	4.7	3.7
UK	7.4	2.5	3.4	2.9	2.5
Eurozone	5.5	2.4	2.1	2.7	2.1
Germany	6.0	2.3	2.2	2.6	2.2
France	4.9	2.0	0.9	1.7	1.3
Italy	5.7	1.0	1.5	2.7	2.1
Spain	3.6	2.8	2.7	3.1	2.1
Central and Eastern Europe	11.2	3.9	5.0	4.4	3.3
Poland	11.6	3.8	3.8	3.0	2.6
Russia	5.9	8.4	8.7	7.0	7.0
Türkiye	53.4	60.0	35.2	31.2	22.2
Asia-Pacific	3.1	2.2	1.3	2.3	2.1
China	0.2	0.2	0.1	1.0	1.2
Japan	3.3	2.7	3.2	2.4	2.2
India	5.7	5.0	2.2	4.5	4.0
Middle East	16.5	10.6	13.8	17.8	10.2
Saudi Arabia	2.5	1.5	2.0	2.5	2.4
Africa	17.0	15.1	8.7	10.2	11.4
South Africa	5.9	4.5	3.2	4.3	3.9

Sources: National, Allianz Research

quarters, core prices will accelerate as supply-chain disruptions from the conflict in the Middle East have pushed up the cost of key industrial inputs. Moreover, despite energy prices retreating, corporates have already passed on part of the rise in oil and gas prices onto their selling prices. Consumer prices typically exhibit downward rigidity so we would not expect lower energy prices, and a rapid easing of supply-chain disruptions, to weigh much on core prices. However, with GDP growth generally soft and below potential, and central banks cautious, we are not forecasting any meaningful “price-wage” loop second-round effects to materialize. After accelerating or remaining sticky through early 2027, core inflation should start to subside

US and Eurozone labor markets remain resilient at the headline level, but adjustment is increasingly visible beneath the surface. In the US, job creations have picked back up since the beginning of the year but hiring intentions are soft, pointing to a shaky outlook. In the

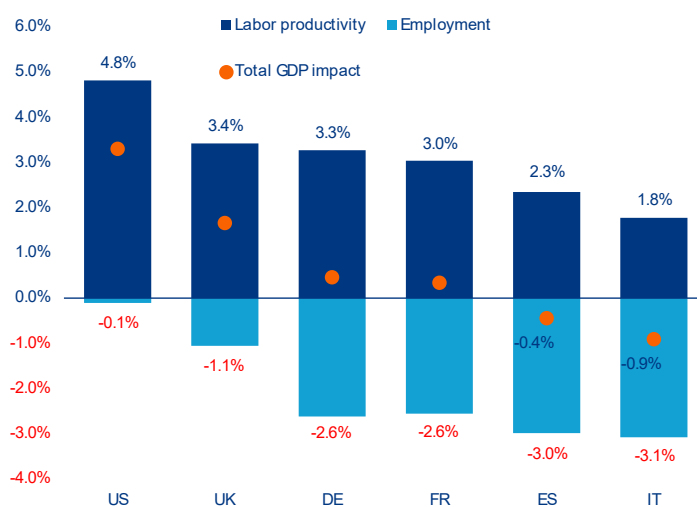
Eurozone, unemployment remains close to historical lows but labor demand has clearly normalized, with declining vacancies, softer hiring intentions and moderating wage growth. Crucially, this is not a repeat of 2022: Labor shortages have largely eased and firms are no longer hiring aggressively, reducing incentives for labor hoarding as growth weakens. Instead, adjustment is occurring primarily through slower hiring rather than layoffs, delaying but not eliminating the risk of rising unemployment. Risks are also becoming more sectoral. Persistent weakness in Eurozone manufacturing – driven by weak demand, elevated uncertainty and still-high energy costs – contrasts with more resilient services, pointing to growing labor-market divergence. Structurally weak productivity growth, despite accelerating AI adoption, adds further pressure on margins and hiring decisions. While a sharp deterioration remains unlikely, labor markets on both sides of the Atlantic are entering a phase of thinner buffers, with Europe’s exposure now industrial rather than energy-driven.

AI is set to transform labor markets over the next decade, with the adjustment likely to come mainly through workforce reallocation rather than mass layoffs. In the near term (one to three years), AI is expected to reshape up to one-quarter of jobs across major advanced economies through task reorganization, automation and augmentation. Exposure is greatest in the US (27.5% of jobs), followed by Germany (18.5%) and France (16.9%), while Italy remains the least exposed (10.0%). Most of those AI-exposed jobs will be reorganized around a human in the loop, while AI automates some roles and augments others. Early signals of the transformation are already emerging before headline employment weakens: slower hiring of junior talent, weaker wage growth and growing polarization between routine cognitive workers and those in AI-complementary roles. In the medium term, disruption is likely as jobs disappear faster than workers can be reskilled, meaning long-term productivity gains will come at the price of medium-term labor-market pain. Over the next decade, we expect a moderately negative employment effect across all major economies, ranging from -0.1% in the US (-170k jobs) to -3.1% in Italy (-740k jobs). The US as the hub of AI production should benefit more from new AI-related job creation and a stronger competitive position, while Europe remains more exposed because of its sectoral employment mix and slower adoption. However, shrinking working-age populations should cushion the blow across much of Europe as AI helps ease demographic labor

shortages: By 2036, the workforce is set to decline by 5.6mn workers in Germany, 4.2mn in Italy and 2.3mn in Spain. At the same time, AI should lift labor productivity across all major developed markets, with cumulative gains ranging from +1.8% in Italy to +4.8% in the US by 2036, further widening the transatlantic productivity and growth gap. Overall, AI should substantially lift potential growth, with inflationary effects in the short term but disinflationary effects over the longer term.

We expect the Fed to deliver a 25bps interest rate hike in September to re-affirm its anti-inflation credentials amid sticky core inflation. Although declining energy prices are now pulling down US inflation, the Fed is not out of the woods yet. To limit the pass-through of higher input costs onto consumer prices and wages, as well as to firmly anchor inflation expectations, we expect the Fed to hike interest rates by 25bps at the September meeting. The fact that the Fed has not hit its 2% target since 2021, as well as signs that spare capacity in the economy has decreased recently, will add further impetus to convince a majority of the FOMC voting members to act after increasingly delivering hawkish comments. Chairman K. Warsh stressed that the Fed has a lot of work to do on the inflation front during the June press conference, while other FOMC members have been increasingly vocal on inflation risks. After hiking in September, we expect the Fed to keep interest rates in the 3.75-4.00% bound for an extended period – through

Figure 4: Cumulative 10-year impact of AI on labor markets and growth



Sources: LSEG Datastream, Bloomberg, Allianz Research

mid-2027 – to keep a sufficiently restrictive stance and oversee a reduction of core price pressures. We expect two rate cuts in H2 2027, pushing the Fed interest rate towards their neutral level of 3.25-3.5%.

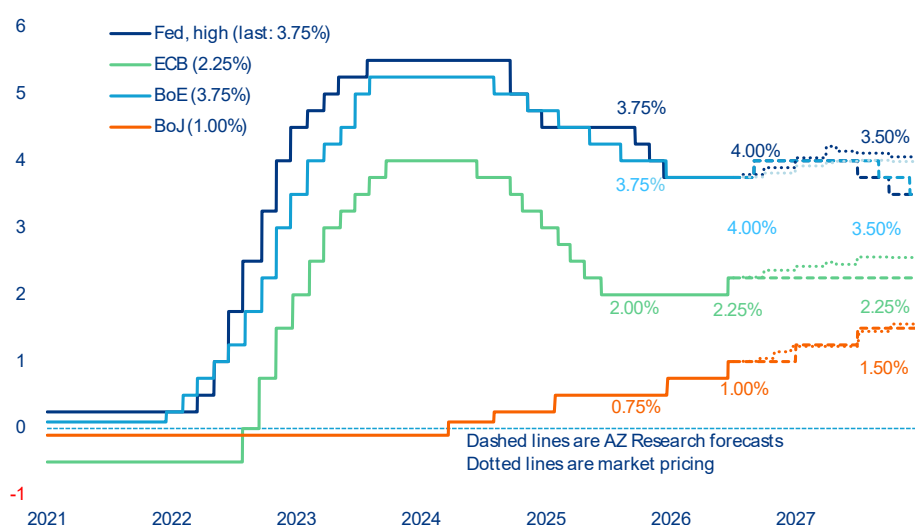
The ECB is done for now after one insurance hike and inflation undershooting expectations. The ECB raised its policy rate to 2.25% on 11 June, reacting to inflation surpassing 3% and fears of a prolonged energy-price shock. With the notable de-escalation in the Middle East and oil prices having dropped to pre-war levels thereafter, inflation risks have largely dissipated. At the same time, the economy is still weak and operating below potential. Taken together, we do not see any reason for the ECB to hike further from here; it is instead likely to keep rates at current levels going forward. The current level could be seen as sitting at the upper range of the self-proclaimed neutral rate range of 1.75% to 2.25%, which is a good spot to be both now and going forward. Inflation is expected to hit the target in 2027 but at the same time growth should rebound mildly, justifying this policy stance. At the same time, quantitative tightening is continuing at full run-off speed – keeping pressure on long-term rates and capital markets but not on the real economy.

The Bank of England (BoE) is expected to deliver a 25bps hike in September. Despite continuing sticky wage growth and core inflation predating the Iran crisis, the BoE has so far refrained from increasing interest rates. Admittedly, positive real rates (on both the short and long

end) even after the spike in energy prices have meant that the stance of UK monetary policy has been more restrictive than the Fed or the ECB. However, inflation expectations have started to drift up while wage growth has proved more sticky than expected recently. Business surveys point to UK corporates planning to increase selling prices sharply in the next couple of months. Meanwhile, GDP growth has proved resilient to the Iran shock. Against this backdrop, we expect the BoE to sound increasingly hawkish through the summer, and to deliver a 25bps rate hike in the September meeting, pushing the bank rate to 4%. Amid persistent core price pressures, we see the BoE on hold throughout September 2027, before cutting the bank rate to 3.5% by end-2027.

The Bank of Japan (BoJ) delivered a 25bps hike to 1% in June, as expected, supported by resilient activity despite disruptions from the Middle East conflict. However, inflation has remained subdued so far thanks to government subsidies and cap on energy prices even as underlying price pressures build. We expect inflation to pick up over the second half of the year as the pass-through from rising producer prices feeds into the broader economy, pushing 2026 inflation to +2.4%. Looking forward, we expect the BoJ to maintain its current pace, deliver the next hike in Q1 2027 and reach a final policy rate of 1.5% by end-2027.

Figure 5: Monetary policy key rates, %



Sources: LSEG Datastream, Bloomberg, Allianz Research



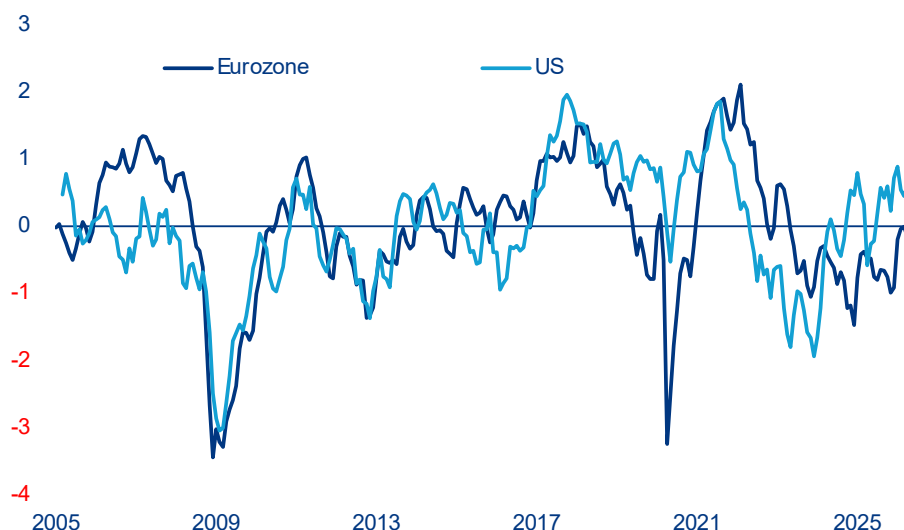
Developed markets: Energy shock manageable

US growth is expected to be soft, at +2.1% in 2026 and +2.0% in 2027, with AI capex spending and consumer spending shaping the outlook. The spike in inflation in Q2 2026 has pushed US households' real income growth into negative territory. Yet, they have mostly "looked through" the temporary dip in purchasing power and continued to spend on non-energy items by reducing their savings, although expenditures have weakened noticeably since year. The savings rate fell to historical lows in April-May. Elevated equity prices have contributed to resilient consumption spending and a lower savings rate. While a sharper pullback of consumer spending represents one of the most significant near-term risks, surveys point to continued softness but no significant slowdown in the next months. Lower inflation should support spending heading into 2027, but we expect households to rebuild savings buffers, which should keep the consumption pick-up moderate. US GDP growth in 2026 should continue to be powered by strong AI-related investment spending, but we expect this momentum to fade in 2027 as hyperscalers consolidate their balance sheets and supply-chain strains constrain capex expansion. Meanwhile, fiscal policy is expansionary in 2026 but will turn slightly restrictive in 2027, with the federal fiscal impulse turning from +0.7% GDP to -0.1%. Tax cuts have helped support consumer spending this year, while tariff refunds have benefitted

corporates. On trade, the fading momentum of the AI capex cycle should lead to a slowdown in imports growth in 2027.

Eurozone growth is set to slow in 2026 before recovering toward potential the following year.

Ireland's outsized volatility – despite accounting for only ~4% of Eurozone GDP – is distorting the aggregate picture: its Q1 2026 contraction, reflecting the reversal of 2025's pharma-export boom, pulled headline Eurozone GDP into negative territory, even as the bloc ex-Ireland still expanded. We expect Eurozone growth ex-Ireland at a decent +0.9% in 2026 (+0.3% including Ireland), picking up to +1.2% next year. The bloc has also borne the brunt of the Hormuz shock, facing weaker growth and higher inflation as a net energy importer. The shock hit industry hard during the second quarter via input costs and margin pressure, while services activity – more exposed to confidence effects from the Iran conflict – also suffered. Providing some offset, Germany's defense and infrastructure stimulus is delivering near-term tailwinds, while Southern Europe continues to benefit from NGEU inflows.

Figure 6: Capex intentions, net balance, z-score

Sources: Philadelphia Fed, European Commission, Allianz Research

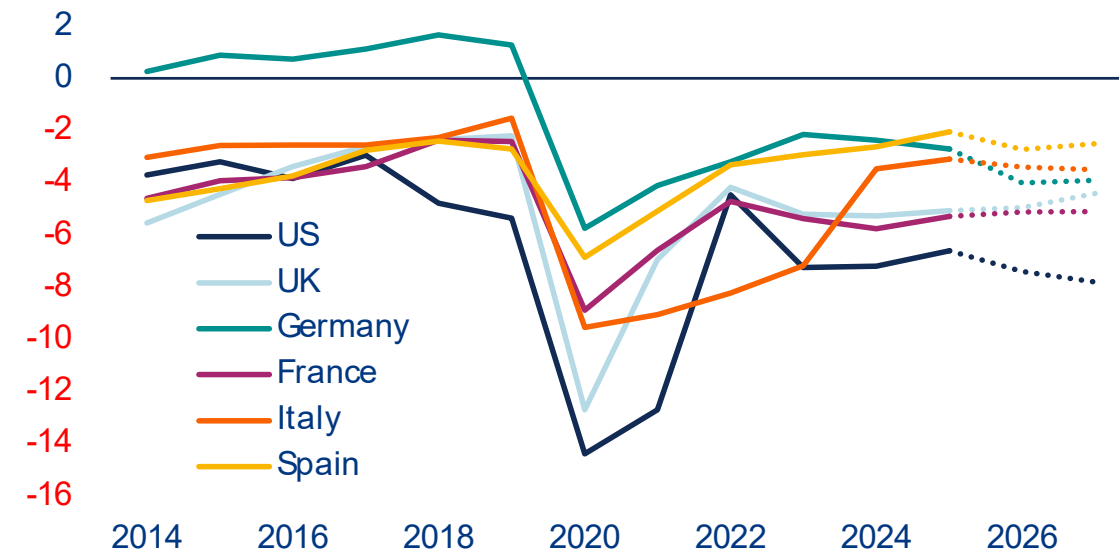
Germany's economic outlook remains fragile. GDP growth is forecast at +0.7% in 2026 amid geopolitical uncertainty, tariffs and elevated energy and input costs despite the fiscal bazooka, before improving to +1.2% in 2027 as trade tensions ease and fiscal support continues. Inflation is set to rise to +2.6% in 2026 due to increased energy and input prices, but moderate to +2.2% in 2027 as energy prices fall, though fiscal spending and wage growth will keep it above target. Fiscal policy is expansionary, with higher defense spending, a EUR500bn infrastructure fund and tax incentives supporting demand, while structural reforms and pension changes aim to address ageing-related pressures. The fiscal deficit will widen to around 4% and thereby surpass that of Italy and Spain in the next two years while and debt will rise to about 66% of GDP.

In France, GDP growth should be muted this year (+0.5%) amid tight fiscal policy and low consumer confidence. French economic momentum has taken a significant hit from the Iran crisis, with GDP growth expected to dip to a meagre +0.5% this year (after +0.9% in 2025). Despite elevated savings buffers, French households have slowed down their spending in the face of declining purchasing power. Furthermore, elevated input costs and a rise in mortgage rates are weighing on the housing market. Fiscal policy is tight, with the

government firmly committed to reining in the deficit, which limits the degree of support to households and corporates' balance sheets and aggregate demand. The industrial sector and exports have been a bright spot, benefiting from strong demand for capital goods and transportation equipment. We expect a pick-up of exports in 2027.

The energy shock has likely weighed on Italy's activity, given its structurally high dependence on imported energy. Still, Italy kicked off 2026 with a degree of quiet momentum, surprising again on growth in Q1 (+0.3% q/q), helped by a positive contribution from the Winter Olympics. Remaining NGEU inflows are helping buffer both structural and cyclical weaknesses, while broader growth remains constrained by geopolitical tensions and global trade shifts. Overall, we expect GDP to grow by +0.6% in 2026 and +0.7% next year. Inflation has already eased to +3.0% y/y in June from May's +3.2%, with the fuel price cap helping cushion the pass-through of higher energy prices. Fiscal metrics will only be mildly affected by these temporary, untargeted measures, but 2027 looks tougher: NGEU grants – which partly count as revenue – wind down just as election-year spending pressures build.

Figure 7: Government fiscal balance, % of GDP



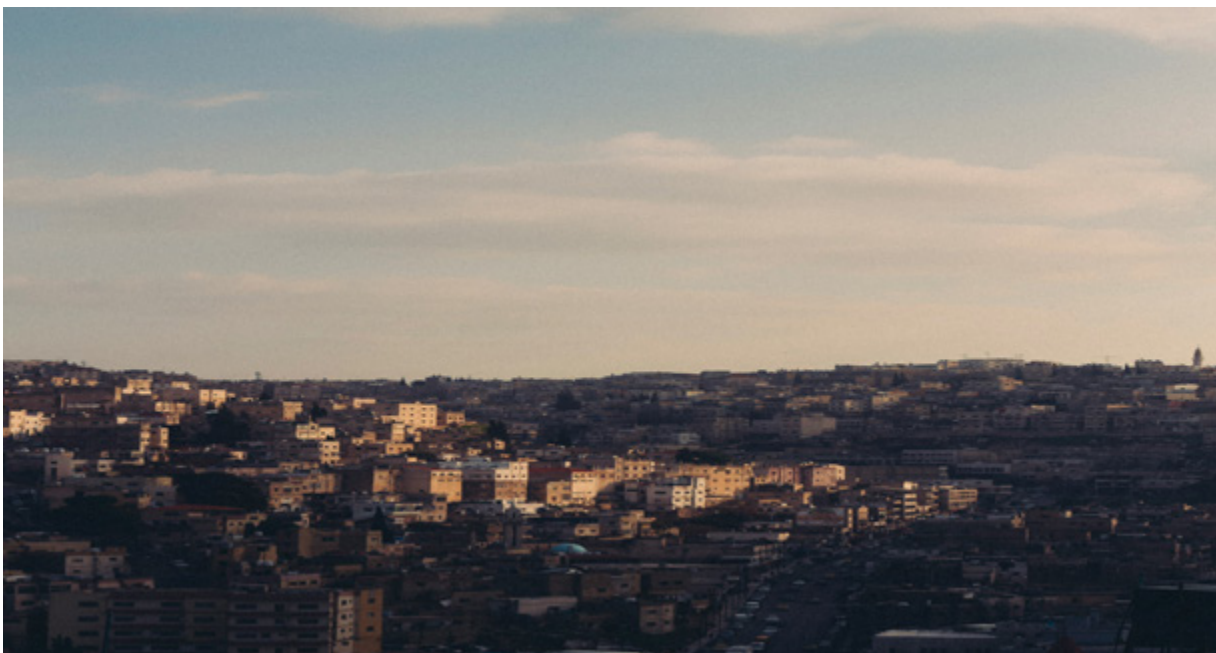
Sources : LSEG Datastream, Allianz Research

Spain continues to post above-Eurozone-average growth despite multiple challenges. GDP expanded by a still-solid +0.6% q/q in Q1, providing a strong carry-over into 2026; we expect economic growth to remain robust but to moderate from +2.8% in 2025 to +2.2% in 2026 and +1.7% in 2027. Crucially, Spain remains better insulated from the energy shock than most Eurozone peers: A structural shift toward renewables has meaningfully reduced exposure to volatile fossil fuel prices. In addition, the government has intervened with one of Europe's most ambitious crisis packages (a EUR5bn fiscal package), which has helped keep inflation anchored at +3.2% for three consecutive months. With that package set to expire just as a new round of measures is being negotiated, the durability of this inflation buffer is itself becoming a near-term risk to watch. Also, political fragility remains a key theme, with approaching elections compounding the challenge of sustaining reform and policy support without a stable parliamentary majority to pass it.

The UK economy has held up, thanks to supportive public policy, with GDP expected at +1.0% this year. While a concern for the BoE, elevated wage growth is supporting households' real income and consumption in the near term. Strong wage growth has been driven by the public sector, with hefty pay hikes decided by the government. As a result, we expect households'

spending to grow at about the same rate as last year (+1.0%). Meanwhile, the strong government push for infrastructure spending is providing some impetus for the sector, though private activity remains weak. The residential sector is expected to dip sharply this year, hit by the double whammy of elevated input costs and higher mortgage rates. British exports should continue to be soft, constrained by trade frictions with the EU, though key products should outperform (pharmaceuticals, transportation equipment).

The Japanese economy remained resilient despite headwinds from geopolitical tensions and we expect GDP growth at +0.9% this year. Growth remains supported by exports linked to global AI demand and robust consumer demand, on the back of strong wage gains secured over the spring and government subsidies cushioning household energy costs. Fiscal policy continues to be a tailwind, with the Takaichi administration maintaining its expansionary stance and supporting near-term demand.



Emerging markets: The Middle East takes a hit

Exposure to energy imports continues to determine the impact of the US-Iran war on emerging markets, while stronger fundamentals have supported resilient performance. The latest shock has served as a live test of increased EM central bank credibility, restarting a hiking cycle or pausing cuts. Southeast Asian countries, together with South Africa, the Czech Republic and Türkiye were first movers, hiking between 25 to 100bps amidst the energy price pressures, while only a handful of EMs continued with their easing cycles (Argentina, Brazil, Mexico and Hungary). Moving forward, the most exposed countries, mainly in Asia, are forecasted to continue hiking through H2 2026, before returning to easing in 2027, while LATAM and CEE countries will begin and continue their easing cycles through the second half of 2026.

The Chinese economy faced headwinds from the disruptions related to the Middle East conflict, but we maintain our +4.7% forecast for 2026. The disruptions ranged from higher oil prices to lower external demand, but strong Q1 results with growth at +5%, driven by strong exports and robust AI demand, provided enough cushion to offset them. The key risk to the economy remains weak domestic demand, rooted in structurally high household savings and low consumer confidence.

Following the normalization in the Middle East, we expect inflationary pressures to peak in Q3 before easing in Q4, though price pressure should remain relatively contained, with consumer inflation at +1% in 2026, well below the +2% upper target. In the short term, Chinese firms are likely to pass most of the rising PPI burden through to export prices rather than absorb it, modestly eroding their competitiveness.

While the Asia-Pacific region was among the most exposed to the Iran conflict, the growth outlook holds firm at +4.3% in 2026, powered by the AI boost. However, gains are far from evenly shared, with a group of winners emerging, including countries such as Taiwan, Singapore and South Korea, while countries such as the Philippines and Indonesia lag as domestic turmoil compounds exposure to the conflict. Despite recent developments towards normalization in the Gulf, we expect inflation to remain elevated in the near term as second-round effects from elevated fuel, energy and fertilizer prices keep weighing on the region. This leaves central banks at different stages of their tightening cycles: We now expect the Philippines and Indonesia, which started earliest, to conclude, while India and South Korea still have hikes ahead.

The Middle East remains the most negatively impacted region, with GDP expected at -2.1% in 2026 as the direct war impact and closure of the Strait weighs negatively on the economies. Growth is expected at +5.0% in 2027. In the Gulf, the drop in oil exports and collapse of tourism and trade will be the main contributors to negative growth, especially in small economies locked by the Strait, such as Bahrain, Kuwait and Qatar. Saudi Arabia, the UAE and Oman are expected to perform better, given that they could bypass the Strait.

In Africa, growth is expected to remain above +4% in 2026 and 2027, even though the war weighed negatively, with increased inflation pressures. The AI investment cycle is also contributing to a surge in exports of minerals and metals in several African economies, especially in the South and Central areas.

After a promising start to the year, Central and Eastern Europe has slowed as the Iran shock and weak German demand weighed on activity, triggering downward growth revisions across the region. We expect regional growth of +2.2% in 2026 before a modest pickup to +2.4% in 2027 as the Middle East disruption fades. Dispersion is wide: Poland remains the growth engine on sizeable EU-funds inflows, with the NBP looking through near-term price pressures, on hold at 3.75% and tightening only from end-2026. Romania is the outlier and the only economy set to contract as front-loaded fiscal consolidation bites and inflation accelerates toward +11% before favorable energy base effects pull it sharply lower from mid-year. Hungary and Slovenia, by contrast, have proved more resilient. The common channel is the consumer: Higher commodity prices and inflation fears have dented household confidence, threatening private consumption. Regional inflation should average +3.8% in 2026, normalizing toward +2.7% in 2027. The energy shock has left most central banks in wait-and-see mode. Czechia is the hawkish exception, having hiked to 3.75% on core pass-through, while Hungary has resumed its easing cycle on a firm forint and contained inflation. We forecast GDP growth of +3.4% in 2026 for Türkiye, reflecting resilient domestic demand despite restrictive financial conditions and elevated external uncertainty. Inflation is expected to remain stubbornly high at around 28-30% throughout 2026, delaying a more meaningful

easing cycle, while the Turkish lira continues to appreciate in real terms, weakening business competitiveness and household consumption.

Latin America is expected to benefit from the normalization of inflation, allowing central banks to continue or resume monetary easing through H2 2026 despite heightened domestic uncertainty. The region is relatively insulated from the direct impact of higher energy prices, with several major economies benefiting from commodity exports and diversified export bases, although weaker global demand and tighter financial conditions will continue to weigh on activity. Growth should therefore remain moderate rather than accelerate, with external demand subject to continued pressure due to potential trade agreement reviews and Asian demand. Weather-related risks associated with a potential return of El Niño could create localized disruptions to agricultural and electricity production as well as food prices, while political uncertainty has generally receded following the completion of the region's main electoral cycle, with Brazil standing out as the principal exception as fiscal concerns and the election outlook continue to weigh on investor sentiment. Despite the expected monetary easing, corporate insolvencies are forecast to increase rather than decelerate, particularly in Brazil and Colombia, as the cumulative effects of higher-for-longer interest rates, slowing economic growth and still-tight financing conditions continue to weigh on corporate balance sheets.



Corporates: Resilience on borrowed time

Corporate earnings across Europe and the US showed remarkable resilience through the first quarter of 2026.

Most sectors held their ground and defied expectations of an immediate derailment. In Europe, the overwhelming majority of sectors reported positive EPS growth in Q1, with no broad-based collapse in profitability that might have been feared given the scale of the geopolitical shock. This is partly because the full economic pass-through of the disruption takes time to materialize and partly because commodity prices, while elevated, did not spike to catastrophic levels. In the US, the story was shaped above all by the technology sector, which continued to provide the primary driver of earnings momentum: AI-related capex remained robust, sustaining corporate profitability across the US market even as other headwinds intensified. The macro backdrop hitting Europe and the Middle East disproportionately harder than other regions had not yet fully percolated through to corporate income statements by the end of Q1. This timing gap between the geopolitical event and corporate earnings impact is a recurring feature of major supply shocks: the transmission channels through energy prices, shipping costs and consumer sentiment tend to work with a lag, meaning that Q1 2026 results largely reflected business conditions shaped before the worst of the disruption took hold. The headline story for Q1 2026 is therefore one of contained damage: companies entered the crisis with resilient balance sheets and, for at least one quarter, managed to absorb or defer the costs of the conflict.

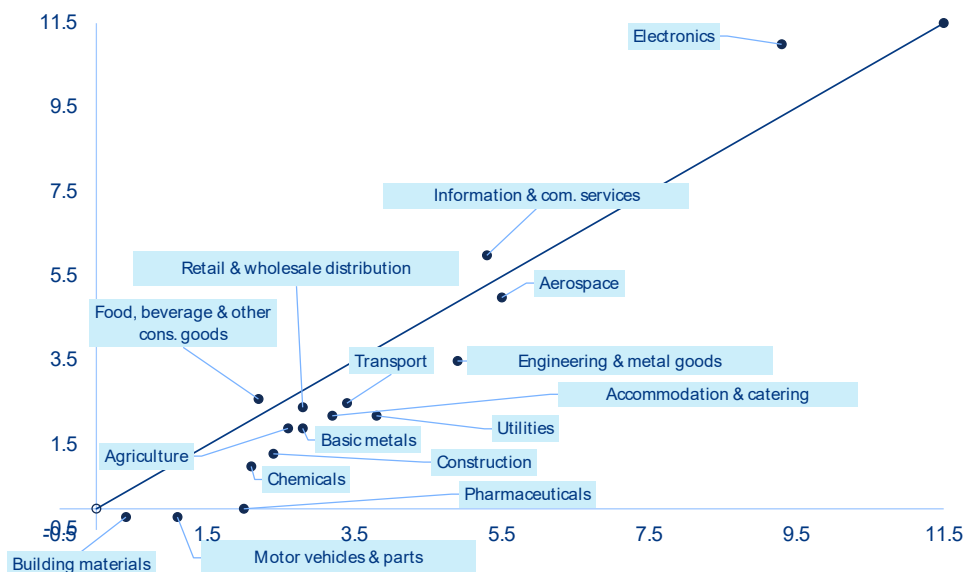
Yet the resilience of Q1 should not be mistaken for immunity: a profitability squeeze might be on the horizon. Several sectors face delayed but very real consequences from the Hormuz disruption that are likely to materialize as early as Q2 2026. Firms across Europe are already signaling rising cost pressures through elevated selling-price expectations, while the broader outlook for corporate profitability is on the downside everywhere, a warning clearly visible in forward-looking business surveys. These signals suggest that the cost shock is working its way through supply chains and input prices even as top-line revenues have, so far, held up. The automotive sector stands out as a particular vulnerability, especially in Europe: Already under structural pressure from Chinese competition, auto manufacturers and parts suppliers carry elevated net leverage and are therefore acutely exposed to the higher short-term interest rates that have accompanied the inflationary consequences of the war. With central banks holding rates above pre-war levels to contain the inflationary shock, the debt servicing burden for highly leveraged sectors is set to intensify in coming quarters. There are, however, some pockets of relief. Jet fuel prices have eased ahead of the summer season, providing a tailwind for transport and aviation sectors during the peak travel period. The net picture for the remainder of 2026 is one of a two-speed corporate world: sectors with high pricing power and limited cyclical exposure could weather the storm, those with the opposite profile face a reckoning that Q1 figures only deferred.

2026 turnover growth forecasts have been revised down for 12 out of 16 sectors. The deepest revisions have fallen on sectors most directly exposed to energy cost pass-through and demand weakness (see Figure 8). Pharmaceuticals suffered the steepest cut, with 2026 turnover growth revised down by 2.0pps to 0% (from +2.0% in March) as the sector contends with both higher input costs and softening consumer purchasing power. Utilities saw their growth estimate reduced by 1.6pps to +2.2%, reflecting the overhang of elevated energy prices on industrial and household demand. Motor vehicles and parts fell 1.3pps to -0.2%, tipping into negative territory and confirming the sector's particular vulnerability. Other notable downward revisions include construction (-1.1pp to +1.3%), chemicals (-1.1pp to +1.0%). Against this broadly downbeat picture, three sectors bucked the trend and received upward revisions. Electronics stands out most sharply, upgraded by 1.7pps to +11.0%, powered by the continued acceleration in AI hardware investment and data center build-out that has proved resilient to the macro environment. Information and communication services also moved higher, up 0.7pp to +6.0%, while food, beverages and other consumer goods received a modest upward revision of 0.4pp to +2.6%, supported by the normalization of agricultural commodity prices following the initial war-driven spike. Looking into 2027, the revisions picture is more balanced, with growth broadening across a wider range of sectors as the Hormuz disruption fades and supply chains normalize.

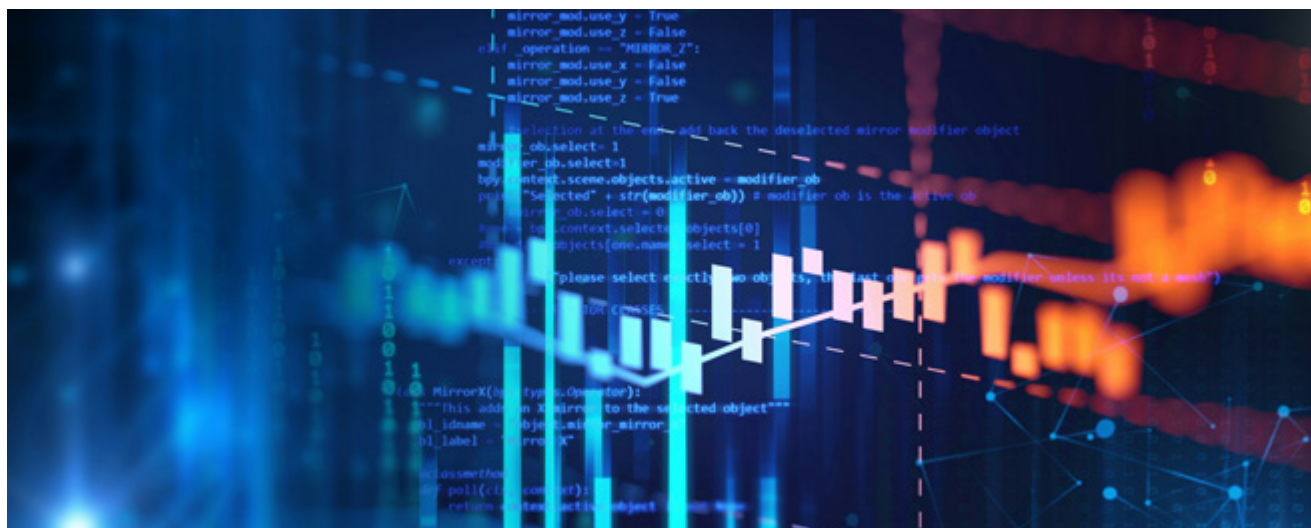
Business insolvencies are projected to increase by +4% globally in 2026 (-2pps vs. April forecasts). Notably, the war in Iran and its various spillovers have not triggered a wave of additional insolvencies. Q1 marked a 15th consecutive quarter of year-on-year increases, slightly

above +5% globally, following a strong +10% in Q4 2025. All regions contributed positively – North America (+19%), Asia (+3%, +11% excluding China) and Western Europe (+6%, now in its 20th consecutive quarter of y/y growth). The exception is Central and Eastern Europe, which posted its first decline (-10% y/y) after 14 quarters of increases. Over the last four quarters, the global trend stands at approximately +7%. Early Q2 data does not point to any exceptional rebound or surge relative to pre-war expectations. Most countries posted y/y declines in April and May, though fewer working days may partly explain this outcome. Year-to-date, a slim majority of countries are on the downside – most often from already elevated levels. Yet our headline indicator still points to +5% y/y globally, as the largest markets (US, Brazil, Japan, Germany, India, South Korea, France) see insolvencies stable at high levels or on a prolonged uptrend. China remains the major exception, with fewer cases than anticipated. We continue to expect a delayed profitability squeeze in the coming months, bringing greater differentiation across countries and sectors depending on their exposure to war-related spillovers – and conversely, their ability to benefit from global tailwinds (AI, defense). Overall, the global rise in insolvencies is expected to continue in 2026 (+4%) before stabilizing in 2027, marking five consecutive years of increases. North America would be the largest contributor, with US insolvencies projected to extend their rebound (+9% in 2026). Asia will see contrasting trends, with most countries prolonging their rise except Australia, Singapore and Hong Kong. Western Europe is expected to post a modest increase (+2%) in 2026, followed by a slight decline (-3%) in 2027, with most countries (nine out of 17) within the -4%/+4% range, indicating quasi-stable insolvency levels, often high by historical standards.

Figure 8: Turnover growth forecast change (%), June forecast vs March forecast)



Sources: Oxford Economics, Allianz Research



Capital markets: Cautiously optimistic

Capital markets navigated a volatile second quarter, dominated by the Iran conflict and its aftershocks. Oil prices, which had surged past USD120 a barrel on the closure of the Strait of Hormuz, tumbled back to pre-war levels following the mid-June MoU between Iran and US, while equities staged a sharp recovery: All major indices more than erased their war-related losses with the MSCI world even surpassing double digit performance year-to-date. Yet the relief has come with caveats. The Federal Reserve, under new chair Kevin Warsh, struck a more hawkish tone at its June meeting, flagging the risk of renewed hikes and dampening investor hopes for easy monetary policy. Markets have also chosen to look through the plethora of risks ahead, buoyed by resilient growth and the ongoing AI boom. We expect this constructive backdrop to persist but narrow. Rates should gradually decline as inflation pressures fade, equities remain underpinned by strong earnings and AI-driven demand despite stretched valuations. Credit should keep delivering solid carry even as issuance nudges spreads wider from historically tight levels. Within private markets, dispersion remains the defining theme, with AI-linked infrastructure and high-quality credit set to outperform.

The Fed and the ECB are trading different macro tapes, yet the rates market could end up in a surprisingly similar place. Both central banks are seen in a hawkish stance in the short-run to contain a common energy-price shock but their macro environment diverges. In the US, the monetary-policy outlook tightened against a backdrop of resilient activity and a starting disinflation trend. In the

Eurozone, the ECB is expected to further tighten into an economy operating below potential and where inflation recedes more slowly. This divergence is paradoxically equally supportive for duration, notably in the belly of the curve. We see markets shifting to a bull flattening in the US and a bull steepening in the Eurozone. The US curve will gradually move back toward a late-cycle profile. The front end might remain volatile due to fragile monetary policy expectations and residual inflation risk, the improving price dynamics and decreasing positive output gap argue for a lower long end of the curve with a 10y yield at 4.35% at year-end and 4.1% in 2027. At current yield levels, intermediate and long maturities offer an attractive mix of positive real returns and convexity. In the Eurozone, we expect the stagflation narrative to fade as price pressures ease and concerns over real activity become the main concern. We expect a bull steepening of Euro curve as markets will reassess the ECB's hawkish leeway on the short end of the curve while the long end remains anchored by a stable term premium keeping the 10y Bund yield around 3%. The belly is therefore the sweet spot to benefit from this curve move in a three to 12 months horizon.

Table 3: Capital market forecasts

EMU	Last*	Unit	2023	2024	2025	2026f	2027f
Government Debt							
ECB deposit rate	2.25	%	4.00	3.00	2.00	2.25	2.25
10y yield (Bunds)	2.94	%	2.03	2.36	2.85	2.90	3.00
10y EUR swap rate	3.00	%	2.48	2.39	2.93	2.90	3.00
20y EUR swap rate	3.23	%	2.51	2.39	3.23	3.00	3.10
Italy 10y sovereign spread	79	bps	168	117	65	80	80
France 10y sovereign spread	79	bps	53	83	71	90	60
Spain 10y sovereign spread	48	bps	97	70	44	40	40
Corporate Debt							
Investment grade credit spreads	76	bps	135	101	78	80	90
High-yield credit spreads	262	bps	395	311	270	280	315
High-yield credit spreads ex-CCC	216	bps	333	239	219	230	250
Equity							
Eurostoxx (total return p.a.)	14 ytd	%	19	10	25	14	11

US	Last*	Unit	2023	2024	2025	2026f	2027f
Government Debt							
Fed Funds rate (high)	3.75	%	5.50	4.50	3.75	4.00	3.50
10y yield (Treasury)	4.47	%	3.87	4.57	4.16	4.35	4.10
Corporate Debt							
Investment grade credit spreads	75	bps	104	82	79	80	90
High-yield credit spreads	272	bps	334	292	281	290	325
Equity							
S&P 500 (total return p.a.)	11 ytd	%	26	25	18	13	11

UK	Last*	Unit	2023	2024	2025	2026f	2027f
Government Debt							
BoE rate	3.75	%	5.25	4.75	3.75	4.00	3.50
10y yield sovereign (Gilt)	4.80	%	3.54	4.57	4.47	4.80	4.60
Corporate Debt							
Investment grade credit spreads	80	bps	134	91	80	90	100
High-yield credit spreads	388	bps	515	364	420	430	450
Equity							
FTSE 100 (total return p.a.)	9 ytd	%	8	10	26	9	8

Emerging Markets	Last*	Unit	2023	2024	2025	2026f	2027f
Government Debt							
Hard currency spread (vs USD)	238	bps	384	325	253	245	270
Hard currency spread ex-CCC (vs USD)	165	bps	215	202	163	170	185
Local currency yield	6.09	%	6.19	6.39	5.87	6.20	6.10
Equity							
MSCI EM (total return p.a. in USD)	24 ytd	%	10	8	34	29	12

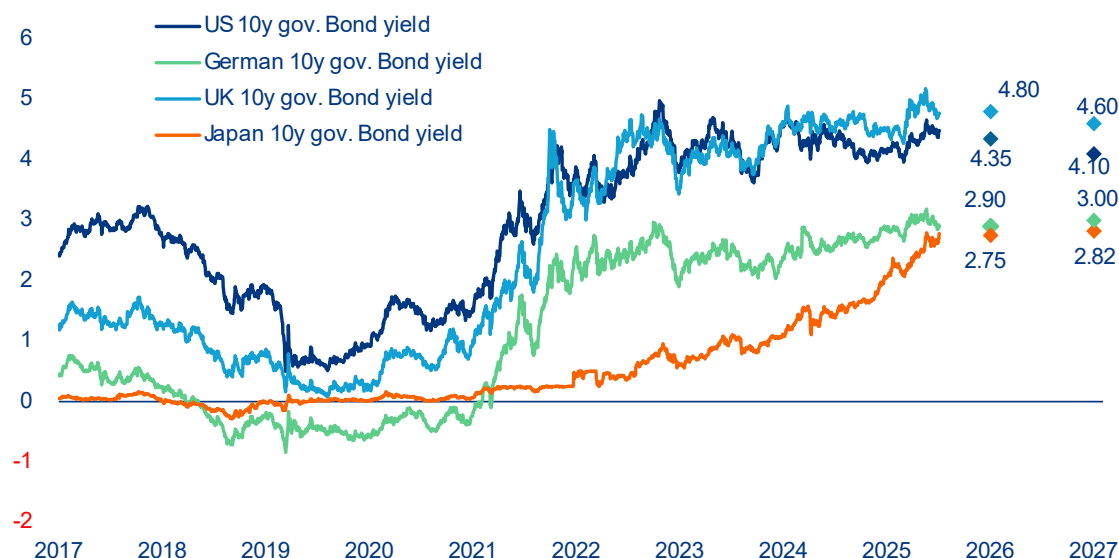
FX & Commodities	Last*		2023	2024	2025	2026f	2027f
EUR USD	1.14	\$ per €	1.10	1.04	1.17	1.15	1.15
Oil (Brent)	72	\$ per bl	78	75	61	75	67
Natural gas (Dutch TTF)	44	€ per MWh	32	49	28	41	32

Sources: LSEG Datastream, Bloomberg, Allianz Investment Management

Notes: Year end figures

* As of 6Jul 2026

Source: LSEG Datastream, Allianz Research

Figure 9: 10y government bond yields including forecasts, %

Sources: LSEG Datastream, Allianz Research

Among Eurozone sovereign bonds, valuations are broadly fair. The special support for Italian BTPs from NGEU flows is fading, while French OATs are likely to trade temporarily wide to Italy as political risk premia are repriced ahead of the presidential election. Our central scenario remains a centrist victory; if that holds, OAT spreads should re-compress and may again move below Italy once the political risk premium normalizes. German Bunds will maintain their safety premium versus the EUR swap curve. This should keep front-to-belly swap spreads positive. The long end stays capped by term premium and a weak duration demand. Overall, German swap spreads should stay tight and broadly range-bound.

The EUR/USD exchange rate traded within a range of 1.14–1.18 during the last quarter, broadly the same range that has prevailed over the past year. Movements within this range continue to be driven by a combination of opposing factors. On the positive side for the euro, persistent policy uncertainty in the US has provided support. On the negative side, geopolitical risks such as the conflict with Iran, although currently fading, have weighed on the currency. In addition, the AI investment theme remains heavily US-centric and continues to support the US dollar. More recently, lower-than-expected inflation figures in the Eurozone have provided additional support for the euro, bringing EUR/USD back towards the

lower end of its trading range around 1.14. Overall, we view the current exchange rate as being close to fair value and therefore expect broadly flat performance going forward.

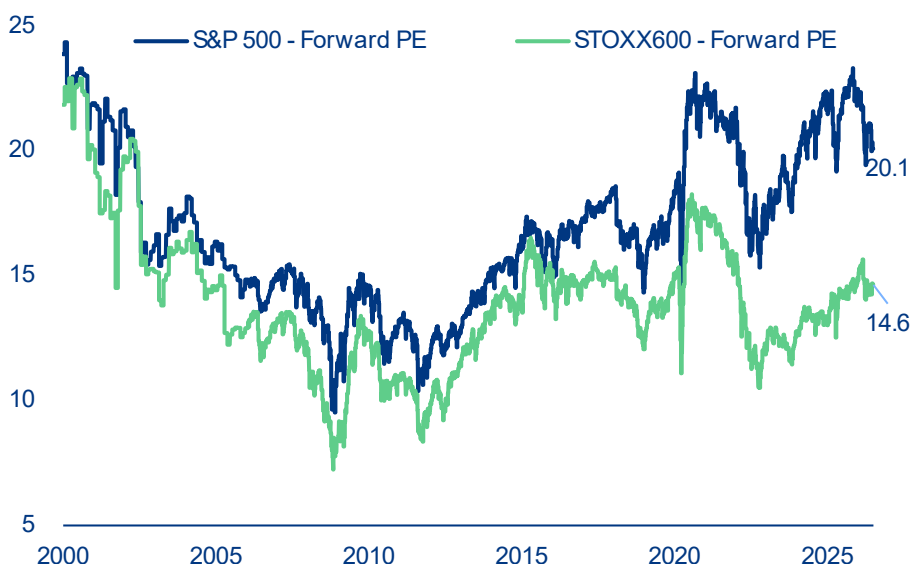
Global equity markets are entering Q3 2026 on solid footing. The S&P 500 has risen by +11% year-to-date, and the Eurostoxx has risen by +14% (total return). Markets have also shown welcome resilience to the conflict in Iran. The Eurostoxx has now recovered its war-related losses and is trading at a level that is 1.5% higher than before the conflict began, even though the situation remains prone to flare-ups following the memorandum to end the war that was signed in mid-June. This resilience supports a positive medium-term outlook for equities, although the distribution of near-term outcomes still carries a downside bias related to the durability of the ceasefire. A confirmed, lasting stand-down would further bolster European equities, given the region's sensitivity to energy prices and geopolitical risk premia. However, renewed strikes or ceasefire violations, as witnessed in the aftermath of the signing, would curtail this potential upside before it translates into earnings. Valuations remain split. The S&P 500 trades at 20.4x forward earnings, compared to 14.7x for the STOXX 600. This difference is largely justified by expected earnings growth: 29.8% for the S&P 500 and 13.5% for

the STOXX 600. This leaves further upside for US equities capped at the level of actual earnings, while Europe's relative discount continues to offer a cushion rather than a catalyst. There is room for either mean reversion higher should the ceasefire hold or compression of the price-to-earnings ratio should it not. Beyond absorbing equity market liquidity, IPOs can create additional selling pressure as venture capital investors receive listed shares that no longer fit their mandates and may reduce positions after the lock-up period. This could weigh on AI stocks and challenge market sentiment, particularly as current valuations appear to rely more on confidence in future growth than on present-day financial metrics. Looking ahead, we forecast total returns of 13% for the S&P 500 in 2026 (easing to 11% in 2027) and 14% for the Euro Stoxx (easing to 11%), with a selective approach favoring quality earnings over growth optionality being advisable until the geopolitical landscape stabilizes further.

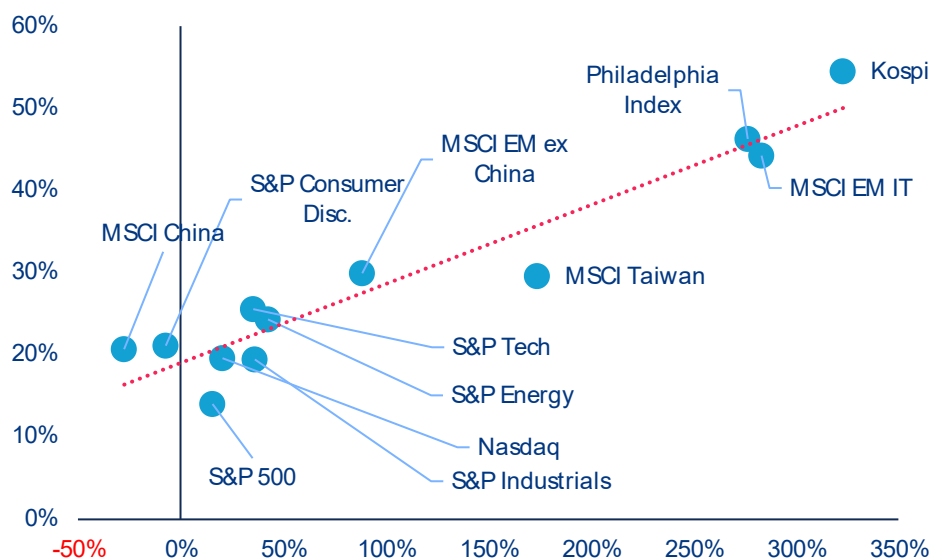
Semiconductors remain the dominant driver of both returns and volatility across global equity markets, and the cycle is not yet exhausted. The Kospi and the Philadelphia Semiconductor Index sit at the top of the

return-risk frontier with annualized returns above 250% – substantially outpacing broader indices like the S&P 500 and Nasdaq, which cluster in the 10-30% return range. Our cyclical analysis of South Korean semiconductor export prices – one of the most reliable leading indicators of the global semi cycle – shows the cyclical component had not yet peaked as of late June 2026, with the reading spiking sharply to its highest level since the early 2000s. Historical episodes show the cycle typically overshoots before reversing, and on that basis the trade still has meaningful room to run. This matters for index-level positioning: Given how heavily semi-exposed markets like South Korea and Taiwan have contributed to recent return dispersion, the durability of this cycle is a first-order input into both regional allocation and sector-level conviction. The risk, as ever, is that a confidence shock in AI-linked capex spending – the primary demand driver behind the current upcycle – could turn the cycle faster than the price signal alone would suggest.

Figure 10: Forward price-earnings ratio for US and European equities



Sources: LSEG Datastream, Allianz Research

Figure 11: Risk-return analysis of selected equity indices, 2026 ytd

Sources: LSEG Datastream, Allianz Research

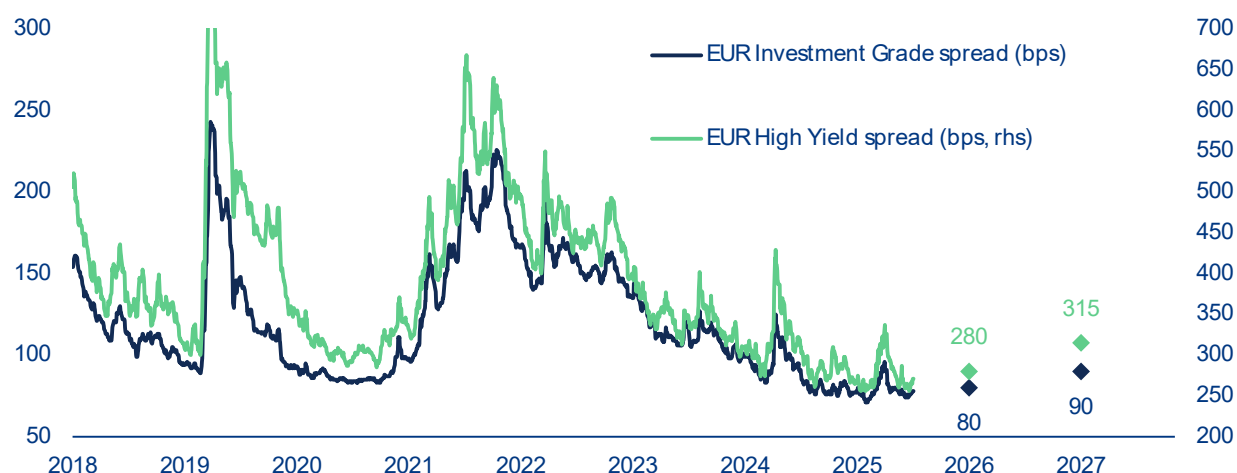
Notes: Annualized returns (x-axis) vs annualized volatility (y-axis)

Corporate credit markets proved resilient through the first half of 2026 despite private debt concerns, the software selloff and the Iran shock. Investment-grade spreads settled around 76–77bps in both Europe and the US, slightly tighter than at the start of the year. Spreads are tight but elevated all-in yields still offer a cushion to total return. With steady rates and growth, faded tail risk and supportive technicals, carry – the income earned from holding credit – was and will continue to be the main source of return. That said, today's tightness is largely demand-driven and therefore less durable: yield-seeking buyers are price-sensitive and could pull back quickly if the rates outlook shifts.

Looking forward, in Europe, spreads should hold around current tight levels – near 80bps for investment grade and 280bps for high yield – through 2026, supported by a mild macro backdrop and steady demand for yield against limited net supply. In 2027 we expect a moderate widening to 90bps for IG and 315bps for HY, anchored by fundamentals rather than a sharp correction: credit metrics are slipping slightly below average and the net upgrade trend is losing momentum. US investment-grade follows a similar path, while high-yield spreads are expected to drift toward 290bps, wider than Euro high yield due to lower rating quality. The ratings drift is slightly weaker in the US credit market with downgrades

prevailing, but the downgrade trend has started to ease. Fundamentals – especially leverage ratios, lifted by the growing share of AI-related financing – warrant a closer look.

EM debt benefits from strong fundamental background and constructive sentiment. However, valuations are stretched for hard currency, while local currency tends to underperform under Fed tightening cycles. The structural growth advantage of emerging markets remains intact, with EM growth continuing to outpace that of advanced economies by a meaningful margin. This growth differential is expected to widen further, supporting sovereign creditworthiness and reducing default risk, thereby providing a solid fundamental anchor. Demand dynamics have been exceptionally strong, driven by sustained inflows into the asset class and increasing participation from crossover investors. As a result, EM sovereign spreads remain at multi-year tight levels, reflecting both robust fundamentals and supportive technicals. Also, spreads offer no premium relative to US corporates. At the same time, spread volatility remains low and does not fully capture prevailing geopolitical uncertainties. From the current levels, spreads are expected to widen modestly to around 270bps by 2027. With hard currency spreads offering no additional pick-up, EM local currency assets provide relatively better

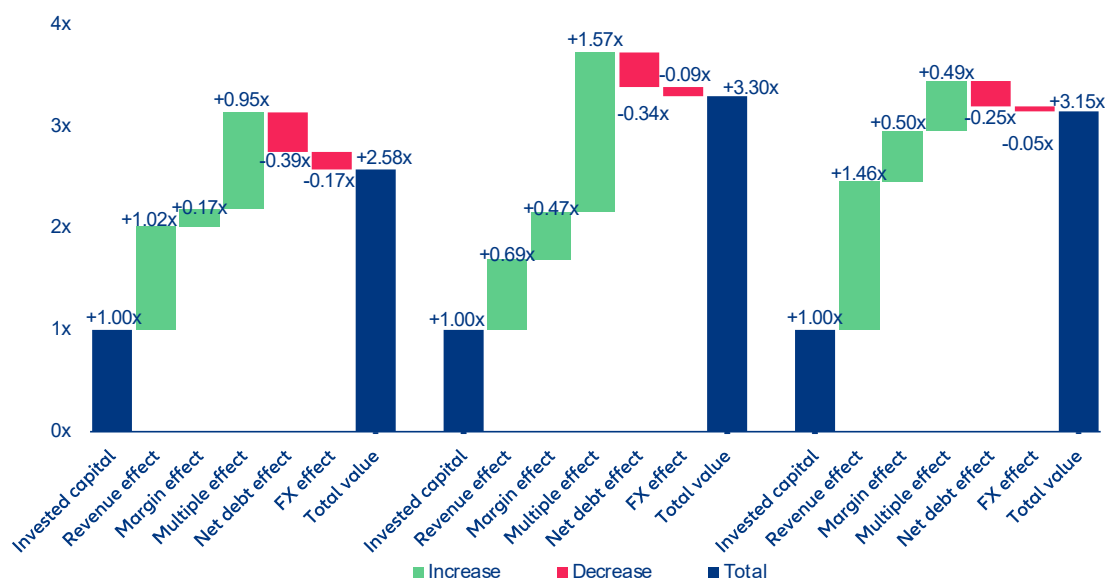
Figure 12: European corporate credit spreads, bp

Sources: LSEG Datastream, Allianz Research

value at current yield levels. Considering their correlation with US Treasury yields and ongoing EM central bank cycles, LC yields are expected to move broadly sideways. Even though the valuation looks better for EM LC, it has historically underperformed during Fed tightening cycles, which warrants a neutral view in the current environment.

Across private markets, a pair of forces is doing most of the sorting: interest rates that stay relatively high and AI defining the line between winners and losers. The same dichotomy bears on each asset class differently. Private equity is stuck at the exit, with a record backlog of unsold companies and fundraising starved to match. Private debt still earns its elevated coupon but stress is gathering in the weakest, AI-exposed borrowers. Infrastructure is the clear standout, carried by the data-center build out at the heart of the AI supercycle. Real estate is recovering, if unevenly, as cheaper financing slowly filters through, but AI keeps a lid on the office recovery. The common thread is dispersion, capital is rewarding contracted, well-covered, AI-aligned assets and shunning the rest, and the gap between the two is the widest in years. But that dispersion is as much an opportunity as a warning. Thematically aligned pockets are still expected to perform through the cycle (i.e. defense, AI etc.), but generalist strategies might see their returns compress.

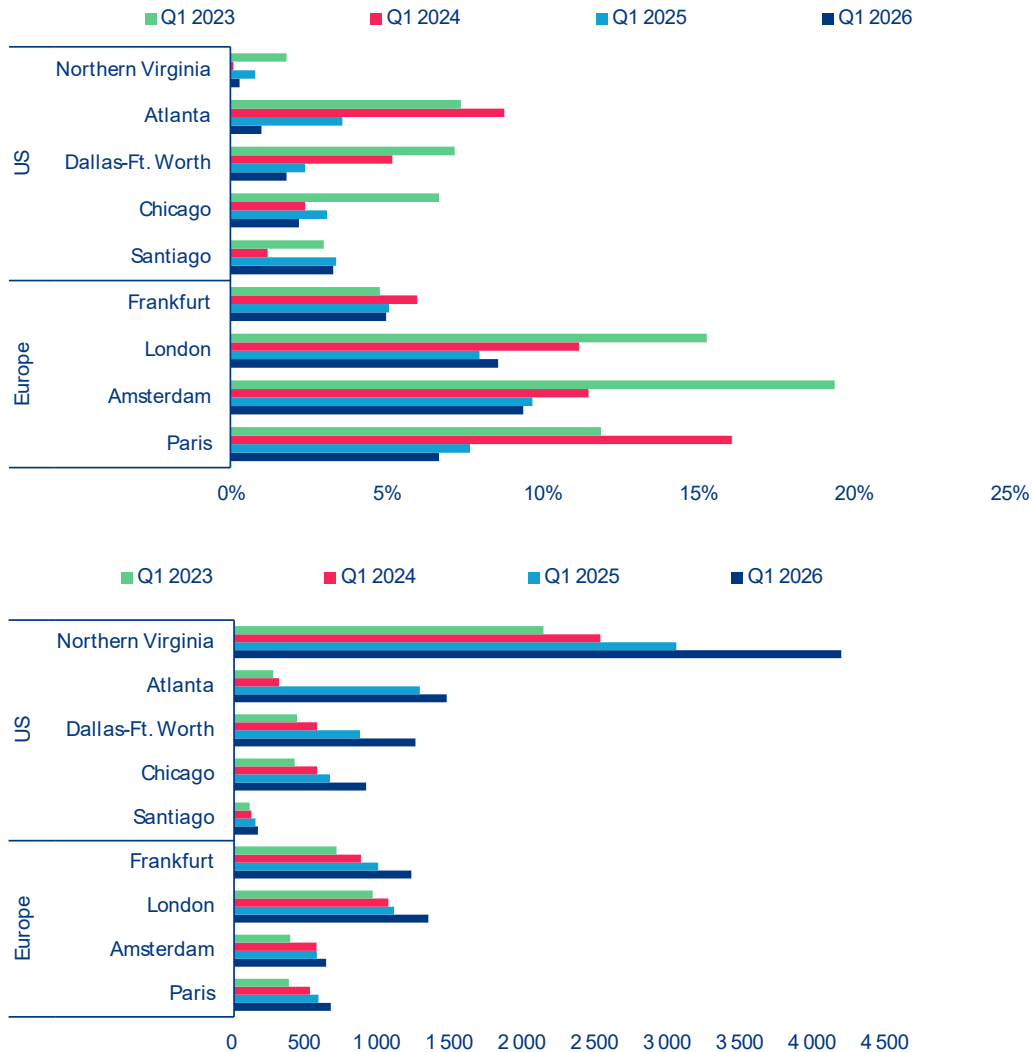
Private equity is at a turning point. Its basic model – buy a company, improve it, then sell it on at a profit – has slowed sharply at the exit, leaving a striking backlog of unsold companies. US buyout funds now hold more than 13,300 of them, about a decade's worth of sales at the current pace. Most were bought in 2020-21, when money was cheap, valuations rich and software allocation was high so they are now hard to sell at those prices. With little cash flowing back, investors stay reluctant to commit fresh capital and most of what is raised pools into the largest funds. Even so, the conditions for a thaw are quietly building as rates stabilize which should lead to a timid gradual recovery rather than a sudden snapback of exits. The more important shift is in where returns come from. The cheap re-rating that drove the 2020-21 peak is over; from here, value has to be earned operationally, through revenue growth and wider margins rather than rising multiples or cheap leverage. That rewards managers who can genuinely improve the businesses they own and leaves the rest behind. Taking all this together, and with the "Softwarepocalypse" still mostly ahead, we expect private equity to return around 7-9% this year, with a stronger pickup towards 11-12% in 2027 as private equity markets normalize (Figure 13).

Figure 13: US private equity value bridge

Data as of Q2 2025. Pooled value bridge across a subset of exited buyout holdings, weighted by invested capital.
Sources: MSCI Private Capital Universe, MSCI Private Asset and Deal Metrics data

Private debt entered the year as one of the better positioned private market sub-asset classes, and the shock it was bracing for has now partly arrived. Much of the alarm bundles four distinct worries (retail-vehicle redemptions, loan quality, software concentration and wrapper design) into one, overstating the systemic threat. If anything, the macro supports the income case, with the Fed holding near 3.75–4.00% into mid-2027, relief comes only after the 2027–29 maturity wall. As a result, floating rate coupons stay high, a tailwind for income but a headwind for the weakest borrowers, whose coverage is squeezed and where payment-in-kind is spreading, now around 8% of BDC investment income. Also on valuations, marks flatter the divergence. Private NAVs continue to hold while listed vehicles trade near 17% discounts. Positioning turns on two points. The asset class straddles the AI story, a legacy software book exposed to disruption versus a fast growing book financing the datacenter build out. Along these lines, capital is rotating accordingly, toward asset-based finance, infrastructure, opportunistic strategies, Europe and closed-end structures. In this junction, direct lending still offers income and seniority, but the prudent posture is selective, not directional, by favoring the senior, well covered and well disclosed books, and treating transparency and capital structure, not headline yield, as the binding constraint. We expect spreads to remain compressed around the 500-550bps range as the pipeline has a quality bias due to software having fallen out of favor and demand remaining high.

Infrastructure is expected to sustain strong performance, underpinned by AI buildout as a growth lever. Data centers continue to dominate the deal market this year. Despite strong new-build across major regions, demand keeps outstripping supply, holding vacancy at record lows (Figure 14) while rents keep growing strongly. With power, permitting, and land constraints sustaining the scarcity, the economics stay attractive for ready-to-use data centers contracted long-term to investment-grade counterparties. Two tailwinds are strengthening the case. First, the token economics are getting more compelling: usage keeps surging despite significant efficiency gains as workloads shift from chatbots to agentic, reasoning-heavy models and the paradigm grows more compute-intensive. Token prices are stabilizing at the frontier even as serving costs keep falling, supporting provider margins. Hyperscaler AI revenue is now beginning to cover the capex depreciation, pointing to a more sustainable cycle. Second, rising data sovereignty demands across regions add a further durable tailwind. We continue to see infrastructure as a strong play in the AI supercycle – the foundation on which the whole AI stack is built – with tangible, contracted earnings rather than speculative upside.

Figure 14: Despite strong new-build (up), data center demand keeps outstripping supply, driving record-low vacancy rates (down)

Sources: CBRE, Allianz Research

Real estate continues to recover in 2026, though along another bumpy and uneven path. The recent energy-led inflation spike and a hawkish near-term policy stance have weighed on pricing and decision-making, but the macro environment ahead is expected to turn more supportive. With the Iran conflict de-escalating and energy pressures easing, we expect the Fed to deliver a final hike before pivoting to cuts, and the 10-year yield to drift lower. This combined with shrinking new supply and resilient occupier demand should underpin a progressive, income-led

recovery. Capital is returning selectively, with the widest dispersion in years between prime and secondary assets and between structural winners (data centers, industrial) and laggards (trophy office), driven by the structural tailwinds and disruptive risks of AI.

A photograph showing a group of diverse hands of various skin tones stacked on top of each other, resting on a thick, textured tree branch. The background is a lush green forest with sunlight filtering through the leaves. The text "Our team" is overlaid on the image, with "Our" in white and "team" in yellow.

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Credit risk relocated, not removed: Newton's cradle transmitting risk across BBB, HY, and private credit

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In Summary

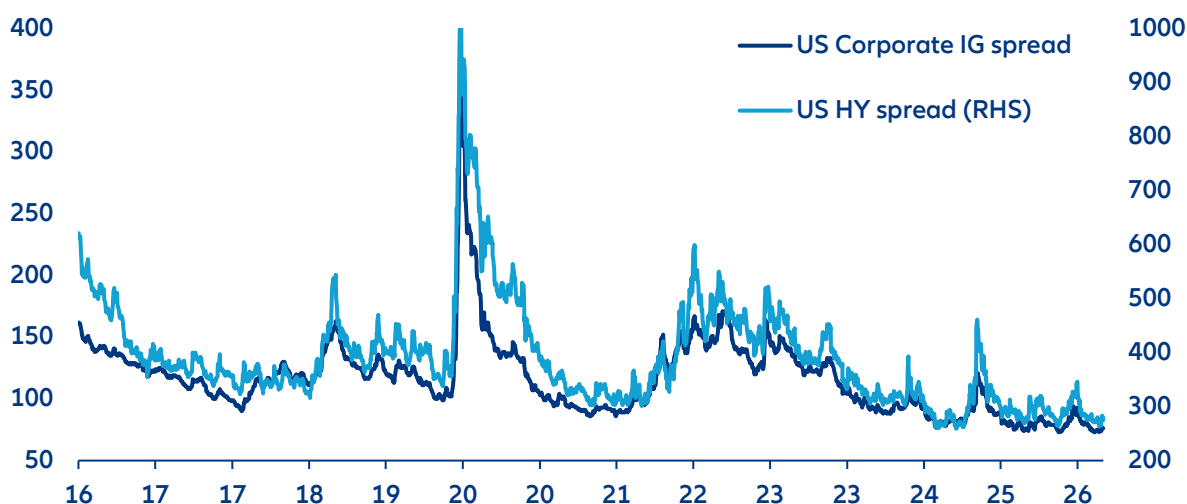
- **US credit looks calm on the surface – spreads tight, headline growth cheerful – but the risk has simply migrated.** Investment grade (IG), high yield (HY) and private credit are one continuous risk-transfer chain and at every link price has decoupled from fundamentals through a different mechanism: spread compression in IG, weakest links bypassing public HY and net asset value (NAV) smoothing in private credit.
- **Softened IG fundamentals and fallen-angel risk expose up to 1/3 of spread.** US IG spreads sit at ~75bps – tighter than their pre-Iran-war level – even as credit quality has softened (interest coverage ~6.3x, net leverage ~2.7x), held tight by price-insensitive all-in-yield demand rather than by fundamentals. The specific danger sits at the bottom of IG: a fallen-angel downgrade forces index-constrained holders to sell into a smaller, less liquid HY market, producing spread moves that far exceed the change in credit quality. In stress case like 2020, downgrades eroded 25bps of excess return, which would be equivalent to ~1/3 of current spread. The net upgrade rate has run below zero for roughly three years and actual fallen angels are ticking up, with historically tight spreads leaving no cushion to absorb the loss.
- **HY offers a more defensive risk/reward profile than BBB – but largely due to a composition effect, as the weakest names have migrated out to private credit.** HY leverage (~3.8x) and coverage (~2.8x) have held firmer than a deteriorating BBB, helped by shorter duration (3yr vs 7yr for IG) and a cleaner index (BB now >50%, CCC near a 20-year low). But much of that strength is a composition effect: the weakest borrowers have migrated out to leveraged loans and private credit. The risk was not removed – it was relocated.
- **The same decoupling runs quietly underneath private credit, via NAV smoothing.** The migrated borrowers are structurally weaker (leverage 5–7x, coverage 1–2x), and because positions are appraisal-valued, NAVs stay smooth while coverage erodes. Stress therefore surfaces not in price but in the plumbing: Payment-In-Kind has roughly doubled to 8.9% of interest income from a 4.3% trough in early 2023, and lender takeovers reached USD39.4bn across 2025–26 – about three times the prior three years combined, though still low single digits against a ~USD1.75trn book. The fallen-angel analog here is a gate or restructuring, not a downgrade – lumpy and deferred – concentrated in software/AI-exposed names (~25% of portfolios) and the 2021–23 vintage.
- **For investors, carry rewards holding but requires selectivity as vulnerabilities concentrate in the lower-rated buckets – BBB in IG, CCC in HY, and the 2021–23 private-credit vintage.** All-in USD IG yields near 5.2% can largely offset a 30–50bps spread widening, while the pure excess return offers only a thin cushion for 10bps widening – basically at the mercy of normal volatility.
- **Above all, in a stress event the risk that left public HY can return to it.** Crowded private-credit positions are hard to exit quickly, so when investors need to de-risk they sell the most liquid instrument first – public HY – regardless of its own fundamentals. Public HY therefore becomes the involuntary shock absorber, and its spreads widen as a liquidity and basis effect rather than a credit-quality one.

The decoupling of fundamentals and price

US corporate credit has shown extraordinary resilience in 2026, but there is an uncomfortable truth lying beneath the calm surface. Spreads of non-financial corporates currently sit at 73bps – 12bps tighter than their pre-war level. Even at the height of wartime fears, the widening was strikingly contained, peaking at just 92bps, only 12bps above where the year began (Figure 1). Yet strong Q1 earnings have not translated into stronger credit quality. Revenue growth in the credit universe is not as strong as in equity because of sector compositions – lower tech exposure but higher utility exposure in credit than equity (Figure 2). And at the balance-sheet level, EBITDA growth has not delivered the lower leverage or higher interest coverage that ultimately determine an issuer's capacity to service its debt – coverage gauges how easily earnings cover interest payments (higher is safer), and leverage measures debt relative to earnings (lower is safer).

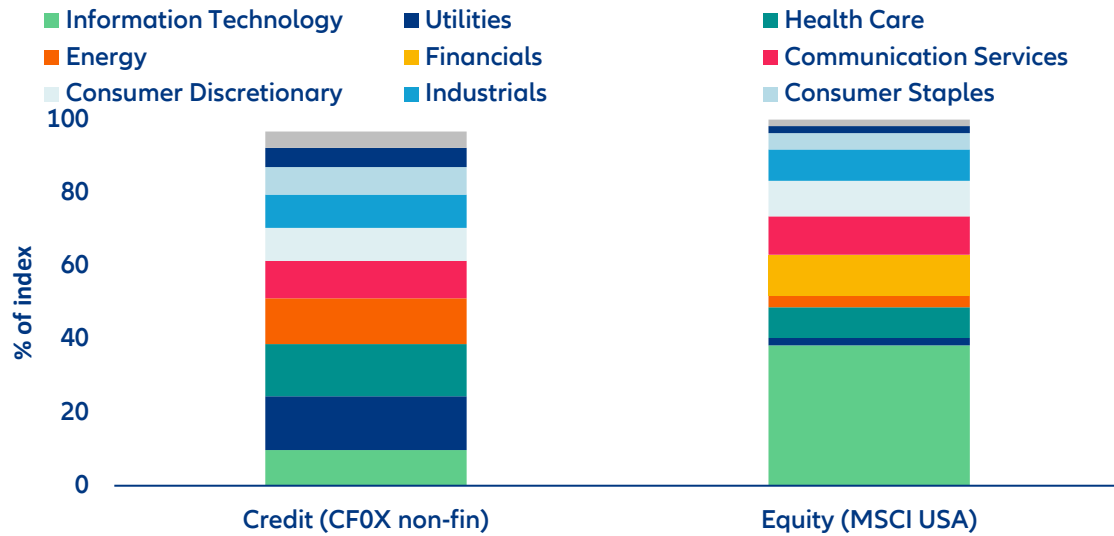
The operative transmission mechanism between revenue growth and debt-service capacity runs through interest coverage (EBITDA/net interest expense), not EBITDA in isolation. Since the US Federal Reserve started its tightening cycle in early 2022, the cost of refinancing legacy fixed-rate debt has risen sharply, and the progressive rolling of maturities into prevailing coupon levels has placed sustained upward pressure on gross interest expense. As a consequence, median EBITDA interest coverage for US investment-grade corporates fell to roughly 6.3× by Q2 2024 and has hovered around that level ever since (Figure 3). Taken in isolation the decline looks benign – coverage stood at 6.6× in Q4 2019, the pre-Covid benchmark – but the comparison is precisely the point: In late 2019, with the Fed funds target at 1.75% and coverage essentially where it sits today, IG spreads traded around 110bps, fully 40bps wider than the current 73bps. Investors are therefore accepting less compensation for a broadly comparable level of fundamental credit quality. Meanwhile, net leverage – net debt/EBITDA – has risen modestly but steadily across the IG universe – from roughly 2.5× net debt/EBITDA in 2022 to about 2.7× today – reflecting a cautious re-leveraging that EBITDA growth has only partly offset.

Figure 1: US Credit spreads at a historical low



Sources: LSEG Datastream, Allianz Research

Figure 2: Credit has lower tech exposure but higher energy and utilities exposure compared to equity



Sources: Bloomberg, Allianz Research

Figure 3: US IG interest coverage ratios

Trimmed mean (bottom/top 10%), EBIT / Interest expense, x

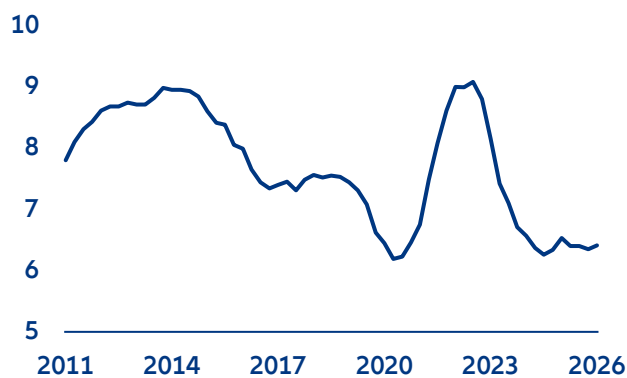


Figure 4: US IG credit net leverage ratios

Trimmed mean (bottom/top 10%), Net debt / EBITDA, x



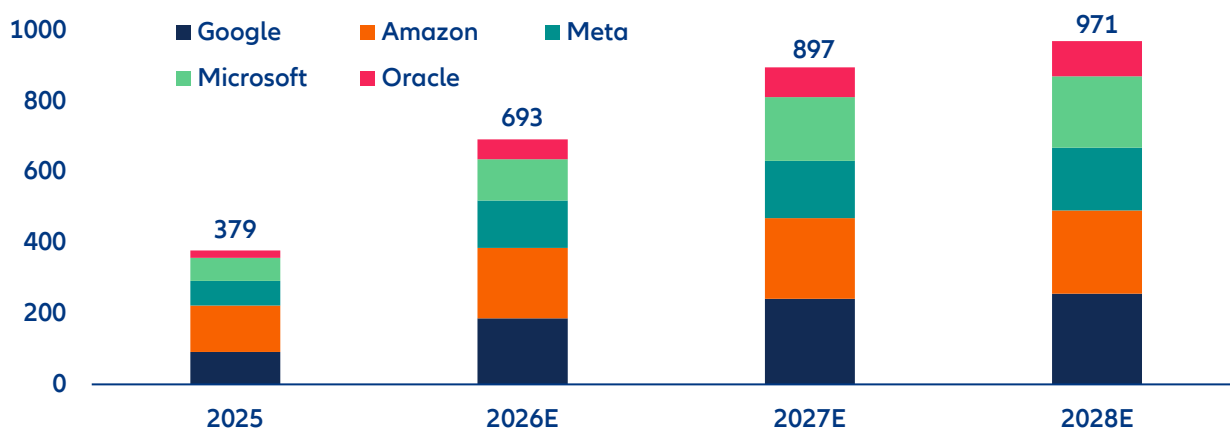
Sources: Bloomberg, Allianz Research.

The explanation for this divergence lies in the structure of demand for investment-grade credit rather than in any fundamental reappraisal of issuer creditworthiness. Positive technicals – strong demand and limited net supply – have dominated spread compression since 2023. The reset to higher all-in yields has drawn a steady, price-insensitive bid into IG credit from yield buyers such as insurers, pension funds and sovereign wealth vehicles. At the same time, heavy gross issuance has been absorbed by a large maturity and redemption calendar, leaving net supply contained. Together, this demand-supply imbalance has provided a consistent bid that has compressed the risk premium well below what fundamentals alone would justify.

Looking ahead, EBITDA growth is the key to credit metrics. The Fed shows little sign of easing soon so interest-expense pressure is unlikely to fade. Higher rates have not yet fully fed through and as legacy debt refinances at today's coupons, gross interest expense can hardly drop. We expect leverage to drift modestly higher even as spreads trade at their most compressed levels, driven by AI-related capital expenditure, the M&A cycle boom and a more hawkish Fed, which pulls borrowing forward as corporates term out debt to lock in funding before rates climb further. Provided EBITDA growth continues to offset the rising interest burden – our base case – these

pressures will remain manageable. However, slower growth and growing debt loads coinciding would mean a gradual deterioration in credit metrics.

Figure 5: Expected hyperscaler capex



Sources: Bloomberg, Allianz Research

The credit fundamentals have softened only moderately and would not yet be a problem. The real risk lies in the mismatch between pricing and fundamentals and that gap rests on technical support rather than on any genuine strengthening of credit quality. Those technicals are less durable – net supply looks set to rise as AI-related capital expenditure and hyperscaler issuance accelerate, while the yield-driven inflows are price-sensitive at the margin and could reverse quickly if the rate outlook shifts. Beyond valuation risk, the downgrade risk triggered by fundamental deterioration should also not be overlooked by credit investors.

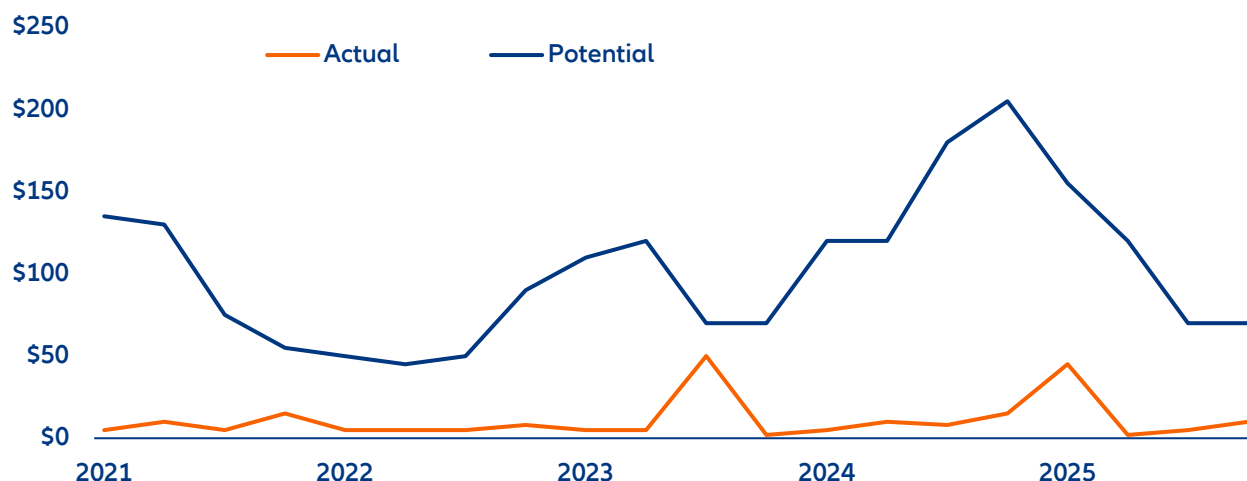
The fallen angel risk

Beneath the moderate deterioration in aggregate credit metrics, downgrade risk is building quietly at the individual-issuer level – and it is concentrated at the bottom of the investment-grade ladder, in the BBB segment, where so-called ‘fallen angels’ are born. A fallen angel is an issuer downgraded from investment grade into high yield and its market impact is determined not solely by the change in credit quality but by the mechanical consequences of index reallocation. When a BBB issuer is downgraded below investment grade, the following sequence is triggered:

- 1) Investment-grade-constrained mandates, esp. ETFs and index-oriented mandates, are required or at least strongly incentivized by tracking error to reduce exposure to the downgraded securities, often within a contractually specified window. This creates a supply of forced sellers with no price sensitivity on the downside.
- 2) The high-yield market, which operates with a smaller asset base and lower marginal liquidity than the IG market, is required to absorb an issuer whose notional size typically exceeds the capacity of the natural HY buyer base to digest in an orderly manner. The supply shock produces spread-widening in the receiving market that exceeds fundamental repricing.
- 3) Once fallen angels begin to outpace rising stars, the cumulative forced selling erodes the carry that anchors investment-grade portfolios and interrupts the hold-to-maturity logic underpinning much of the IG bid – a shift capable of turning the broader spread cycle.

This mechanism makes the spread adjustment far exceed the underlying change in credit quality – which is precisely why fallen-angel risk has become a leading source of idiosyncratic loss and a persistent alpha challenge for investment-grade-mandated asset managers. The effect is material in severe downgrade periods like in 2020: Losses from downgrades eroded 25bps from the expected excess return. BBB-rated debt constitutes approximately 45% of the investment-grade universe by notional value. This concentration magnifies the systemic impact of any BBB deterioration, making the segment worth watching now.

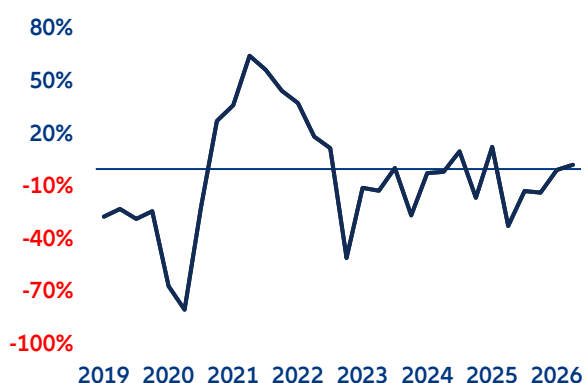
Figure 6: US fallen-angel debt: actual vs potential (USD bn, quarterly)



Sources: Bloomberg, Allianz Research

The fallen-angel wave peaked in the first and second quarters of 2025 so the pool of potential candidates and debt has been shrinking (Figure 6). The downgrade trend has slowed but not turned yet: The net upgrade rate for US IG non-financial issuers has run below zero for roughly three years (Figure 7) and as that pressure persists, actual fallen angels have been ticking up recently. Moody's counted six in the first quarter of 2026, up from three last quarter. Investors should therefore not overlook fallen-angel risk, particularly when historically tight spreads leave no cushion to absorb the losses a downgrade brings. Once slower EBITDA growth meets a higher interest burden, the pool of issuers sitting close to the cut-off will grow sharply.

Figure 7: US IG non-financial issuers rating drift (net upgrade rate)



Sources: Bloomberg, Allianz Research. *Net upgrade rate = (Upgrade# – Downgrade#) / Total rating actions

High Yield: The counterintuitive bright spot

When the economy slows, High Yield (HY) is usually the first cause for concern, with lower ratings and higher default probability. But that instinct is misleading in this cycle. US HY actually offers a more defensive risk/reward profile, especially when compared with the BBB segment. As shown in Figure 8&9 below, the trajectory of HY fundamentals has held more firmly than BBB's over the last decade. Figure 8 captures the divergence in leverage:

After spiking through the 2020 pandemic peak, HY net leverage fell sharply from 4.2x to 3.6x and has since re-leveraged only modestly to 3.8x as of last quarter. Meanwhile, BBB leverage has ground steadily higher, increasing from 2.75x in 2021 to 3.1x in Q1 2026. The interest coverage trajectory tells a similar story – both peaked in 2022 as rates climbed, but HY coverage has broadly stabilized at 2.8x since 2024 while BBB coverage has continued to slide.

The reason might be structural. HY issuers are smaller firms with weaker fundamentals, so they deleveraged heavily through the pandemic and cautiously re-leveraged thereafter due to high interest rate environment. The cautious capital expense and limited financing access has therefore left HY credit metrics in a relatively good shape.

Shorter maturities reinforce the case. The duration of the HY universe is 3 years, much shorter than IG paper (seven years). Shorter duration means less price sensitivity so for any given move in spreads, their prices fall by less. Shorter maturities also improve visibility: the nearer the repayment date, the easier it is to judge whether an issuer can refinance or repay, leaving less time for fundamentals to deteriorate before the bond comes due. Together, these features cushion HY against the sharper spread swings that today's rate environment can produce.

Figure 8: US high yield net leverage vs BBB

Trimmed Mean (Bottom/Top 10%), Net Debt to EBITDA

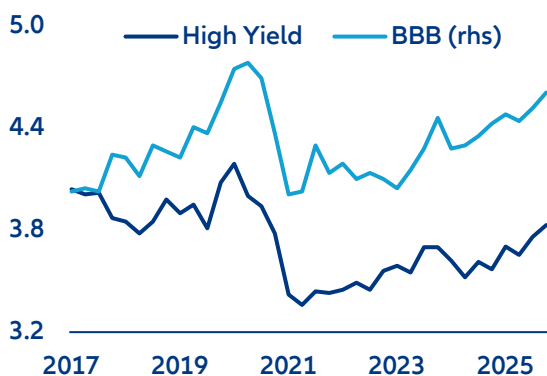
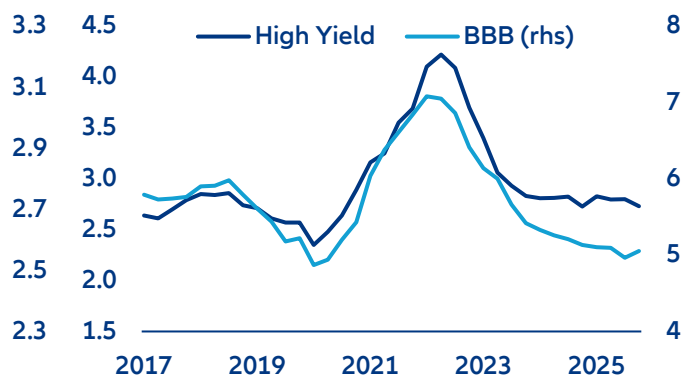


Figure 9: US high yield interest coverage vs BBB

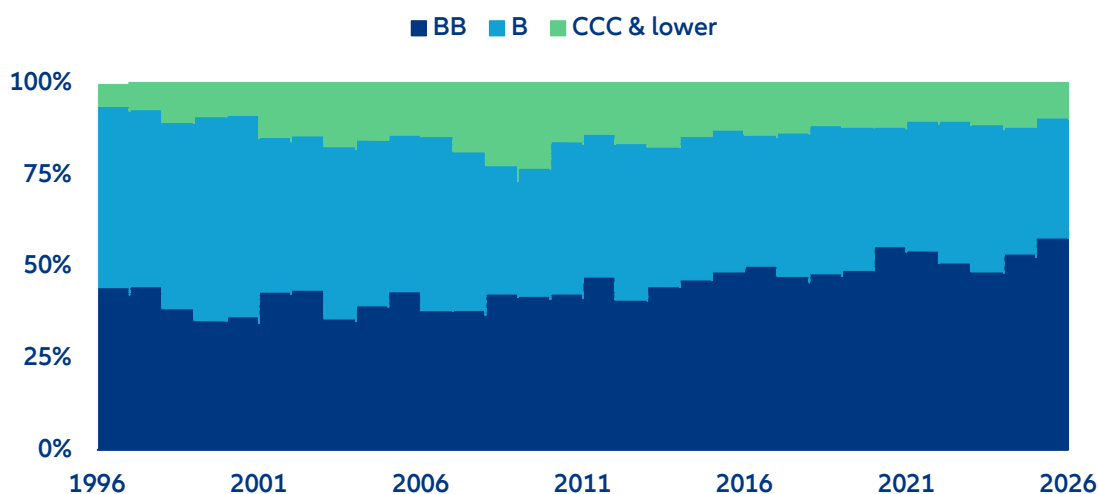
Trimmed Mean (Bottom/Top 10%), EBITDA / Interest expense



Sources: Bloomberg, Allianz Research

Part of the improvement is also a composition effect: The public HY index has become a cleaner, higher-quality universe because the weakest marginal borrowers are increasingly migrating to other alternatives. BB-rated issuers are over 50% of the US HY market versus a single-B majority in 2000, and the CCC-and-below cohort has shrunk to roughly 10%, near a 20-year low. This mix is a result of a two-direction migration. With the rates shock in 2022, net HY issuance turned negative and clearly skewed toward higher-rated BB names. At the lower end, weaker, more levered borrowers have limited market access, especially under the higher-interest-rate environment. And they have largely migrated into leveraged loans and private credit. With both ends removed, what remains is a narrower middle, structurally less representative of below IG risk. Public HY sees the benefit of a higher-quality index mix, while other leveraged lending areas inherit the counterpart smaller, more levered, less liquid borrowers dependent on sponsor support. That said, invest should also remain risk-aware: High yield is still highly sensitive to macro shocks and industry disruption, which have historically triggered most credit losses. Three things are worth watching: whether economic growth and interest rates stay resilient, how AI disruption affects individual businesses and whether investor demand for high yield holds up – above all, appetite for newly issued bonds – given that today's tight spreads are held down by strong demand rather than by genuinely stronger fundamentals.

Figure 10: US high yield index has become a higher-quality universe, BB represent over 50% of the index.



Sources: Bloomberg, Allianz Research

The borrowers leaving public high yield do not move straight into private credit. Leveraged loans are the intermediary step, which both links the two markets and keeps some distance between them. That link still matters for liquid investors: many have added private credit allocations they cannot readily exit, so an eventual credit cycle can transmit stress back into liquid markets and high yield – the liquid sleeves that, at reduced allocations, would need to give way first to keep the overall portfolio within its risk limits. For that reason, a close read of private-credit dynamics matters even for investors who hold none directly.

Private credit: Where the credit went

The issuer migration narrative carries an immediate corollary: if public HY has been cleansed of its weakest borrowers, those borrowers did not disappear, they relocated. That market is, in part, private credit; now a USD1.75trn direct lending market with a further USD315bn of institutional dry powder. The analytical frameworks that apply to public IG and HY translate there only imperfectly, and sometimes not at all.

The first translation difficulty concerns the assumption of size separation. Private credit was built around a structural gap of companies too small or too idiosyncratic to access public syndicated markets, where minimum deal size, rating requirements and investor base breadth create a de facto floor. The data still broadly reflects this origin. The median private credit borrower carries EBITDA of USD30–50mn, roughly ten times smaller than the median HY bond issuer (USD568mn EBITDA) or leveraged loan borrower (USD522mn EBITDA). In that respect, the original logic holds: private credit occupies the space below IG and HY by borrower size, not by credit quality, and the analytical frameworks governing public credit markets apply to it at most by analogy.

But that clean separation has been eroding since 2022 and at the upper end of the market, it has broken down materially. The 2022 rates shock that froze the broadly syndicated market for leveraged buyout (LBO) financing for the better part of 18 months pushed large-cap sponsors into private credit as the only available execution route for transactions that, by borrower size, would previously have been routed to the BSL market. The result was the emergence of "jumbo" direct lending – bilateral or small-club transactions providing debt packages above USD1bn, a territory previously the exclusive preserve of syndicated finance. That shift has not fully unwound: By Q1 2026, private credit was financing approximately 80% of all LBO deals by count, and the average enterprise value of new third-party buyouts exceeded USD1bn.

The spread data confirms how far the competitive frontier has moved. Upper middle market unitranche deals (borrowers with EBITDA above USD100mn) now price at approximately 516bps, barely above where they stood at year-end 2024, reflecting the arbitrage compression that occurs when private lenders and a reopened BSL market compete directly for the same transactions. By contrast, sub-USD50mn EBITDA deals, the core historical private

credit market, price at 505–511bps with no equivalent convergence pressure, because the syndicated market has no structural interest in that size tier.

The analytical implication is that private credit today encompasses two distinct populations operating under the same label. The core middle market — the USD30–100mn EBITDA borrower for whom direct lending was designed, remains structurally separated from public markets by size. The upper middle market tier sits in active competition with public syndicated instruments and inherits the pricing and structural dynamics of both markets simultaneously. What unifies both populations, and what makes the translation from public credit frameworks genuinely difficult regardless of borrower size, is the absence of price discovery. In public credit, the gap between market pricing and underlying fundamentals is legible in basis points. In private credit, it accumulates underneath in amendment activity, Payment in Kind, maturity extensions and sponsor infusions, bilateral and outside any mechanism that forces recognition. The risk has changed venue. The question is whether the venue remains capable of containing what it has absorbed.

The credit quality underneath

The aggregate credit quality of private credit borrowers is structurally weaker than the public HY index that the migration narrative implicitly invites comparison with. Borrowers in direct lending portfolios carry leverage in the range of 5–7× and interest coverage of 1–2×, placing them broadly in line with low single-B to CCC public equivalents, not with the BB-heavy HY index that public market composition data now reflects. The comparison that the HY section draws, between public HY's 3.8× leverage and stable 2.8× coverage versus a deteriorating BBB universe, does not carry across to private credit, whose fundamental starting point is considerably more stretched.

Table 1: Leverage and interest coverage across private credit, leveraged loans, and high yield (Q1 2026)

Category	Subcategory	Debt/EBITDA			EBITDA/Interest Expense		
		Overall	Avg Across Sectors	Range Across Sectors	Overall	Avg Across Sectors	Range Across Sectors
Private Credit (as of Q1 2026)	Fitch	5.9x	5.7x	4.5x–7.1x	2.0x	2.1x	1.5x–2.7x
	S&P	7.0x	7.3x	6.1x–8.6x	1.8x	1.8x	1.4x–2.3x
	KBRA	4.7x	4.7x	4.3x–5.2x	2.3x	-	-
	Lincoln International	5.2x	5.2x	4.8x–5.8x	1.9x	-	-
MS Fundamental Dataset (Public Filers) Q1 2026	BSL BB	3.3x	3.4x	2.6x–4.3x	5.2x	5.1x	4.2x–5.6x
	BSL Single B	5.1x	5.3x	4.2x–8.9x	2.5x	2.8x	1.4x–4.1x
	BSL CCC	10.1x	-	-	1.1x	-	-
	HY BB	3.2x	3.3x	1.4x–4.9x	5.6x	6.1x	3.6x–9.9x
	HY Single B	4.1x	3.9x	2.2x–5.4x	3.6x	3.8x	2.6x–5.4x
	HY CCC	8.7x	-	-	1.2x	-	-
SP Publicly Rated Universe (Public + Private Reporters) Q4 2025	B+	4.3x	-	-	3.2x	-	-
	B Flat	5.4x	-	-	2.3x	-	-
	B-	7.2x	-	-	1.5x	-	-
	CCC+	10.4x	-	-	1.1x	-	-

Source: Fitch, S&P, KBRA, Lincoln International VOG, Bloomberg, FactSet, Morgan Stanley, Allianz Research.

Note: Fundamental data as of Q1 2026, except the data on public credit; the asset universe is “Morgan Stanley Fundamental Dataset”, thus the levels are not directly comparable to the credit metrics we talked above; “average (range) across sectors” refers to average (min and max) of the sector averages or medians. KBRA data for leverage and coverage are for new deals within the quarter.

The trajectory matters as much as the level. Outside the software sector, private credit leverage has been broadly stable and interest coverage has actually edged higher from its 2023 trough, reaching approximately 1.84× by early 2026, a genuine improvement from the 1.52× recorded at end-2023. The aggregate picture therefore echoes the HY story in one respect: cautious re-leveraging after the rate shock, with borrowers that could not refinance having

largely worked through or exited the system. The pool that remains is, on average, more resilient than 2021-vintage underwriting suggested it would be at this point in the rate cycle.

The concentrated vulnerability is sector-specific rather than broad, but with widening dispersion bears watching. Software and AI-exposed borrowers constitute roughly 25% of aggregate direct lending portfolios and their credit profile is materially weaker than the broader universe: median leverage in the software cohort sits at approximately 8.3x against 7.0x for all private credit sectors, with cash interest coverage of just 1.4x – the lowest of any major sector in the S&P private credit universe. These numbers were underwritten in a world of near-zero rates and peak growth multiples; the companies now carry that legacy debt into an environment where AI disruption threatens both their revenue visibility and their enterprise value, the latter being the primary lever through which sponsor support decisions are made.

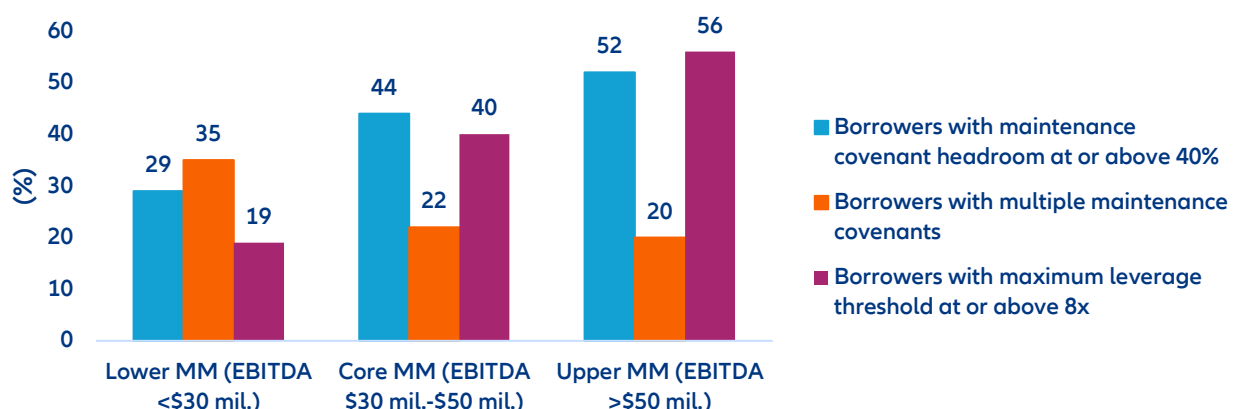
Structural mitigants: genuine but limited

The structural case for private credit has always rested on three pillars: tighter covenants, direct lender relationships and first-lien security. All three genuinely differentiate direct lending from broadly syndicated alternatives. The differentiation has proven real in the current cycle, but it is less uniform than the asset class's promotional narrative suggests.

Recovery rates illustrate the point. Structurally, direct lending's smaller lender groups, tighter documentation and out-of-court resolution pathways ought to support meaningfully higher recoveries than broadly syndicated loans, where covenant erosion over the past decade has degraded lender leverage in workouts. S&P data show nearly a 10pps recovery differential between covenanted and covenant-lite term loans in default. Yet implied recovery from BDC marks at default for 2025 placed direct lending recoveries at around 45 cents on the dollar (KBRA) – 8 points below the comparable BSL figure – dragged lower principally by the PIK-heavy, software-concentrated portfolios where the weakest underlying credits cluster. The forward-looking projection from Morgan Stanley – that recoveries should rise to 70–80% on direct lending defaults over the next 12-18 months, driven by out-of-court resolution and structural protections – is plausible, but rests on the assumption that the remaining stress is concentrated in assets where lender control remains meaningful. If AI-disruption defaults involve impaired enterprise values rather than cash-flow shortfalls, that structural leverage shrinks considerably.

Covenant quality, too, has been eroded in the upper-middle market segment specifically. An analysis of over 2,000 private credit agreements shows that more than a third of borrowers with maintenance covenants carry headroom exceeding 40% – not because they have deleveraged, but because covenants were set wide at origination to win mandates in a competitive deployment environment. This is the private credit equivalent of the loosening structural protections in public leveraged loans, concentrated in the USD500mn-plus deal segment where direct lenders competed most aggressively with the BSL market. The core middle market – broadly, borrowers with EBITDA below USD50mn – has retained tighter covenant terms and genuine lender control and it is this segment that institutional investors are now refocusing on as a consequence.

Figure 11: Loose maintenance covenants have permeated the private credit market



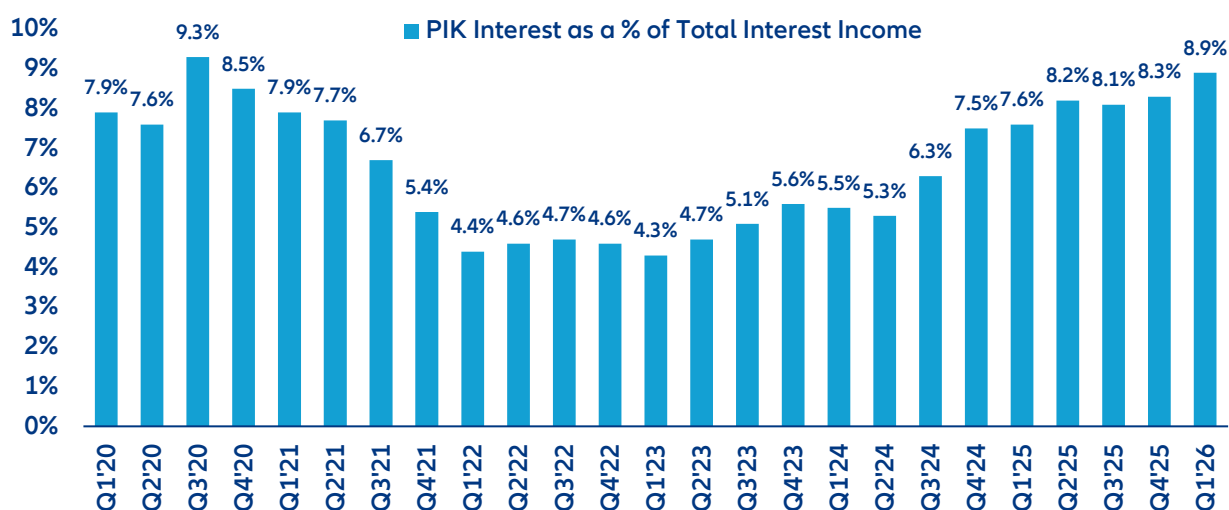
Source: "Loose Maintenance Covenants Permeate Private Credit," 2025 S&P Global Ratings.

The decoupling analog

Just as public spreads are compressing even as interest-coverage ratios fall, private credit runs the same decoupling through a quieter channel. Because positions are appraisal valued rather than marked to market, the price signal is smoothed: NAVs can hold broadly stable while underlying coverage erodes. The fundamental-deterioration signal therefore does not appear in price, it appears in the plumbing, and the plumbing now shows it clearly.

The absence of daily marks and index-driven forced selling, the features that made private credit's Q1 2025 experience so much calmer than public market equivalents, also define the principal risk that the asset class now carries. Stress in private credit accumulates and is resolved bilaterally, which means it remains invisible to aggregate market indicators until it cannot be managed that way. Payment-In-Kind (PIK) interest as a percentage of total interest income in the broader direct lending portfolio reached 8.9% in Q1 2026, up from 8.3% in Q4 2025 and a trough of 4.3% in Q1 2023 (Lincoln. See Figure 12), and roughly 20% of PIK loans are now marked below 90 cents, at or above post-Covid highs depending on the measurement basis. Lender take-overs, where lenders have moved to foreclose on equity and take operational control of borrowers, reached USD39.4bn across 2025 and 2026 combined, against just USD13.6bn over the preceding three years in total. Almost three-quarters of those transactions involved 2021 and 2022 vintage deals, precisely the cohort underwritten at peak leverage, peak valuation multiples and near-zero rates.

Figure 12: Private credit stress is surfacing in the plumbing — rising PIK use increases leverage



Source: Lincoln, Allianz Research. Note: Calculations are based on dollar amount of interest paid (not based on the number of investments that have a PIK component).

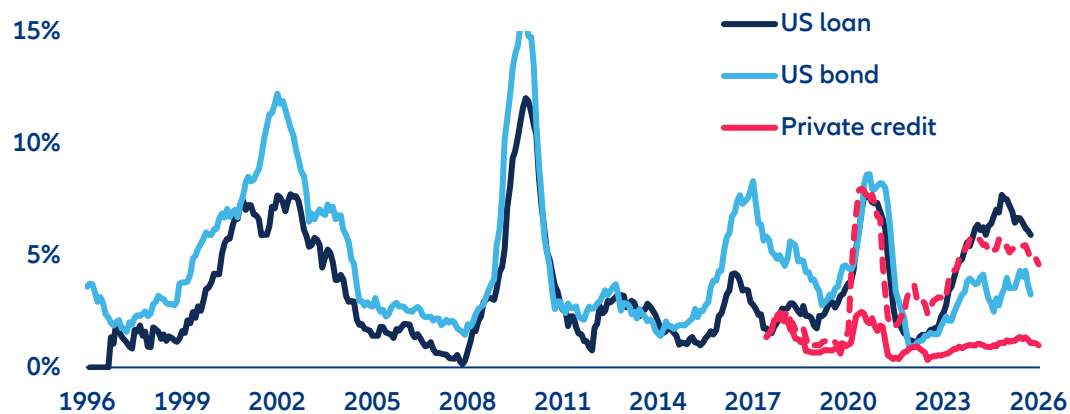
The decoupling between public market pricing and public credit fundamentals examined earlier rests in part on a composition effect: the weakest marginal credits have migrated out of the indices against which spread levels are benchmarked. That migration has genuinely improved the average quality of what remains in public markets. But it has not extinguished the underlying credit risk – it has deferred and concentrated it in an asset class that resolves stress through amendment and negotiation rather than market pricing, that carries limited liquidity for redemptions at scale and that has grown to a size where idiosyncratic failures can no longer be assumed to be individually contained.

The fallen angel analog

In private markets, the fallen-angel analog has a different trigger. There is no rating boundary to cross and no index to be ejected from. The repricing instead arrives as covenant amendment or restructuring. The asymmetry is sharper, not softer, and the default data now sizes it. S&P credit estimates reported a post pandemic 4.37% against leveraged loans of 3.3% in Q1 2026. The populations are not comparable, S&P portfolio skews to smaller issuers and methodologies differ, but the FSB frames the same point cleanly: outright private-credit defaults sit near 1%

but rise to roughly 5% once selective defaults and distressed exchanges are included¹. In public markets, a holder facing a downgrade can still sell into a continuous, if widening, market; the loss is realized quickly and observably. In private markets there is no such exit at the moment it is needed, the liquidity that would cushion the adjustment is exactly what is withdrawn. The downside is gated, lumpy and deferred until unavoidable, realized against a NAV that may itself prove optimistic.

Figure 13: Slowing growth, sticky inflation and software sector distress keep default risk elevated



Source: Private Credit S&P Global Ratings. Loans index Morningstar LSTA US Leveraged Loan Index. Note: Dotted line includes distressed debt exchanges as default events.

None of the above is a verdict on the asset class; it is a verdict on the average deal and the marginal manager. The upside relative to public credit is genuine and it rests on the one structural feature public bondholders lack: position and control. Direct lenders sit at the top of the capital structure – typically 95%+ senior secured with weighted-average loan-to-values below 50% – with real covenants, information rights and a seat at the workout table. This is why higher default frequency can coexist with lower realized loss: the lender can intervene early, restructure or convert before value leaks, compressing loss-given-default in a way a dispersed high-yield holder cannot. Loss-adjusted spread captures this advantage as readily as it exposes the laggards; it cuts both ways, and for disciplined origination it lands clearly positive.

What does this mean for investors?

Technicals have not merely supported fundamental pricing this cycle – they have displaced it. The same mismatch identified in public credit extends to private markets. Where public spreads compress against weakening coverage, private valuations stay smooth while coverage erodes beneath them; where the public repricing event is a downgrade, the private one is a gate, an amendment or a forced secondary sale below NAV. The asymmetry is not escaped by moving private – it is relocated and made less visible.

Three implications follow:

Total return is more resilient than excess return under current market conditions. The near-term technical bid – price-insensitive yield demand – remains in place and carry is meaningful. The distinction between total return and excess return is central: total return combines spread and base rate so all-in IG yields of around 5.2% represent a substantial income component, and a 30–50bps spread-widening could be largely offset by carry. Excess return isolates the spread and with US IG OAS at a historical low of roughly 75bps that component offers only a thin cushion.

The vulnerabilities are concentrated in the lower-rated buckets, not broad yet. Within investment grade, the BBB segment carries the fallen-angel risk, where forced index selling makes the spread move far exceed the change in

¹[FSB warns on private credit vulnerabilities - Financial Stability Board](#)

credit quality. Within high yield, the CCC-and-below cohort carries the genuine default risk as slower EBITDA growth meets a higher interest burden. Within private credit, the 2021–23 vintage is the analog – deals originated at the peak of competition on the loosest terms (covenant-lite, Payment in Kind) carry the deferred risk, in contrast to newer senior-secured primary lending at the other end of that spectrum.

Technically driven turns can be fast. Because pricing rests on technicals rather than fundamentals, the adjustment can be rapid – yield-driven inflows are price-sensitive at the margin and net supply is set to rise as AI-related and hyperscaler issuance accelerates. In that setting, liquidity and flexibility carry a premium and the publicly-priced leveraged-loan / BSL market serves as the liquid, transparent proxy for private-credit stress.

Above all, under market stress, the risk that left public HY could return to it. Crowded private-credit and HY-overweight positions cannot be exited quickly. So when investors need to de-risk, they tend to sell the most liquid public HY first, regardless of its own fundamentals, and its spreads widen. In that sense, the near-term risk in public HY is more a liquidity and basis effect than a credit-quality one. Spreads can move beyond what the improved composition would justify – simply because HY is the proxy being sold for positions that cannot be.

These assessments are, as always, subject to the disclaimer provided below.

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The statements contained herein may include prospects, statements of future expectations and other forward-looking statements that are based on management's current views and assumptions and involve known and unknown risks and uncertainties. Actual results, performance or events may differ materially from those expressed

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(v) persistency levels, (vi) particularly in the banking business, the extent of credit defaults, (vii) interest rate levels, (viii) currency exchange rates including the EUR/USD exchange rate, (ix) changes in laws and regulations, including tax regulations, (x) the impact of acquisitions, including related integration issues, and reorganization measures,

and (xi) general competitive factors, in each case on a local, regional, national and/or global basis. Many of these factors may be more likely to occur, or more pronounced, as a result of terrorist activities and their consequences.

NO DUTY TO UPDATE

The company assumes no obligation to update any information or forward-looking statement contained herein, save for any information required to be disclosed by law.

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Low earth, high stakes: The LEO satellite race between AI demand and geopolitical fragmentation

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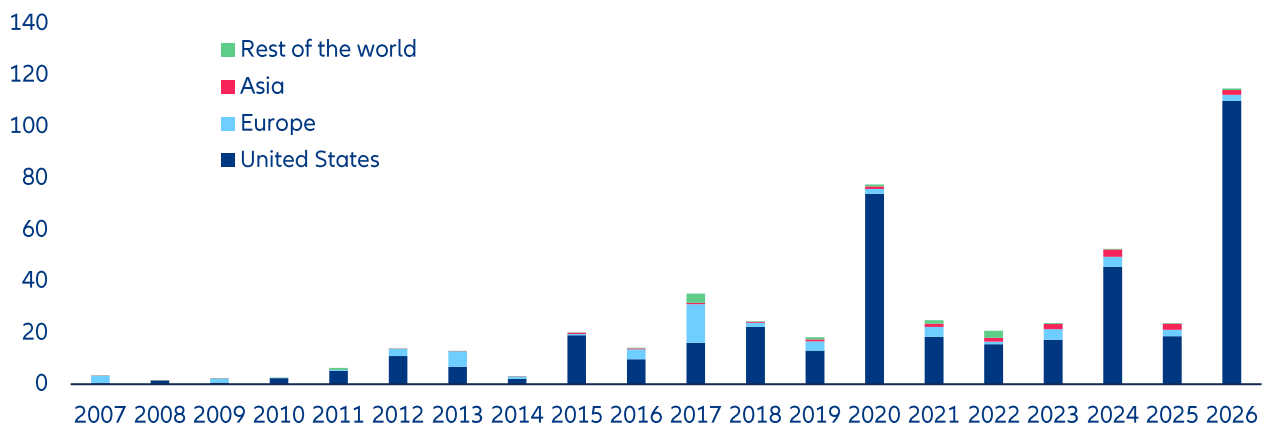
In Summary

- **SpaceX's record-breaking IPO puts a public price on a conviction that private investors already hold: space is a durable investment vertical, not a passing cycle.** Private investment in space tech hit a record USD12.4bn in 2025, up 48% year-on-year, even as venture funding cooled elsewhere, with the US capturing 60% (about USD7.3bn). Late-stage deals reached 41.3% of space-tech VC transactions, signaling capital concentrating on proven, revenue-generating businesses rather than speculative bets.
- **The next wave of the AI economy requires real-time connectivity that terrestrial networks cannot deliver. LEO satellites are the only viable solution.** Autonomous vehicles, industrial robotics, drone logistics and AI-driven grid management all depend on continuous, low-latency connectivity, which 5G cannot meet since 40% of global land surface sits outside reliable coverage.
- **The LEO services market is on track for a 7-8x expansion this decade, within a broader space economy set to triple.** Valued at about USD16bn in 2025, LEO services are projected to reach USD120bn by 2030, driven primarily by enterprise and industrial demand. The broader space economy, currently accounting for about USD600bn, is projected to exceed USD1.8trn by 2035.
- **The US holds a commanding lead through SpaceX's vertical integration and Amazon's hyperscaler logic, but China is closing the gap with state capital and sovereign ambition.** SpaceX's over 9,000 active satellites in orbit and its successful IPO position it as the default connectivity backbone of the AI economy; Amazon has invested over USD10bn in Project Kuiper targeting the same enterprise segment. Meanwhile, China's credible near-term pipeline stands at about 28,000 satellites across state-backed operators – expected to represent close to 40% of effective LEO satellites by 2030 – replicating the model used in EVs, with state funding eliminating the capital constraints that limit commercial rivals.
- **Emerging markets are the strategic battleground where the long-term balance of digital infrastructure power will be decided.** Fixed broadband penetration remains below 40% in Africa and 80% in Asia-Pacific and Latin America (vs. over 90% in Europe and North America). China has moved earlier and more deliberately than Western competitors, positioning LEO as the connectivity layer of Belt and Road's digital extension and replicating the Huawei playbook – below-market pricing, technical lock-in and conversion of infrastructure dependency into geopolitical leverage. India, Brazil and Gulf sovereigns have enough scale or strategic assets to negotiate terms; the remaining 150-plus economies face straightforward dependence.
- **Europe's best strategy is to own a chokepoint in the value chain rather than compete on constellation scale – the ASML model applied to launch.** With Eutelsat OneWeb's about 650 satellites financially constrained and IRIS²'s 280-satellite sovereign constellation limited by design, Europe cannot match the US-China pipeline of over 100,000 satellites. The credible path is deepening Arianespace's position as an indispensable launch provider – already validated by Amazon's Kuiper contract on Ariane 6 – while closing the reusability gap, building toward Ariane Next in the 2030s.

Investors are buying into the space race

Follow the capital, not the rhetoric. Where private money flows, and how much of it, is the clearest evidence that space has become a durable investment vertical rather than a passing enthusiasm. Private investment in space tech reached a record USD12.4bn in 2025, up +48% on the prior year and past the previous peak set in 2021, even as venture funding cooled across most other sectors. The money is also concentrating geographically with the US capturing roughly 60% of the global total, around USD7.3bn, propelled by launch and defense programs including the Pentagon's Golden Dome; Asia, led by China, added some USD2bn and Europe grew from a far smaller base, still lagging both. That concentration is the story the chart below tells; appetite is real but not evenly distributed, and the gap between the leaders and everyone else is widening.

Figure 1: Capital invested in space tech by global region (EUR bn)



Sources: Pitchbook, Allianz Research

Composition over headline: the deal mix confirms this paper's thesis. Commercial launch funding has eased as the segment matured into commoditized infrastructure that no longer commands venture multiples, while satellite communications held firm and geospatial intelligence rose. Investors are paying for the part of the stack closest to recurring revenue – connectivity and data – rather than the hardware that merely reaches orbit.

A maturing market, not a retreating one. Late-stage deals reached 41.3% of space tech VC transactions in 2025, up from 39% in 2024 and the highest share in a decade, a sign that capital is flowing to proven, revenue generating businesses rather than speculative bets. Exits are reviving too, primarily through acquisition as incumbents and hyperscalers buy capabilities cheaper than they can build them, the signature of an industry consolidating around a few winners rather than one still searching for a model. The fundamentals underwrite the conviction, the satellite communications market alone was worth roughly USD25bn in 2025, with LEO deployment, 5G integration and IoT as its named growth drivers.

The Starlink IPO is not an isolated event, but the public-market confirmation of a capital base that has already concluded that satellite connectivity is the segment with the clearest path to durable revenue. A listing of that scale would not open the door to broad-scale, massive space investing, but the capital is already flowing so much as to put a public price on a conviction that private investors reached some time ago.

Terrestrial networks cannot deliver the AI economy

The real-data economy is outgrowing terrestrial infrastructure. The first wave of the AI economy ran on data that already existed – text, images and historical records fed into training pipelines. The next wave is different. Autonomous vehicles navigating live traffic, industrial robots coordinating across factory floors, drone logistics

networks, precision surgical systems and AI-driven grid management all share a single dependency: continuous, low-latency, high-bandwidth connectivity to real-time data, at any location, with no tolerance for coverage gaps. This is precisely what current terrestrial networks cannot provide in a reliable way. 5G offers low latency and high throughput, but only where infrastructure has been deployed (i.e. dense urban zones in wealthy economies). Roughly 40% of the global land surface, including most of the oceans, remains outside reliable 5G coverage. For applications that move (e.g. vehicles, drones, ships, aircraft etc.) this gap is not a minor inconvenience but a significant operational constraint.

Satellite connectivity is not new, but the economics and physics of the previous generation made it unsuitable for real-time applications. Geostationary satellites orbit at approximately 36,000 km, introducing round-trip latency of 500-600 milliseconds – workable for broadcast and slow data transfer but entirely unsuitable for autonomous systems requiring sub-20ms response times. Low Earth Orbit (LEO) changes this equation fundamentally: at 550-1,200 km altitude, LEO constellations achieve latencies of 20-50ms, comparable to terrestrial broadband, while providing global surface coverage through overlapping orbital shells. The technological shift from GEO to LEO is therefore the architectural precondition for deploying AI-intensive applications at global scale.

Table 1: Technical differences between LEO and geostationary satellites

Attribute	LEO (low Earth orbit)	GEO (geostationary)
Altitude	~160–2,000 km (Starlink ~340–570 km)	~35,786 km, fixed over the equator
Round-trip latency	~20–50 ms (fiber-like)	~500–600+ ms
Satellites for global coverage	Hundreds to thousands (mega-constellation)	~3 cover most of the globe (poles excepted)
Satellite unit	Small, mass-produced, low unit cost	Large, bespoke, expensive
Footprint per satellite	Small	Very large (~1/3 of Earth)
Operational life	~5–7 yrs	~15 yrs
Terminal & link	Lower path loss → smaller, cheaper terminals	Higher path loss → larger antennas/power

Source: Allianz Research

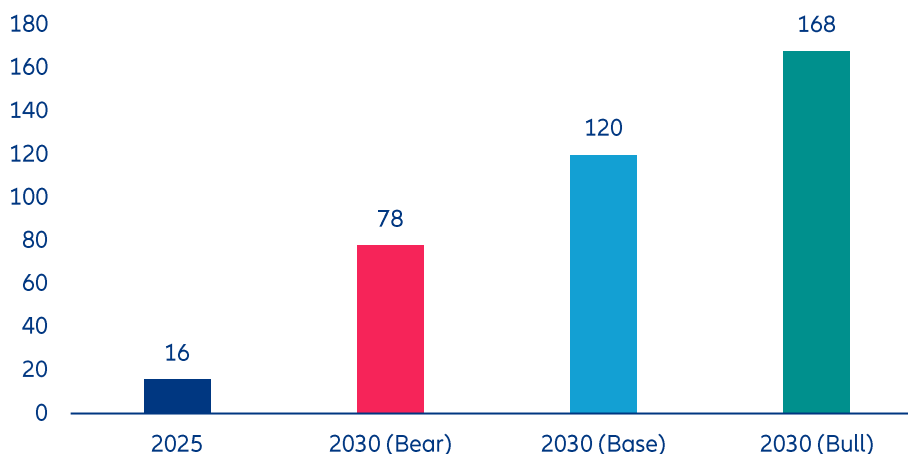
Table 2: Business applications of LEO satellites services

Application	What LEO unlocks
Rural & enterprise broadband	Fiber-like speed/latency where terrestrial build-out is partial
Direct-to-cell (sat-to-smartphone)	Unmodified phones connect in dead zones, erasing mobile not-spots
Mobile/5G backhaul	Connects cell towers and extends 5G where fiber can't reach
IoT / asset tracking / M2M	Ubiquitous low-power links for logistics, agriculture, energy, supply chains
Earth observation & geospatial analytics	High-resolution imagery for climate, urban planning, insurance, defense ISR
Maritime & aviation connectivity	Low-latency in-flight Wi-Fi and vessel comms
Resilience & sovereign/defense comms	Backup connectivity in case of climate events / secure government/military links
Latency-sensitive niches	Use cases GEO couldn't serve

Source: Allianz Research

Demand sizing: from niche to infrastructure. The high expansion potential of the LEO satellite market relies on its own characteristics. Small footprints mandate thousands of satellites, short lifespans make replacement recurring and low unit costs make that perpetual replenishment pay – a self-sustaining demand engine, not a one-and-done build. The services market was valued at approximately USD16bn in 2025. Current forecasts project growth to USD120bn by 2030, a 7-8x expansion in under a decade, driven primarily by enterprise and industrial demand rather than consumer broadband. This sits within a broader space economy currently valued at around USD600bn, which is projected to triple to over USD1.8trn by 2035, with LEO services and defense-related space infrastructure displaying the highest growth rates.

Figure 2: Estimated revenue scenario of LEO satellites services market by 2030



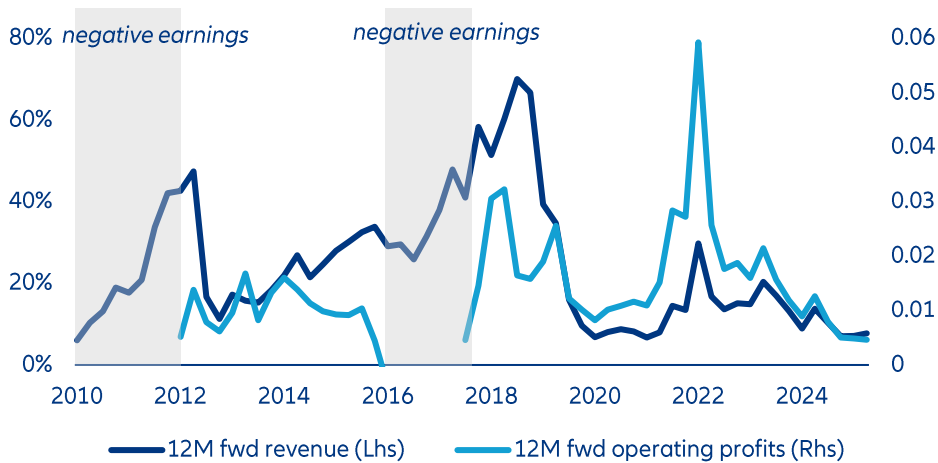
Source: Allianz Research

The geopolitical dimension compounds this demand. The drone strikes on three Amazon-linked data center facilities in the Gulf region in early 2026, and the resulting pause in data center investment decisions across GCC markets, illustrate a structural shift: physical terrestrial infrastructure is increasingly exposed to conflict risk. Distributed LEO constellations, by contrast, present no single point of failure and no fixed geographic target. For enterprises and governments operating in geopolitically unstable regions, satellite connectivity is becoming a resilience requirement, not merely a coverage solution.

The US is well ahead in the race, but China is running hard

SpaceX and the IPO as a market signal. Starlink, SpaceX's satellite internet division, has no comparable competitor today. With over 9,000 satellites already in orbit and regulatory approval filed for up to one million, it operates at a scale that compresses unit economics in ways no rival can currently replicate. The vertical integration is equally important: SpaceX designs the satellites, manufactures them at scale, launches them on its own Falcon 9 and Starship vehicles and operates the ground network. This eliminates the margin leakage that plagues operators dependent on third-party launch providers and means that SpaceX can deploy, replace and upgrade its constellation at a cost structure competitors cannot match. Its IPO converted this operational dominance into a public market signal. At a listing valuation of USD1.75trn, Starlink's implied revenue multiple is aggressive, reflecting not just current commercial contracts but the embedded option on becoming the connectivity backbone for the global AI economy. For investors, the relevant question is not whether SpaceX is worth USD1.75trn or so but what the IPO reveals about where capital is flowing and what it is pricing in. IPO valuations are rarely a reflection of current fundamentals; they are a measure of investors' conviction in a company's ability to dominate future markets. Tesla proved that markets can assign extraordinary value to a perceived pioneer long before revenues and profits justify it, pricing in opportunities from robotaxis, humanoids and other disruptive technologies. Would SpaceX benefit from the same premium? Perhaps as the group is the leader in an industry in the early phase of development. Yet the higher the expectations, the narrower the margin for error. When valuation is built on future promises rather than present fundamentals, execution risk becomes the ultimate determinant of success.

Figure 3: Tesla case study – historical disconnection between market value and fundamentals is at its climax



Source: Allianz Research

Amazon's Project Kuiper is the most significant near-term competitive development. With approximately 3,200 satellites authorized and the first commercial launches beginning in 2026 (carried in part by Arianespace's Ariane 6), Kuiper is not a standalone business. It is Amazon Web Services' hedge against dependence on SpaceX for the edge connectivity that enterprise cloud customers will increasingly require. Amazon has already invested over USD10bn in Kuiper development. The strategic logic is identical to the reasoning behind AWS itself in 2006: build the infrastructure that your customers will need before they are forced to buy it from a competitor. This matters for the competitive landscape because Kuiper's primary market is not consumer broadband, it is enterprise and government connectivity, the same market where Starlink is making its most aggressive commercial moves. Two well-capitalized, vertically-oriented operators competing for the same enterprise segment will compress service pricing over the medium term, which benefits end users but narrows the commercial moat for pure-play satellite service operators.

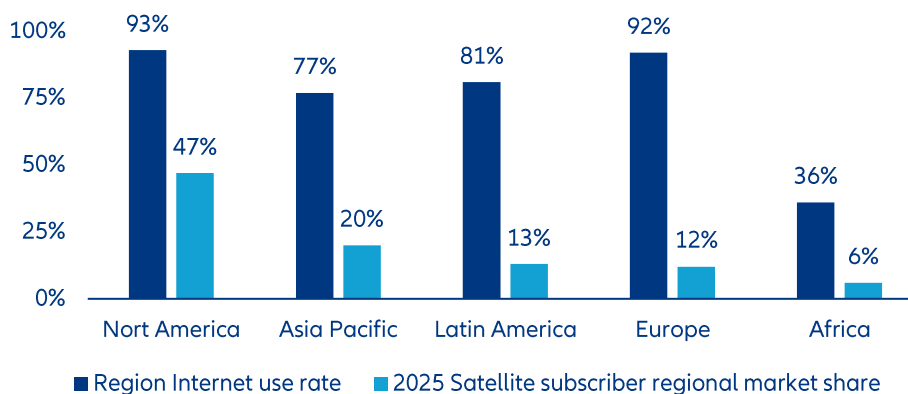
China's satellite ambitions are substantial and systematically underappreciated in Western market analysis. China SatNet, the state-backed operator, has filed for 13,000 satellites, Shanghai Spacecom Satellite Technology (SSST) has filed for a further 15,000. A broader accounting of filed Chinese constellations - against less than 1000 currently in orbit - puts the pipeline around 30,000 satellites across multiple operators, though execution timelines vary widely. The Chinese program differs from US operators in its strategic logic. Where SpaceX and Kuiper are primarily commercial ventures with sovereign dimensions, China's LEO build-out is sovereign infrastructure with commercial dimensions. State funding removes the capital constraint that limits most commercial competitors, and the domestic manufacturing base, now extending to satellite bus production at mass scale, replicates the cost-compression playbook that made China competitive in solar panels and electric vehicles. Spectrum and orbital slot allocation, governed under ITU first-come-first-served rules, creates an additional urgency: slots occupied by Chinese constellations are slots unavailable to future US or European entrants. The net result is an emerging bipolar structure in LEO: US commercial operators with a current lead, and Chinese state-backed operators with the capital and political mandate to close the gap regardless of short-term economics. Every other actor is operating in the margin between these two gravitational forces.

Emerging markets are the battleground

The case for LEO satellites is strongest where terrestrial infrastructure is weakest. Fixed broadband penetration remains below 40% in Africa and at or near 80% threshold in Asia Pacific and Latin America regions - against over 90% in both Europe and North America - with some noticeable discrepancy between large and small countries. Rural populations in these regions are not waiting for fiber rollout, they are waiting for connectivity that may never arrive via terrestrial means within a commercially viable timeframe. The infrastructure economics simply do not

close: low population density, difficult terrain and limited purchasing power make laying cable or building cell towers unattractive for private operators and fiscally difficult for governments. LEO satellites change the economics in the same way mobile phones did in the 2000s. A single orbital shell provides blanket coverage regardless of ground infrastructure, at costs that are declining rapidly as constellation scale increases and with it the range of services provided based on market structure and demand. The applications that follow (e.g. precision agriculture, remote diagnostics, e-government services, financial inclusion, supply-chain connectivity for commodity industries etc.) could be structural economic enablers for economies where the productivity gap between connected and unconnected populations is large. Emerging markets are therefore a large untapped demand pool. The question is not whether these economies will depend on LEO satellite connectivity but whose infrastructure they will depend on, and on whose terms.

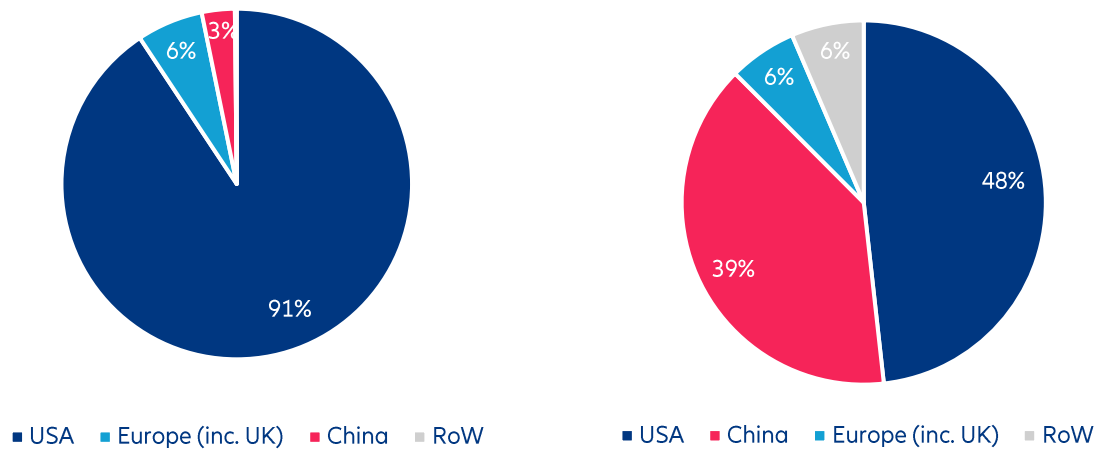
Figure 4: Internet user rate (as % of total population) and regional market share of internet broadband services via satellites



Sources: ITU DataHub, Allianz Research

China understood this dynamic earlier and more strategically than its Western competitors. Chinese state-backed satellite operators are being explicitly positioned as the satellite connectivity layer of the Belt and Road Initiative's digital extension and expect to substantially ramp-up their position in that segment, which has so far been broadly given to US firms. Governments across Africa, the Middle East and Southeast Asia that have accepted BRI infrastructure financing are natural targets for preferential access to Chinese LEO services, bundled with ground station equipment, network management systems and data-handling arrangements that create lasting technical and political dependency. The model replicates the playbook used with Huawei 5G equipment: offer below-market pricing backed by a state subsidy, establish technical standards and integration points that are difficult to unwind and convert infrastructure dependency into geopolitical leverage over time. In the satellite context, the dependency runs deeper – a government whose critical communications infrastructure, agricultural monitoring systems and government data flows run through a foreign-operated constellation has limited practical ability to renegotiate terms after the fact. This trajectory already visible in the terrestrial digital infrastructure decisions made across BRI partner economies over the past decade, now extending into orbital infrastructure to keep control over the whole technology on which its AI infrastructure is based. Accounting for almost 40% of LEO satellites to become effective by 2030, China displays a double ambition : 1) reducing external technology reliance and 2) increasing its digital footprint over high-growth-potential data. For Western policymakers and investors assessing the geopolitical architecture of the LEO race, the emerging market connectivity gap is a key arena where the long-term balance of power in global digital infrastructure will be decided.

Figure 5: Current and expected regional distribution of LEO satellites fleet



Sources: Corporate public announcement, Allianz Research

Not all emerging economies are passive recipients. A handful of larger markets have the domestic capability, market size and political will to negotiate terms or develop partial autonomy. India is the most consequential case. The Indian Space Research Organization has established commercial launch credentials and is actively developing indigenous satellite constellation capabilities. The Indian government has so far blocked or heavily conditioned Starlink's market access, pending domestic regulatory frameworks that ensure data localization and sovereign oversight. With over 1.4bn people and a rapidly expanding digital economy, India's connectivity market is large enough that operators need it more than it needs any single operator, a negotiating position no smaller emerging economy can replicate. Brazil holds a structural geographic advantage that gives it disproportionate leverage in the launch segment: equatorial launch sites reduce the fuel cost of reaching LEO by a material margin, and Brazil's Alcântara launch base is among the best-positioned in the world. This asset has been underexploited. Although a 2019 technology safeguards agreement with the US opened the facility to US commercial operators, launches remain low. But as launch economics tighten and competition intensifies, Alcântara could become a genuine strategic asset. The UAE and Saudi Arabia are pursuing sovereign space capabilities – including domestic satellite manufacturing and launch ambitions – as part of broader economic diversification programs. Their capital depth and tolerance for long investment horizons makes them credible entrants in the sovereign LEO infrastructure segment. However, for the remaining 150-plus economies without the scale, capital or technical base to negotiate on equal terms, the LEO connectivity future is one of dependence: on US commercial operators, on Chinese state-backed alternatives or on the institutional frameworks that multilateral bodies like the ITU and the UN Office for Outer Space Affairs can realistically enforce. None of those options guarantees equitable access or protects against infrastructure leverage.

Europe's narrow window: Specialize or be left behind

Europe is not absent from the LEO landscape. But its position reflects a pattern familiar from the cloud development era: strong institutional intent, inadequate commercial execution. Two programs define Europe's current exposure. Eutelsat OneWeb, formed from the merger of French operator Eutelsat and UK-based OneWeb, operates a constellation of approximately 650 LEO satellites and serves enterprise and government connectivity markets. It is the only European operator with a functioning LEO constellation at meaningful scale, but its financial position is constrained (the merger was partly a rescue operation) and as a result its business strategy is more qualitative than quantitative-centric (440 satellites to be deployed by 2027-2030, mostly replacements of old models) compared with US or Chinese competitors. IRIS², the pan-European institutional constellation backed by the EU and European Space Agency, targets 280 satellites (260 LEO, 20 GEO) with a 2030 operational target. It is designed primarily for government and critical infrastructure connectivity, offering encrypted communications resilient to conflict disruption. Eleven companies form the industrial consortium. IRIS² is a genuine strategic achievement, a coordinated European response to the sovereignty vulnerabilities exposed by dependence on non-European satellite operators, but its scale is deliberately limited and its commercial ambitions are constrained by its sovereign mandate. Against a US-China combined pipeline running well above 100,000 satellites, Europe's credible commercial constellation capacity – including replacement of old models – should remain below 1,000 level. No European operator has the capital depth of Blue Origin, the launch verticality of SpaceX or the hyperscaler integration of Amazon Kuiper.

The ASML analogy: own the chokepoint. The history of European industrial competitiveness in technology-intensive sectors suggests a different model: identify a structural bottleneck in the value chain, invest deeply in the capabilities required to dominate it and leverage that position to capture value regardless of which end-product competitors win. ASML's dominance in extreme UV lithography is the benchmark case. ASML does not compete in the semiconductor end-market, it supplies the tool that every leading chipmaker requires to produce advanced semiconductors. Its market position is not the result of having the largest chip factory, it is the result of owning an irreplaceable step in the production process. Arianespace occupies an analogous position in launch operations. Europe's launch infrastructure, Ariane 6, with the Vega-C small launcher alongside, is not the cheapest or remains constrained by its non-reusable model, but it is an independent, reliable, sovereign-controlled capability. Critically, Amazon has contracted Arianespace to launch a significant tranche of Kuiper satellites on Ariane 6. This is not a minor commercial deal: it is validation that a US hyperscaler investing in its own constellation has chosen European launch capacity as a component of its supply chain. But with around 10 flights per year compared with over 100 for its US rivals SpaceX, Europe is today not yet really sized to compete with the US and China, at least on the commercial segment. Next-generation initiatives point to a credible response: MaiaSpace, an ArianeGroup subsidiary, is developing the partially reusable Maia mini-launcher (maiden flight expected from 2026–27), drawing on the Prometheus engine and the Themis reusable first-stage demonstrator, whose first hop test is slated for 2026. These building blocks are intended to feed a reusable first stage for Ariane Next in the 2030s, marking the path by which Europe could close the reusability and cost gap and consolidate its position across the LEO supply chain. The commercial proof-of-concept for the “ASML model” already exists. Rather than attempting to build a commercial constellation at US or Chinese scale – an effort that would require capital and political coordination that is not available – European industrial policy should concentrate on making Arianespace's launch capability indispensable across operators regardless of nationality, in the same way that ASML's tools are used by the world's leading chip manufacturers.

Table 3: key technical differences between main LEO satellite launchers

Criteria	Arianespace	SpaceX	Blue Origin	United Launch Alliance
LEO payload	22 t	23 t Falcon 9 / >100 t Starship	45 t	27 t+
Reusable	No	Yes	Yes	No
Fairing volume	Large	Moderate	Very large	Large
Current cadence	Low (~5-10/yr)	Very high (>100/yr)	Very low	Moderate
Constellation experience	OneWeb, Kuiper	Starlink leader	Kuiper-focused	Kuiper
Sovereign access	European	US	US	US
Vertical integration	No	Very strong	Strong	No
Cost competitiveness	Medium	Excellent	Potentially excellent	Weak

Source: Allianz Research

These assessments are, as always, subject to the disclaimer provided below.

FORWARD-LOOKING STATEMENTS

The statements contained herein may include prospects, statements of future expectations and other forward-looking statements that are based on management's current views and assumptions and involve known and unknown risks and uncertainties. Actual results, performance or events may differ materially from those expressed

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(v) persistency levels, (vi) particularly in the banking business, the extent of credit defaults, (vii) interest rate levels, (viii) currency exchange rates including the EUR/USD exchange rate, (ix) changes in laws and regulations, including tax regulations, (x) the impact of acquisitions, including related integration issues, and reorganization measures,

and (xi) general competitive factors, in each case on a local, regional, national and/or global basis. Many of these factors may be more likely to occur, or more pronounced, as a result of terrorist activities and their consequences.

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Code, carbon, kilowatts: AI's hidden toll and the race to green the grid

30 June 2026

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Appendix

Executive Summary



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- **Data-center investment reached USD580bn in 2025, putting AI on track to become one of the world's fastest-growing sources of electricity demand.** Installed capacity is expected to double by 2030, with AI workloads already accounting for 15–20% of data-center electricity use and potentially approaching 40% by the end of the decade. Yet the sector's environmental footprint remains underestimated as most analyses focus only on operational electricity use. This analysis takes a broader systems view across 26 countries (+93% of global capacity), adding lifecycle emissions, water use and AI's growing resource demand.
- **Identical workloads can generate up to 24 times more emissions depending on the emission intensity of the grid, making location as decisive as demand growth.** Fossil-dependent grids in Indonesia, India and Malaysia exceed 600 gCO₂/kWh, compared with under 30 gCO₂/kWh in Norway and Sweden. The US and China, which host the largest data-center clusters, sit in between at 384 gCO₂/kWh and 526 gCO₂/kWh, respectively, giving Europe's cleaner power mix a structural advantage for low-carbon AI growth. These disparities are amplified by transmission and distribution losses of 10–15% in some markets, while less reliable grids raise electricity needs and dependence on backup generation.
- **At 286 MtCO₂ in 2025, the true carbon footprint of data centers is 57% larger than IEA estimates suggest.** Electricity consumption (Scope 2) accounted for 76% of this footprint, at 218 MtCO₂, with hardware manufacturing and construction (Scope 3) contributing a further 66 MtCO₂, or 23%, and direct Scope 1 emissions remaining negligible (<1%). Emissions are also heavily concentrated geographically, with the US and China alone accounting for roughly 70% of the global total. AI accounts for an estimated 43–60 MtCO₂ of today's emissions, and this is set to climb steeply as deployment widens and computing demand grows.
- **Without grid decarbonization, global data-center emissions would more than double to 643 MtCO₂ by 2030, leading to an estimated USD154bn in annual climate damages (up from USD68bn today).** AI workloads already account for an estimated USD13bn in climate damages annually and could exceed USD50bn by 2030. By contrast, an ambitious decarbonization pathway would hold emissions to around 329 MtCO₂ despite continued growth in computing demand, and keep climate damages at USD79bn. This makes the pace of power-sector decarbonization the primary determinant of whether AI growth can be decoupled from emissions in the near term. Even under ambitious decarbonization, however, the footprint does not vanish but moves up the supply chain: as Scope 2 falls from more than 70% of the footprint today to around half by 2030, embodied emissions from servers, semiconductors and infrastructure become the binding constraint, approaching 50% of the total. Achieving genuinely low-carbon AI will therefore require not only cleaner power, but also closer attention to emissions embedded in digital infrastructure supply chains.

- **Deployed across the economy, AI could cut global CO₂ emissions by around 1.4 Gt a year by 2035, more than offsetting the emissions generated by its own infrastructure and creating net savings of roughly 750 MtCO₂.** According to the IEA, these reductions would result from efficiency gains, optimization and improved resource management across sectors such as energy, industry, buildings and transport, and are equivalent to around 2.6% of current global emissions. However, this outcome is not guaranteed. With most AI applications still at an early stage of deployment, its ultimate climate impact will depend on whether these economy-wide benefits can scale faster than the infrastructure required to support them.
- **Data centers consumed 814bn liters of water in 2025 and could require 1.3–1.8trn liters by 2030, comparable to Switzerland’s annual consumption, making water the overlooked resource constraint of AI.** Most of this footprint is indirect, with roughly three-quarters originating from electricity generation and the remainder from on-site cooling and semiconductor manufacturing. This ties water use closely to the energy transition, since fossil and nuclear plants require substantial cooling water while wind and solar use little or none in operation, lowering both the carbon and water footprints of a cleaner grid. Although power-sector decarbonization can help moderate future water demand, water-related risks are becoming increasingly concentrated in water-stressed regions such as South Korea, India, Mexico and parts of China, where rapid data-center growth is colliding with existing pressure on local water resources, raising the risk of access constraints and community or regulatory opposition to new capacity.
- **Realizing „green AI“ will depend less on making data centers marginally more efficient than on transforming the energy systems that power them.** Unlocking AI’s environmental potential will require a broader policy framework, combining clean-power expansion, greater transparency on resource use, stronger incentives to price environmental costs, and faster deployment of AI applications that reduce emissions across the wider economy.



Setting the scene: Power demand growth and impact on energy investments

Data center investment reached USD580bn in 2025 and installed capacity is set to double again by 2030, but the sector's environmental costs and geographic scope remains poorly understood.¹ In 2024, the IEA estimated that global data center emissions from electricity use stood at around 180 MtCO₂ (0.3% of global emissions), comparable to the total emissions of Kuwait and sufficient to place the sector among the top quarter of global emitters if it were a country. The carbon intensity of individual AI models illustrates how rapidly this trajectory is accelerating: Training AlexNet in 2012 produced an estimated 0.01 tCO₂ equivalent, whereas training Grok 4 in 2025 generated approximately 72,816 tons, exceeding the lifetime carbon output of more than 1,000 average cars.² Looking ahead, the IEA projects that emissions from data-center electricity use could rise from 180 MtCO₂ today to 320 MtCO₂ by 2030 in its baseline scenario, and to as much as 500 MtCO₂ under a high-growth scenario (0.6-0.9% of global emissions).³ Importantly, these projections account only for operational emissions associated with electricity consumption, leaving a substantial gap in our understanding of the sector's broader environmental footprint.

To address these gaps, this analysis adopts a broader systems perspective on the environmental implications of data-center growth. It focuses on 26 countries that together account for more than 93% of global data-center capacity and represent the principal geographies for both digital infrastructure and AI deployment. In addition to operational emissions from electricity use, the analysis considers lifecycle emissions across the different emission scopes, assesses the sector's water footprint and evaluates the increasingly important contribution of AI workloads to future resource demand. Together, these dimensions provide a more comprehensive assessment of the environmental consequences of rapid data-center expansion.

A first step in understanding these impacts is to examine how data-center capacity and electricity demand are evolving across major markets. Estimates of electricity demand are derived from installed computing infrastructure, adjusted for regional power usage effectiveness (PUE) and utilization rates averaging around 49%, with country-specific variations in line with IEA region estimates. Future demand is projected to 2030 using announced capacity pipelines and official forecasts where available, providing a baseline assessment of likely sectoral growth should currently planned developments materialize.

¹ [2026 Data Center Marketplace Report | Colliers](#)

² [2026 AI Index Report \(Stanford HAI\)](#)

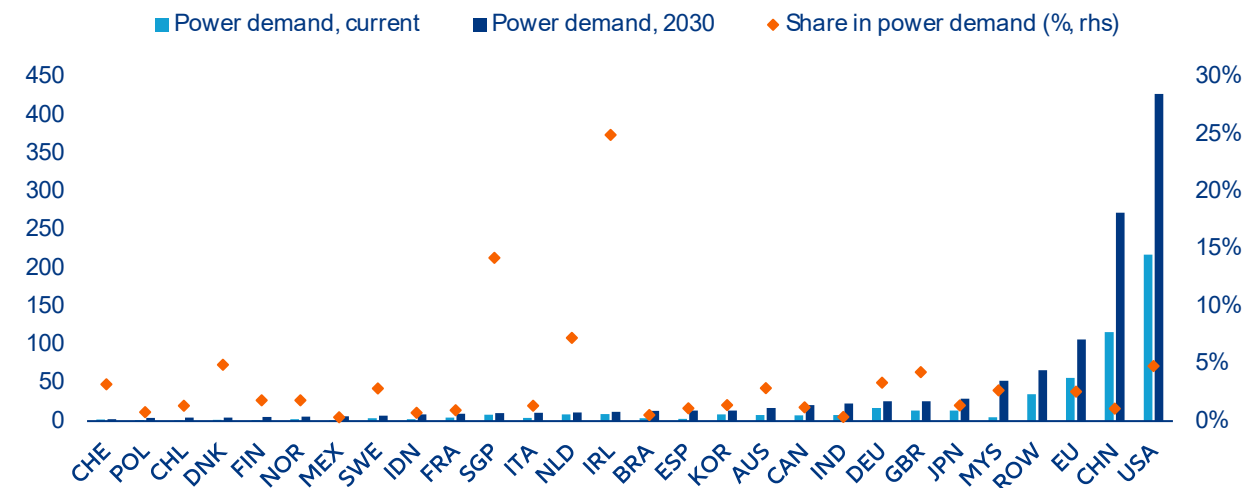
³ [AI and Energy 2025 \(IEA\)](#)

Based on this approach, data-center electricity demand globally is estimated to have reached approximately 515 TWh in 2025 and could more than double to around 1,110 TWh by 2030 (Figure 1). Such growth would make data centers one of the fastest-growing sources of electricity consumption and an increasingly important driver of power-demand growth in several major economies. While expansion is occurring across most regions, growth remains heavily concentrated in the sector’s two largest markets. The US is projected to account for the largest absolute increase in demand, adding approximately 49 GW of data center load, equivalent to around 210 TWh of annual electricity consumption by 2030. China follows with an additional 38 GW, or roughly 155 TWh. Together, the two countries are expected to retain approximately 63% of global data center demand by the end of the decade, only

marginally below their current share of 64%, highlighting the continued concentration of digital infrastructure despite the emergence of new regional hubs.

Beyond the two largest markets, some of the fastest relative growth is expected in emerging data-center hubs. Electricity demand could increase by a factor of three to eight in Malaysia, Mexico and Spain, reflecting strong investment pipelines and their growing role as alternative locations for digital infrastructure, driven by competitive energy costs, supportive policy frameworks and strategic connectivity to adjacent markets. However, these estimates remain subject to considerable uncertainty as a substantial share of announced capacity has yet to reach final investment decisions, secure grid connections or obtain long-term power supply agreements.

Figure 1: Data-center power demand in 2025 by country (TWh, left) and share in total demand (% , right)⁴



Sources: Allianz Research based on EMBER, IEA, LBNL

⁴ For specific country level power capacity and demand estimates, see Table A1 in the Appendix

The system-level implications of this expansion are already visible in several smaller and highly digitalized economies. Ireland stands out as the most exposed, with data centers accounting for approximately 25% of total electricity demand. Significant impacts are also evident in Singapore at around 14%, while the Netherlands has reached approximately 7% and both Denmark and the US around 5%. At these levels, data-center growth is increasingly shaping national demand trajectories and raising constraints around grid capacity, permitting and decarbonization planning.

A key uncertainty is the extent to which artificial intelligence is driving this demand growth. While attribution remains complex, current estimates suggests AI workloads are becoming increasingly material: The IEA reports that AI-accelerated servers accounted for around 15% of total data-center electricity demand in 2024, while Gartner projects the share of AI-optimized servers could rise from 21% of power demand in 2025 to 44% by 2030. Similarly, Goldman Sachs estimates that AI workloads currently account for around 13% of demand and could rise to 27% by 2027.⁵ Taken together, these estimates suggest AI-related electricity consumption of roughly 67–77 TWh today, rising to approximately 300–489 TWh by 2027–2030, although some of these figures likely understate total AI-related demand, as a share of inference also runs on conventional infrastructure.

The rapid expansion in data-center electricity demand is already having a measurable impact on power-sector investment. This is occurring against a broader structural shift in global energy capital allocation, with electricity increasingly becoming the dominant destination for energy investment. Its share of total energy investment has risen from around 25% in 2015 to roughly 50% in 2026, with annual electricity supply and infrastructure investment reaching approximately USD1.6trn.⁶ Within this shift, technology-related demand is playing an increasingly important role, with the tech sector accounting for around 40% of corporate power-purchase agreements globally.

Against this backdrop, data-center-driven electricity demand is emerging as a material force shaping investment decisions across generation, grids and flexibility assets. According to the IEA World Energy Investment 2026 analysis, total spending on data center-related IT equipment and energy reached approximately USD565bn in 2025, with investment in associated energy infrastructure, including grid upgrades, power equipment and onsite generation, estimated at around USD105bn. This alone exceeds total annual energy investment across the entire African continent, highlighting the scale of capital mobilized directly or indirectly by digital infrastructure expansion.

These pressures are increasingly translating into concrete investment decisions in electricity systems. Grid expansion and storage investment are rising across most major regions, including China, Europe, the US and parts of Asia, as utilities respond to the need to connect and serve large, concentrated data-center loads alongside broader electrification trends. While renewable investments are projected to decline slightly in 2026, driven by falling technology costs, policy shifts in the US and China and a smaller construction pipeline, their shorter build times and lower lifetime costs make them a critical supply component for meeting data-center demand growth. At the same time, gas-fired generation has also seen renewed investment, although this trend is largely concentrated in the US, where approximately USD24bn of gas turbine final investment decisions over the past five quarters have been directly associated with data-center demand, with onsite generation increasingly preferred for large-scale facilities requiring high reliability and fast deployment.

Overall, these developments suggest that data centers are becoming a structural driver of power-sector investment. While they are accelerating investment in grids, renewables and network reinforcement, they are also contributing to renewed demand for dispatchable fossil generation, particularly natural gas, with important implications for the pace and credibility of long-term decarbonization pathways.

⁵ IEA, Gartner and Goldman Sachs

⁶ World Energy Investment 2026 (IEA)



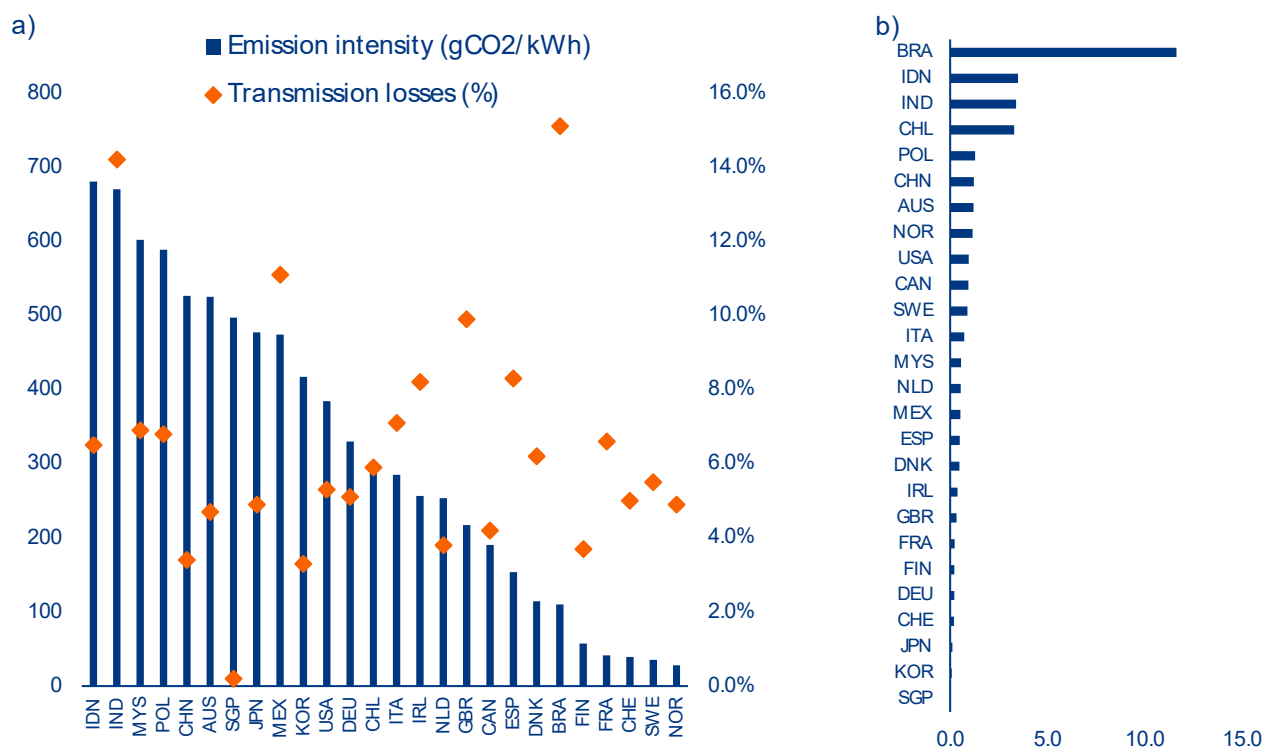
How well are countries prepared for green AI?

The environmental footprint of AI-driven data-center expansion is not determined solely by electricity demand growth, but also by the characteristics of the power systems supplying that demand. Three dimensions are particularly important in assessing how well countries are prepared for “green AI”: the carbon intensity of electricity generation, the efficiency of transmission and distribution networks and the operational reliability of the grid. Together, these factors determine the extent to which incremental data-center demand translates into emissions, system losses and reliance on backup generation. In addition, technology-specific factors such as power usage effectiveness (PUE) and utilization rates influence the efficiency of data centers themselves, although these are likely to evolve more gradually over the short-term horizon considered in this analysis.

There is substantial heterogeneity across countries in the carbon intensity of electricity supply. Eight of the ten best-performing countries in the sample are located in Europe, led by Norway, Sweden and Switzerland, where hydropower- and nuclear-dominated generation mixes result in the lowest emissions per unit of electricity

consumed (Figure 2 a). In contrast, several emerging data-center markets exhibit significantly higher grid emission intensities. Indonesia, India and Malaysia all exceed 600 gCO₂/kWh, implying lifecycle emissions per unit of electricity that are around 20 to 24 times higher than in Norway. These differences imply that identical increments of data-center demand can have vastly different climate impacts depending on location.

System efficiency further amplifies some of these disparities. Transmission and distribution losses are highest in Brazil, India, Mexico and the UK, where they increase effective electricity supply requirements by approximately 10–15% (Figure 2 a). By contrast, losses remain below 3% in Singapore, South Korea and China, reflecting more efficient and tightly managed grid infrastructures. These differences are particularly relevant for data centers, which are highly electricity-intensive and sensitive to marginal changes in delivered power requirements.

Figure 2: AI preparedness of the power system: a) emission intensity vs transmission losses and b) average annual grid outage duration (hours)

Sources: Allianz Research, EMBER, World Bank

Grid reliability adds a further layer of variation.

Among the most data-center–exposed regions, reliability is generally high: the US, China and most European economies experience annual outage durations of less than two hours per customer (Figure 2 b). However, several emerging markets, including India, Indonesia and Brazil, exhibit higher outage levels, with Brazil reaching approximately 11 hours per year. While this remains lower than the roughly 19 hours or more observed in many other developing economies, it nonetheless increases the likelihood that data centers rely on on-site backup generation or behind-the-meter capacity to ensure operational continuity, which in turn

can contribute to higher direct on-site emissions during grid disruption events.⁷ In addition, increasing exposure to extreme weather events may place upward pressure on outage durations in several regions, reinforcing the importance of backup generation and distributed reliability solutions. This is already evident in the US, where major natural catastrophe events have increased outage durations to around 6.5 hours per year on average, peaking at approximately 11 hours in 2024, highlighting the extent to which extreme weather can materially alter reliability outcomes in advanced grid systems.

⁷ IEA and World Bank



Sizing the footprint: data center emissions today

To assess the carbon footprint of data centers comprehensively, three emission levels are of interest.

First, there are direct on-site emissions from fuel combustion and fugitive gases (Scope 1). Second, indirect emissions from purchased electricity, which represent the dominant term in most facility assessments (Scope 2). Third, upstream value chain emissions embedded in the manufacture of hardware and construction of infrastructure (Scope 3). While the relative contribution of each varies with grid conditions and facility design, a complete assessment requires all three, particularly as grids decarbonize and the balance between operational and embodied emission shifts.

Direct on-site (Scope 1) emissions are negligible relative to the total carbon footprint of a data center. They primarily consist of emissions from three main factors: diesel combustion in backup generators during routine testing and grid outages, fugitive refrigerant leakage from cooling systems and increasingly on-site (behind-the-meter) natural gas generation used to bypass slow grid connections. For simplicity, behind-the-meter (BTM) generation is abstracted from this analysis, resulting in all electricity-related emissions being attributed to Scope 2.⁸ The remaining two components are largely independent of operational load and grid conditions

as their magnitude is primarily determined by system design parameters and local grid reliability. To quantify their emissions, we follow the methodology of Schneider Electric (2023)⁹, who estimate Scope 1 at 0.2–0.5% of total lifecycle emissions for a representative 1 MW enterprise data center. Applied to approximately 89 GW of current global installed capacity, this yields an estimate of 1.8 MtCO₂/year, equivalent to only 0.6% of total data center emissions in 2025 (Figure 3).

More than 70% of global data-center emissions can be attributed to indirect emissions from purchased electricity (Scope 2), making the power sector the single most important lever for decarbonizing digital infrastructure in the near term. To quantify these emissions, country-level electricity demand is combined with grid-specific carbon intensity factors, capturing the emissions associated with each kilowatt-hour consumed. As a result, the carbon footprint of an otherwise identical data center can differ substantially across locations depending on the electricity sources supplying the grid. Facilities operating in coal- and gas-intensive power systems therefore face significantly higher emissions than those located in regions with extensive renewable, hydro or nuclear generation.

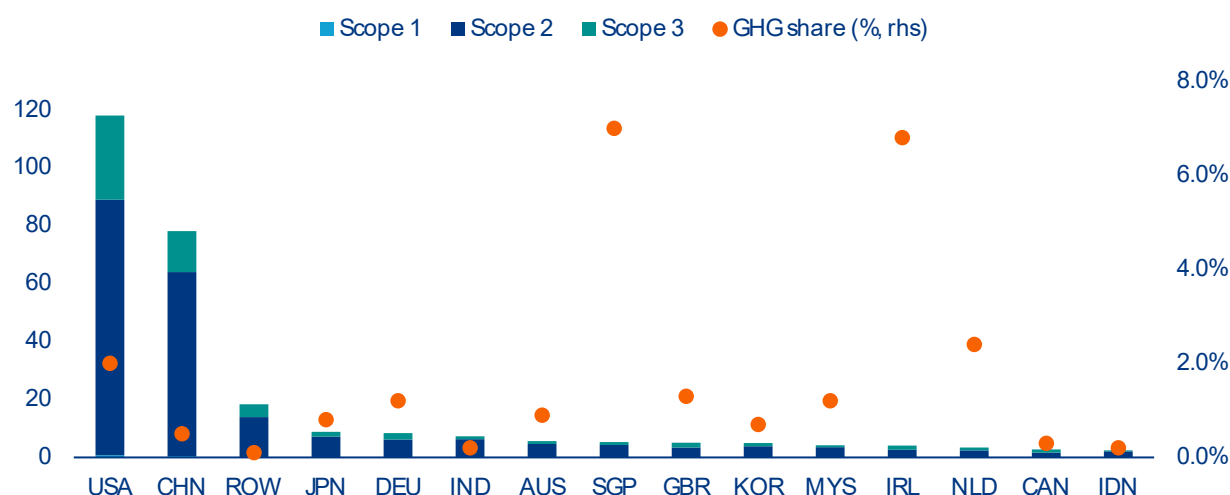
⁸ This implicitly assumes that BTM generation carries a similar carbon intensity as the local grid average. While this may lead to over- or underestimation of emission results in theory, the practical error is likely limited given that BTM capacity of approximately 2 GW currently represents roughly 2% of total global data centre capacity (Cleanview)

⁹ Schneider Electric (2023)

Globally, Scope 2 emissions are estimated at approximately 218 MtCO₂ in 2025. These emissions are highly concentrated geographically, with the US and China accounting for around 88 MtCO₂ and 64 MtCO₂ respectively, together representing roughly 70% of the global total. The European countries covered in this analysis host around 15% of global data center capacity but generate only 8% of Scope 2 emissions, reflecting the comparatively low carbon intensity of their electricity

systems. Conversely, Asia-Pacific countries account for roughly one-third of global capacity but around 44% of emissions, indicating a significantly more carbon-intensive power supply. These differences highlight that the future climate impact of data centers will depend not only on the growth of computing demand, but equally on the pace of power sector decarbonization in the regions where new capacity is deployed.

Figure 3: Data-center emissions for top 15 global hubs by emission scope (MtCO₂, left) and share in regions total CO₂ emissions (% , right)¹⁰



Sources: Allianz Research based on IEA, Schneider Electric, EMBER and LBNL

Scope 3 emissions capture the upstream carbon footprint embedded in the construction and operation of data centers. Unlike Scope 1 and Scope 2 emissions, which are linked directly to on-site fuel use and electricity consumption, Scope 3 spans a broad value chain that includes building construction, servers and networking equipment, cooling systems, electrical infrastructure, transport, waste and other supporting activities. Among these categories, embodied emissions from capital goods are by far the most material, reflecting the carbon-intensive production of semiconductors, servers and facility infrastructure.

Applying the Schneider Electric (2023) lifecycle framework to estimated country-level data center capacity yields Scope 3 emissions of approximately 66 MtCO₂ in 2025, equivalent to around 23% of the sector's total carbon footprint. The estimate combines embodied emissions from IT equipment, building

construction („core and shell“), transformers, backup generators, chillers and other supporting infrastructure, scaled to the installed capacity of each market. Capital goods and upstream fuel-cycle emissions together account for more than 84% of total Scope 3 emissions. While these emissions remain smaller than those associated with electricity consumption today, their relative importance is likely to increase as power systems decarbonize and the deployment of AI infrastructure drives demand for increasingly hardware-intensive computing systems.

Global data-center emissions are estimated at approximately 286 MtCO₂ (0.5% of total global emissions) in 2025. This is around 57% higher than the latest IEA assessments, reflecting the inclusion of Scope 3 emissions, updated capacity estimates and the explicit accounting of system components such as transmission losses. Despite this upward revision, data centers

¹⁰ For current country level emission estimates, see Table A2 in the Appendix

account for only around 0.5% of global greenhouse gas emissions in 2025. At the national level, their contribution generally remains below 2%, although significantly higher shares are observed in concentrated hubs such as Singapore (around 7%) and Ireland, where data-center electricity demand represents a materially larger share of domestic power consumption.

Artificial intelligence is emerging as a rapidly growing driver of electricity demand and associated emissions. Based on an estimated 15–20% share of global data center electricity consumption attributable to AI workloads, AI-related emissions are estimated at approximately 43–60 MtCO₂ in 2025, broadly consistent with recent literature (De Vries & Gao, 2025). Within AI-related emissions, inference accounts for around 80%, reflecting the rapid scaling of deployed AI applications and the increasing integration of model use into search, productivity tools, and enterprise systems. These estimates highlight an important structural dynamic: future emissions trajectories will depend less on incremental efficiency improvements within data centers and more on the pace of electricity system decarbonization and the geographic location of new capacity. In the absence of sufficiently rapid clean power expansion, continued growth could reinforce fossil-based generation in certain regions, creating potential lock-in effects that delay broader decarbonization trajectories.





Can AI growth be decoupled from emissions?

The future emissions trajectory of data centers will depend critically on whether power-system decarbonization can keep pace with rapidly rising electricity demand. To assess this, we model four scenarios that differ in their assumptions about future grid carbon intensity. A no-improvement scenario assumes emissions factors remain at 2025 levels, while a linear trend scenario extrapolates historical decarbonization rates observed over the past decade. These are complemented by two NGFS-based pathways: a below-2°C transition scenario and a scenario aligned with current Nationally Determined Contributions (NDCs). Across all scenarios, electricity demand growth from data centers is held constant, isolating the effect of power-sector decarbonization on emissions outcomes.

In addition to grid decarbonization, assumptions on power usage effectiveness (PUE) and utilization rates are allowed to evolve over time but are kept constant across scenarios. While improvements in cooling efficiency, workload optimization and next-generation chip design are expected to reduce energy intensity at the margin, these effects are highly heterogeneous across facility types and vendors and are relatively small compared to changes in Scope 2 emissions over the 2025–2030 horizon. For this reason, they are not varied across scenarios.

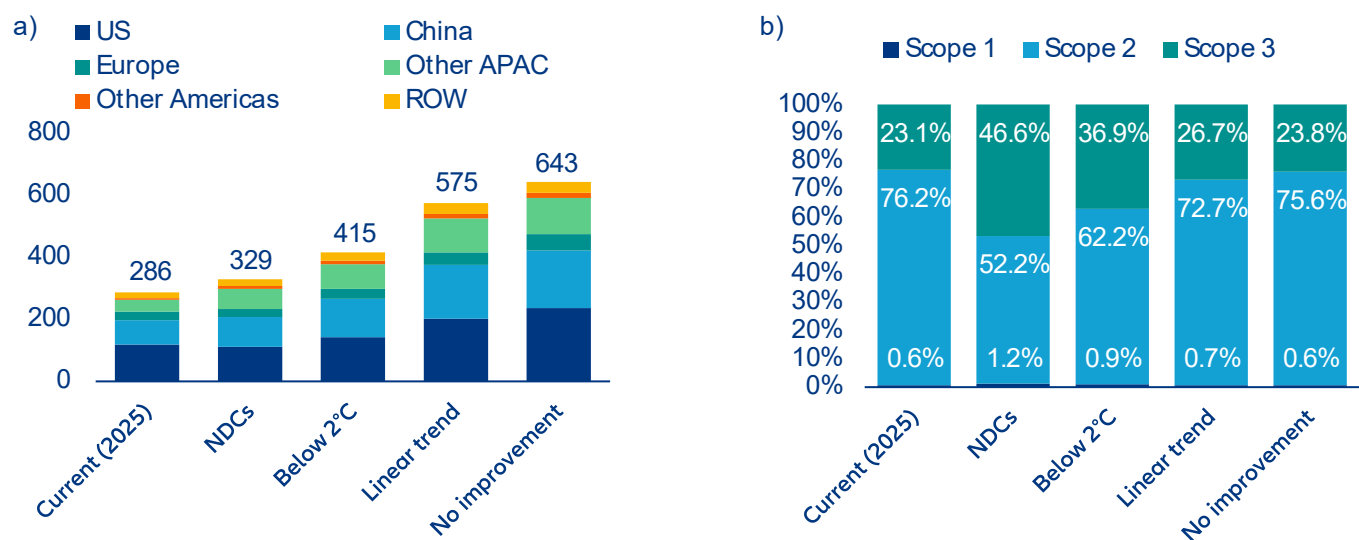
Under these assumptions, global data-center emissions are projected to diverge significantly across scenarios. They range from approximately 643 MtCO₂ in the no-improvement case to around 329 MtCO₂ under the most ambitious decarbonization pathway by 2030 (the NGFS NDC scenario) (Figure 4 a). This wide range highlights that the emissions trajectory of digital infrastructure is driven less by uncertainty in demand growth than by differences in the pace of power sector decarbonization. Achieving the lower bound of this range requires a rapid reduction in grid carbon intensity, equivalent to an average decline of around 54% over the next five years. This would imply particularly steep reductions of 70–80% in major data-center markets such as the US and parts of Europe. Given current trends in electricity demand growth, continued expansion of gas-fired generation capacity, and grid integration constraints in several regions, such a trajectory appears highly challenging under present policy and investment conditions.

More moderate decarbonization pathways yield more incremental but still material improvements.

Under a continuation of historical trends in power sector emissions intensity, global data-center emissions would still rise substantially, but by around 70 MtCO₂ less than in the no-improvement case. This reflects an average reduction in grid carbon intensity of approximately -26% over the same period, underscoring that even partial decarbonization of electricity systems can significantly moderate the emissions impact of rapid data-center expansion.

A more ambitious 2°C-aligned power sector transition would further reduce global data-center emissions by approximately 160 MtCO₂ relative to the no-improvement case. Under this pathway, several economies with more moderate data center expansion profiles, including Germany, the Netherlands and Ireland, could even experience absolute declines in emissions compared to current levels, reflecting the combined effect of cleaner electricity systems and relatively contained demand growth.

Figure 4: Data-center emissions to 2030: a) absolute emissions by region (MtCO₂) and b) composition by emission scope (%), across decarbonization scenarios¹³



Sources: Allianz Research based on IEA, NGFS, Schneider Electric, EMBER and LBNL

The implications for national decarbonization targets are therefore highly uneven. In countries such as Brazil, China, Poland and Italy, data center emissions remain below 2% of total national emissions across all scenarios, suggesting limited implications for overall economy-wide mitigation pathways. By contrast, in high-growth and high-exposure markets such as the US, Ireland, Singapore and Malaysia, the choice of decarbonization pathway becomes materially relevant, with scenario outcomes diverging by around 2–5pps of national emissions. For example, in Malaysia, the data-center

emission share in total national emissions ranges from roughly 7% to 12% depending on the decarbonization pathway, up from 1.2% today. This highlights that data center emissions are becoming more material in national emissions profiles in a subset of countries, and that the choice of decarbonization pathway can have a meaningful impact on their overall emissions outcomes.

Beyond differences in aggregate emissions levels, decarbonization pathways also fundamentally reshape the structure of the data-center carbon footprint. In

¹³ For more detailed country level emission projections, see Table A3 in the Appendix

today's system, emissions are dominated by electricity consumption, with Scope 2 accounting for approximately three-quarters of total emissions (Figure 4 b). However, as power systems decarbonize, this dominance gradually weakens even in scenarios where total emissions remain broadly stable or only moderately decline. Under an ambitious NDC-aligned pathway, the share of Scope 2 emissions declines to around 52% by 2030, reflecting the decarbonization of electricity supply across major data center markets. At the same time, Scope 3 emissions, which are largely driven by embodied carbon in servers, semiconductors and construction materials, increase in relative importance from around 23% today to almost half of total emissions.

The effect is not uniform across countries. In already low-carbon electricity systems such as Scandinavia, Scope 2 emissions account for only 16–28% of the total footprint today, limiting the potential for further reductions through further grid decarbonization. In these systems, mitigation increasingly shifts toward supply-chain and hardware-related emissions, with Scope 3 becoming the main remaining lever. By contrast, in higher-carbon systems such as Germany or Italy, electricity decarbonization still delivers substantial absolute reductions in operational emissions and remains the dominant near-term mitigation priority. However, even in these countries, progress in grid decarbonization alone is insufficient to offset the rising importance of Scope 3 emissions as capacity expands. Overall, this implies that the balance of mitigation priorities differs systematically by stage of power system decarbonization: electricity supply dominates in less advanced systems, while embodied emissions become increasingly binding in mature low-carbon systems.

The environmental implications of data center expansion can also be expressed in monetary terms through the social cost of carbon, which estimates the economic damages associated with an additional ton of CO₂ emissions. Applying a range of carbon-cost estimates from the literature to current emissions levels suggests that the annual climate-related cost of global data-center emissions amounts to between USD23bn and USD141bn in 2025 (Table 1). Using the mid-range estimate proposed by Renner (2022), this corresponds to an annual climate externality of approximately USD68bn.

Looking ahead, these external costs are expected to increase further as data-center demand expands, although the magnitude of the increase depends heavily on the pace of power sector decarbonization.

Across the scenarios considered, annual climate costs rise by between 15% and 125% by 2030. In the absence of further improvements in grid carbon intensity, annual climate damages would increase to approximately USD154bn. By contrast, under the NGFS NDC pathway, costs rise only modestly to around USD79bn despite continued growth in electricity demand. While achieving this trajectory globally appears challenging, the comparison illustrates the extent to which power sector decarbonization influences the environmental cost of digital infrastructure.

¹⁴ [Understanding climate damages: Consumption versus investment - ScienceDirect](#)

Table 1: Estimated social cost of data center emissions globally (in USD bn; 2026)¹⁴

Scenario	Nordhaus (2023) SCC: ~USD81/tCO ₂	EU ETS (2026) SCC: USD92/tCo ₂	Renner (2022) SCC: USD239/tCO ₂	EPA high (2023) SCC: USD492/tCO ₂
Current	23.2	26.3	68.4	140.8
No improvement	52.1	59.1	153.6	316.2
Linear trends	46.6	52.9	137.3	282.7
Below 2°C	33.6	38.2	99.2	204.2
NDCs	26.6	30.2	78.6	161.7

Sources: Allianz Research, Nordhaus (2023), Renner (2022), EBA (2023), LSEG Workspace; Note: Carbon prices from the respective primary data sources have been adjusted to current 2026 values

Artificial intelligence represents an increasingly

important share of these externalities. Assuming AI-related workloads account for around 20% of data-center electricity demand today and increase to approximately 40% by 2030, the climate cost attributable to AI rises from an estimated USD13bn currently to roughly USD54bn by 2030 under the linear trend scenario, assuming a carbon cost of USD239/tCO₂. While still a minority of total data-center emissions today, AI-related workloads are likely to become one of the fastest-growing sources of climate externalities within the digital economy. As AI deployment scales, the environmental challenge therefore shifts from managing a niche source of emissions to ensuring that rapidly growing digital demand remains aligned with broader climate objectives. Achieving this will require continued decarbonization of electricity systems, stronger incentives to improve the energy and material efficiency of AI infrastructure and a more systematic internalization of the environmental costs associated with digital growth.

¹⁴ For country level social cost of carbon estimates of current emissions, see Table A4 in the Appendix



Beyond the data center: AI as a decarbonization tool

The environmental impact of AI hinges not only on its own footprint but also on its potential to accelerate decarbonization across the wider economy. While growing attention has focused on the energy, emissions and water requirements of AI infrastructure, the technology could also contribute to reducing resource consumption and emissions in sectors such as industry, buildings, transport and energy. The International Energy Agency (IEA) identifies a broad range of applications through which AI could support decarbonization, including methane leak detection, optimization of industrial processes and energy consumption, predictive maintenance, more efficient operation of power plants and electricity grids and improvements in transport and logistics.¹⁵ Under a scenario of widespread deployment, the IEA estimates that AI-enabled applications could reduce global final energy consumption by up to 13 EJ by 2035 and lower annual CO₂ emissions by around 1.4 Gt. This would more than offset even our high-end estimate of AI-related emissions growth of 643 MtCO₂, implying a net reduction of approximately 750 MtCO₂. Put differently, AI-enabled emissions savings could exceed the technology's own emissions footprint by a factor of more than two.

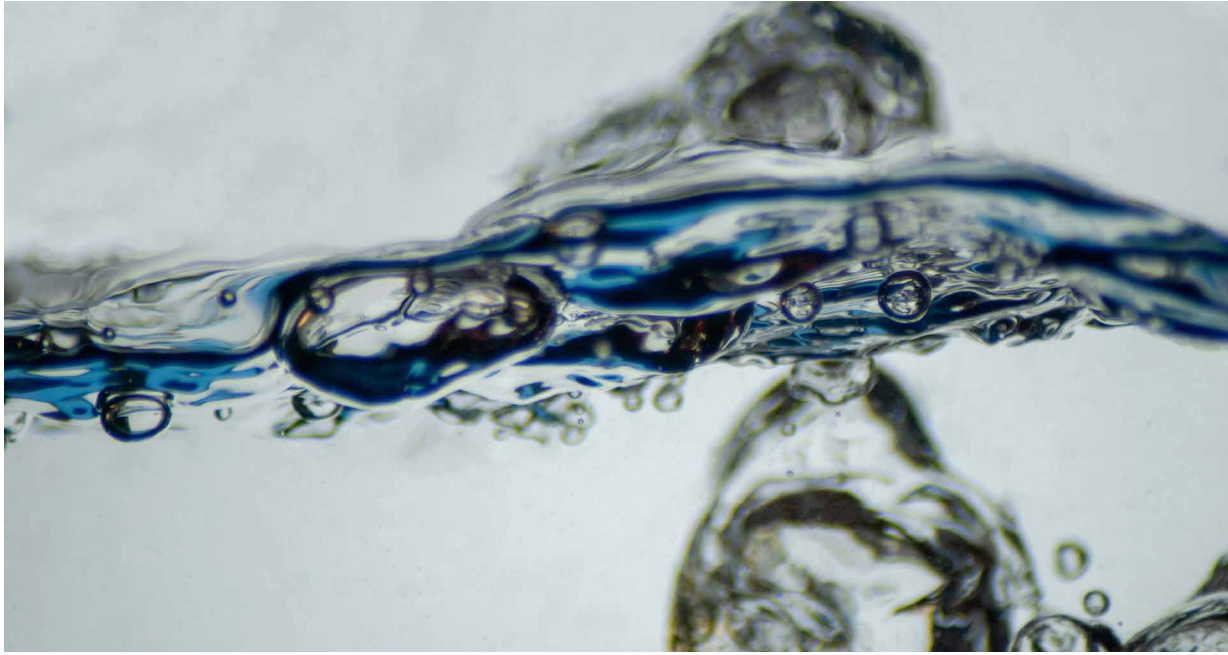
Emerging evidence from the corporate sector suggests that some of these benefits may already be materializing. According to recent Bearingpoint research, around two-thirds of technology leaders expect AI and digital technologies to reduce business emissions by between 6% and 30% through operational optimization, improved resource efficiency and lower waste generation.¹⁶ At the same time, however, most organizations remain in the early stages of adoption, with AI deployment often confined to isolated use cases rather than enterprise-wide transformation programs.

As a result, the gap between technical potential and realized environmental benefits remains considerable.

The IEA notes that there is currently insufficient momentum to ensure widespread adoption of many AI-enabled decarbonization applications. Consequently, the environmental case for AI depends not only on improvements in data-center efficiency but also on the successful diffusion of productivity-enhancing and emissions-reducing applications throughout the real economy. While AI's direct footprint is becoming increasingly visible, its ultimate environmental impact will be determined by the balance between these costs and the potentially much larger, but still uncertain, efficiency gains it can unlock elsewhere in the economy.

¹⁵ IEA

¹⁶ WEF



The blue cost: water consumption in data centers

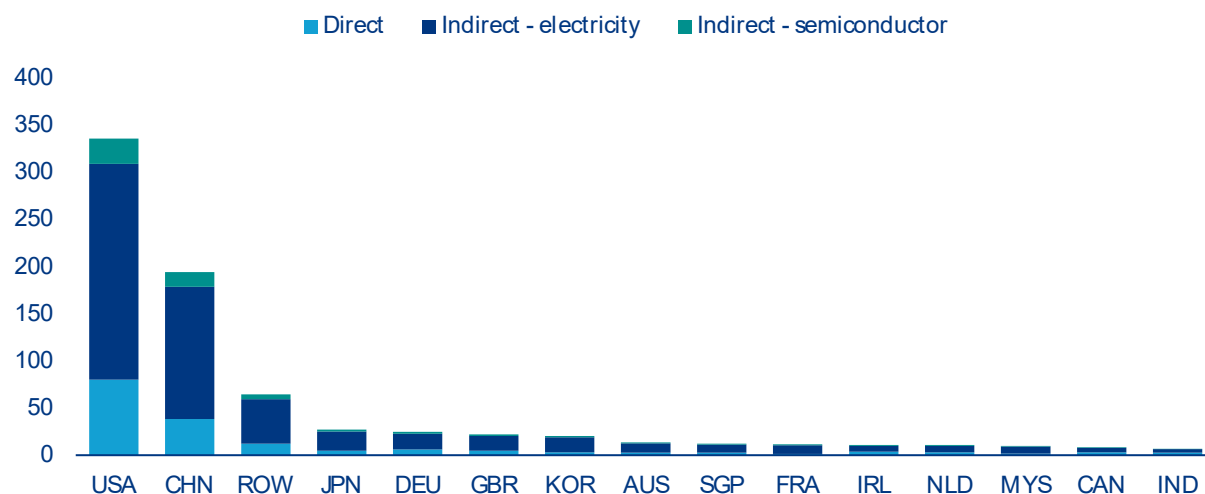
Beyond emissions, the primary environmental footprint of data centers stems from water consumption. This footprint can be decomposed into three components: direct on-site consumption for cooling and humidification, indirect consumption associated with electricity generation and upstream consumption embedded in semiconductor manufacturing. The total water footprint is therefore shaped both by the efficiency of on-site systems and by the water intensity of the electricity system supplying the facility.

Direct on-site cooling is the most visible component of data-center water consumption. Data centers generate substantial heat through continuous server operation and must remove it to avoid hardware failure, with water-based cooling systems being the dominant solution at scale. Water use depends on cooling system efficiency, measured by Water Usage Effectiveness (WUE), defined as liters of water consumed per

kilowatt-hour of IT energy use. WUE varies across facility types, averaging around 0.32 l/kWh in smaller facilities and 0.62 l/kWh in mid-sized and colocation data centers. Given the absence of consistent country-level reporting, a uniform value of 0.48 l/kWh is applied, based on European Commission and LBNL estimates.¹⁷

Applying these parameters to country-level IT load data, global data centers consumed approximately 180bn liters of water on-site in the past year (Figure 5). The US accounts for around 44% of this total and China for a further 21%, reflecting the concentration of global capacity in these markets. This figure could more than triple by 2030 under projected capacity expansion, unless substantial improvements in WUE are achieved. While direct cooling accounts for around 22% of total water consumption on average, its share varies significantly across countries, from 12% in France to 87% in Norway, highlighting the dominant role of indirect water use in many electricity systems.

¹⁷ European Commission and LBNL.

Figure 5: Data-center water consumption by country (current, bn liters)¹⁸

Source: Allianz Research, Jin et al (2019), IEA

Indirect water consumption from electricity constitutes the lion's share of total data-center water use, yet remains the least visible part of the footprint.

Much like Scope 2 carbon emissions, this consumption occurs off-site at power plants but is nonetheless attributable to the data center through its electricity demand, as most forms of generation require water for cooling during the production process. To quantify this, we apply grid water intensity factors expressed in liters per kilowatt-hour, capturing the weighted average water consumption of each country's electricity mix.

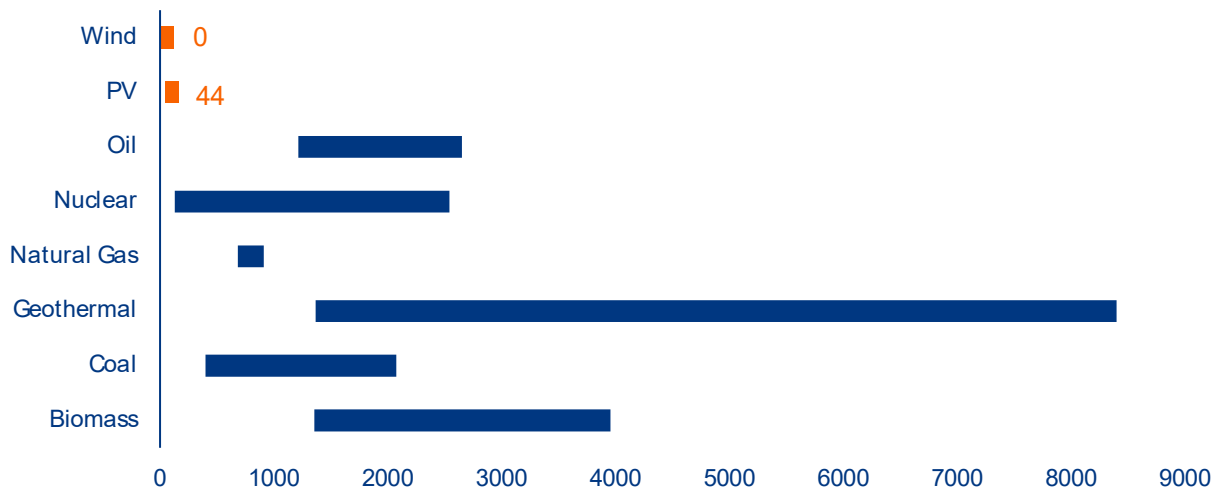
As with carbon emissions, grid water intensity varies substantially across generation technologies.

Renewable sources generally have the lowest operational water requirements, with wind power exhibiting effectively no on-site water use, making renewable penetration a key determinant of indirect water intensity (Figure 6).¹⁹ However, the relationship between carbon and water intensity is not one-to-one. Nuclear power, while carbon-free in operation, is

relatively water-intensive due to cooling requirements in reactor systems. As a result, countries with nuclear-heavy grids such as France may exhibit low carbon intensity but comparatively high grid water intensity, helping to explain part of the gap between France and Norway in indirect water consumption (around 80% versus 5% of total data-center water use, respectively). The other main driver of this difference is methodological: reservoir evaporation losses from hydropower are excluded due to substantial uncertainty in their estimation. This is particularly relevant for Norway, where hydropower accounts for around 89% of the electricity mix, implying that indirect water consumption in highly hydro-dependent systems such as Norway, Brazil and Canada may be understated.

¹⁸ Country level estimates of the current water consumption in data centers can also be found in Table A5 in the Appendix.

¹⁹ In Life-cycle assessments wind power has a median water footprint of 43l/MWh, associated with turbine production, but this is abstracted from in this analysis

Figure 6: Operational water consumption by power generation technology (l/MWh)

Source: Jin et al (2019)

Despite this caveat, indirect water consumption from electricity supply is estimated at more than 560bn liters per year, making it roughly three times larger than direct water consumption associated with on-site cooling. However, this component is expected to grow more slowly than electricity demand, increasing by around 35–61% to 2030 as power systems continue to decarbonize and shift towards lower water-intensity renewable generation, thereby reducing the average grid water intensity over time.

The second component stems from semiconductor and microchip manufacturing, which following the IEA's sectoral breakdown is assumed to account for roughly 8% of total data center water consumption. In our assessment this translates into around 65bn liters annually, reflecting water use across chip fabrication processes, particularly ultra-pure water required for wafer cleaning and rinsing, which represents the dominant share of manufacturing-related water demand. This component is expected to grow in importance as demand for high-performance and AI-accelerated chips increases.

In total, our estimate puts data-center water consumption today at around 814bn liters per year, equivalent to the annual domestic drinking-water use of approximately 17mn people in Germany.²⁰ This is around 45% higher than the IEA estimate of 560bn liters for 2023, reflecting updated capacity figures and

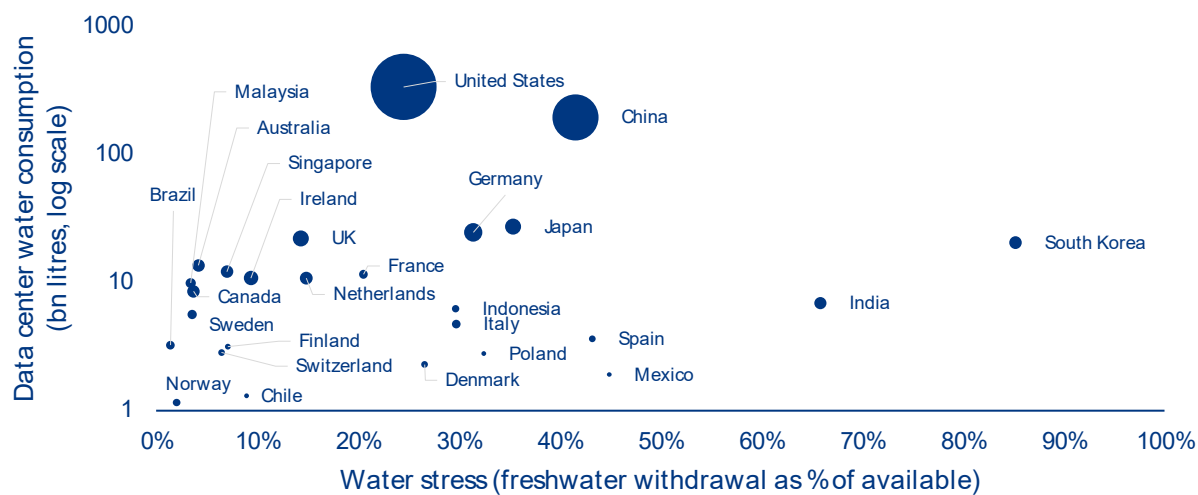
differences in underlying assumptions across direct and indirect components. Looking ahead, total water consumption is projected to increase to between 1,286bn liters under a decarbonization pathway aligned with national targets (NDC scenario) and 1,791bn liters under a scenario in which today's electricity mix is held constant. This range highlights the extent to which power sector decarbonization can simultaneously reduce both carbon emissions and the water intensity of data center operations. At the upper end of this range, global data-center water consumption would exceed that of more than 80 countries and be comparable to the annual water use of Switzerland, underscoring the growing importance of water as a constraint in digital infrastructure expansion.²¹

Besides the sheer amount of water consumed, the location of withdrawals is also critical. In water-scarce regions, additional demand can intensify competition between households, agriculture and industry, while also placing pressure on local ecosystems and biodiversity. These trade-offs can be illustrated by comparing data-center water consumption with national water stress levels (Figure 7). Most countries in our sample exhibit low to moderate water stress, with freshwater withdrawals accounting for less than 30% of available resources. However, South Korea, India, Mexico, Spain and China experience considerably higher stress levels, exceeding 40%, with South Korea reaching as high as 85%. While this country-level view does not capture local conditions

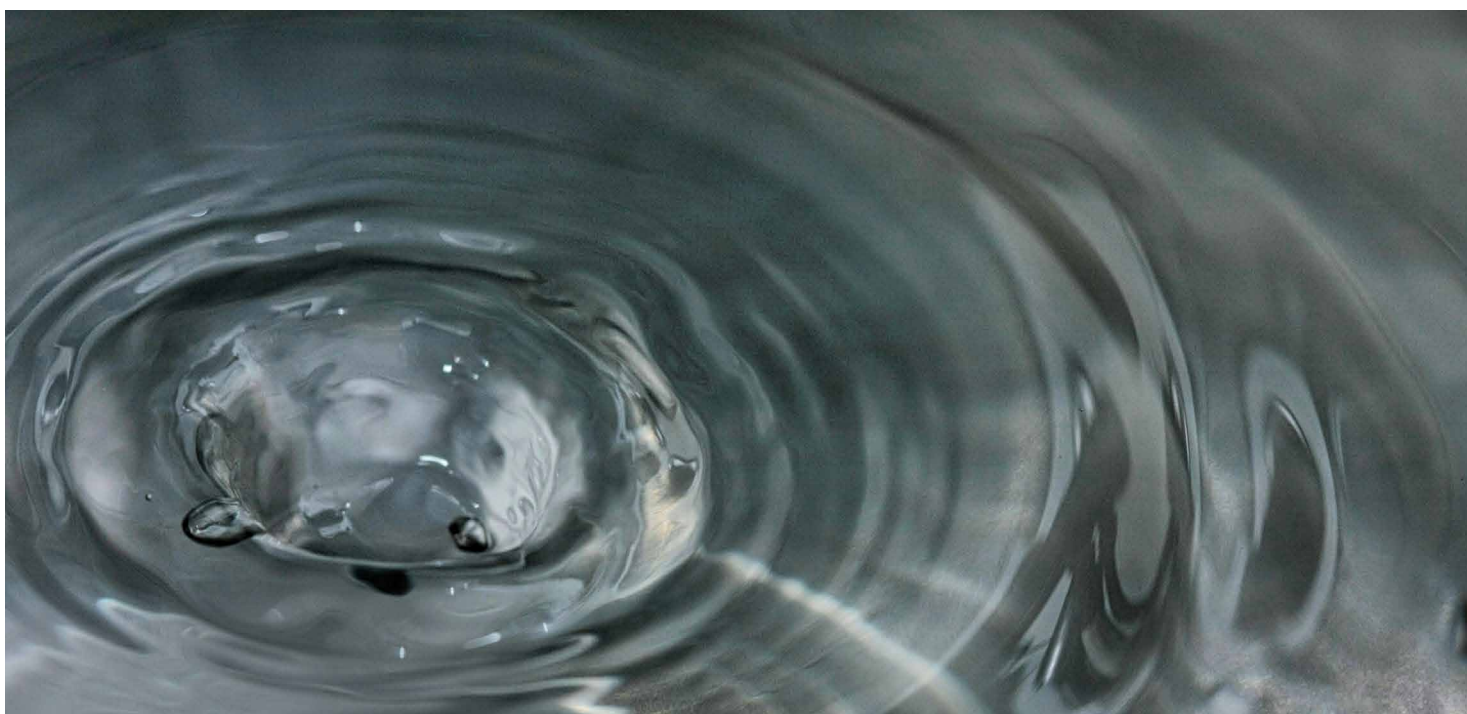
²⁰ This estimate is based on UBA assuming a daily water consumption of 126l/person²¹ World Population Review

at individual data-center sites, it underscores the importance of accounting for water consumption impacts in already water-stressed regions when planning data-center deployment.

Figure 7: Data-center water consumption (bn liters) vs water stress level (%; 2023)



Sources : Allianz Research, FAO Aquastat



What do we need to realize green AI?

The environmental footprint of AI is shaped less by improvements in model or data-center efficiency alone than by the broader energy and infrastructure systems in which it is embedded. Across emissions and water use, renewables-led power system decarbonization emerges as the dominant determinant of future outcomes, while embodied emissions and supply-chain impacts gain importance as electricity systems become cleaner.

At the same time, a gap remains between private investment decisions and the full societal costs of digital infrastructure. While operators face direct costs for electricity, hardware and construction, a significant share of climate and water impacts is not fully reflected in market signals. Closing this gap will require cleaner electricity systems, better information on resource use and stronger alignment between private incentives and environmental costs.

Key policy priorities are as follows:

- **Greening the grid remains the central priority.**
Expanding renewable generation is the most effective lever for reducing both the carbon and water footprint of AI-driven data center growth. Grid expansion, faster permitting and investment in storage and flexibility are essential to ensure that rising electricity demand can be met with low-emission supply.
- **Improve transparency and environmental reporting.**
Standardized disclosure of electricity consumption, emissions, water use, utilization rates and AI-specific workload intensity is needed to improve comparability across operators and geographies. This would enable more consistent benchmarking and better policy targeting of high-impact segments.

- **Internalize climate and water externalities.** Current investment decisions do not fully reflect the social costs of emissions and water consumption. Strengthening carbon pricing, integrating environmental costs into electricity market design, and developing pricing mechanisms for water use in stressed regions would help align private incentives with societal costs. For AI-related applications, usage-based environmental charges (for example per token or compute use) could in principle help reflect their environmental footprint more directly, but would require better transparency and reliable measurement of energy use and emissions.

- **Address embodied emissions across the AI supply chain.**
As electricity systems decarbonize, upstream emissions from semiconductors, servers and construction materials will become increasingly important. Lifecycle disclosure requirements, low-carbon procurement standards and supply-chain decarbonization policies will be key to managing this shift.

- **Maximize AI-enabled decarbonization opportunities.**
AI applications in energy systems, industry, transport and buildings offer substantial potential for emissions reductions. Policy support should focus on accelerating deployment in high-impact areas such as grid optimization, industrial efficiency and logistics.

Appendix

Table A1: Data-center capacity and power demand by country (current & 2030)

Region	Country	IT capacity (GW)		PUE		Capacity Factor		Power Demand (TWh)	
		Current	2030	Current	2030	Current	2030	Current	2030
North America	United States	39.0	81.0	1.30	1.23	0.49	0.49	217.6	427.7
	Canada	1.4	4.0	1.30	1.24	0.49	0.49	7.9	21.0
	Mexico	0.2	1.2	1.30	1.24	0.49	0.49	1.2	6.5
Europe	Germany	3.0	4.8	1.39	1.29	0.48	0.48	17.5	26.0
	France	0.7	1.8	1.55	1.29	0.48	0.48	4.6	9.7
	Spain	0.4	2.5	1.66	1.29	0.48	0.48	3.1	13.8
	UK	2.3	4.8	1.42	1.29	0.48	0.48	13.8	26.0
	Italy	0.7	2.0	1.46	1.29	0.48	0.48	4.2	11.0
	Netherlands	1.5	2.2	1.39	1.29	0.48	0.48	8.8	11.7
	Poland	0.2	0.8	1.55	1.29	0.48	0.48	1.4	4.4
	Denmark	0.4	1.0	1.21	1.21	0.48	0.48	2.0	5.0
	Finland	0.3	1.2	1.17	1.17	0.48	0.48	1.6	5.7
	Sweden	0.8	1.5	1.17	1.17	0.48	0.48	3.9	7.4
	Norway	0.5	1.2	1.18	1.18	0.48	0.48	2.5	6.0
	Ireland	1.9	2.5	1.18	1.18	0.48	0.48	9.4	12.3
	Switzerland	0.4	0.5	1.37	1.29	0.48	0.48	2.1	2.9
APAC	China	19.0	49.0	1.46	1.35	0.48	0.47	116.6	272.4
	India	1.3	4.2	1.46	1.35	0.48	0.47	8.0	23.3
	Japan	2.3	5.3	1.46	1.35	0.48	0.47	14.1	29.4
	South Korea	1.4	2.5	1.46	1.35	0.48	0.47	8.8	13.8
	Australia	1.4	3.1	1.46	1.35	0.48	0.47	8.3	17.2
	Malaysia	0.9	9.6	1.46	1.35	0.48	0.47	5.4	53.1
	Indonesia	0.5	1.6	1.46	1.35	0.48	0.47	2.8	9.1
	Singapore	1.4	1.9	1.46	1.35	0.48	0.47	8.6	10.6
South America	Brazil	0.7	2.4	1.67	1.50	0.43	0.44	4.1	13.6
	Chile	0.2	0.8	1.67	1.50	0.43	0.44	1.2	4.7
ROW	ROW	6.0	12.3	1.38	1.29	0.49	0.48	35.5	66.7
World		88.8	205.6	1.38	1.29	0.49	0.48	515.2	1,110.9

Sources: Allianz Research based on EMBER, IEA and LBNL

Table A2: Estimated data center emissions by country (current)

Region	Country	Emission intensity (GCO ₂ /kWh)	Scope 1 (M tCO ₂)	Scope 2 (M tCO ₂)	Scope 3 (M tCO ₂)	Total Emissions
North America	United States	384.40	0.75	88.35	29.07	118.17
	Canada	190.61	0.03	1.57	1.05	2.65
	Mexico	474.02	0.00	0.64	0.16	0.80
Europe	Germany	329.65	0.06	6.09	2.24	8.39
	France	41.44	0.02	0.21	0.53	0.75
	Spain	153.60	0.01	0.51	0.33	0.85
	UK	217.41	0.05	3.34	1.73	5.11
	Italy	284.78	0.01	1.29	0.51	1.81
	Netherlands	253.56	0.03	2.32	1.12	3.47
	Poland	588.60	0.00	0.91	0.16	1.07
	Denmark	114.41	0.01	0.25	0.30	0.55
	Finland	57.47	0.01	0.10	0.24	0.34
	Sweden	35.26	0.01	0.15	0.60	0.76
	Norway	28.11	0.01	0.07	0.37	0.46
	Ireland	256.38	0.03	2.63	1.41	4.07
	Switzerland	39.22	0.01	0.09	0.27	0.37
APAC	China	526.49	0.40	63.59	14.16	78.15
	India	670.13	0.03	6.23	0.97	7.23
	Japan	477.26	0.05	7.09	1.71	8.85
	South Korea	417.06	0.03	3.81	1.07	4.92
	Australia	525.18	0.03	4.57	1.01	5.60
	Malaysia	601.97	0.02	3.47	0.65	4.14
	Indonesia	680.25	0.01	2.04	0.34	2.39
	Singapore	497.09	0.03	4.28	1.04	5.36
South America	Brazil	109.95	0.01	0.53	0.55	1.10
	Chile	289.49	0.00	0.38	0.15	0.53
ROW	ROW	363.86	0.12	13.77	4.46	18.36
World		363.86	1.78	218.24	66.21	286.23

Sources: Allianz Research based on IEA, Schneider Electric, EMBER and LBNL

Table A3: Estimated data center emissions by country by 2030 scenario (MtCO2)

Region	Country	No improvement	Linear Trends	Below 2°C	NDCs
North America	United States	235.47	202.15	142.33	111.24
	Canada	7.20	6.90	4.46	3.79
	Mexico	4.40	4.04	2.49	1.85
Europe	Germany	12.71	10.18	6.24	4.90
	France	1.79	1.76	1.79	1.62
	Spain	4.24	2.56	2.82	2.36
	UK	9.95	7.32	5.52	4.55
	Italy	4.92	4.39	2.54	2.03
	Netherlands	4.73	2.36	2.84	2.22
	Poland	3.38	2.94	1.48	1.03
	Denmark	1.37	0.86	1.04	0.89
	Finland	1.23	0.96	1.20	1.04
	Sweden	1.42	1.40	1.42	1.34
	Norway	1.10	1.08	1.10	1.10
	Ireland	5.31	4.02	2.88	2.36
	Switzerland	0.52	0.51	0.52	0.52
APAC	China	185.94	172.85	123.83	95.63
	India	21.44	20.95	13.57	10.49
	Japan	18.83	17.27	13.90	11.33
	South Korea	7.85	7.11	5.91	4.98
	Australia	11.87	9.85	9.23	7.64
	Malaysia	41.65	39.55	27.52	22.85
	Indonesia	7.84	7.72	5.05	4.17
	Singapore	6.71	6.69	4.48	3.78
South America	Brazil	3.56	2.83	2.91	2.46
	Chile	2.05	1.41	1.53	1.16
ROW	ROW	35.26	34.96	26.44	21.38
World		642.70	574.60	415.00	328.70

Sources: Allianz Research based on IEA, NGFS, Schneider Electric, EMBER and LBNL

Table A4: Estimated current social cost of data center emissions by country (in USD bn; 2026)

Region	Country	Nordhaus (2023) SCC: ~USD81/tCO ₂	EU ETS (2026) SCC: USD92/tCo ₂	Renner (2022) SCC: USD239/tCO ₂	EPA high (2023) SCC: USD492/tCO ₂
North America	United States	9.57	10.87	28.24	58.14
	Canada	0.21	0.24	0.63	1.30
	Mexico	0.07	0.07	0.19	0.39
Europe	Germany	0.68	0.77	2.00	4.13
	France	0.06	0.07	0.18	0.37
	Spain	0.07	0.08	0.20	0.42
	UK	0.41	0.47	1.22	2.52
	Italy	0.15	0.17	0.43	0.89
	Netherlands	0.28	0.32	0.83	1.71
	Poland	0.09	0.10	0.26	0.53
	Denmark	0.04	0.05	0.13	0.27
	Finland	0.03	0.03	0.08	0.17
	Sweden	0.06	0.07	0.18	0.37
	Norway	0.04	0.04	0.11	0.22
	Ireland	0.33	0.37	0.97	2.00
	Switzerland	0.03	0.03	0.09	0.18
APAC	China	6.33	7.19	18.68	38.45
	India	0.59	0.66	1.73	3.56
	Japan	0.72	0.81	2.12	4.35
	South Korea	0.40	0.45	1.17	2.42
	Australia	0.45	0.52	1.34	2.76
	Malaysia	0.34	0.38	0.99	2.04
	Indonesia	0.19	0.22	0.57	1.17
	Singapore	0.43	0.49	1.28	2.63
South America	Brazil	0.09	0.10	0.26	0.54
	Chile	0.04	0.05	0.13	0.26
ROW	ROW	1.49	1.69	4.39	9.03
World		23.18	26.33	68.41	140.82

Sources: Allianz Research based on Nordhaus (2023), LSEG Workspace, Renner (2022) and EPA (2023)

Table A5: Data center water consumption by country (current, bn liters)

Region	Country	Direct	Indirect - electricity	Indirect - semiconductor
North America	United States	80.4	229.4	27.06
	Canada	2.9	5.0	0.7
	Mexico	0.4	1.3	0.2
Europe	Germany	6.1	16.7	2.0
	France	1.4	9.3	0.9
	Spain	0.9	2.5	0.3
	UK	4.7	15.7	1.8
	Italy	1.4	3.0	0.4
	Netherlands	3.0	6.9	0.9
	Poland	0.4	2.1	0.2
	Denmark	0.8	1.3	0.2
	Finland	0.7	2.3	0.3
	Sweden	1.6	3.6	0.5
	Norway	1.0	0.1	0.1
	Ireland	3.8	6.2	0.9
	Switzerland	0.7	1.9	0.2
APAC	China	38.3	140.8	15.7
	India	2.6	3.7	0.6
	Japan	4.6	20.5	2.2
	South Korea	2.9	16.0	1.7
	Australia	2.7	9.8	1.1
	Malaysia	1.8	7.4	0.8
	Indonesia	0.9	4.8	0.5
	Singapore	2.8	8.4	1.0
South America	Brazil	1.2	1.8	0.3
	Chile	0.4	0.9	0.1
ROW	ROW	12.3	47.2	5.2
World		180.9	568.2	65.5

Sources:: Allianz Research based on Jin et al (2019) and IEA

A photograph showing a group of diverse hands of various skin tones stacked on top of each other, resting on a thick, textured tree branch. The background is a lush green forest with sunlight filtering through the leaves. The text 'Our team' is overlaid in the center, with 'Our' in white and 'team' in orange.

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Allianz Research encompasses Allianz Group Economic Research and the Economic Research department of Allianz Trade.

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Pension Reform Survey 2026: Everyone knows reform is needed, few expect it to happen



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Appendix

Executive Summary



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- Citizens recognize the need for social security reforms, but have far less confidence that reforms will actually be delivered.** Our survey of more than 8,000 respondents across Austria, France, Germany, Italy, Poland, Spain, the UK and the US explores public attitudes towards social security reform, pension sustainability and retirement preparedness. Recognition of the need for reform is overwhelming, exceeding 80% in most countries. Yet confidence in governments' ability to implement the reforms needed to ensure long-term sustainability is much weaker. The latter ranges from just 36% in Italy and 42% in Germany to 63% in Poland and 66% in the US. Views diverge more sharply when it comes to how the adjustment burden should be shared. Taking greater personal responsibility through additional saving is the most widely accepted option, while higher taxes and social contributions receive little support. Importantly, while respondents differ on how adjustment should be achieved, only a minority rejects all proposed measures outright. The challenge for policymakers is therefore no longer convincing citizens that reform is needed, but demonstrating that reforms can be delivered and designing politically viable reform packages.
- Older generations are not blocking reform.** Recognition of the need for reform increases with age, but trust that governments will successfully implement the necessary reforms declines. While 61% of respondents aged 18 to 34 express confidence that the necessary reforms will be implemented, this falls to 54% among those aged 35 to 49 and to just 44% and 43% among respondents aged 50 to 64 and 65 to 79, respectively. The result is a striking paradox: the older cohorts are most convinced that reform is necessary, while younger respondents attach less urgency to the issue, suggesting that pensions suffer from a salience gap despite their long-term importance. Contrary to common stereotypes, respondents aged 50 to 64 are also among the most willing to accept measures such as working longer, postponing retirement or receiving lower public benefits.
- Responsibility for retirement is increasingly shifting onto individuals, yet many lack the knowledge to prepare for it.** Only half of respondents expect the public pension system to provide the majority of their retirement income. As occupational and private pension provisions gain importance, pension and financial literacy are becoming essential pillars of retirement income adequacy and pension system sustainability. Yet only one in six respondents shows a high level of financial literacy (18%), while one in four scores low (26%). The knowledge gaps have concrete implications: only one-third of respondents correctly identified how long they are likely to spend in retirement, and only half state that they have a clear picture of their expected financial situation in retirement. These findings suggest that the financial literacy gap is also a pension literacy gap.
- More funded pension provision – as envisaged in Germany – would be a win-win for households and the wider economy.** Stronger occupational and private pension pillars would improve retirement security through greater private wealth accumulation while simultaneously providing long-term capital to financial markets. This could support investment in innovation, infrastructure and the green transition. Asset-backed pension arrangements amount to more than 200% of GDP in Denmark, 145% in the US and 78% in the UK, compared with just 6% in Germany and 13% in France. If Austria, France, Germany, Italy, Poland and Spain had reached the UK's level of pension assets relative to GDP in 2025, an additional EUR8.8trn of long-term investment capital would have been available.



Most agree reform is needed – few believe it will be delivered

Ageing populations are putting social security systems under growing pressure. With government spending on social protection already amounting to more than 25% of GDP in many countries and demographic change pushing old-age dependency ratios even higher, public pension and healthcare systems urgently need to be reformed¹. Against this backdrop, we surveyed more than 8,000 respondents across Austria, France, Germany, Italy, Poland, Spain, the UK and the US to explore public attitudes towards social security reform, pension sustainability and retirement preparedness.

How citizens view the need for reform, the likelihood that reforms will be delivered and their own role in the adjustment process matters greatly for reform success. Public support can facilitate reform, while a lack of trust or perceived unfairness can undermine even well-designed policy proposals. The survey therefore examines attitudes towards social security reform and pension sustainability from several perspectives, including perceptions of reform need, trust in implementation, retirement preparedness, pension and financial literacy as well as individual preferences regarding different reform options.

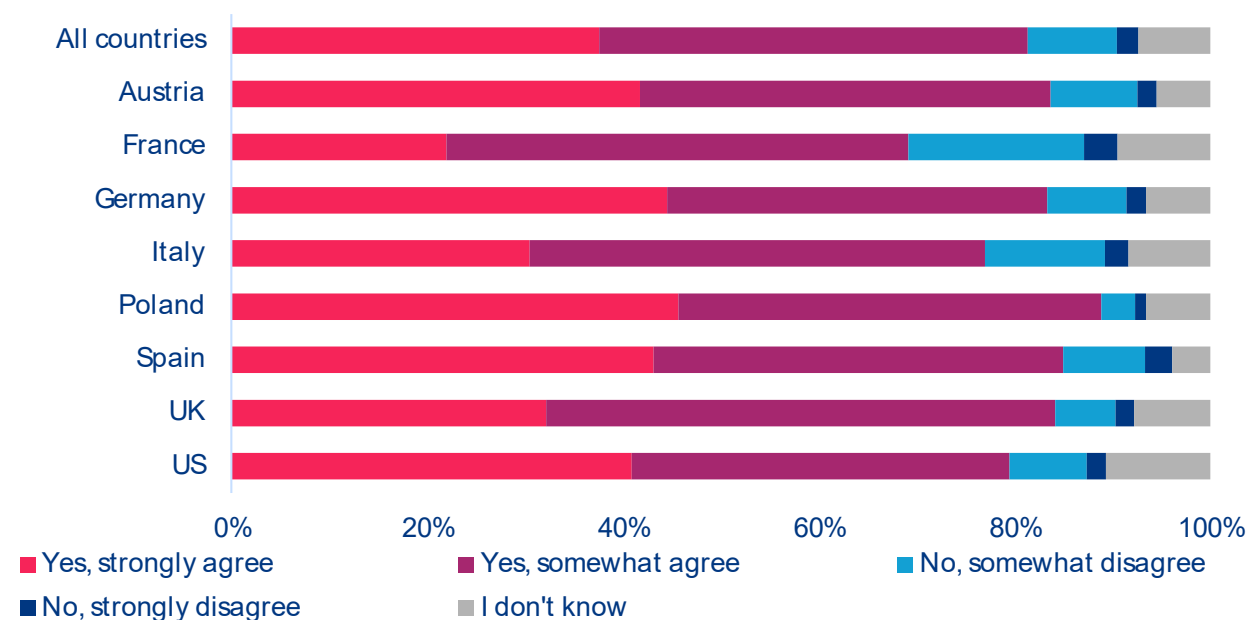
Nearly half of respondents in Poland and Germany recognize the need for reform, compared to only 22% in France. Across all eight countries, more than 80% of respondents agree or strongly agree that social systems must be reformed to remain sustainable. The share of respondents who strongly agree is highest in Poland (46%)

and Germany (45%), followed by Spain (43%), Austria (42%) and the US (41%). In the UK and Italy, around one-third of respondents express a strong need for reform. France is the clear outlier, with only 22% strongly agreeing and 21% either somewhat or strongly disagreeing (Figure 1).

Support for reform is broad across all age groups, although older respondents express slightly stronger conviction. The share of respondents who strongly agree that reform is needed ranges from 36% among young adults aged 18 to 34 to 39% among those aged 35 to 49. The relatively limited sense of urgency among younger respondents may reflect a salience and financial-literacy gap. For many younger people, topics such as pensions and healthcare remain too distant to register as immediate concerns, despite the fact that they will be most affected if governments fail to improve the sustainability of social security systems. The survey also reveals gender differences: Men express stronger conviction that reform is needed than women, with 39% strongly agreeing compared with 36% of women (Figure 2).

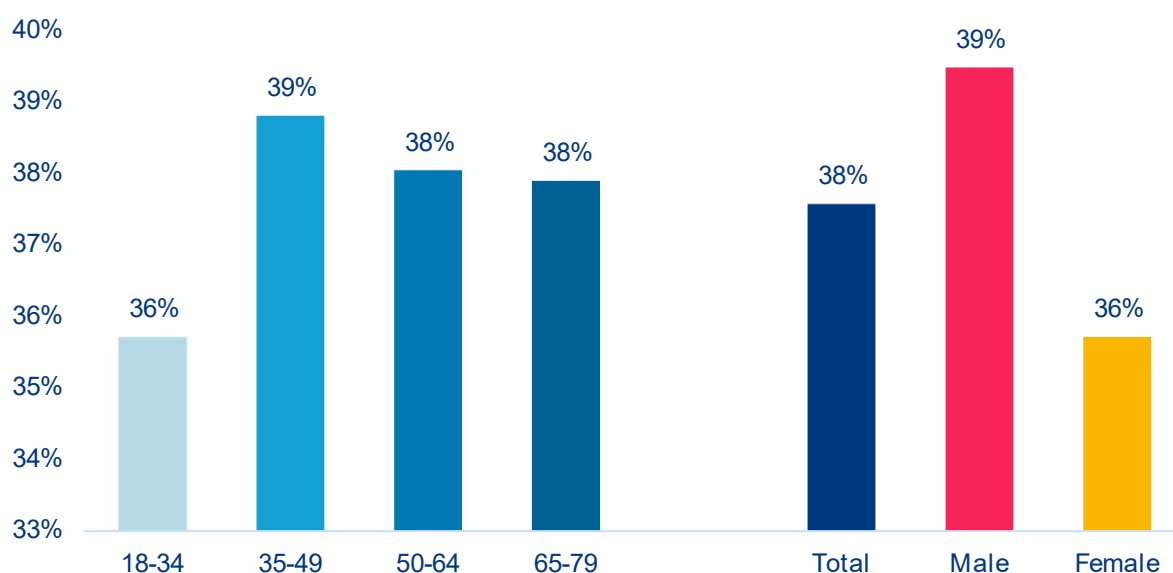
¹ In the eight countries covered by our survey, public social expenditure as a share of GDP in 2024 stood at: Austria 31.6%, France 30.6%, Germany 27.9%, Italy 27.6%, Spain 25.9%, Poland 23.1%, UK 23.0% (2023), US 19.8%. Source: OECD, Public social spending, [Social spending | OECD](#).

Figure 1: To what extent do you agree with the following statement: “Current social systems (e.g., social security, pensions, healthcare) will need to be reformed to remain sustainable.” Share of answers, in percent of all answers, by country



Source: Allianz Research.

Figure 2: To what extent do you agree with the following statement: “Current social systems (e.g., social security, pensions, healthcare) will need to be reformed to remain sustainable.”, share of respondents who “strongly agree”, by age group and gender (in percent of respective population group)



Source: Allianz Research.

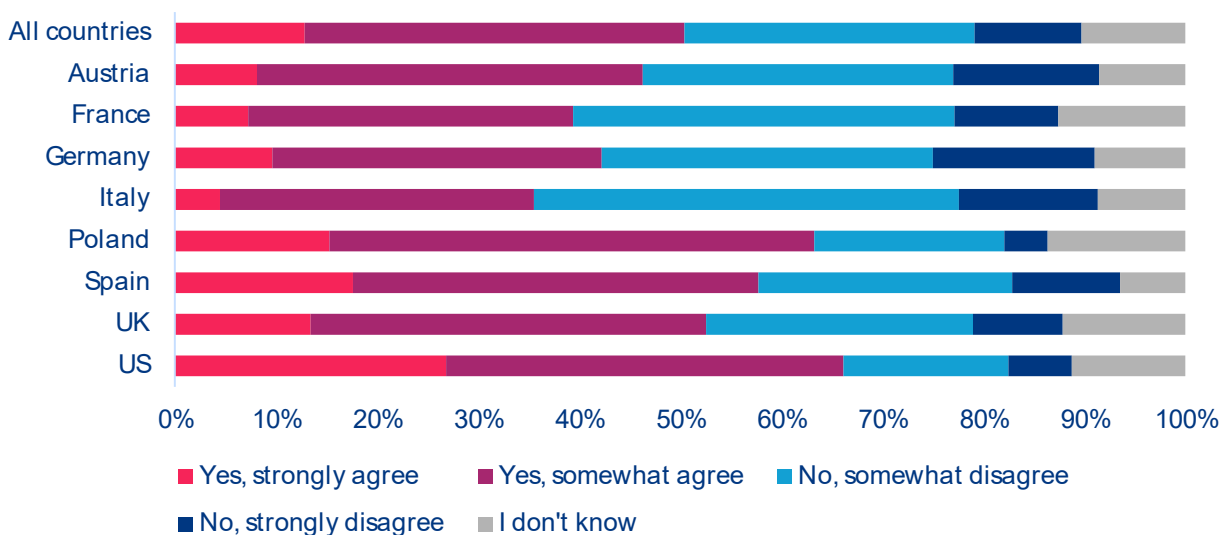
Surprisingly, nearly four in five retirees believe reform is necessary even though they are already drawing benefits from the current system. While the intensity of reform demand varies somewhat across subgroups, support remains broad throughout the sample. It is strongest among university graduates (85%) and lowest, though still substantial, among respondents with lower educational attainment (71%). Employment status shows a similar pattern. Recognition of reform need is highest among the employed (85%) and self-employed (82%), suggesting that those who bear social contributions most directly are particularly aware of the pressures facing the system. The more striking finding, however, comes from retirees. While the level of support is at 78% somewhat lower than among the workforce, nearly four in five still agree that reform is necessary despite already drawing benefits from the current system. Recognition of reform need is therefore not confined to future contributors; it is also shared by many who already depend on the system today.

While more than 80% of respondents in most countries agree that social systems require reform, confidence that governments will follow through on the reforms needed is far lower. Across all eight countries, only around 50% of respondents believe that the necessary reforms will

actually be implemented. Confidence is highest in the US, where around two-thirds of respondents believe reforms will be delivered, and relatively strong in Poland (63%). By contrast, confidence is markedly lower in Germany at 42% and lowest in Italy, where only 36% believe the necessary reforms will be implemented (Figure 3). The challenge therefore appears less one of recognizing the need for reform than of building confidence that policymakers will follow through on reform commitments.

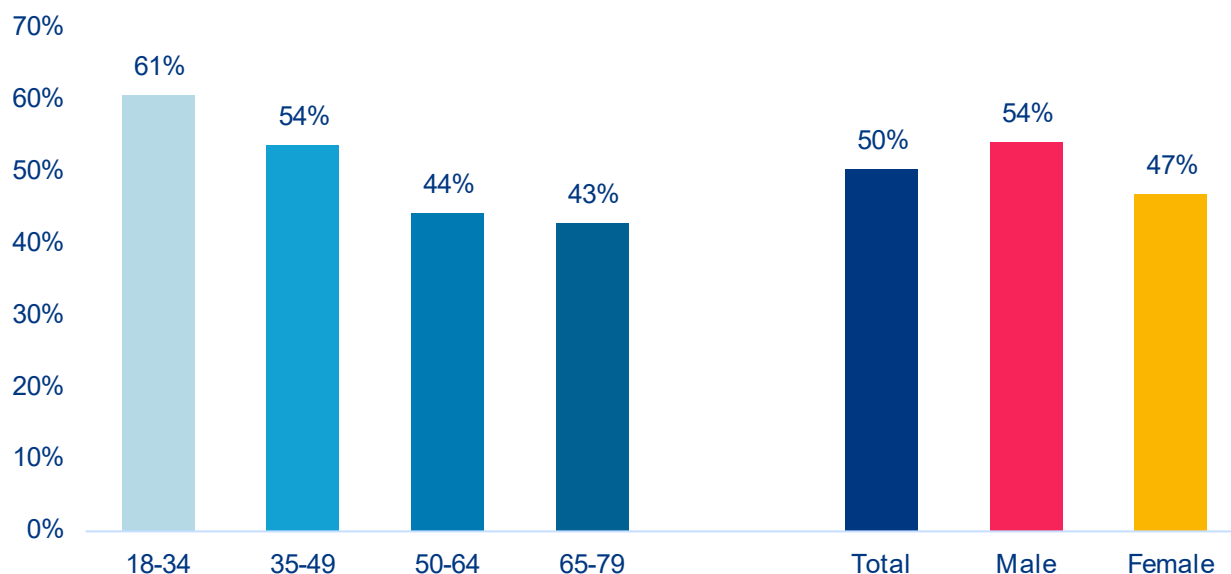
The generational pattern is the mirror image of reform need. While recognition of reform need tends to rise with age, confidence that reforms will actually be implemented declines. Younger respondents remain considerably more optimistic that governments will follow through on reform commitments, whereas older respondents combine the strongest support for reform with the deepest skepticism that it will happen. This divergence between reform demand and reform confidence is one of the most striking findings of the survey and helps explain why broad support for reform so often fails to translate into expectations of action (Figure 4).

Figure 3: Do you think your country is capable of implementing the reforms needed to ensure long-term economic sustainability? Share of answers in percent of all answers, by country



Source: Allianz Research

Figure 4: Do you think your country is capable of implementing the reforms needed to ensure long-term economic sustainability?, approvers by age group and gender (in percent)

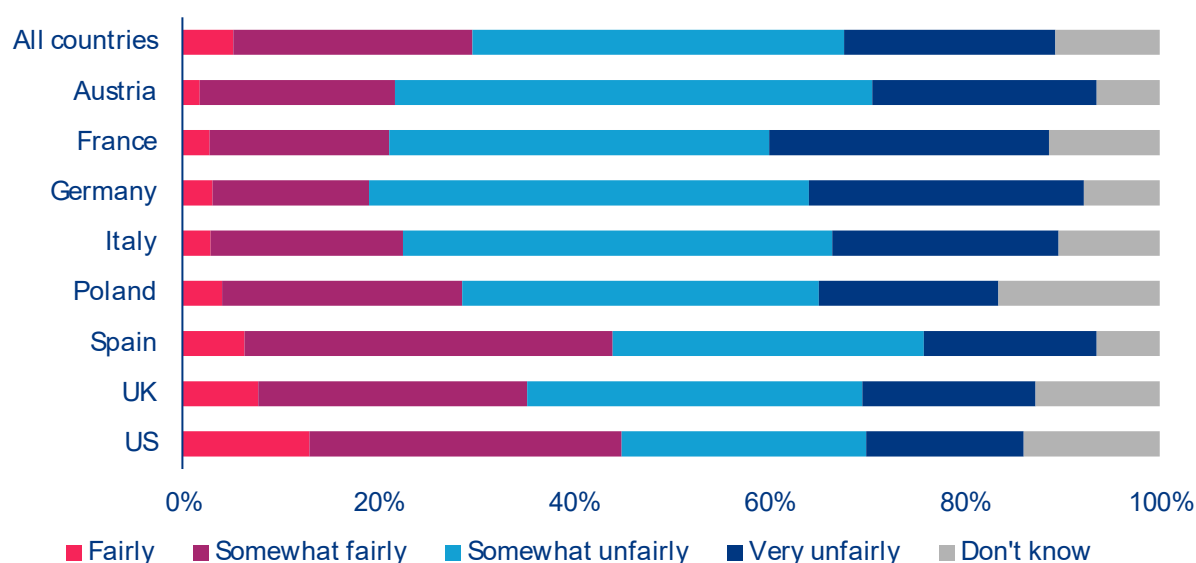


Source: Allianz Research.

Many citizens doubt not only that reforms will happen, but also that they will be fair. Across the eight countries, **only** around 30% of respondents expect the costs and benefits of reform to be distributed fairly, while almost 60% expect an unfair outcome (Figure 5). This skepticism is particularly pronounced in continental Europe and may

help explain the broader trust deficit identified above. Citizens do not merely question whether governments will implement reform; many also question whether they will do so in a fair and balanced way. Reform credibility is therefore as much a question of legitimacy as of policy design.

Figure 5: How fairly do you think the costs and benefits of reforms will be distributed across society?, share of answers in percent of all answers, by country



Source: Allianz Research



Pension reform is not on the top of the list

Pension reform is widely recognized as necessary, but it is not at the top of the public's reform agenda.

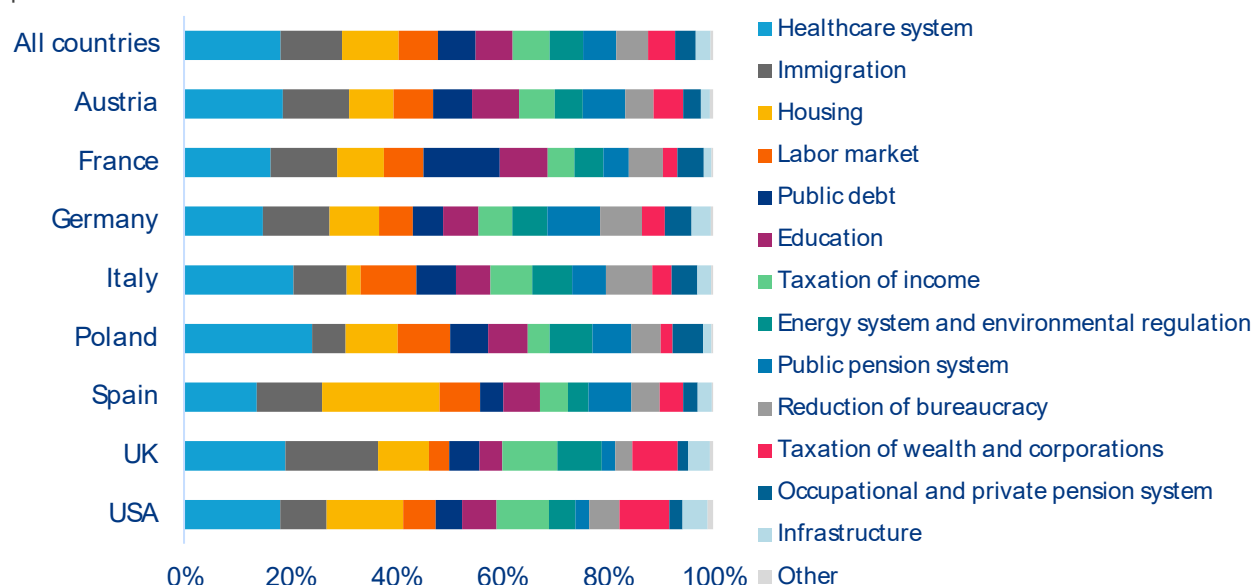
When asked where reforms are most urgently needed, respondents consistently rank healthcare, immigration and housing at the top, far ahead of public, occupational and private pension systems (Figure 6). This may help explain why pension reform is so frequently postponed despite broad recognition of the underlying challenges. The issue is therefore not a lack of awareness, but a lack of urgency.

the broader survey finding that perceptions of reform need increase with age. The pattern points to a salience effect: as retirement moves from a distant prospect to an imminent reality, concern about the sustainability of pension systems rises markedly. Younger generations, despite having the greatest long-term stake in the sustainability of pension systems, tend to view retirement as a distant concern.

The older the wiser: the strongest demand for pension reform comes from those closest to retirement.

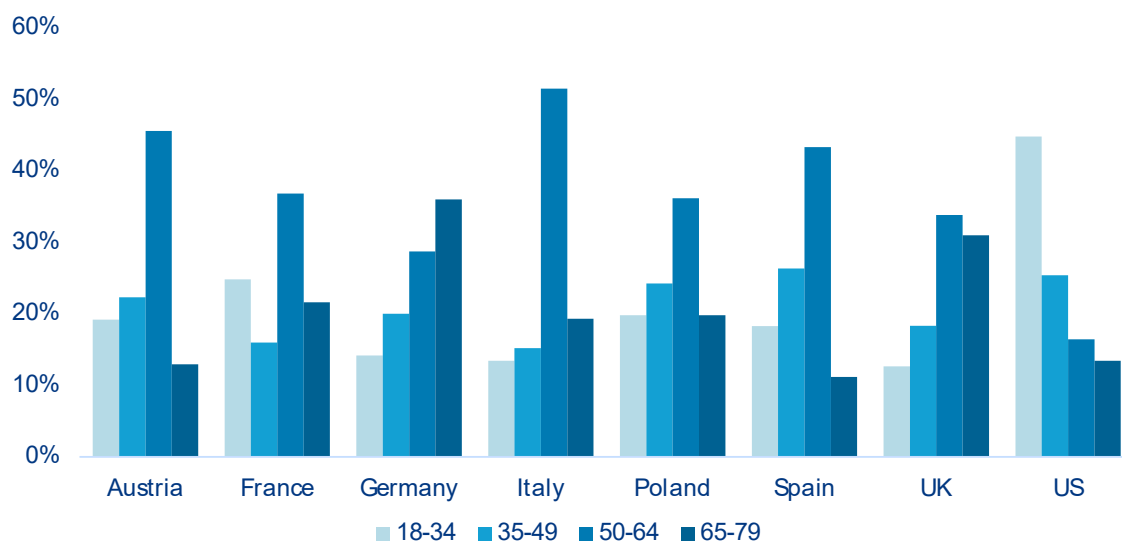
Respondents aged 50 to 64 are most likely to identify pension reform as a policy priority (Figure 7), mirroring

Figure 6: In which areas do you think reforms are most needed in your country? Select max. 3, share of answers in percent of all answers



Source: Allianz Research

Figure 7: In which areas do you think reforms are most needed in your country? Select max 3. - Public pension system, approvers by age group (in percent)



Source: Allianz Research

Demographic pressures are accelerating and the window for gradual reform is narrowing. With the baby-boomer generation reaching retirement age, the increase of the old-age dependency ratio (OADR) – the number of people aged 65 and over relative to every 100 people of working age – is set to accelerate sharply across most advanced economies over the next decade before levelling out on a comparatively high level. The countries covered in our survey are no exception. By mid-century, in Italy and Spain, there are projected to be around 75 retirees

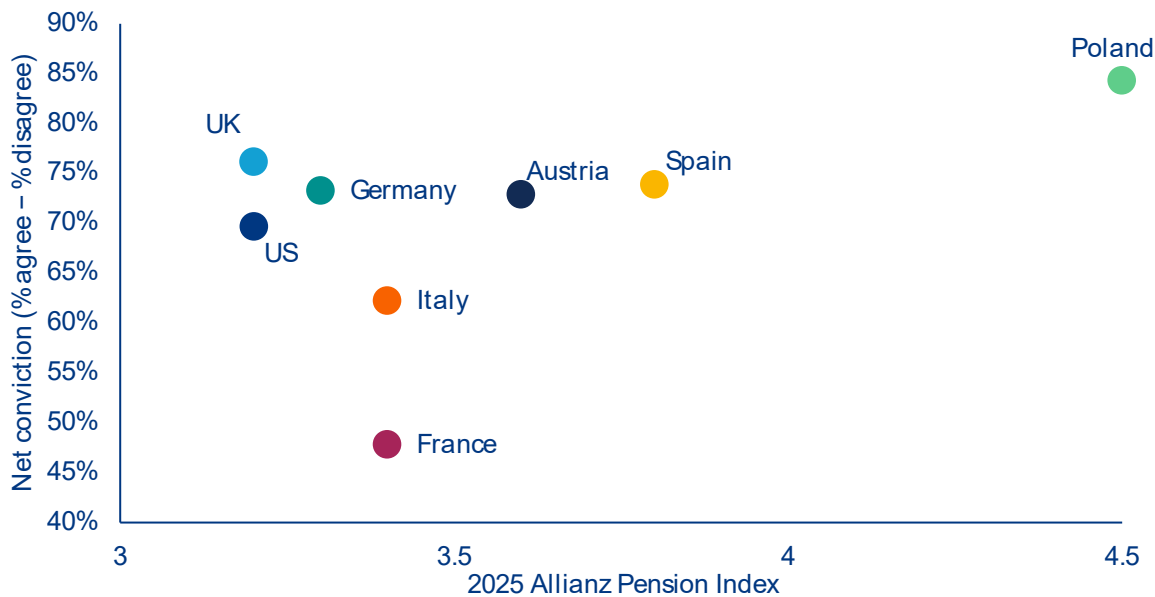
per 100 people of working age², while in Poland the ratio will exceed 60%. In Austria and Germany it is expected to stabilize at around 58%, and even in France it is projected to surpass 50%. Only in the UK and the US is it expected to remain below 40%.

Figure 8: Allianz Pension Index 2025

	API 2025	Basic Conditions	Sustainability	Adequacy
<u>Surveyed Countries:</u>				
UK	3.17	3.40	3.54	2.70
US	3.24	3.10	3.45	3.11
Germany	3.26	3.80	3.58	2.67
Italy	3.35	4.34	3.32	2.89
France	3.39	3.47	3.70	3.05
Austria	3.57	4.14	4.08	2.78
Spain	3.77	4.18	4.43	2.90
Poland	4.51	4.39	4.92	4.16
<u>Reference Countries:</u>				
Denmark	2.34	3.17	2.29	1.98
Netherlands	2.60	2.98	3.33	1.68
Sweden	2.62	2.93	2.94	2.15

1 = no need for reforms 7 = high reform pressure

Source: Allianz Research

Figure 9: Net convictions about reform need vs. Allianz Pension Index 2025, by country

Source: Allianz Research

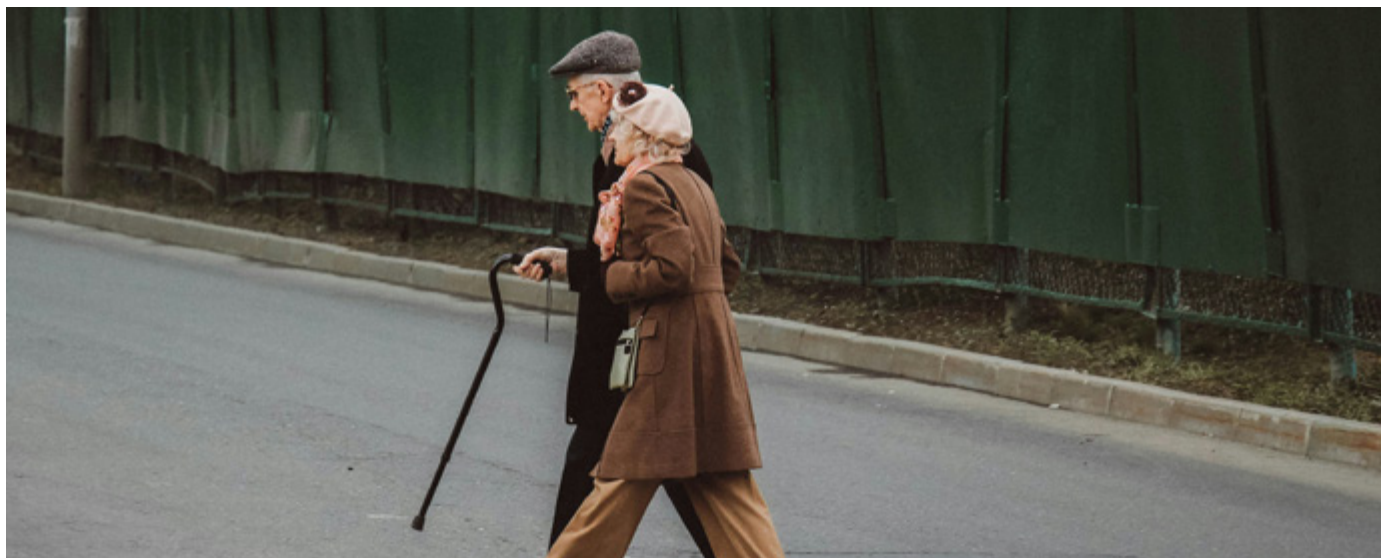
Demographic pressures are accelerating and the window for gradual reform is narrowing. With the baby-boomer generation reaching retirement age, the increase of the old-age dependency ratio (OADR) – the number of people aged 65 and over relative to every 100 people of working age – is set to accelerate sharply across most advanced economies over the next decade before levelling out on a comparatively high level. The countries covered in our survey are no exception. By mid-century, in Italy and Spain, there are projected to be around 75 retirees per 100 people of working age², while in Poland the ratio will exceed 60%. In Austria and Germany it is expected to stabilize at around 58%, and even in France it is projected to surpass 50%. Only in the UK and the US is it expected to remain below 40%.

Public perceptions of reform need broadly mirror the structural pressures facing pension systems. The stronger the reform need identified by the Allianz Pension Index³ (API), which assesses the adequacy and sustainability of pension systems across countries, the stronger the perceived need for reform revealed in our survey (Figure 8). Citizens appear remarkably adept at recognizing where pension systems face the greatest strain, suggesting that

awareness is not the main obstacle to reform. Poland tops both rankings (API: 4.5; net conviction: 84%), with Spain (3.8; 74%) and Austria (3.6; 73%) not far behind. At the other end, the UK (3.2; 76%) and the US (3.2; 70%) combine comparatively good API scores with somewhat lower reform urgency. The deeper divide, however, is not national but institutional. The perceived and actual need for reform tends to be strongest in countries that rely heavily on pay-as-you-go pension systems, while countries with stronger funded pension pillars are less exposed to demographic pressures (Figure 9). France (3.4; 48%) and Italy (3.4; 62%) remain notable exceptions, where the perceived need for reform appears more subdued than the underlying structural pressures would suggest.

² Here aged between 20 and 64. Source United Nations, Department of Economic and Social Affairs, Population Division (2024).

³ The Allianz Pension Index (API) is based on three sub-indices – starting points, sustainability and adequacy – and takes into account 40 parameters. Each parameter is rated on a scale of 1 to 7, with grade 1 signaling no need for reforms and 7 a high reform pressure.



The reform options are known — but there is no one size fits all

The inventory of possible reforms is well known, implementation remains the real challenge. The broad contours of pension reform look remarkably similar across countries: increasing the retirement age, encouraging longer working lives, strengthening funded occupational and private pensions, improving financial literacy and adjusting contribution rates and benefit levels. What differs across countries is the mix. Pension systems vary widely in their reliance on public transfers, occupational pensions and private savings, as well as in their demographic pressures and labor-market conditions. Successful reform therefore requires adaptation rather than imitation. The challenge is not a lack of solutions, but turning well-known reform options into policy action and implementing the right combination of measures before demographic pressures force more abrupt and costly adjustments.

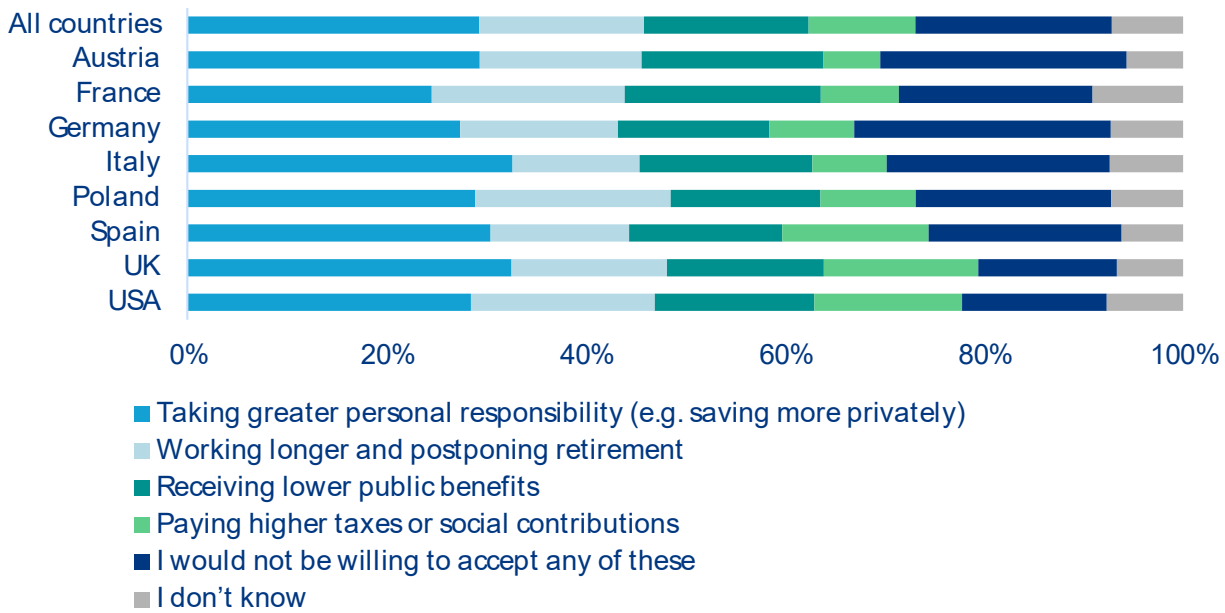
Support for pension reform is broad, but there is much less agreement on which reform measures should be implemented. When asked which measures they would personally be willing to accept to improve the long-term sustainability of public finances, respondents paint a more nuanced picture than their broad recognition of the need for reform might suggest. The most widely supported option is taking greater personal responsibility through additional private saving (29% of all responses). However, the second most common response is a refusal to accept any of the proposed measures (22%). Working longer and postponing retirement ranks third, followed by accepting lower public benefits. The results highlight a central

challenge of pension reform: while many agree that change is necessary, there is far less consensus on how the adjustment burden should be shared (Figure 10).

Taking greater personal responsibility is the most widely accepted reform option. In most countries, support for additional private saving is strongest among older age groups (Figure 11). This likely reflects greater financial flexibility later in life, when major household expenditures such as mortgages and child-related costs have often declined. Across all countries, however, support for this measure is consistently higher among men than among women.

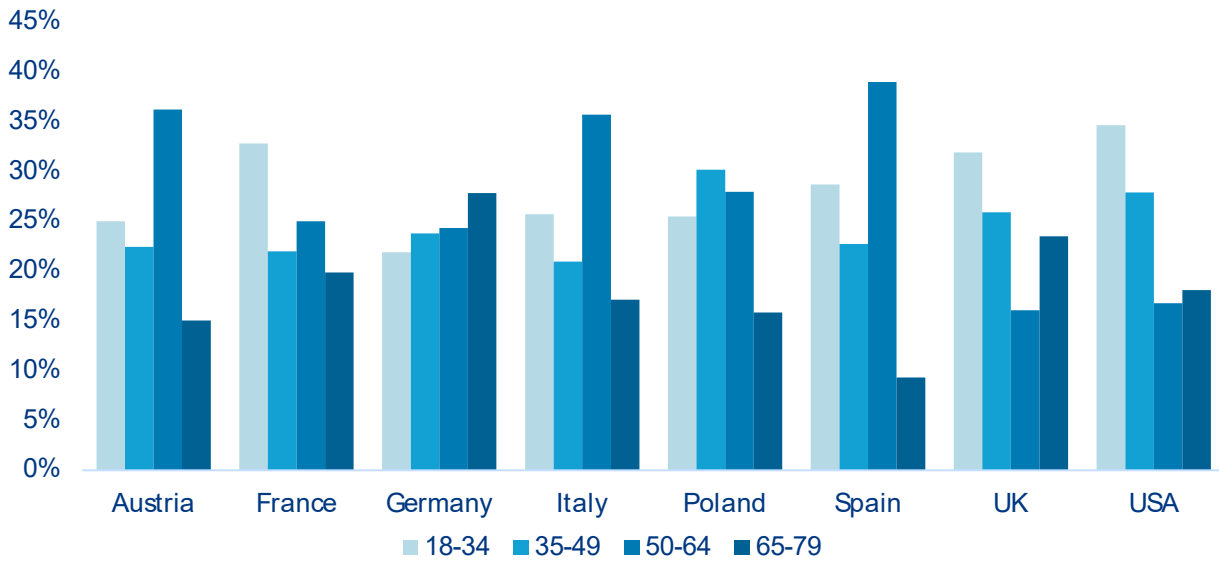
The stereotype that older workers are unwilling to work longer finds little support in the data. In Austria, Italy and Spain, more than one-third of respondents supporting later retirement are aged between 50 and 64. In France, support is distributed relatively evenly across age groups. Germany presents a more nuanced picture: only 21% of supporters are aged between 50 and 64, compared with 25% aged 65 to 79. One possible explanation is that extending working lives may appear less realistic in an environment characterized by economic weakness, corporate restructuring and relatively high long-term unemployment among older workers. By contrast, support for working longer is more broadly distributed in the UK and the US, where labor-force participation at older ages is already higher and funded pension systems play a larger role (Figure 12).

Figure 10: Which of the following would you personally be willing to accept to ensure long-term sustainability of public finances? share of answers in percent of all answers



Source: Allianz Research

Figure 11: Which of the following would you personally be willing to accept to ensure long-term sustainability of public finances? Taking greater personal responsibility, approvers by age group (in percent)

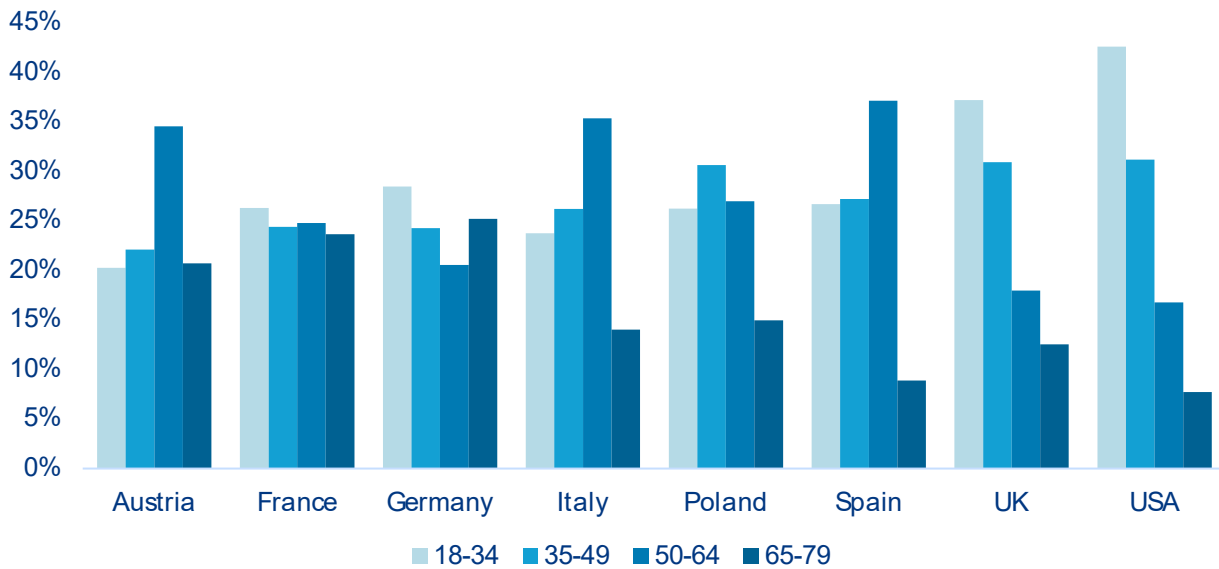


Source: Allianz Research

Support for lower public benefits remains limited, but is not confined to younger generations. Around one-quarter of respondents would consider lower public benefits as part of a reform package. In Austria, Italy and Spain, where public pension replacement rates remain comparatively high, more than one-third of supporters are aged between 50 and 64. In most countries, however, support is lower among those approaching or already in retirement than among younger cohorts, reflecting the shrinking time horizon available to compensate for benefit reductions through additional private saving (Figure 13).

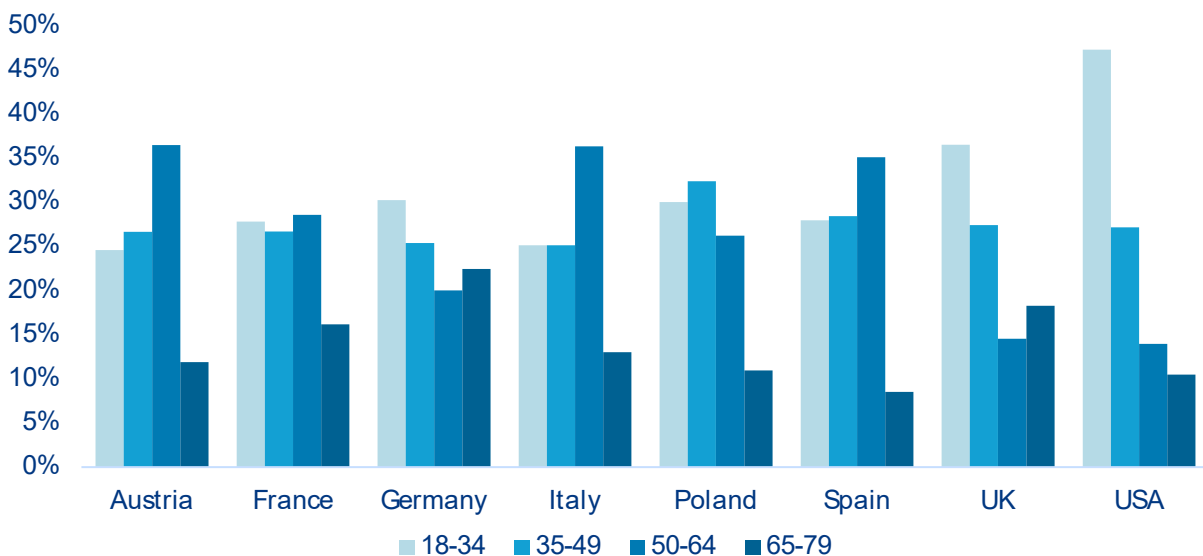
Disagreement over reform measures should not be confused with opposition to reform. While respondents differ on how adjustment should be achieved, only a minority reject all proposed measures outright. The challenge for policymakers is therefore less one of overcoming universal resistance than of combining measures in a way that distributes the adjustment burden credibly and fairly. The survey results suggest that citizens may disagree on the preferred path, but many recognize that some form of adjustment will be necessary to ensure the long-term sustainability of pension systems.

Figure 12: Which of the following would you personally be willing to accept to ensure long-term sustainability of public finances? Working longer and postponing retirement, approvers by age group (in percent)



Source: Allianz Research

Figure 13: Which of the following would you personally be willing to accept to ensure long-term sustainability of public finances? Receiving lower public benefits, approvers by age group (in percent)



Source: Allianz Research



Retirement preparedness starts with knowing the numbers

Limited knowledge of key pension parameters makes retirement planning more difficult. Pension reform ranks below healthcare, immigration and housing on the public policy agenda, and this lower salience is reflected in respondents' understanding of pension systems. Across countries, many respondents struggle to estimate key parameters such as the statutory retirement age, the number of years they are likely to spend in retirement and the level of income they can expect from the public pension system. Yet these are precisely the variables individuals need to understand in order to assess how much additional saving toward retirement will be required.

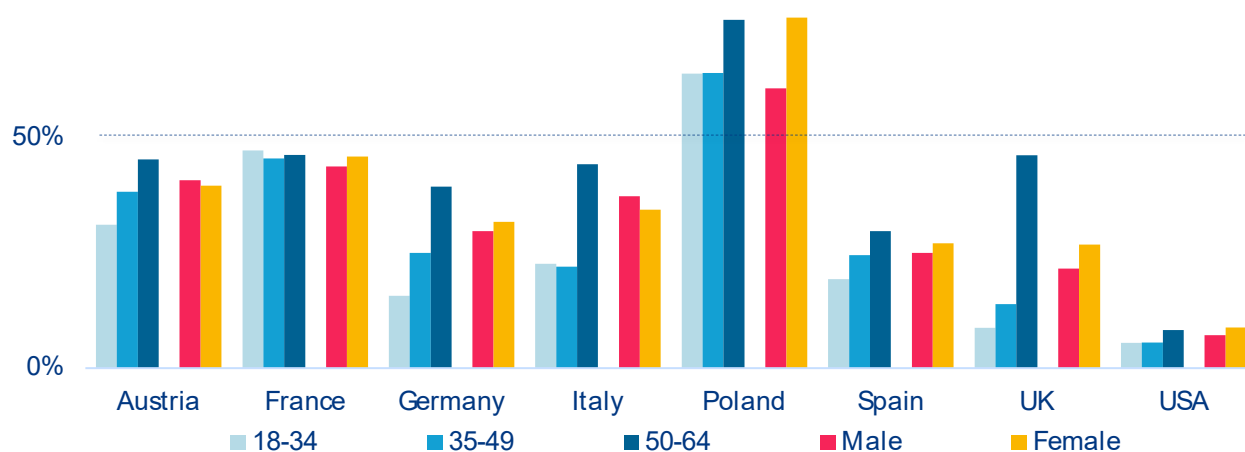
The knowledge on retirement options increases with age. More than half of respondents in every country except Poland plan to retire before the current or already legislated retirement age. However, as respondents approach retirement age and become more familiar with the options and financial implications of retiring early or

later, their expected retirement age converges towards the statutory retirement age. Furthermore, the answers reflect debates about or actual reforms of the retirement age in recent years. Respondents aged 50 to 64 were generally the most likely to plan retirement in line with the statutory retirement age, with the exception of France. Only in Poland, more than 50% of respondents plan to retire at the statutory retirement age, which has remained unchanged over the past decade (Figure 14).

Most respondents expect to retire between the ages of 65 and 69. France is the notable exception, where many respondents expect to retire as early as possible by making use of the possibility to retire at age 63, or age 64 in the future, once the required contribution periods have been met (Figure 15).

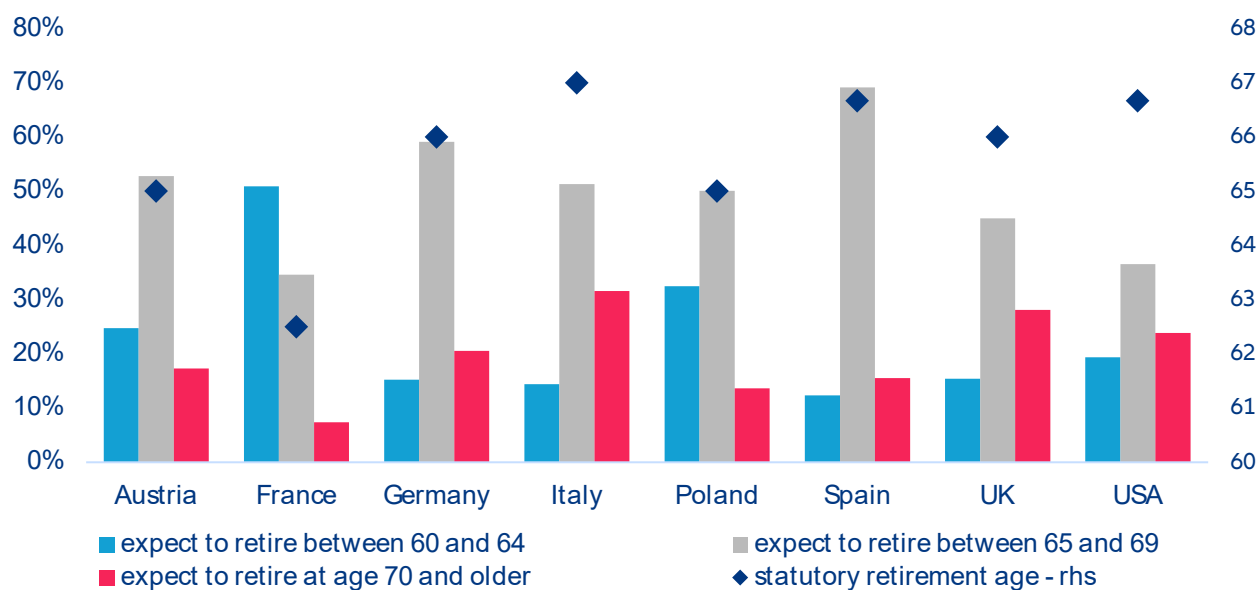
Figure 14: At what age do you realistically expect to retire?, share of respondents who gave the actual statutory current or after reform statutory retirement age (in percent of respective population group).

100%



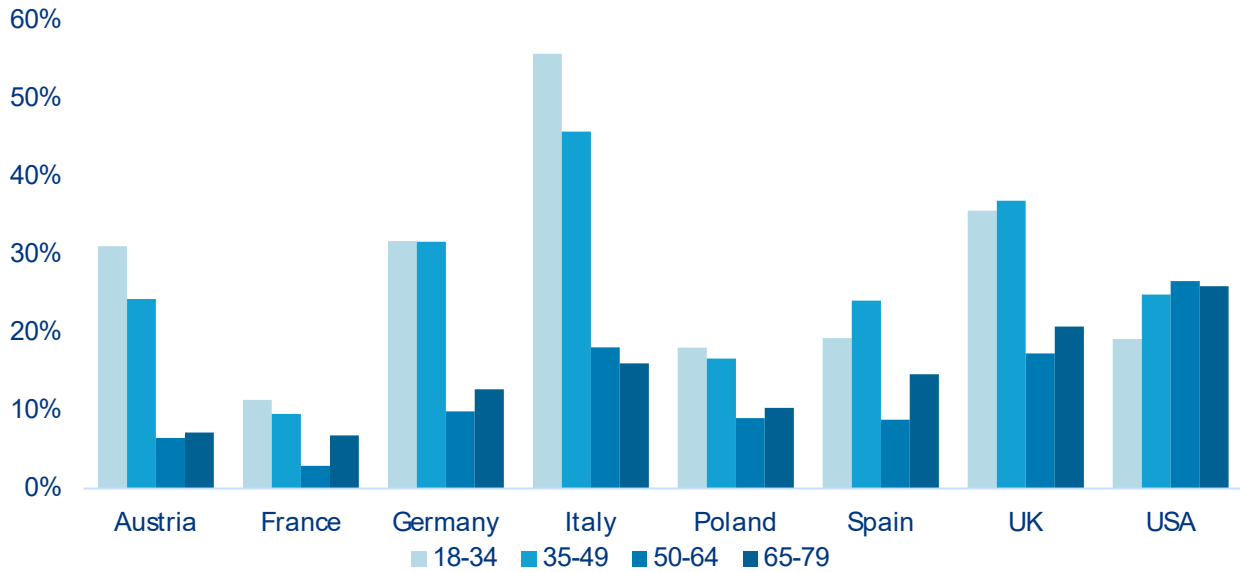
Source: Allianz Research

Figure 15: At what age do you realistically expect to retire?, age brackets (share of respondents in percent)



Source: Allianz Research

Figure 16: At what age do you realistically expect to retire? – At the age of 70 and older (in percent of the respondents in the respective age class)

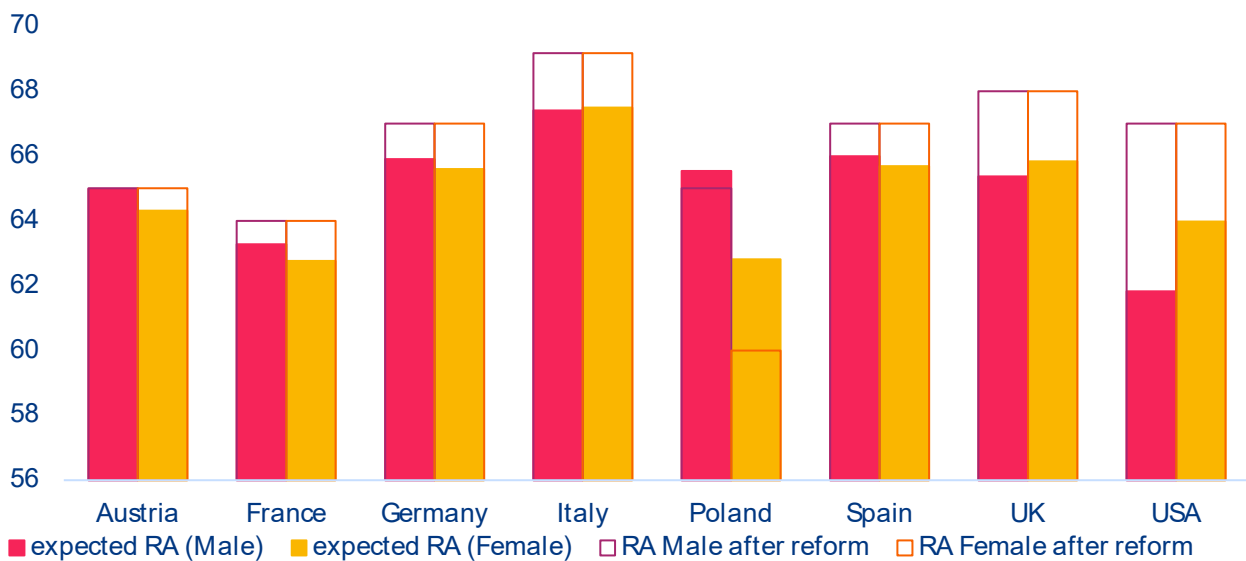


Source: Allianz Research.

Younger generations in particular expect to work beyond the age of 70. In most countries, fewer than 25% of respondents expect to retire at age 70 or later. Only in Italy and the UK does this share reach around one-third. In both countries, future increases in the retirement age have already been linked to rising life expectancy. Since these changes are being implemented gradually, it is particularly younger respondents aged 18–34 and 35–49 who expect to retire at age 70 or beyond (Figure 16).

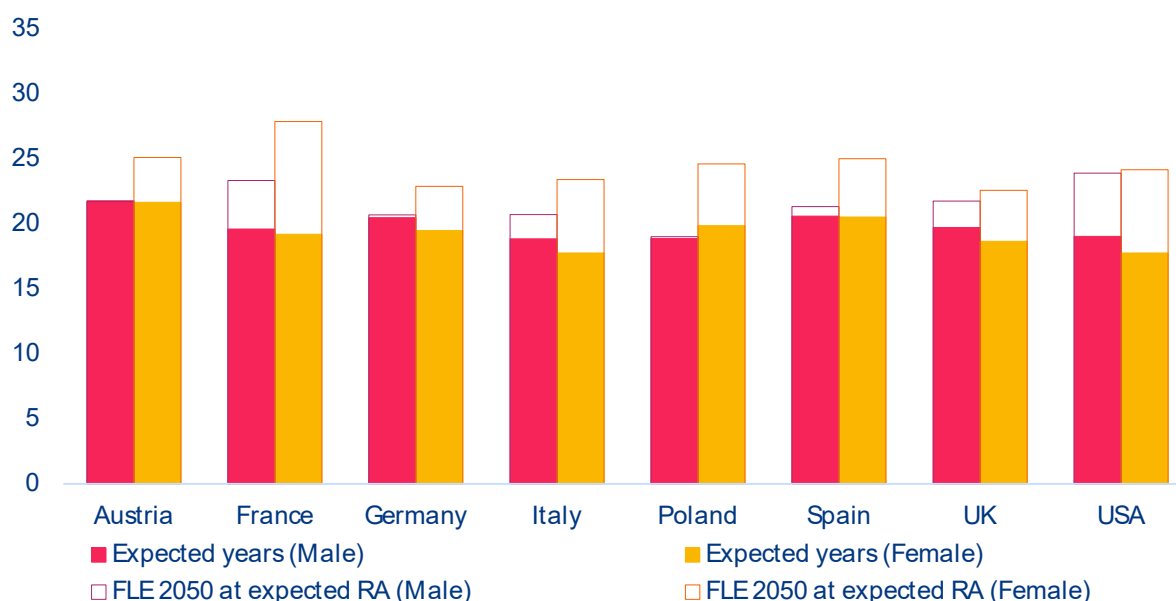
Many respondents expect to retire up to two years earlier. Ranging from 62 years in the US to 67 years in Italy, the average expected retirement age remains between one and two years below the future statutory retirement age already agreed upon in legislation (Figure 17) in most countries. Poland is the main exception, partly because many women choose to work beyond the statutory retirement age of 60 in order to receive a higher pension. Another explanation may be the expectation that the retirement age will be increased in the long run.

Figure 17: At what age do you realistically expect to retire?, average expected retirement age compared to the statutory retirement age after reforms (in years)



Source: Allianz Research.

Figure 18: How many years do you expect to spend in retirement?, expected average time spend in retirement compared to the average further life expectancy at retirement age (in years)



Sources: UN Population Division (2024), Allianz Research.

Respondents are even less informed about how long they are likely to spend in retirement. Of course, it depends on the individual situation, but the average further life expectancy at the age of retirement is a rather good proxy for the time one can expect to spend in retirement. In most surveyed countries, only around one-third of respondents correctly estimated average life expectancy at their expected retirement age. Women in particular tend to underestimate the number of years they are likely to spend in retirement. The largest gap emerges in France, where female respondents expect to spend around 19 years in retirement on average, while the average further life expectancy of a woman retiring at age 63 is currently around 26 years and is projected to increase to 28 years by 2050 (Figure 18).

Underestimating longevity can lead to a significant underestimation of retirement savings needs. Rising life expectancy poses a challenge not only for public pay-as-you-go pension systems but also for individual retirement planning. Linking retirement ages more closely to life expectancy could help preserve intergenerational fairness by maintaining a more stable relationship between working life and retirement duration across generations. The implications for individual saving behavior are equally important. Taking the example of an average French woman aiming for a retirement income equal to 70% of her final wage, annual savings of 3.7% of earnings would

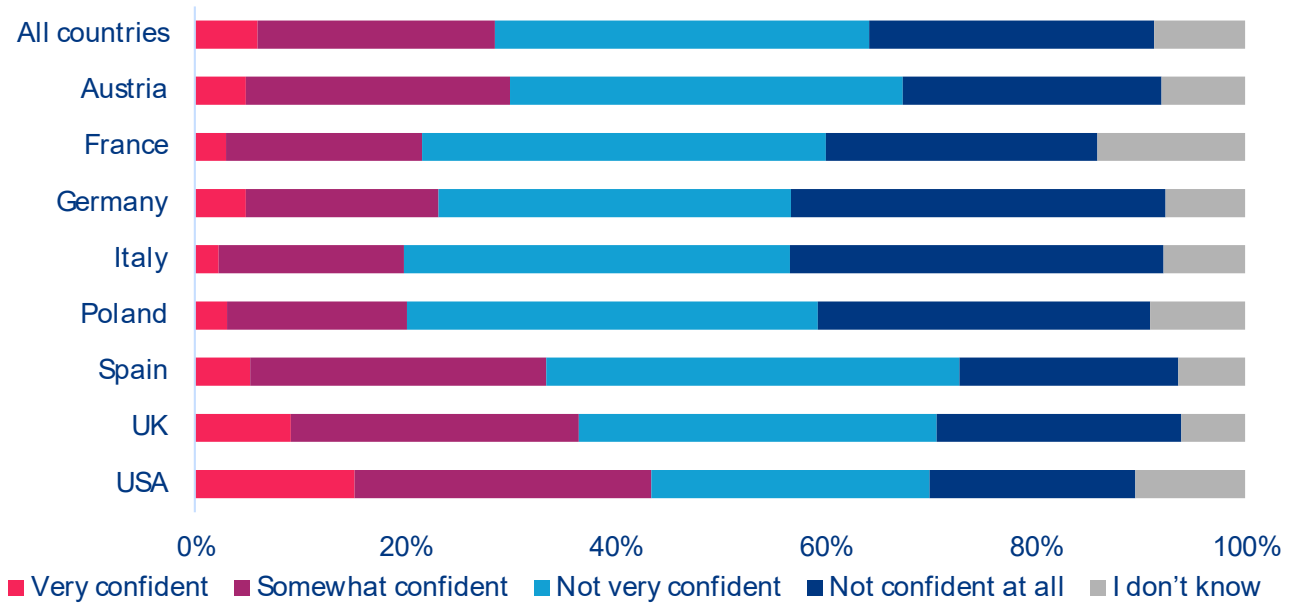
be sufficient if retirement lasted only 19 years. However, if retirement lasts closer to 28 years, the required savings rate rises to 5.0% of earnings, or alternatively requires accepting a lower retirement income of around 67%⁴.

The pension and financial literacy gap therefore has direct consequences for retirement preparedness.

Understanding retirement age, longevity and expected pension income is essential for estimating future pension gaps and determining how much supplementary saving will be required. Yet confidence in the ability of public pension systems to provide an adequate retirement income is already limited. Only in the US do more than 40% of respondents report being very or somewhat confident that the public pension system will provide a decent standard of living in retirement. In Spain this share falls to 37%, in the UK to around one-third, while in France, Germany, Italy and Poland fewer than 25% of respondents express such confidence (Figure 19). The challenge is therefore not a lack of awareness that additional retirement provision may be needed, but a limited understanding of how large the resulting savings gap may be.

⁴ Furthermore, we assumed an average inflation of 2%, an annual increase of the nominal wage of 2%, an average annual yield of 5.0% during the savings phase and of 3.0% during the payout period.

Figure 19: How confident are you that the public pension system in your country will provide you with an adequate income when you retire? share of answers in percent of all answers, by country

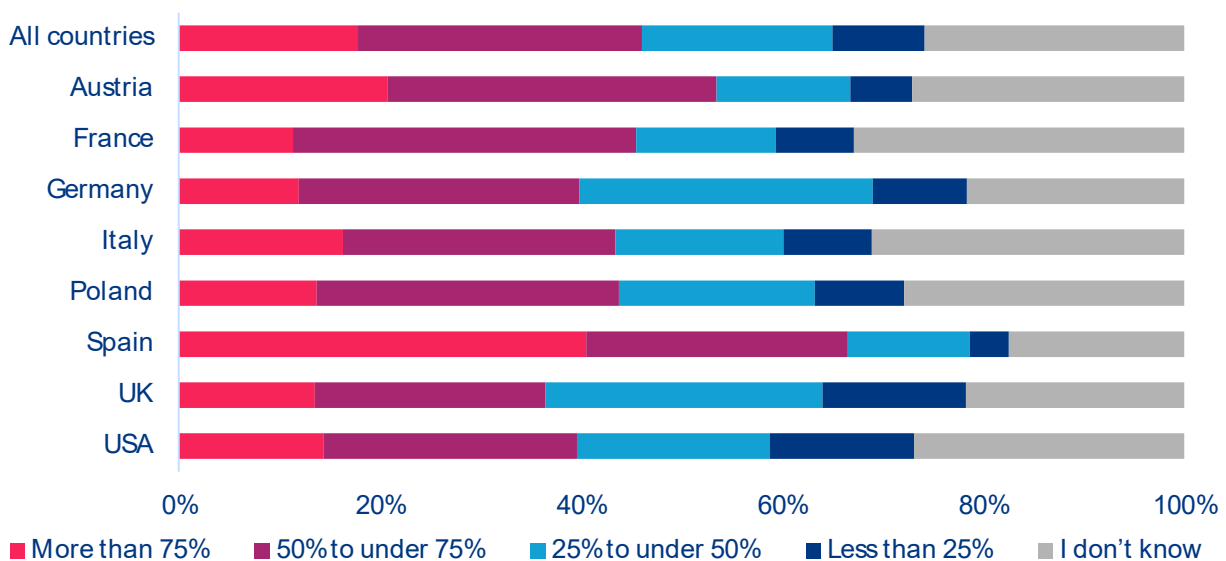


Source: Allianz Research

Many respondents expect public pensions to remain their main source of retirement income despite limited confidence in their adequacy. In Austria and Spain, 53% and 64% of respondents respectively expect the public pension system to provide more than half of their retirement income. In the remaining countries, the share ranges from less than 40% in the UK and the US to 45% in Italy (Figure 20). The results reveal a striking tension: many respondents expect to rely heavily on public pensions, yet relatively few are confident that these systems will provide an adequate standard of living in retirement.

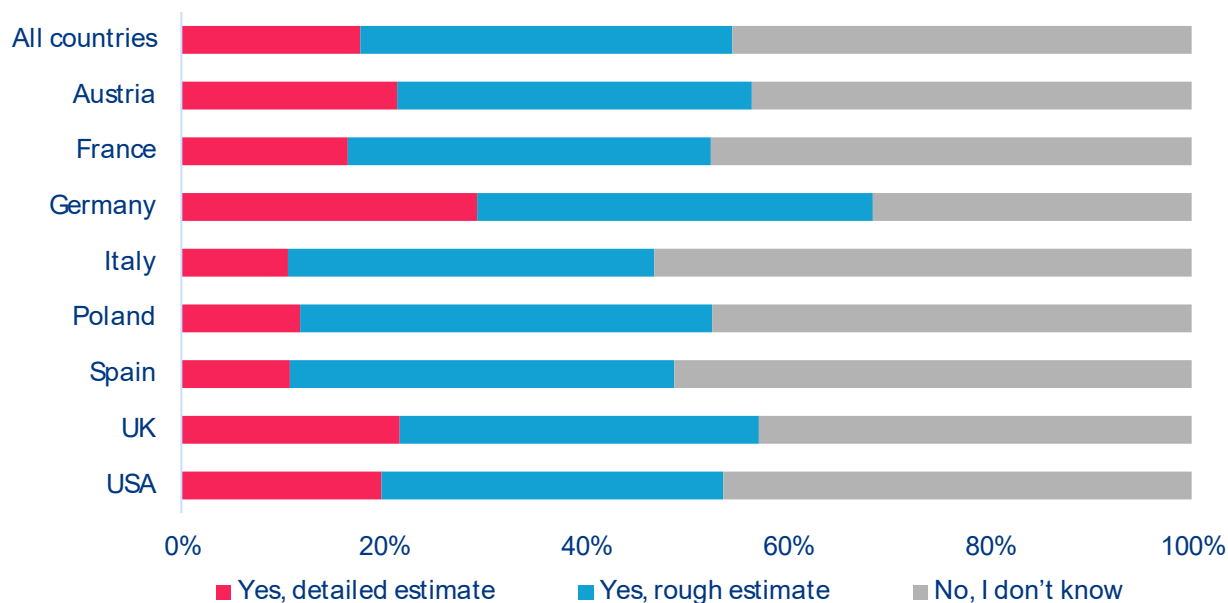
Knowledge of expected pension income remains limited. In most countries, fewer than half of respondents report having a detailed or rough estimate of their future public pension income. Across all countries, only 11% of respondents have a detailed estimate of their expected public pension income. Germany is the exception at 29%, likely reflecting the annual pension statements provided by the public pension provider. Between 34% of respondents in the US and 41% in Poland report having at least a rough estimate. At the same time, an average of 45% do not know how high their future public pension income will

Figure 20: What percentage of your retirement income do you expect to come from the public pension system? share of answers in percent of all answers, by country



Source: Allianz Research

Figure 21: Do you know what your expected public pension income will be?, share of answers in percent of all answers, by country



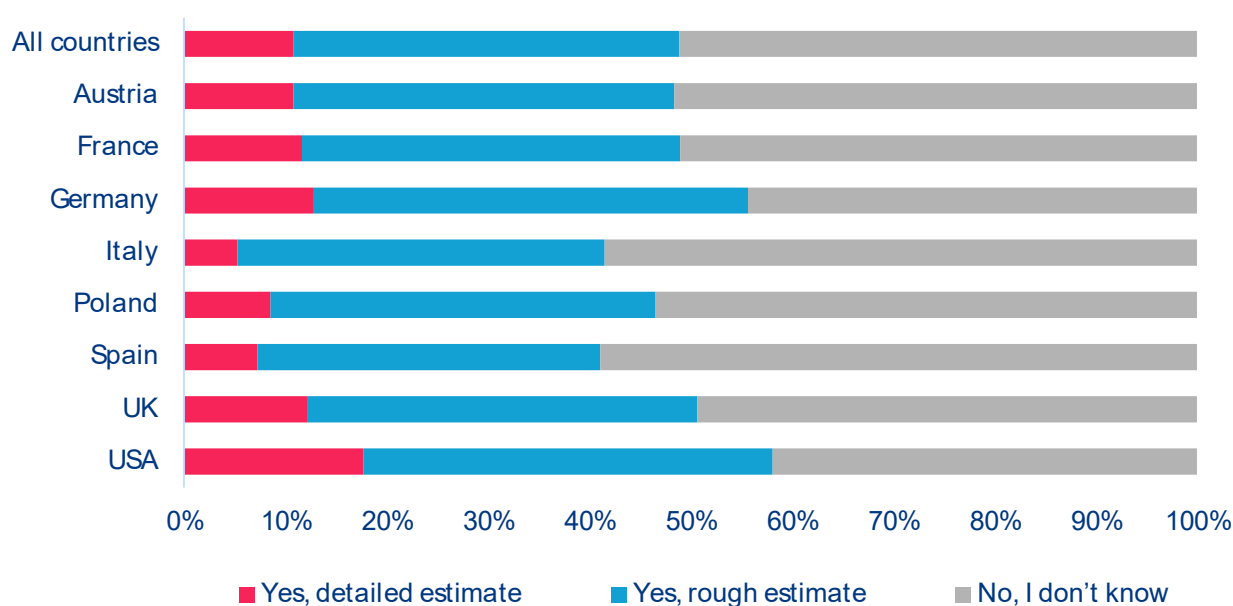
Sources: Allianz Research

be, ranging from 32% in Germany to 53% in Italy (Figure 21). For many households, retirement planning therefore begins without a clear understanding of expected retirement income.

Many respondents have not estimated how much income they will need in retirement. With the exception of the US and Germany, where the share is close to 60%, around half of respondents across countries have never

estimated their required retirement income (Figure 22). Women are particularly likely to report not knowing how much income they will need. Surprisingly, the largest share of respondents without such an estimate is often found among those aged 50 to 64, i.e. the group approaching retirement most rapidly. The results suggest that retirement planning remains incomplete even among individuals nearing the end of their working lives.

Figure 22: Have you ever estimated how much income you will need in retirement?, share of answers in percent of all answers, by country



Source: Allianz Research

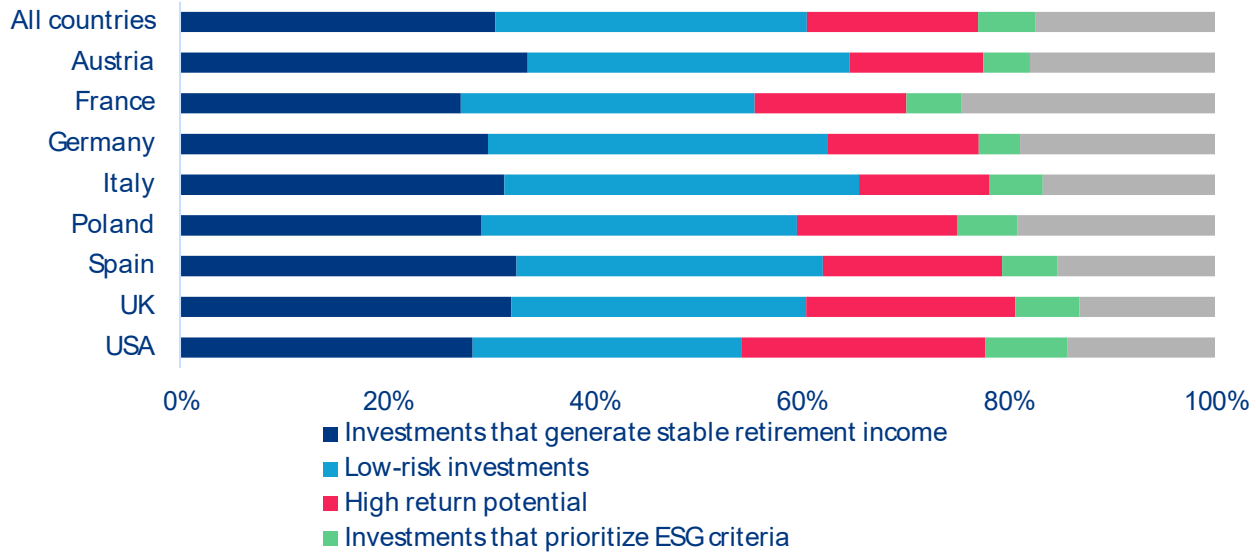


Financial literacy, a growing imperative for pension policy

Financial literacy is becoming an increasingly important pillar of retirement planning. As trust in public pension systems remains limited and funded occupational and private pensions become more important, individuals will need to take greater responsibility for retirement planning. A solid financial education becomes essential to ensuring the maintenance of the desired living standard in old-age. Across the eight countries surveyed, only one in six respondents shows a high level of financial literacy (18%), compared with 57% of respondents with medium and 26% with low financial literacy. The low levels of financial literacy translate into savings and investment behaviors that favor suboptimal low-risk investments. When asked about their investment preferences, most respondents favor low-risk investments and products that provide a stable retirement income. By contrast, fewer than 10% identify ESG characteristics as their primary consideration when saving for retirement (Figure 23).

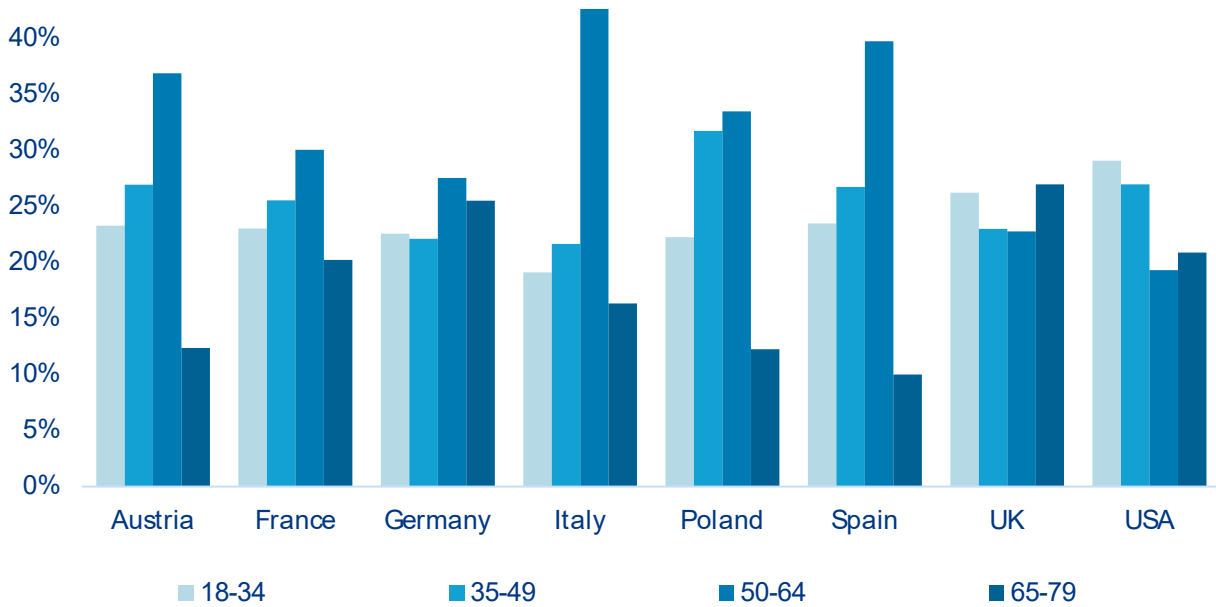
Investment preferences vary systematically across age groups. Respondents preferring low-risk investments are predominantly aged 50 to 64 (Figure 24), reflecting a shorter investment horizon and greater sensitivity to market fluctuations. By contrast, ESG-related preferences are most common among respondents aged 18 to 34.

Figure 23: When saving for retirement, which characteristics are most important to you?, share of answers in percent of all answers, by country



Source: Allianz Research

Figure 24: When saving for retirement, which characteristics are most important to you? Low-risk investments, respondents by age group



Source: Allianz Research



More capital-funded pension provision is a win-win

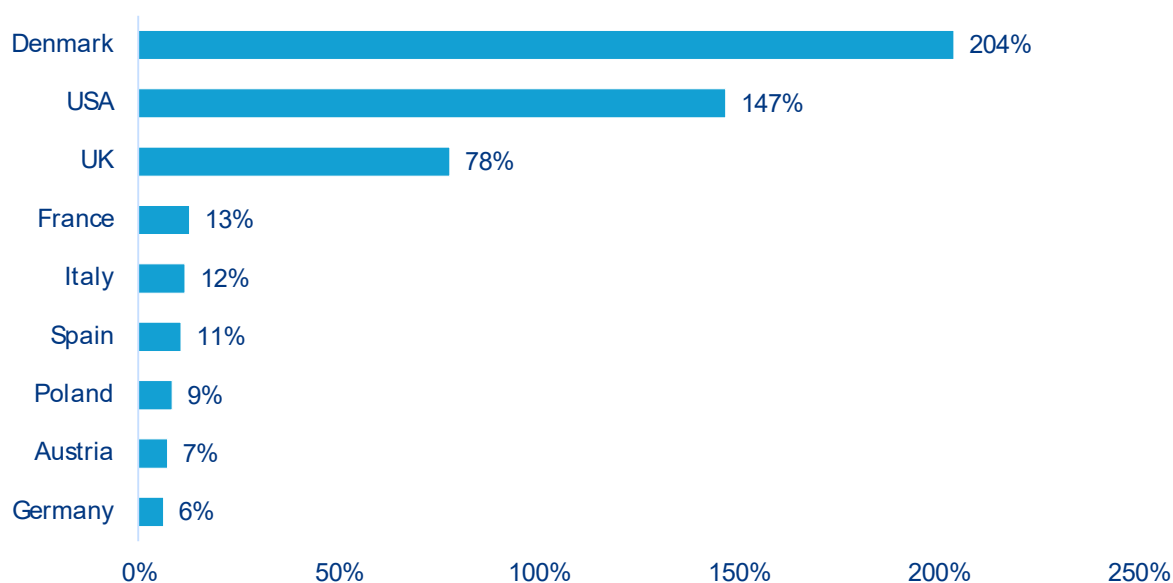
More capital-funded pension provision could create benefits beyond retirement income. Strengthening funded pension pillars could be a win-win for households and the wider economy. It would strengthen retirement security by increasing private wealth accumulation while simultaneously providing long-term capital to financial markets, supporting investment in innovation, infrastructure and projects aimed at mitigating climate change. Countries with strong funded pension pillars tend to exhibit significantly higher levels of household financial wealth. In Denmark, net financial assets per capita amount to EUR191,560, more than twice the level observed in Germany (EUR86,800)⁵. At the same time, pension assets correspond to more than 200% of GDP in Denmark, compared with just 6% in Germany. Comparable figures amount to 147% in the US and 78% in the UK (Figure 25).

To illustrate the effect of strengthening capital-funded pension pillars with respect to availability of long-term investment capital, we conduct the following thought experiment: If pension assets in Austria, France, Germany, Italy, Poland and Spain were to reach the same share of GDP as in the UK today, additional long-term investment capital of around EUR8.8trn could become available, equivalent to approximately twice the free-float market capitalization of the Euro Stoxx 50⁶ (Figure 26).

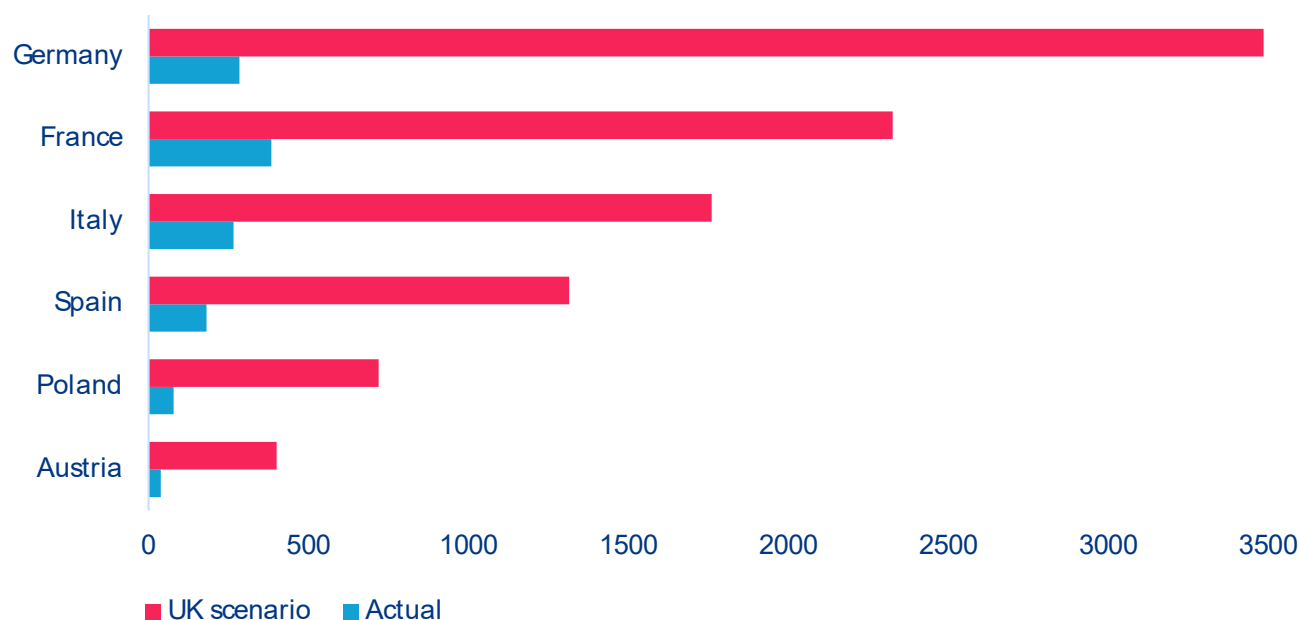
How pension assets are invested is as important as how much is accumulated. Significant differences remain in asset allocation across countries, reflecting differences in regulation and pension-system design. US pension funds invest only 12.5% of assets in bills and bonds and 1.5% in cash and deposits, whereas German pension providers allocate 42.8% to bills and bonds and 2.6% to cash. These differences have important implications for long-term returns, retirement outcomes and the availability of capital for financing economic transformation and innovation. The potential benefits of stronger funded pension pillars therefore depend not only on the amount of assets accumulated, but also on how effectively these assets are invested. Pension reform is thus not only about strengthening retirement security, it is also about mobilizing long-term capital, deepening capital markets and supporting future growth.

⁵ See Allianz Global Wealth Report 2025.

⁶ On May 29, 2026, the free-float market capitalization of the Euro Stoxx 50 was EUR 4.3trn, see Euro Stoxx (2026) factsheet, <https://stoxx.com/index/sx5e/?factsheet=true>

Figure 25: Pension assets (in percent of GDP 2024)

Source: OECD Pension Outlook 2025.

Figure 26: Actual pension assets vs UK scenario (in bn EUR)

Sources: IMF, OECD, Allianz Research.

Appendix: About the survey

The survey covered more than 8,000 respondents across eight countries: Austria, France, Germany, Italy, Poland, Spain, the United Kingdom and the United States. It was conducted as a web-based survey between 22 April and 25 May 2026. Overall responsibility lies with Allianz Research, Allianz Investment Management SE. Sample planning and fieldwork were carried out by Qualtrics.

Respondents were selected using representative quota sampling. Qualtrics was given quotas specifying how many people to survey in each country and which selection criteria to apply, with quotas set in line with official statistics for gender and age groups. A comparison with official statistics confirms that the resulting sample broadly corresponds to the population aged 18 and over in each of the eight countries.

Table 1: Sample statistics

	Total	Austria	France	Germany	Italy	Poland	Spain	UK	USA
Number of respondents	8048	1006	1006	1006	1006	1006	1006	1006	1006
Gender, in % of respondents									
Female	50.6%	50.0%	51.4%	49.9%	50.7%	50.9%	50.4%	50.9%	51.0%
Male	49.2%	50.0%	48.2%	49.9%	49.1%	49.0%	49.4%	48.9%	48.9%
Age groups, in % of respondents									
18-34	24.4%	25.0%	24.9%	23.1%	20.8%	22.7%	22.7%	27.4%	29.1%
35-49	25.0%	25.0%	23.9%	23.9%	21.3%	29.8%	24.8%	25.9%	25.1%
50-64	31.0%	36.7%	30.3%	26.0%	40.1%	30.8%	40.8%	21.8%	21.7%
65-79	18.4%	12.4%	20.4%	25.7%	16.8%	16.3%	11.5%	23.0%	21.1%
Education, in % of respondents									
Primary (less than high school)	4.8%	2.2%	3.7%	2.6%	14.1%	4.0%	2.8%	2.0%	7.2%
Secondary (high school)	52.5%	75.2%	47.9%	69.2%	52.5%	55.2%	36.0%	43.6%	40.1%
Tertiary (university or comparable)	42.7%	22.6%	48.4%	28.2%	33.4%	40.9%	61.2%	54.4%	52.8%

Source: Allianz Research

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The semiconductor premium: EM equity and the concentration risk within

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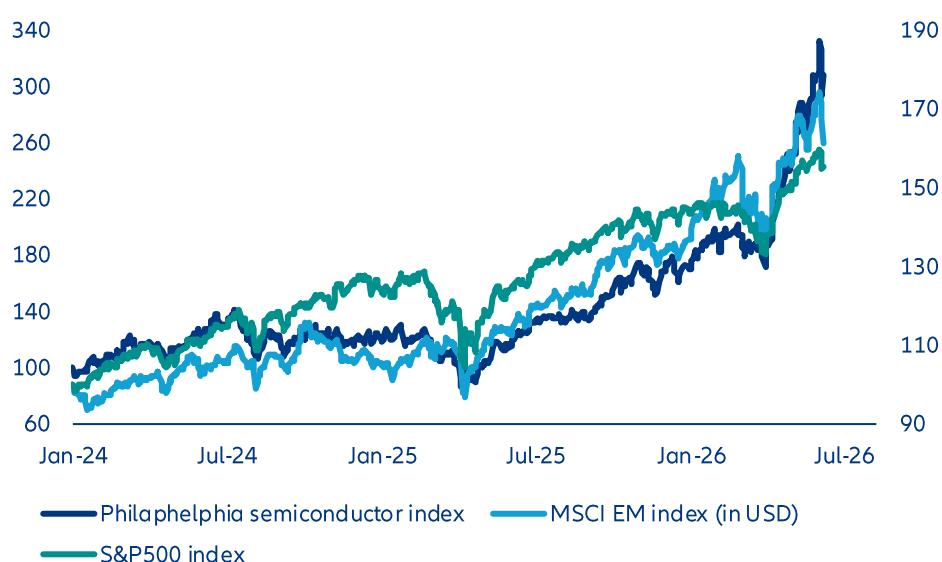
In Summary

- **Semiconductors are the backbone of the 2026 AI rally.** Capital investment is overwhelmingly directed at AI infrastructure, and the data centers doing the heavy lifting run entirely on chips. The numbers speak for themselves: the Philadelphia semiconductor index is up +90% YTD – twelve times the S&P 500's +7.5% - despite a couple of recent sell-off episodes in June. In emerging markets, which concentrate most of the supply chain, markets like South Korea and Taiwan are up +110% and +60% this year.
- **Concentration is the flip side of the boom but fundamentals are strong.** Concentration is most acute in EMs, where dependence on a single industry is now extreme. If the chips sector disappoints, the re-rating could be brutal. But the earnings are real and lopsidedly so: chips deliver 13% of S&P 500 profits on just 5% of sales, half of South Korea's profits on a fifth of its sales and over 70% of Taiwan's earnings and revenue alike. Ferocious demand colliding with supply bottlenecks from Middle East tensions are keeping prices sky-high.
- **The bull case still stands on five pillars.** First, hyperscalers show no sign of easing off. Upward capex guidance in Q1 points to a still-aggressive stance that should lift Asian chipmakers' profits by a remarkable +70% CAGR through 2026-2029. Second, the demand pipeline looks structurally solid: Broader AI diffusion, the corporate shift to agentic platforms and the memory-hungry new chip generation all reinforce one another. Third, the moat is real: capital and expertise are needed to build a modern foundry, keep competition at bay and pricing power intact. Fourth, valuations are compelling as based on solid earnings trajectory, and recent sell-offs offer new entry point on top picks. And fifth, in the over-concentrated EM space, semiconductors ensure geography and supply-chain position diversification, with an attractive trade-off compared to developed markets.
- **But strong convictions do not mean complacency: Four risks could unsettle the rally over the next 12-24 months.** First, a Middle East re-escalation could disrupt energy routed to Asia, forcing production halts – temporary or not. Second, bottlenecks in the Asian supply chain for inputs critical to data-center fleets could lift costs and lengthen lead times, throttling AI capex momentum. Third, a timing mismatch looms: New supply could come online just as capital allocation decelerates, should a fresh rate cycle take hold in the US or Europe. Fourth, competition is sharpening fast – Beijing is piling on funds and pressure to drive chip self-sufficiency (70% target in AI chips in 2026) and a bigger global footprint for its champions. Contrary to what market volatility suggests, we do not see lower inference cost to be a negative driver for memory chip business, but rather a guarantee of a sustainable book order if it comes with stronger AI diffusion and usage.

Semiconductors are driving the 2026 AI equity rally – and concentrating its risk

Equity markets are roaring back in 2026, with semiconductors as the primary engine. A brutal tech sell-off erupted last autumn, fueled by a rotation out of hyperscalers as the "AI bubble" narrative gained traction. The pain extended well into this year as frontier AI models – armed with native Customer Relationship Management (CRM), cybersecurity and data-processing capabilities – triggered a sharp de-rating of traditional software names suddenly fighting for relevance. Yet through the storm, one segment barely flinched: semiconductors. Quarter after quarter, chip makers delivered earnings that kept beating expectations, rewarding investors who held their nerve with relentless outperformance. The Philadelphia Semiconductor Index tells the story plainly, up 90% ytd, dwarfing the S&P 500's modest +7.5% gain and cementing semis as the single biggest beneficiary of the AI equity rally. But the most eye-catching trade of this cycle is unfolding not in the US but across emerging markets: The MSCI EM index has surged +23.2% YTD, and the engine behind that re-rating is, again, semiconductors as Asian chip and advanced packaging names ride the same AI capex supercycle. The investment wave is global and equity markets are pricing it that way.

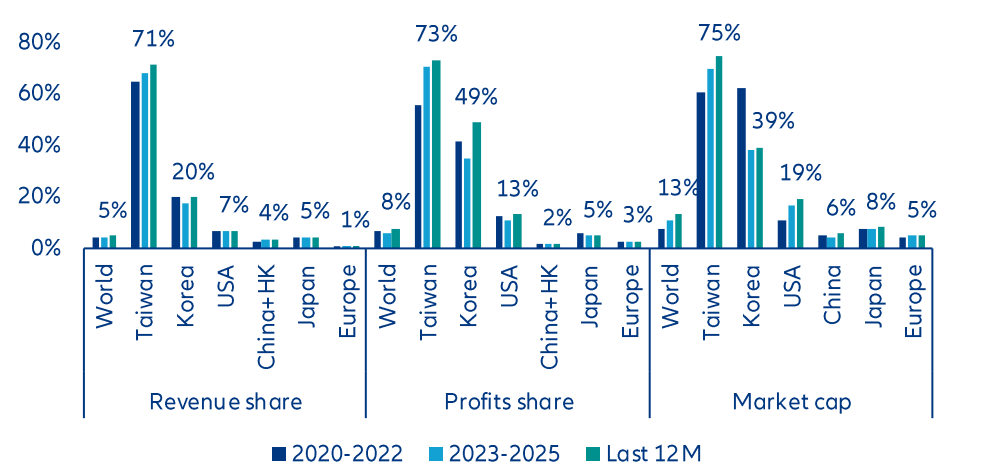
Figure 1: Equity index performance, rebased to 100 on January 2024



Sources: LSEG Datastream, Allianz Research

Semiconductors have quietly become the backbone of global equity markets – and a structural vulnerability at their core. Taiwan and South Korea stand in a league of their own. Between them, they concentrate the world's most critical foundry capacity and their stock markets reflect it starkly – Taiwan's semiconductor sector commands a 75% market cap share of its equity universe, South Korea's a still-commanding 39%, dwarfing every other major index. Yet the paradox lies in the revenue line: Semiconductor revenue shares across most exchanges remain surprisingly moderate, exposing an industry structured around a handful of dominant Asian foundries towering over a fragmented ecosystem of smaller players. Europe and the US have built their digitalization ambitions – and increasingly their AI infrastructure – on an Asian supply chain they do not control, a dependency that geopolitical stress-testing has already proven fragile.

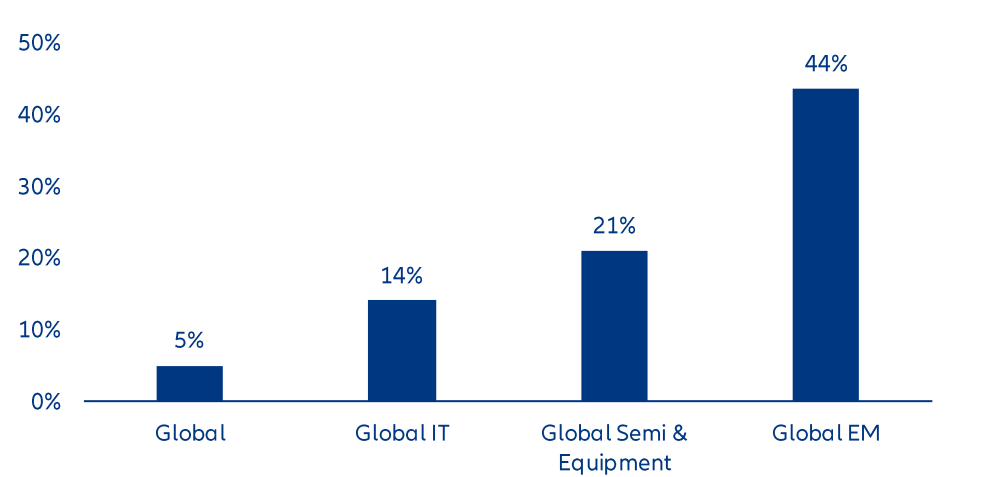
Figure 2: Semiconductor industry performance and footprint in the EM equity universe



Sources: LSEG Datastream, Allianz Research

Concentration is extreme – and the risk is double-layered. Taiwan's semiconductor profit share within its equity universe has vaulted from 55% in 2020-2022 to nearly 75% over the last 12 months. South Korea has mirrored that trajectory, climbing from 40% to close to 50% over the same period. This is not a gradual drift; it is a structural repricing of an entire industry's earnings power, driven by AI-fueled demand and ferocious pricing discipline from a handful of dominant foundries. But that concentration cuts both ways, and for EM investors, the vulnerability is not single-sided. MSCI EM already carries a 37% sector weight in IT and a 45% combined geographic exposure to Taiwan and South Korea alone, meaning the index is structurally pre-loaded with semiconductor risk before a single stock is picked. Stack on top of that the surging profit dominance of semis within those very markets and MSCI EM has effectively become a leveraged proxy on the AI capex cycle – a winning position today but a precarious one if sentiment toward the AI theme softens, earnings surprise cycles turn or geopolitical disruption bites into Asian supply chains. Any de-rating of the semiconductor complex would hit EMs through two channels simultaneously – sectoral and geographic – amplifying drawdown risk well beyond what headline index diversification would suggest. The US tells a more nuanced story: Semiconductor profit share is also at a cycle peak – approaching 20% – yet its broader and deeper equity market provides a natural buffer, diluting the concentration risk that EM investors simply do not have access to. The same industry driving the most exciting equity returns of this cycle is simultaneously the single biggest source of unpriced fragility within it.

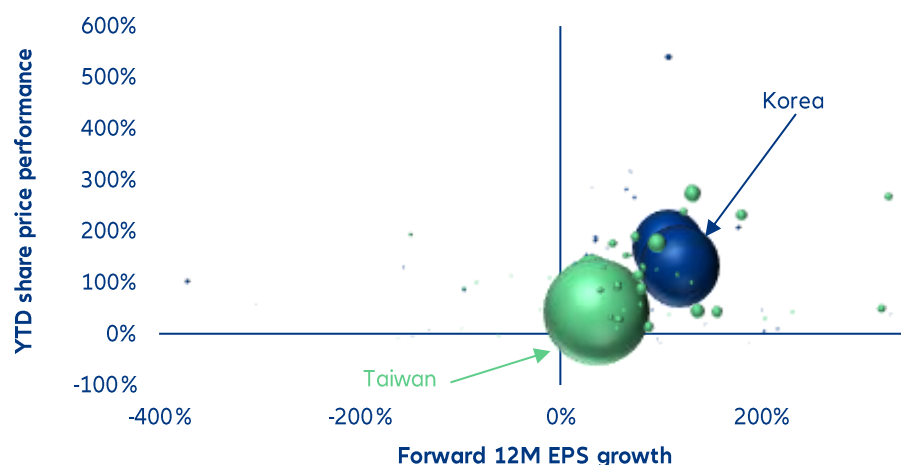
Figure 3: Aggregated market share of South Korea and Taiwan in MSCI index (as of April 2026)



Sources: MSCI, Allianz Research

Hyperconcentration of profits does not mean an absence of opportunity. South Korea dominates High Bandwidth Memory (HBM), the memory architecture at the heart of every AI accelerator, stacking dies vertically to deliver the massive low-latency bandwidth that large language models cannot function without. Taiwan holds what is effectively an unchallengeable near-term monopoly on advanced logic at 2-3nm node, where compute performance, miniaturization and energy efficiency converge – a process gap that Intel and SMIC are years from closing. The positive externalities from these twin centers of excellence ripple across equipment makers, packaging specialists and testing firms, generating an investment universe that extends well beyond the obvious names. That breadth is visible in the numbers: Median forward earnings growth recorded mid-May stood at 52% in South Korea and 42% in Taiwan, up sharply from 35% and 26% at the start of the year, a broad-based acceleration confirming that AI demand is diffusing through the entire supply chain, not just concentrating at its peak.

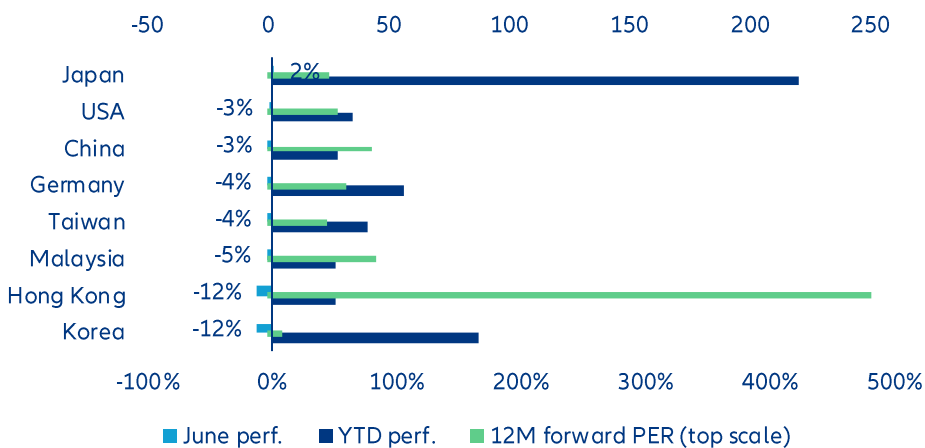
Figure 4: Forward earnings growth vs. 2026 share performance from South Korean and Taiwanese semiconductor firms



*Bubble size refers to market capitalization weight within national semiconductor universe.
Sources: LSEG Datastream, Allianz Research*

Volatility increases when you are under spotlights: go beyond day-to-day analysis. The -12% correction in June in global semiconductor equities was a positioning shock, not a fundamental one. It was caused by a sequence of negative signals that offer a reason for investors to cut their positions and take back their profits after an outstanding run (+82% YTD even after the sell-off). Among main downside catalysts, the disappointment caused by the non-revision of earnings trajectory from a top US chipmakers coupled with rising odds of a rate hike in USA and new concerns toward a weaker demand for HBM chips caused by the release of new Anthropic's Claude 4.8 opus that displays a more efficient inference cost model participate to pinch investor's sentiment toward semiconductor stocks. In reality, the damage was narrow rather than broad-based, concentrated in the two highest-beta markets — Korea, where the KOSPI fell 8.3% and tripped a circuit breaker, and Taiwan, where the largest foundry worldwide posted its largest-ever intraday drop. Korea bore the brunt for structural reasons: index hyper concentration in Samsung and SK hynix, heavy leverage, foreign outflows and a won past 1,560/USD, a mix that turns global tech wobbles into forced-deleveraging cascades. Yet the rally's anchors were not impaired — supply-chain demand kept accelerating and the megacaps rebounded fast — and valuations stay undemanding, with Korea near 10–11x forward earnings, the cheapest of the major Asian markets and North Asia broadly at a discount to the US.

Figure 5: Performance of semiconductor market and forward valuation per region breakdown

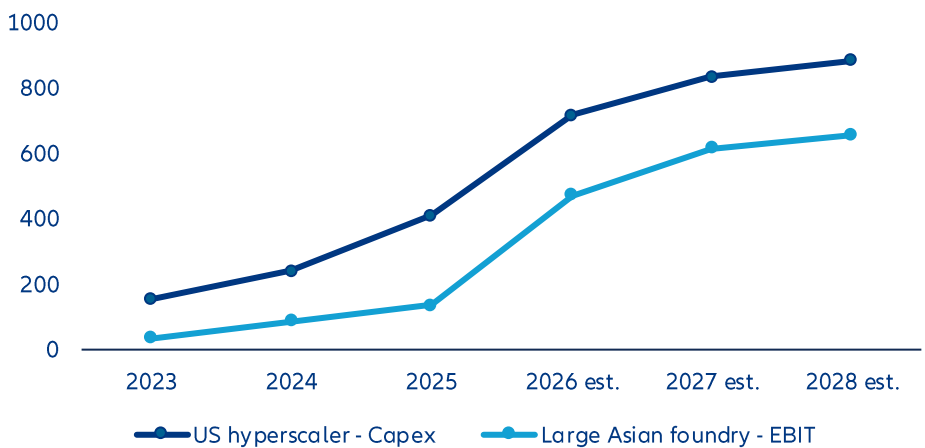


Sources: LSEG Datastream, Allianz Research

Semiconductors remain the purest expression of the AI capex cycle

The current phase of AI development is overwhelmingly infrastructural. Nations, hyperscalers and corporates alike are locked in a race to accumulate the largest possible compute capacity, with digital sovereignty increasingly measured by the volume of cloud space owned and the raw processing power available to run frontier models. Every data center announcement is, in essence, a chip order. US hyperscaler capex has surged from USD150bn in 2023 to over USD400bn in 2025, with guidance pointing to nearly USD900bn by 2028 – a near-sixfold expansion in five years. Large Asian foundry EBIT has tracked that trajectory with striking fidelity, albeit with a one-to-two-year lag that reflects the natural conversion time from capex commitment to wafer demand to earnings delivery: From a negligible USD35bn in 2023, foundry profits are projected to reach USD650bn by 2028. The lag is a feature, not a friction: It means the earnings wave is still building. With no credible sign of hyperscaler capex fatigue, the foundry profit outlook remains extraordinary. The AI-powered world being built is, above all, a chip-powered world.

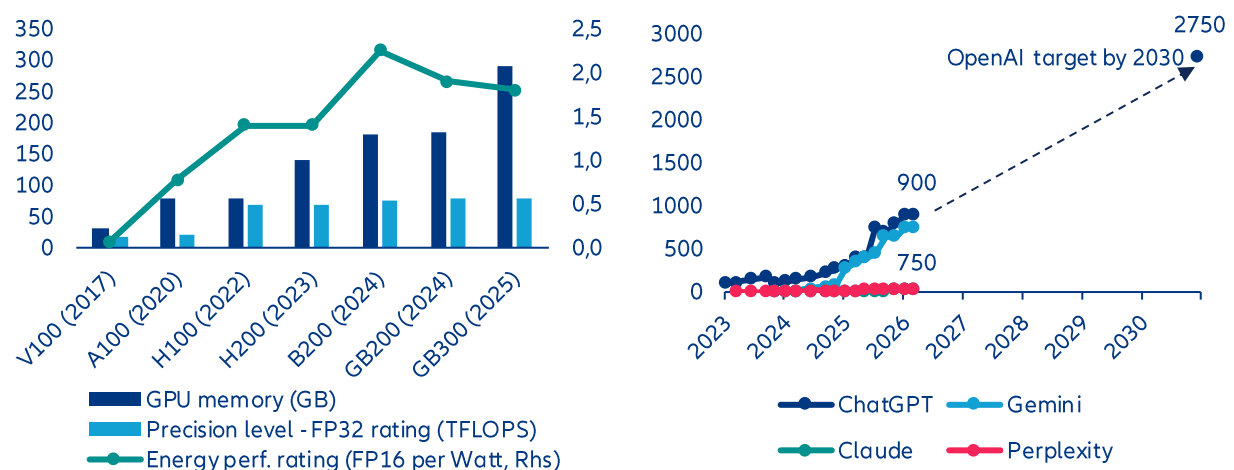
Figure 6: Guidance of US hyperscaler capital expenditure and large Asian foundry EBIT



Sources: LSEG Datastream, Allianz Research

AI diffusion is deepening, not plateauing – and memory demand follows. Each successive Nvidia GPU generation has delivered a step-change in memory capacity: from 32GB on the V100 in 2017 to 280GB on the GB300 in 2025, while energy performance per watt has expanded nearly twentyfold over the same period – a relentless hardware escalation that has made every frontier model generation structurally more chip-hungry than the last. The shift in AI workloads from training toward inference – often cited as a potential headwind for memory demand – is a false concern at the system level: what inference loses in per-query intensity, it more than recovers through sheer volume explosion. OpenAI alone is targeting 2.75bn users by 2030, implying that nearly one in four people on the planet will be regularly querying a frontier model – a demand profile that makes aggregate memory chips Dynamic Random Access Memory (DRAM) and Not-And (NAND) consumption essentially non-discretionary infrastructure spending. Unlike software incumbents facing direct substitution from AI-native capabilities, semiconductor and memory suppliers sit upstream of every model, every query and every data-center expansion – insulated from the application-layer disruption playing out below them. The democratization of AI access is not a risk to the chip cycle; it is its most powerful demand multiplier. The efficiency-bear case that used to come back at our ears at each release of new frontier AI model assumes demand is fixed, but it rarely is: cheaper, leaner inference is precisely what pulls AI into more users, more always-on agentic and multimodal workloads, and more applications that were previously uneconomic — so falling cost per query tends to expand aggregate query volume faster than it compresses the memory intensity of each one. Crucially, the memory call is driven by the scale of deployed infrastructure rather than the elegance of any single model: every incremental accelerator shipped is loaded with HBM and DRAM, and the shift toward inference and long-context, agentic workloads actually raises KV-cache memory requirements, meaning a broader, more efficient installed base de facto consumes more chips in aggregate, not fewer.

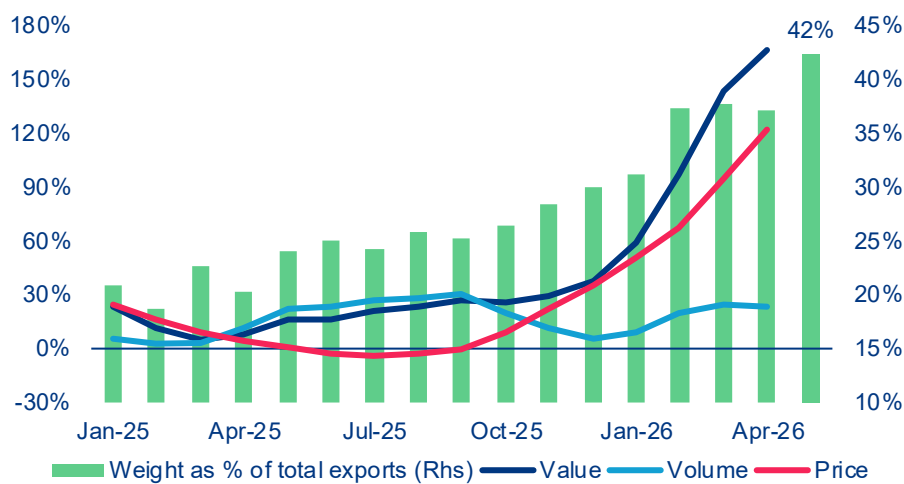
Figure 7 : Memory capacity, performance and power efficiency of GPU (based on Nvidia models) / Number of users of main LLM models, current and estimated (in mn of users)



Sources: Nvidia corporate brochure, Allianz Research

Supply tightness is extending pricing power – and acting as a macro hedge. Supply was already structurally tight before geopolitical stress entered the equation. The Middle East conflict has added an acute dimension: disruption to helium supply chains – a critical purification gas sourced partly from Qatar and indispensable to wafer fabrication – triggered precautionary front-loading across South Korean and Taiwanese buyers, compressing available inventory precisely when AI-driven demand is at its most insatiable. The result is visible in spot prices with brutal clarity: Flash NAND 64GB spot prices have nearly tripled in 2026 alone, surging from below USD5 to nearly USD20 per unit – a pass-through dynamic that semiconductor firms have executed without friction, reflecting a pricing power that few industries can match. When the end-demand justification is AI productivity gains worth trillions in enterprise value, customers absorb chip price inflation with remarkably little resistance. South Korea's semiconductor export value growth has accordingly exploded to +150% year-on-year – a magnitude that dwarfs volume growth and confirms that this is a margin and pricing story as much as a volume one, with the spread between value and volume growth at its widest on record. The slow progress in new capacity build-up postpone in time breakeven point between supply and demand, leading to a likely extension of memory shortage until 2027. As a result, the mix of supply imbalances coupled with extremely-high concentration of the segment ensure a strong pricing power Asian chipmakers that should help them keeping a high margin profile for long.

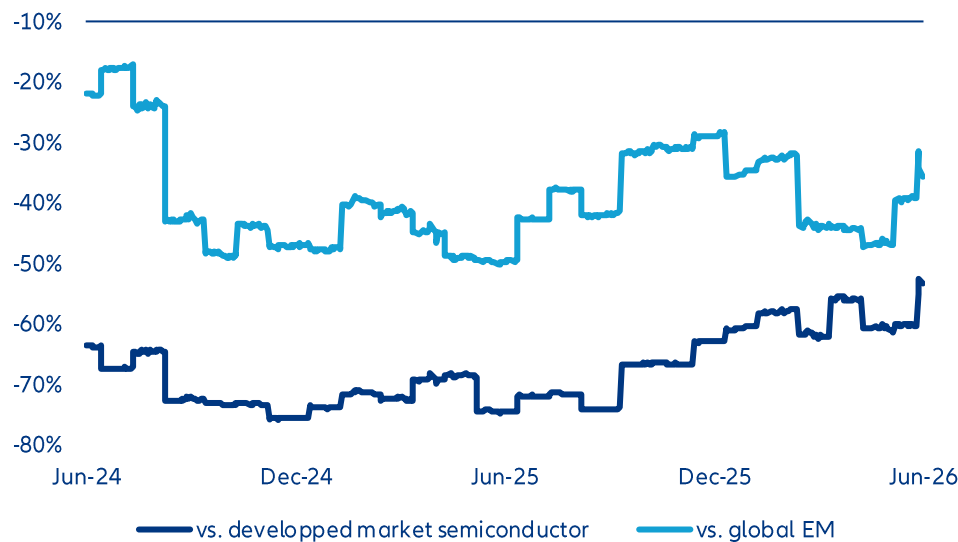
Figure 8: South Korea semiconductor exports trend (price, value and volume) and weight



Sources: Bank of Korea, LSEG Datastream, Allianz Research

Valuations remain compelling despite rally. Even after the year-to-date rally, EM semiconductors remain conspicuously cheap on a market-weighted basis: as of June-26 the industry still trades at roughly a 55% forward-P/E discount to developed market (DM) semiconductor peers and around a third below broad global emerging market benchmark (EM). The rally has admittedly started to dilute that gap — both discounts have compressed off their late-2024 troughs (circa -75% versus global semis, -50% versus global EM), with the narrowing accelerating sharply in the most recent leg, particularly against global EM. Yet the residual discount to DM names sits oddly with the reality that EM players now occupy the most critical, highest-value nodes of the chain — leading-edge foundry and HBM — a positioning that arguably warrants convergence rather than a persistent markdown. Part of the optical cheapness is also a concentration artefact: the aggregate multiple is dragged down by a handful of mega-caps whose steep forward-earnings trajectories compress the denominator, so the headline discount reflects those few names rather than uniform value across the sector. Beyond the pure valuation call, that same skew opens an arbitrage-and-diversification angle — the better risk-reward may sit further along the supply chain in equipment, materials, advanced packaging and testing, rather than in the crowded foundry and pure-memory leaders.

Figure 9: Valuation premium/discount of EM semiconductor industry vs. global EM and semiconductor industry (based on forward 12M price to earnings ratio)



Sources: LSEG Datastream, Allianz Research

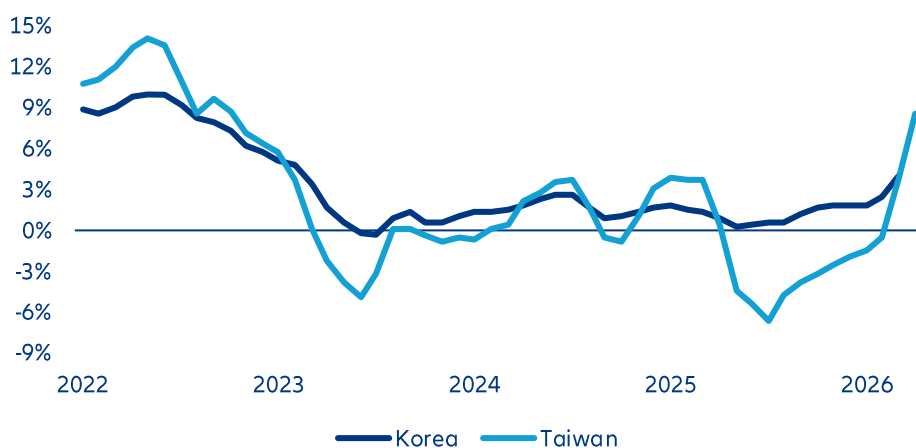
Four risks that warrant active monitoring

Despite the compelling semiconductor narrative, complacency would be a mistake. A clear-eyed view of the next 12-18 months reveals four factors that deserve active monitoring before the next leg of the rally is taken for granted.

Risk 1 - Middle East disruptions: when scarcity becomes a double-edged sword

The same supply tightness that has gifted semiconductor firms extraordinary pricing power now carries a sting in the tail. Helium storage levels have dropped to uncomfortable lows as tensions persist. Compounding the picture, wafer component costs and freight rates have moved meaningfully higher, inflating production costs at a moment when delivery pipelines are already stretched. The structural constraint is unforgiving: New fabrication lines outside Asia take years to reach operational relevance, leaving current capacity with little slack to absorb sustained disruption. Q1 earnings provided some relief – margin guidance pointed only to modest downward adjustments rather than any structural deterioration – and the industry is not defenseless: three to six months of secured helium inventory, combined with recycling protocols put in place as far back as 2022, provide a meaningful buffer against near-term shocks. But that buffer has a clock on it. Should Middle East tensions persist and logistical disruption deepen, the scarcity narrative that has been a profit tailwind could rapidly invert into a production bottleneck – with spot price spikes masking volume shortfalls that ultimately disappoint revenue expectations. The PPI chart captures the inflection already: Taiwan's production cost index has snapped from -7% year-on-year in late 2025 to nearly +9% by mid-2026 – the sharpest reversal in the dataset – while South Korea's is also accelerating, signaling that cost pressures are landing on balance sheets now, not as a future risk. For an industry priced for perfection, that reversal is worth watching with precision.

Figure 10: Production cost index in South Korea and Taiwan (PPI, annual change in %)



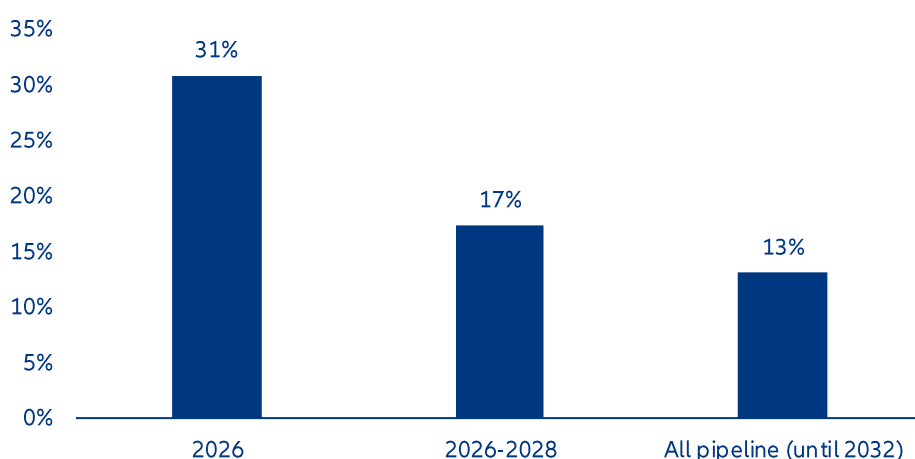
Sources: LSEG Datastream, Allianz Research

Risk 2 - AI capex execution gap: when the infrastructure pipeline meets real-world friction

The AI capex supercycle is the single most powerful demand driver underpinning the semiconductor bull case – which makes any credible threat to its execution timeline a risk that cannot be dismissed. As of end-January 2026, only 31% of US data-center capacity committed for delivery in 2026 was officially under construction – a figure that drops to 17% for the 2026-2028 window and a sobering 13% across the full pipeline to 2032. Behind the relentless drumbeat of hyperscaler capex announcements lies an execution gap that is widening, not closing, driven by planning bottlenecks, grid connection delays, construction labor shortages and permitting friction that no balance sheet commitment can shortcut. A data center that is announced but not built does not pull forward a chip order.

The risk compounds if the macroeconomic backdrop shifts. A renewed inflationary impulse – entirely plausible given rising freight costs, energy price pressure from Middle East tensions and sticky services inflation – could force another monetary tightening cycle, pushing long rates higher at precisely the moment when capital-intensive infrastructure projects are most rate-sensitive. Higher rates do not just raise discount rates on future cash flows, they make banks materially more restrictive in extending project finance to private data-center developers, the fastest-growing segment of the pipeline. Given the near-perfect correlation between hyperscaler capex trajectories and Asian foundry profit guidance, any sustained flattening of investment commitments would transmit rapidly into semiconductor earnings downgrades, eroding pricing power simultaneously as the demand urgency that has allowed cost pass-through begins to dissipate. For a sector priced at a meaningful premium, the margin for disappointment is narrow.

Figure 11: Percentage of the US data-center pipeline officially under construction



Sources: Sightline Climate, Allianz Research

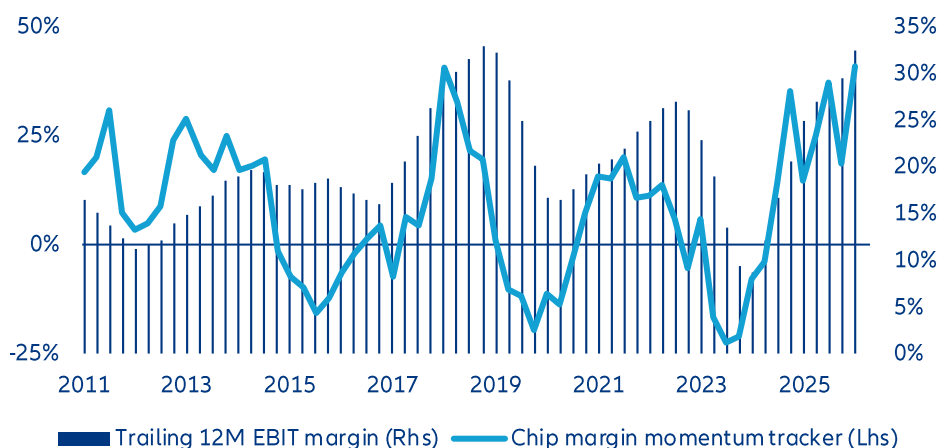
Risk 3 - The capex timing trap: will history rhyme?

The semiconductor industry has a well-documented and costly habit: it invests in new capacity at the top of the cycle, and the production lines come online just as demand rolls over. The pattern is visible with painful regularity in Figure 11: EBIT margins and chip margin momentum have oscillated in violent cycles, with the most brutal compression episodes – 2015, 2019, 2023 – coinciding precisely with waves of new supply hitting markets into weakening end-demand. The post-2022 episode is the most instructive: A surge of capacity investment triggered by pandemic-era shortage hysteria delivered production lines into a demand vacuum, crushing margins to near zero in 2023 and generating inventory write-downs across the industry. The question is whether 2027-2028 will rhyme. The timing risk is concrete and quantifiable. TSMC's Arizona Fab 21 Phase 2 – targeting 3nm and 2nm production – is scheduled for completion by 2027-2028, alongside its ongoing Japan facility expansion in Kumamoto targeting mature nodes. Samsung's P3 and P4 lines in Pyeongtaek – among the world's largest semiconductor complexes – are expected to reach full operational capacity over the same window, adding substantial HBM and advanced logic output to global supply. SK Hynix's M15X facility in Cheongju, purpose-built for next-generation HBM production, is similarly slated for ramp-up through 2027. Collectively, these projects represent the largest synchronized capacity expansion in the industry's history, hitting markets simultaneously, with limited ability to delay or redirect output once fab construction reaches completion.

Trailing 12-month EBIT margins are approaching 30-32%, levels last seen fleetingly around 2018 and never sustained beyond a few quarters. Chip margin momentum, the leading indicator, is already showing the first signs of oscillation at elevated levels, a pattern that historically has preceded compression. The bull case argument – that unprecedented data-center pipeline demand will absorb the incoming supply wave – is plausible but carries the execution uncertainty documented in Risk #2: only 13% of the US data-center pipeline to 2032 is currently under construction. If the demand realization timeline slips by even 12-18 months relative to supply delivery, the industry faces the same inventory build and margin compression dynamic it has experienced in every prior cycle.

Semiconductor profits have been the fuel of the EM equity rally; a cyclical mean-reversion in foundry margins would not just slow the trade, it would reverse it.

Figure 12: EBIT margin and margin momentum index of large Asian foundries



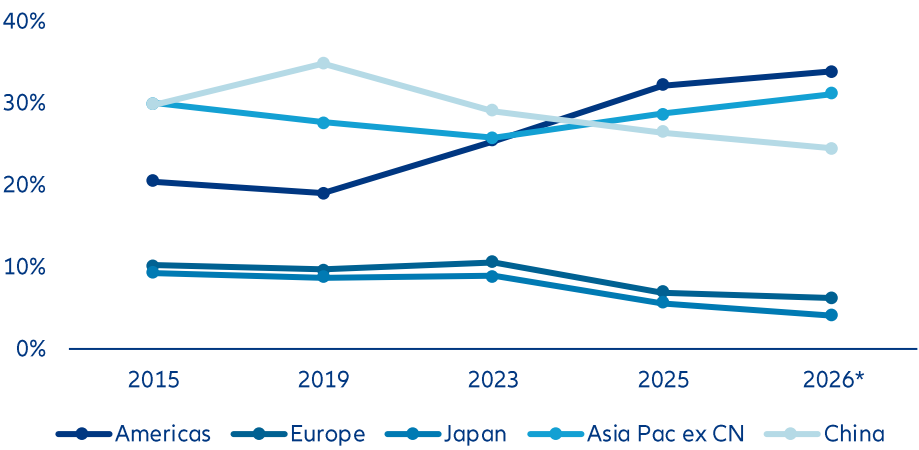
Sources: LSEG Datastream, Allianz Research

Risk 4 - The China wildcard: a competitor in construction, not yet in contention

Figure 13 below frames the competitive landscape with striking clarity: Americas' share of global semiconductor shipments by value has surged from 20% in 2015 to 34% in 2026, while Asia Pacific ex-China has climbed to 31% – the two dominant and widening poles of the industry. China, meanwhile, has seen its share compress from 30% to 24% over the same period, a trajectory that reflects both export controls tightening its access to advanced process technology and a domestic industry still years from self-sufficiency. Europe and Japan have faded to near-irrelevance at 4-6%, a structural marginalization that makes their dependence on Asian supply chains all the more acute. But the direction of Chinese policy ambition is unambiguous and dismissing it entirely would be a mistake. Beijing has set an explicit target of covering 70% of domestic semiconductor needs through indigenous production – a goal backed by the most aggressive state financing program in the industry's history. The Big Fund Phase III alone mobilizes an estimated USD30-50bn, part of a total USD70bn incentive envelope that dwarfs both the US CHIPS Act (USD53bn) and the EU Chips Act (EUR43bn) in raw commitment size. The political signal is equally unambiguous: During President Donald Trump's recent visit, Chinese officials made clear they had no interest in dismantling semiconductor trade barriers. The strategic priority is building domestic capacity, not negotiating market access. Semiconductors have become a theatre of technological sovereignty, not commercial competition.

Yet the gap between ambition and execution remains vast across the three most critical chokepoints. On EUV lithography – the indispensable tool for sub-7nm node production – China has made headline progress, with reports of a prototype developed under a project drawing on expertise from former ASML personnel. But scaling from prototype to mass production is an entirely different challenge: The official target of EUV mass production by 2028 is viewed skeptically by industry experts, with more credible timelines pointing to post-2030 at the earliest for volume output. On High Bandwidth Memory, the process know-how required to compete with SK Hynix and Samsung at HBM3/HBM4 levels demands both multi-billion dollar capex and years of yield optimization that state funding alone cannot shortcut. And on EDA software – the design automation tools that underpin every advanced chip architecture – building competitive capability requires deep, accumulated human expertise that neither subsidies nor industrial policy can manufacture quickly. China is a competitor in construction, not yet in contention. But the 2028-2030 window, when domestic capacity begins reaching meaningful scale, is precisely when Asian foundry supply is also expected to surge. The collision of those two dynamics is a risk the market has not yet begun to price.

Figure 13: Global semiconductor shipments by value - geographic market share



Sources: SEMI, Allianz Research

These assessments are, as always, subject to the disclaimer provided below.

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The statements contained herein may include prospects, statements of future expectations and other forward-looking statements that are based on management's current views and assumptions and involve known and unknown risks and uncertainties. Actual results, performance or events may differ materially from those expressed

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(v) persistency levels, (vi) particularly in the banking business, the extent of credit defaults, (vii) interest rate levels, (viii) currency exchange rates including the EUR/USD exchange rate, (ix) changes in laws and regulations, including tax regulations, (x) the impact of acquisitions, including related integration issues, and reorganization measures,

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Ten Years After Brexit: Resilience Without Revival

22 June 2026
Allianz Research



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Executive Summary



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- **Brexit 10 years on: Neither collapse, nor renaissance.** As the UK looks for its 7th Prime Minister in 10 years, this week also marks a decade since Brexit roiled markets and divided forecasters. Predictions ranged from economic collapse to a renaissance. A decade later, we take a hard look at which projections came through and which failed to pass. The outcome has been more nuanced than many expected, with indicators pointing to both economic strengths and weaknesses — not all of them Brexit related.
- **Britain's Economic Backbone: The knowledge economy, tech and clean energy.** UK ICT exports to the EU have almost doubled since Brexit, demonstrating the continued competitiveness of Britain's knowledge economy. The UK remains the world's second-largest exporter of financial services, accounting for 21% of global exports, while deepening its provision of financial services to EU markets. In private markets, the UK continues to attract more venture-capital funding than any European competitor. Between 2020 and 2026, it raised roughly USD164bn. Since then, London has retained much of its global financial importance. The UK still accounts for nearly 50% of global OTC interest-rate derivatives trading and almost 38% of global foreign-exchange turnover. In clean energy, the UK has emerged a leader within Europe. Wind generation has increased by 130% since 2016, reducing the UK's fossil dependence while enabling a successfully phased out coal in 2024.
- **Burned by Brexit: Dragging growth, trade frictions and lagging Leavers.** Independent studies estimate that GDP would have been 2 to 4% larger without the political instability and trade friction triggered by Brexit. Since 2016, "Leave" regions have generally underperformed compared to the UK as a whole: 59% of the population of the "Leave" areas have seen their regions fall further behind national income per capita average. Since the vote and the formal exit from the bloc's economic structures on 1 January 2021, growth has relied increasingly on foreign-born workers, which has fueled more than half of GDP expansion. Brexit has increased frictions and reduced trade flows. While the EU remains the UK's largest trading partner, structural estimates suggest UK-EU trade is around 21% lower for goods than it would have been without Brexit. New trade agreements and diversified supply chains with the US, China and Commonwealth countries have failed to match the scale of the economic ties previously enjoyed within the EU. UK assets continue to trade at a discount relative to international peers.

- **A lasting lesson of the 2022 mini-budget crisis is that fiscal credibility matters.** Investors now demand a structurally higher risk premium on UK assets against a backdrop of rising fiscal imbalances. Equity markets, where UK stocks have underperformed both US and European peers in the past decade, reflect this premium. In private markets, the magic has faded into a priced-in discount. A decade on, Brexit's imprint reads through the UK's two largest private-capital engines, private equity and venture. UK investments were 12–18% below a no-Brexit path as of 2025, a one-off level shift now cleared into a structurally higher cost of capital. Private equity has proved resilient, but increasingly as a US funded value trade rather than home grown momentum with compressed valuations turning UK companies into take-private targets for US capital.
- **Beyond Brexit: Identifying and implementing solutions for the future.** The UK scores high on business creation, labor-market flexibility, higher education and research, compared with many other advanced economies. This indicates that domestic bottlenecks — exposed by Brexit but not caused by it — are the root of the country's disappointing growth performance. The government has identified many of these issues, but needs to push further. Priority actions should include:
 - Accelerating planning reform,
 - Increasing investment in housing and energy infrastructure,
 - Strengthening support for innovation and strategic industries,
 - Treating NHS reform as an economic priority, and
 - Addressing the slow diffusion of new technologies from frontier firms to the broader economy.
- **A revamp of fiscal rules, combined with reforms to pensions spending and property taxation as well as broadening the VAT base, would help redirect scarce public resources toward deficit reduction and investment, with an eye to defense.** Unlike the EU-27, Britain has staged no distinct post-2022 rearmament inflection. The UK government has committed to increase defense spending to 2.5% of GDP by April 2027, and 3.0% target by 2030, but they are not set in stone. The cooperation map looks busy — AUKUS with the US and Australia, GCAP with Japan and Italy, a privileged-partner track toward the EU's SAFE instrument — yet it reads as a catalog of commitments rather than tangible delivery. Government investment would help attract investors and domestic defense companies to commit capital to UK capacity. Reintegrating more closely with European energy markets would generate meaningful economic benefits. High energy prices caused by the UK leaving the EU Internal Electricity Market, successive energy supply crises and grid bottlenecks linked to the rapid expansion of renewable energy have continued to undermine industrial competitiveness. Divergence between UK and EU carbon markets also risks creating new trade barriers for energy-intensive industries. Reintegrating energy and carbon markets could help reduce friction costs, stabilize power prices and improve supply security.

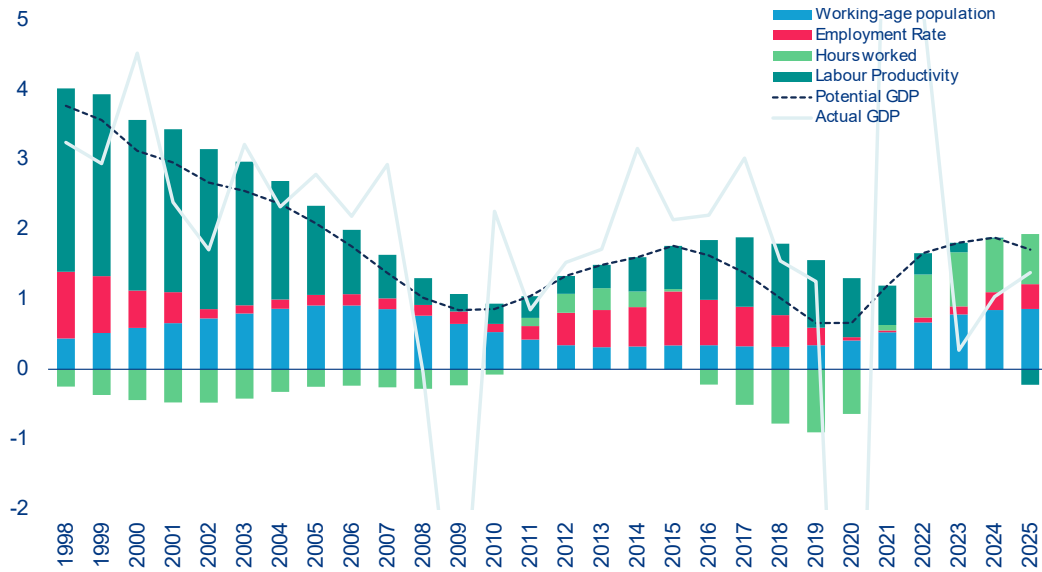


10 years since the Brexit vote: an evolving economic and energy landscape

This month marks 10 years since the Brexit vote and the promises of greater democratic accountability, reduced immigration and independent trade policy. The Brexit referendum took place on 23 June 2016, gathering a high turnout of 72.2%, with 51.9% (17.4mn) votes for leaving the EU against 48.1% (16.1mn) for remaining. The Leave campaign was built around several core themes and promises. First, sovereignty, or "Take Back Control": restoring the authority of the UK Parliament by removing the influence of EU institutions and courts. Second, immigration control: ending the free movement of people from the EU, and allowing the UK to set its own immigration policy. For most Brexiters, this meant reducing the flows of immigration drastically, including from non-EU countries. Third, trade and regulation: gaining the freedom to negotiate independent trade agreements and reducing or removing EU-derived regulations. Fourth, budget contributions: stopping contributions to the EU budget and redirecting some of those resources to domestic priorities (notably providing additional funding to the NHS). Finally, democratic accountability: decisions affecting the UK would be made by elected British politicians rather than EU bodies. In all, Leave supporters generally argued that Brexit would increase political autonomy and economic flexibility, while opponents warned about trade barriers, lower investment and weaker economic growth. The UK effectively left the EU on 31 December 2020.

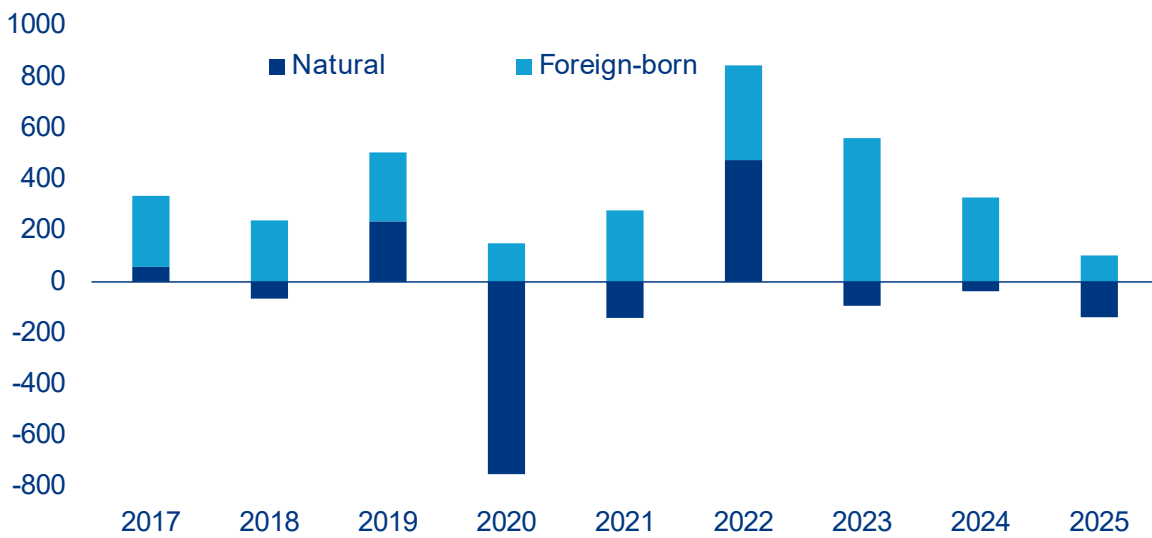
UK politics has been shaken since the 2016 vote. Debates about closer ties with the EU, or even potential re-entry into the EU, have gained momentum recently amid a Labour party leadership contest. The post-Brexit period has been one of unusually high political turnover in the UK. Since the referendum, the UK has had six Prime Ministers (PM), compared with only four PMs during the entire 1997-2016 period. Much of this high turnover was directly linked to Brexit negotiations, disagreements over the UK's future relationship with the EU and subsequent political pressures. Poor fiscal decisions against a backdrop of rising budget imbalances added to the political instability, with the infamous "mini Budget" debacle of unfunded tax cuts by PM Liz Truss and Chancellor Kwarteng in September 2022 spooking financial markets. The recent challenge of PM Starmer's leadership with the ruling Labour Party has sparked a renewed debate over the opportunity to deepen trade and economic ties with the EU. The current government's stance is to rule out rejoining the customs union or the single market or bringing back freedom of movement.

Figure 1: Potential GDP growth decomposition (% y/y)



Sources: LSEG Workspace, Allianz Research

Figure 2: Employment growth decomposition (000s)



Sources: LSEG Workspace, OECD, Allianz Research

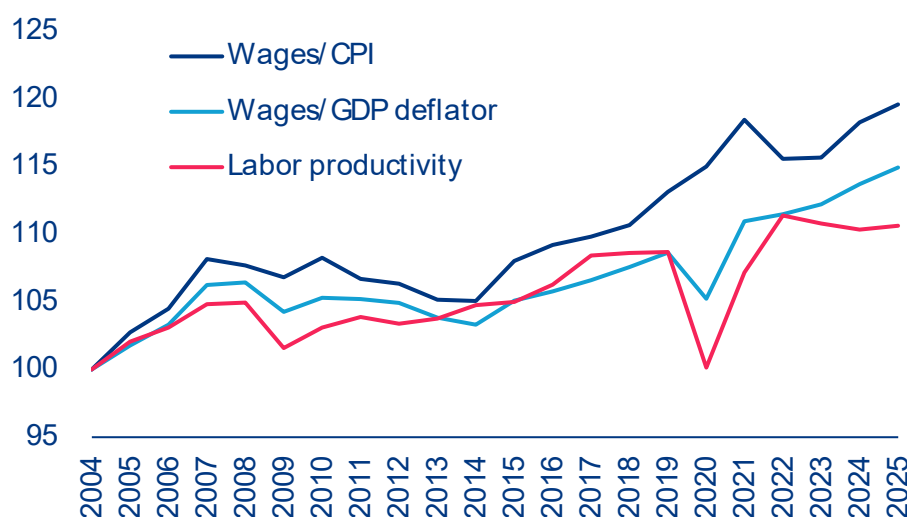
Note: we take into account only permanent migration, which is defined by legal status, not duration of stay by the OECD. It includes migrants with a pathway to remain indefinitely such as refugees, family migrants and some labor migrants, while those without such rights (e.g. students or seasonal workers) are not considered. As per the OECD estimate, we assume that roughly 75% of new immigrants to the UK become employed.

Reduction of regional economic disparities has been disappointing. While Scotland and London strongly supported remaining within the EU, other regions such as the Midlands and North East of England voted firmly to Leave. Reducing inequalities across regions, was a key talking point both during and after the Brexit referendum. ONS data indicates that convergence has generally failed to materialize. We look at the sample of 332 areas in Great Britain representing over 61 million people. 227 areas voted Leave and 105 voted Remain. 59% of the population of “Leave” areas have seen their regions fall further behind national income per capita average since 2016, although some important leave-voting cities such as Blackpool, Doncaster and Wolverhampton have seen an improvement compared to the UK average.

GDP growth has held up since Brexit, but shifted towards more government consumption and less business investment. Higher immigration flows and working hours per worker have offset worsening productivity trends. Despite the challenging environment since Brexit, including the Leave vote, “Truss” moment, political instability, Covid pandemic and energy shock of 2022, UK GDP growth has held up well. Annual GDP growth averaged +1.4% since 2016 — exactly the same

average pace than the previous 10 years (2005-2015). The composition of growth changed, however, with higher growth of government consumption (+2% vs +1.3%) and residential investment (+3.6% vs -1.8%) offsetting lower business investment growth (+2.4% vs +3.7%). Export and import volumes both decelerated at roughly the same pace (-0.7pp to +2.5% for imports, -0.5pp to +2.4% for exports). Household consumption growth was unchanged (+1.3%). From a supply-side perspective, the underlying drivers of potential/sustainable GDP growth¹ changed even more dramatically. The working-age population re-accelerated after 2016 (Figure 1), supporting job creations and GDP. Job creation remained concentrated among foreign-born workers (newcomers) (Figure 2), amid worsening demographics for native and longer-term residents. In all, foreign born workers’ jobs have grown +0.8% per year on average since 2016, contributing to more than 50% of UK GDP growth². Meanwhile, trend growth of hours worked per worker has also accelerated, whilst trend productivity growth faltered. In all, while UK GDP growth has been resilient since Brexit, studies generally estimate that in a counterfactual were Brexit not to happen, GDP would have been between +2% and +4% larger³.

Figure 3: Labor productivity & real wages (indexed 100 in 2004)



Sources: ONS, Allianz Research

¹ We look at potential GDP to strip out cyclical and financial effects that could blur the analysis.

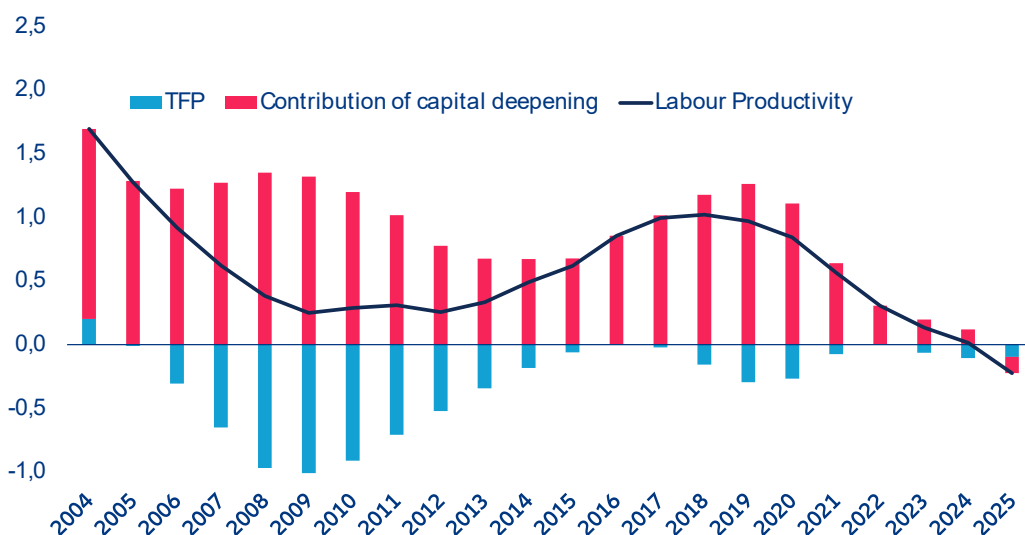
² Immigrants may generally have lower productivity levels than the rest of the population in aggregate (because they tend to work in lower-paid sectors), and therefore weigh on productivity dynamics and GDP growth by this channel. As a result, we take a conservative estimate (50%) lower than headline employment numbers would imply (57%).

³ For instance the Bank of England estimates a cost of -3.5% GDP.

Real wages and income growth has been decent despite an inflation surge, but poor productivity is likely to increasingly weigh on prospects.

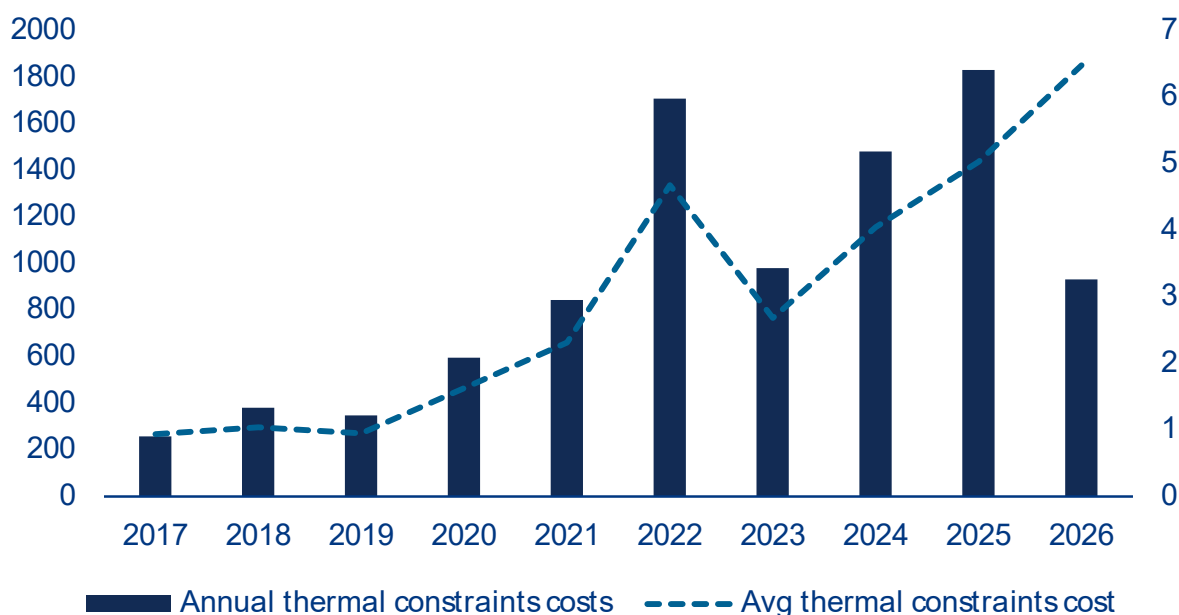
Real income per capita and wage growth have picked up pace since Brexit, despite continued poor productivity performance and elevated inflation. Real income per capita has grown by +0.9% per year on average since 2026, against +0.5% in the ten years prior. Real wage (adjusted by CPI inflation) trends showed even better improvements (+0.9% vs. +0.1%). But this occurred against a backdrop of lingering productivity weakness, which has increased in importance since the pandemic (Figure 3), signaling increasing margin compression for corporates. In other words, decent real wage (and income) growth is unlikely to last unless productivity picks up and/or corporates continue to take the hit on their margins. Very slow investment and capital accumulation has intensified since the pandemic, increasingly weighing on trend productivity growth (Figure 4). The persistent weakness of total factor productivity (TFP), i.e. the underlying efficiency of the economy, reflects many factors that are common to many other developed nations. Recent analysis suggests the slower diffusion of new technologies from frontier firms to the broader economy has been a key impediment to TFP⁴.

Figure 4: Trend productivity growth decomposition (% y/y)



Sources: LSEG Workspace, Allianz Research

⁴ The basic idea comes from a series of OECD papers by Dan Andrews, Chiara Criscuolo and co-authors. They show that productivity growth at the global frontier remained relatively strong during the 2000s and 2010s, but that the gap between frontier firms and the rest of the economy widened. In other words, innovation did not stop; the problem was that innovations spread more slowly to ordinary firms.

Figure 5: Annual thermal grid constraints/congestion costs in the UK power grid (in EUR mn, left) and average daily costs

Sources: LSEG Workspace, Allianz Research

Sector-wise, some services activity (transport and storage, ICT, support services activities, professional services) and public administration output have done well, Manufacturing flatlined but with some bright spots. Structural changes since the pandemic, including global ones, such as increasing Chinese competition on manufacturing, have likely played more of a role in shaping economic outcomes than Brexit itself. The UK manufacturing sector was -1.4% lower in 2025 relative to 2016. However, this masks dramatic shifts within the industrial sector, with energy-intensive sectors shrinking and less energy-intensive sectors doing well. Transportation equipment (+25% between 2025 and 2016), pharmaceuticals (+43%), textiles (+8%), agrifood (+31%), electrical equipment (+20%) and computer & electronics (+20%) saw a sharp increase of production. Meanwhile food/paper (-11.7%), coke/refined petroleum products (-38.4%), chemicals (-10%) and gas steam/electric output (-54%) shrank dramatically.

On energy markets, the UK's departure from the EU Internal Electricity Market (IEM) on 1 January 2021 is one of the most direct and measurable Brexit-related costs for UK energy consumers. Under IEM market coupling, interconnectors used implicit capacity allocation to converge prices efficiently across borders. After Brexit, the shift to explicit auctions reduced trading efficiency and increased transaction costs, resulting in an estimated upward pressure on wholesale prices of between 0.25-0.7% (GBP90-250mn) in 2021⁵. The UK also lost its previous role in EU electricity-market governance and ENTSO-E decision-making structures, although operational cooperation between British and continental system operators remains extensive.

⁵ UK-EU Energy and Climate Cooperation: Why heightened engagement is imperative for Net Zero (Energy UK)

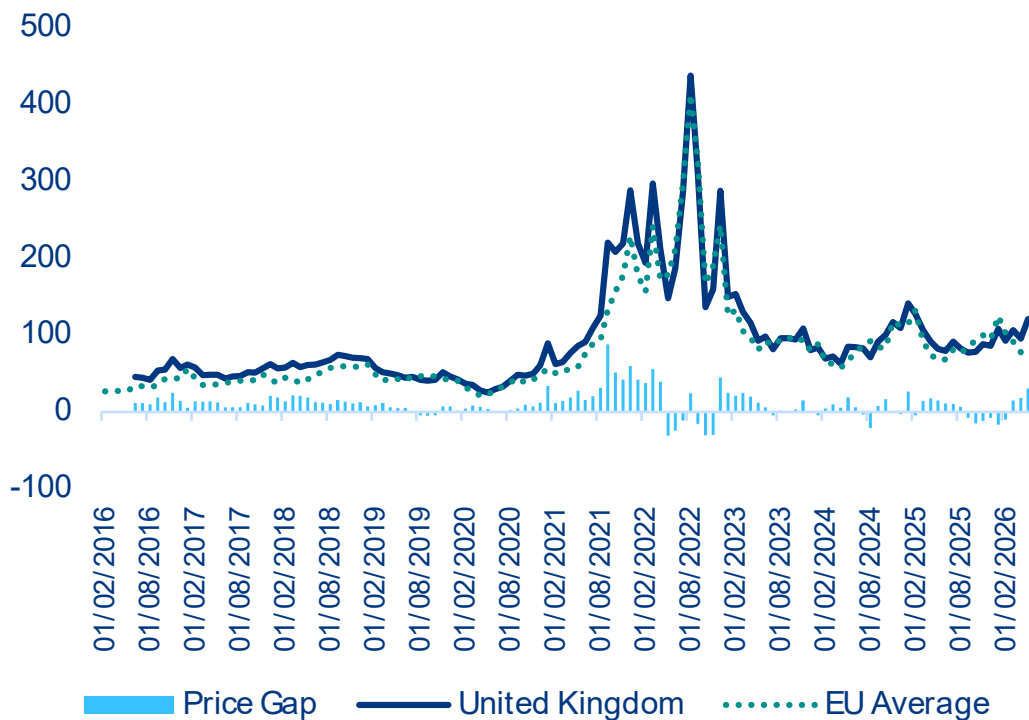
These institutional changes coincided with a broader structural evolution of the UK electricity system. Wind generation has more than doubled since 2016 (+130%), while solar output has increased by 85%, marking a rapid expansion of variable renewable capacity. This transformation enabled a landmark milestone: the UK completed its coal phase-out in September 2024, becoming one of the first major economies to eliminate coal from its electricity system entirely.

However, the rapid expansion of renewable generation has been highly geographically concentrated, particularly in northern Scotland, creating a growing imbalance between generation and demand centers. This has resulted in a severe north–south transmission bottleneck with rapidly escalating system costs. Thermal constraint costs, the cost of managing congestion in the electricity grid when power cannot be transported from where it is generated to where it is needed, rose from GBP258mn in 2017 to GBP1.83bn in 2025 (Figure 5). Based on the H1 2026 run-rate of GBP6.5mn per day, full-year 2026 costs are on track for approximately GBP2.4bn (+29% year-on-year). While these costs are primarily driven by domestic grid constraints and the

spatial distribution of renewable generation, rather than Brexit itself, Brexit remains relevant in a second-order sense. Reduced integration with neighboring electricity markets and slower progress towards coordinated North Sea infrastructure may have limited the system’s ability to alleviate congestion through cross-border balancing and joint offshore grid development.

The UK has materially benefited from cross-border electricity trade, with interconnectors enabling imports of lower-priced continental electricity and reducing the need for domestic redispatch. This is reflected in a close relationship between UK–EU price differentials and physical flows, with the UK typically acting as a net importer when domestic wholesale prices are below those in neighboring EU markets (Figure 6). This mechanism also helps reduce price volatility by allowing continuous arbitrage across markets. In 2022, however, this relationship temporarily reversed: the European energy crisis and reduced French nuclear availability tightened continental supply conditions, leading the UK to become a net exporter as flows adjusted to extreme price differentials.

Figure 6: UK vs EU power price development and gap (USD/MWh)

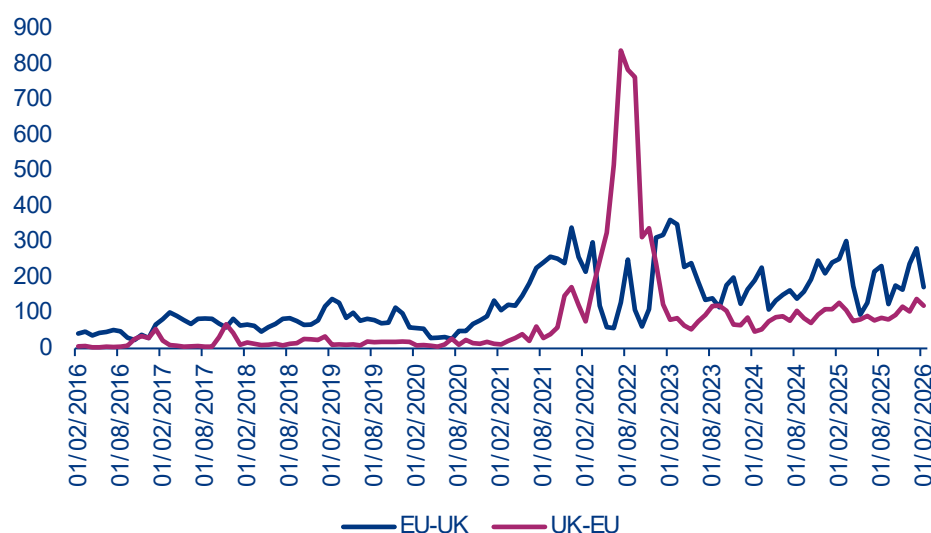


Sources: EMBER, Eurostat, Allianz Research

By contrast, the widening of the UK–EU electricity price gap in 2021–22 reflected a period in which both domestic flexibility and cross-border trading efficiency were reduced. This included the IFA1 interconnector outage, an unusually low wind year in Great Britain, increasing reliance on gas-fired generation and reduced interconnector efficiency following the post-Brexit shift from implicit to explicit capacity allocation. Together, these factors increased the decoupling of UK and EU power prices during the initial post-Brexit period and

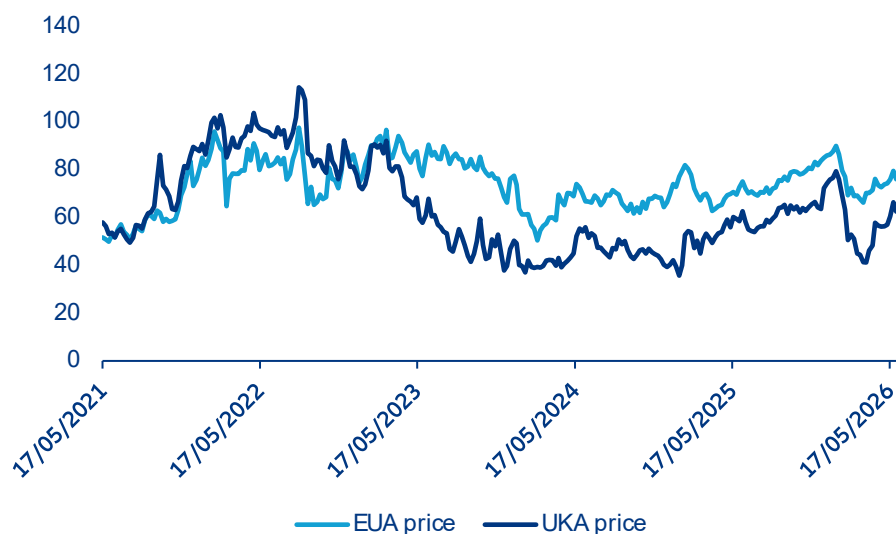
subsequent energy shock. As a result, the post-Brexit UK electricity price premium over the EU rose by 23.6%, peaking between September 2021 and May 2022, when UK wholesale prices averaged EUR52/MWh above the EU average. Although this premium has partially normalized, it remains structurally elevated at EUR15–25/MWh in 2024–25, reflecting persistent integration frictions and a higher reliance on domestic gas-fired generation.

Figure 7: Cross-region power trade (EUR mn, right)



Sources: EMBER, Eurostat, Allianz Research

Figure 8: Co-development of UK and EU ETS allowance prices (in EUR/tCO₂)



Sources: LSEG Workspace, Allianz Research

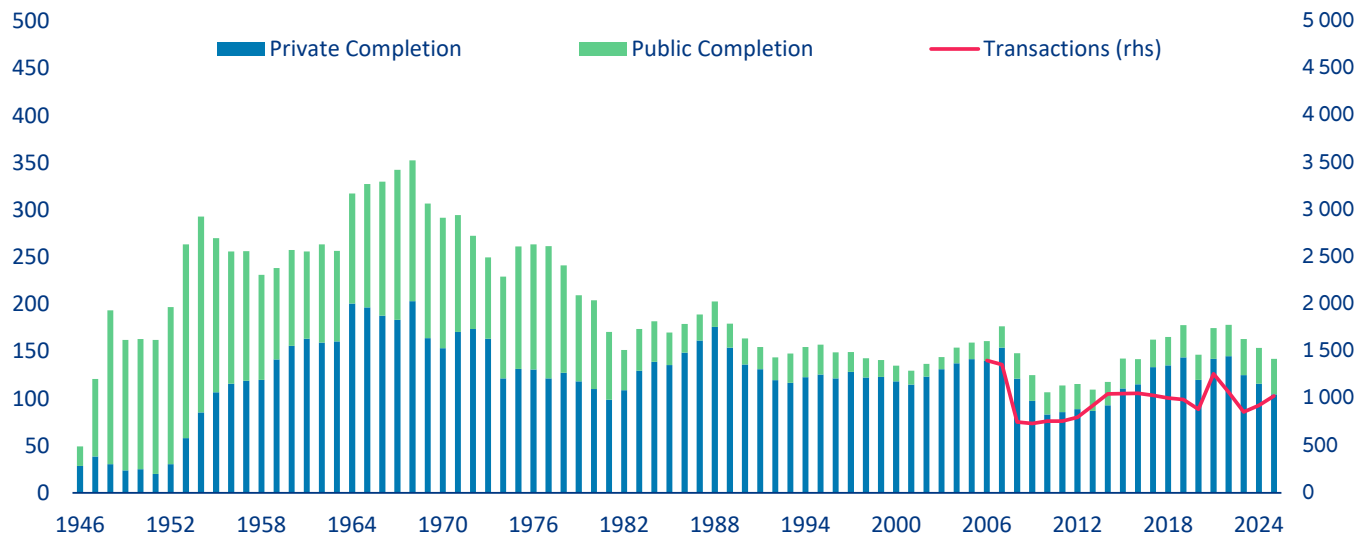
While post-Brexit divergence in energy markets has remained relatively limited, carbon pricing is emerging as a potentially more material channel of separation through its impact on trade-exposed industrial costs.

The UK ETS has traded at a persistent discount to the EU ETS since 2023 (Figure 8). This reflects weaker allowance demand in the UK power sector following rapid decarbonization and greater policy uncertainty around the future tightening path relative to the more clearly signaled trajectory of the EU system. While this reduces compliance costs for UK emitters, it weakens the relative carbon price signal and creates a growing competitiveness gap for energy-intensive sectors exposed to European markets.

This gap becomes more consequential in the context of the EU Carbon Border Adjustment Mechanism (CBAM), which from 2026 will require importers of carbon-intensive goods such as steel, cement, aluminum, fertilizers and electricity to pay the difference between EU and domestic

carbon prices. As a result, the existing UK–EU ETS price differential is increasingly translated into an effective cost wedge on UK exports to the EU, rather than remaining a purely domestic compliance gap. As free allocation in the EU ETS is phased out through 2034, this transmission of carbon price differences into border-adjusted costs is set to strengthen further. Taken together, ETS divergence and CBAM therefore increasingly embed carbon price differentials into trade frictions for energy-intensive UK exports to the EU.

Figure 9: Co-development of UK and EU ETS allowance prices (in EUR/tCO₂)



Sources: LSEG Workspace, Allianz Research



UK economic and energy policies in need of a revamp

A decade after the Brexit referendum, the UK economy remains constrained by a few, but big, structural growth drags: elevated electricity costs and infrastructure bottlenecks.

At the same time, the UK continues to boast many strengths. Productivity growth has been among the weakest in the developed world since the GFC, business investment remains low relative to peer economies, housing supply is chronically insufficient and infrastructure bottlenecks continue to constrain economic activity. Table 1 looks at the performance of the UK on key structural metrics relative to “well performing” developed economies which display elevated levels of income per capita: the US, the Netherlands, Denmark and Sweden. In terms of “output”, the UK is clearly lagging behind in terms of productivity, but hours worked per working person and the employment rate are not particularly low, although the latter has improved less than peers since Brexit. In terms of education proficiency, the UK boasts good scores, particularly in science, and has even improved in country rankings in mathematics and reading proficiency. The UK devotes the largest budget to education (6.1% GDP). In terms of business environment and financial markets integration, the UK also scores well. High electricity costs and a low AI infrastructure index are where the UK really stands out as lagging behind.

In this regard, the current Labour government has correctly identified the economy’s core challenges. Its emphasis on raising public and private investment and accelerating infrastructure delivery reflect a recognition that the UK’s growth problem is fundamentally a supply-side problem. The government’s reform agenda has also correctly highlighted the role played by excessive planning restrictions and regulatory barriers. According to the OECD, the UK’s planning system imposes unusually

high costs and uncertainty on investment projects. Lengthy and unpredictable permitting processes have become an important constraint on housing construction, infrastructure deployment and private investment, contributing to the country’s weak capital accumulation and productivity performance. The consequences are visible in persistently high housing costs, lengthy approval processes and chronic underinvestment in transport, energy and industrial projects. The government’s efforts to streamline planning procedures therefore target one of the most important barriers to long-term growth.

Similarly, the government’s attempts to provide greater strategic direction and higher funding to industrial policy goes into the right direction. The Industrial Strategy identifies priority sectors – including advanced manufacturing, clean energy, defense, digital technologies and life sciences – in which the UK enjoys strong comparative advantages. The creation of the National Wealth Fund and additional support for clean-energy investment represent useful steps towards mobilizing private capital. Recently, the step-up of public funding toward AI hardware companies (GBP1.1bn) represents a welcome development.

Efforts to improve pay and working conditions in essential public services, particularly the NHS, are sound policies to be pursued. The government has well recognized that persistent recruitment and retention difficulties have weighed on labor participation, healthcare outcomes and overall economic performance. Restoring the attractiveness of these occupations is a necessary condition for improving public service delivery and raising the economy’s productive capacity.

Table 1: Economic outcomes and performance of UK economy versus peers and change since Brexit

	Output			Education and Labour				Business		Innovation and AI						Energy	
	Productivity per hour	Hours worked per worker	Employment Rate	PISA Maths Scores Rank	PISA Reading Scores Rank	PISA Science Scores Rank	Educational Institutions Spending (% GDP)	Financial Markets Depth Index	Product Market Regulation	R&D (%GDP)	AI R&D (%GDP)	Total AI Investment (% GDP)	AI R&D Index Rank (per capita)	AI Infrastructure Index Rank (per capita)	AI Investment Index Rank (per capita)	Electricity Cost (\$ per kWh)	Wholesale Electricity Prices (€/Mwh) - Average last 5 years
UK	74,0	1512	74,8	14	13	15	6,1	0,99	0,9	2,7	0,2	0,9	13	27	10	0,4	128,2
US	84,1	1796	71,9	34	9	16	5,8	0,99	1,6	3,5	0,4	1,2	5	7	2	0,2	
Denmark	92,3	1379	77,2	13	15	20	5,3	0,61	1,1	3,1	0,3	1,0	15	12	9	0,4	107,4
Netherlands	82,1	1445	82,3	10	35	25	5,0	0,96	0,9	2,3	0,2	1,5	16	9	13	0,3	117,5
Sweden	81,9	1431	76,7	22	18	21	5,3	0,93	0,8	3,6	0,3	0,8	12	5	3	0,2	66,1
Change since 2015	%	%	%	2018 Rank GDP p.p.				GDP p.p.		GDP p.p.						% change since 2015-2019	
UK	5,0	-0,1	1,9	27	22	15	-0,1			0,4						135,5	
US	14,9	-1,9	4,6	40	24	25	-0,3			0,7							
Denmark	8,5	-2,0	6,8	12	18	21				0,0						223,2	
Netherlands	0,5	-0,3	8,7	11	15	17	-0,4			0,1						186,1	
Sweden	5,0	-2,4	1,6	24	17	28	0,0			0,4						99,0	

Sources: various, Allianz Research

Restoring the attractiveness of these occupations is a necessary condition for improving public service delivery and raising the economy's productive capacity.

However, while the government's diagnosis is broadly correct, implementation has often fallen short of its ambitions. Despite the government's emphasis on industrial policy, funding commitments remain modest relative to the scale of the challenge. Public support for strategic sectors remains fragmented across numerous programs and falls short of the large-scale industrial policies pursued by competitors such as the US, China and increasingly the EU. The result is a mismatch between ambitious rhetoric and limited fiscal resources. Similarly, some of the initial ambition of planning reform has been diluted in response to political opposition, raising concerns that housing supply and infrastructure delivery may not improve sufficiently to alter the UK's long-term growth trajectory.

Questions also arise regarding the credibility of the fiscal strategy. The increase in employer National Insurance (NI) Contributions in 2025 raises the cost of labor at a time when the economy requires stronger hiring, investment and labor demand. From an efficiency perspective, greater reliance on broad-based consumption taxes such as VAT would likely have generated similar revenues while imposing fewer distortions on employment decisions. The government has also relied heavily on relatively narrow

tax measures targeting capital gains, carried interest and non-domiciled taxpayers, despite the fact that such revenue sources are often uncertain and susceptible to behavioral responses. Table 2 depicts the key public expenditures and revenues ratios for the UK against the "well performing" peer countries. Figures from the OECD (for cross-country comparisons purposes) are only from 2024, but highlight that the UK has a relatively low tax take on goods & services consumption (essentially VAT), and this take has decreased since Brexit (at 10.2% GDP vs over 11% in Sweden and Denmark). VAT is one of the least distortive form of taxation (ie with limited negative effect on potential growth⁶). Social security contributions, while starting from a relatively low level, have increased, and the NI contributions hike in 2025 would have increased the ratio further. Meanwhile, a substantial share of the planned fiscal consolidation is backloaded towards the latter years of the Parliament. This reduces confidence that the measures will ultimately be implemented. Markets typically discount future fiscal tightening when political incentives may change before implementation occurs. This contributes to uncertainty regarding the medium-term fiscal outlook and may increase the risk premium demanded by investors, leading eventually to higher funding costs for the private sector, weighing on business investment – a key constraint on growth.

⁶ VAT has several appeals over some other forms of taxation: it does not directly discourage work, it is difficult to avoid, it raises a lot revenue

Table 2: Selected public expenditures & revenues ratios

	Fiscal Policy				Healthcare and Social Spending		
	Firms' Income, Profits and Capital Gains Tax	Goods and Services Taxes	Social Security Contributions (% GDP)	Personal Income and Capital Gains Taxes (% GDP)	Healthcare Spending (% GDP)	Public social expenditure (% GDP)	Public expenditure on pensions (% GDP)
UK	3,5	10,2	6,1	10,8	11,1	23,0	7,1
US	2,2	4,1	6,0	10,3	17,2	19,8	7,3
DK	4,2	12,3	0,1	25,2	9,4	26,4	7,5
NL	4,3	11,1	12,0	9,8	10,0	18,9	6,4
SW	3,9	11,2	9,0	11,3	11,3	26,1	8,0
Change since 2015	GDP p.p.	GDP p.p.	GDP p.p.	GDP p.p.	GDP p.p.	GDP p.p.	GDP p.p.
UK	1,19	-0,71	0,11	1,96	1,2	-3,7	-1,2
US	0,14	-0,20	-0,17	-0,23	0,8	0,3	0,4
DK	1,15	-2,23	-0,02	-0,07	-1	-1,3	-0,4
NL	1,63	-0,05	-1,74	2,33	-0,2	-0,7	-0,1
SW	0,93	-0,98	-0,61	-1,33	0,4	1,3	-0,3

Sources: OECD, Allianz Research

A durable growth strategy requires a fundamental overhaul of the UK's fiscal rules and fiscal framework.

On top of poor fiscal policy choices, credibility is also undermined by poorly designed fiscal rules — a subject often overlooked. The current fiscal framework relies on two principal fiscal rules: public sector net debt must be falling as a share of GDP in the fifth year of the forecast horizon, while the current budget must be in balance over the same period. These targets are problematic because both depend heavily on forecasts produced by the Office for Budget Responsibility (OBR) and are assessed at a moving five-year horizon. As a result, governments can often satisfy the rules through relatively small forecast revisions rather than genuine improvements in underlying fiscal sustainability. The framework is therefore complex, opaque and poorly understood outside specialist circles. A simpler framework combining a medium-term expenditure growth rule with a deficit ceiling would offer substantial advantages. Spending rules are transparent, easier to monitor and less vulnerable to forecast manipulation. They also allow automatic stabilizers to operate during downturns, thereby supporting macroeconomic stabilization without requiring repeated discretionary interventions. Such frameworks have generally delivered stronger fiscal discipline and greater credibility in several northern European economies.

Reducing deficits and creating additional fiscal space will require confronting the rapid growth of age-related spending.

Although UK public pension spending remains moderate by international standards (at around 7% GDP, see Table 2), the triple lock creates a source of spending growth that is difficult to justify in the context of mounting demographic pressures. By guaranteeing annual increases based on the highest of inflation, earnings growth or 2.5%, the mechanism tends to raise pension incomes faster than required to preserve either purchasing power or relative living standards. As the population ages, this implies a steadily rising claim on public resources. Reforming the triple lock would therefore not address an immediate pension crisis, but rather prevent future spending pressures from crowding out public investment in areas such as housing, infrastructure, innovation and healthcare that are more likely to support long-term growth.

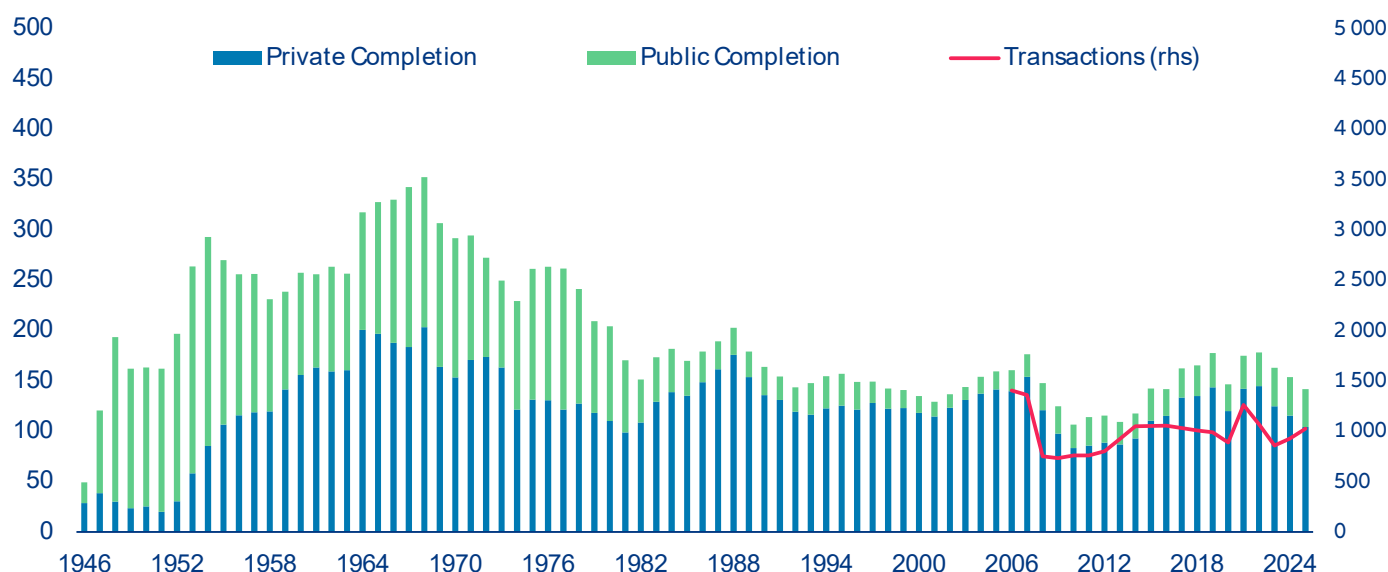
Fiscal consolidation should rely on a more efficient tax mix. A broader VAT base (reduced exemptions and zero rates), continued fiscal drag through temporarily frozen income-tax thresholds and reform of the pension triple lock would raise significant revenues while imposing relatively limited costs on growth. Part of the proceeds could be directed towards lower-income households. At the same time, the government should eliminate distortions within the tax system, notably the effective 60% marginal tax rate faced by individuals earning between GBP100 000 and GBP125 000. Additional savings could be generated through improvements in public-sector productivity, procurement reform and greater administrative efficiency. The government should consider reforming property taxation, which remains based on decades-old valuations and imposes fewer distortions on growth than taxes on labor or business investment⁷. The resulting fiscal space could be directed toward defense, housing, infrastructure, healthcare and innovation, strengthening both the sustainability of the public finances and the economy's long-term growth potential.

Doubling down on housing policy and the NHS should occupy a central place in the government's growth strategy. While planning reform is essential, it is unlikely on its own to generate the scale of construction required to close decades of underbuilding. The public sector therefore has an important role to play both through direct investment and by supporting local authorities and housing associations. From 1950 to 1970, the building of public housing was behind the surge in housing construction, while the private sector's role never really took off. Increasing housing supply would generate benefits extending far beyond the construction sector. Lower housing costs would raise real disposable incomes, reduce welfare expenditures linked to housing support and improve labor mobility by allowing workers to relocate more easily to areas with stronger employment opportunities. More affordable housing would also reduce cost pressures on employers and improve the allocation of labor across the economy. Meanwhile, addressing the NHS' core challenges should be a top priority, rather than focusing on cutting disability benefits, where the current government faced stiff opposition from backbench Labour MPs on its welfare reform. The government's objective

should not simply be to reduce benefit expenditure, but to address the underlying drivers of inactivity through improved healthcare provision, faster treatment, stronger occupational health support and more effective incentives to remain connected to the labor market. Long waiting times for treatment delay the return of workers to employment, increase absenteeism and weaken productivity. Improving healthcare capacity, investing in preventative care and addressing workforce shortages within the NHS would therefore yield economic returns beyond the healthcare sector itself. A healthier workforce is more productive, participates more actively in the labor market and imposes lower fiscal costs on the state. In an ageing society facing persistent labor shortages, improving population health may represent one of the most effective ways of expanding the economy's productive capacity.

One often overlooked feature of the UK's productivity problem is that it does not primarily stem from a lack of innovation. The UK hosts world-class universities, a vibrant start-up ecosystem and globally competitive firms in sectors such as finance, life sciences and artificial intelligence. Rather, the challenge lies in the diffusion of innovation across the wider economy. OECD research shows that productivity growth at frontier firms has remained robust, while the gap between leading and lagging firms has widened over time. In the UK, a relatively small number of highly productive firms and regions have continued to innovate, but many smaller firms have struggled to adopt new technologies, invest in digitalization and upgrade workforce skills. As a result, the benefits of innovation have not spread sufficiently across sectors and regions. Improving technology diffusion through stronger investment incentives, skills policies, business support programs and closer collaboration between universities and firms could therefore represent one of the most effective ways to raise economy-wide productivity growth

⁷ The UK's main recurrent residential property tax, Council Tax, is still based on property values assessed in England in 1991 (2003 in Wales). As a result: a £500,000 property and a £5 million property may face similar tax bills, effective tax rates fall as property values rise. The system is therefore regressive with respect to property wealth. Moreover, the UK taxes transactions heavily instead of ownership.

Figure 9: Housing completions & housing transactions (000s per year)

Sources: LSEG Workspace, Capital Economics, Allianz Research

On energy policy, the Brexit-related energy costs point to three priorities, two of which are already the subject of active negotiations. The most immediate is completing IEM reintegration: the post-Brexit shift to explicit capacity auctions has contributed to a persistent UK wholesale price premium over continental markets, with efficiency losses that grow as electricity import dependence increases. Following the May 2025 Lancaster House Summit, the EU Council authorized the Commission in March 2026 to open negotiations on UK participation in the Internal Electricity Market. These talks are time-sensitive: delays prolong consumer costs and investment uncertainty around interconnector capacity, while the potential efficiency gains remain material at system scale.

A second priority is linking the UK and EU ETS. The UK carbon price discount has evolved from a domestic pricing divergence into a direct trade cost via CBAM for energy-intensive exports, an effect that strengthens as EU free allocation is phased out through 2034. The May 2025 summit established a political agreement in principle to pursue linkage, with CBAM exemption forming the central commercial rationale. Translating this into a binding agreement is the most direct near-term lever to limit carbon-related trade frictions.

The third focus should lie on deepening North Sea infrastructure cooperation. The “Great Grid Upgrade” remains essential to address the UK’s internal north–south transmission bottleneck by connecting Scottish renewable generation to demand centers in the south. In parallel, deeper coordination under the North Seas Energy

Cooperation framework offers scope to reduce system costs through shared offshore grid planning and hybrid interconnector development, particularly in the southern North Sea where congestion pressures and offshore wind expansion most directly overlap. While these initiatives would not substitute for domestic grid investment, they can partially ease balancing constraints and improve cross-border optimization in an increasingly interconnected regional system.

The UK’s productivity challenge ultimately requires a broader investment strategy. Energy costs remain significantly higher than in many competing economies, reducing the attractiveness of the UK as a destination for manufacturing and capital-intensive investment. Accelerating grid expansion, nuclear deployment and domestic energy production should therefore be viewed as industrial policy as much as energy policy. At the same time, the UK should expand support for innovation through a combination of enhanced R&D tax incentives and mission-oriented public investment. While institutions such as ARIA, Innovate UK and the National Wealth Fund move in this direction, they remain considerably smaller than programs such as France 2030 or the industrial initiatives introduced under the US Inflation Reduction Act and CHIPS Act. A more ambitious innovation strategy, combined with housing and infrastructure investment, would help address the UK’s longstanding productivity shortfall and strengthen the economy’s capacity to generate sustainable growth in the post-Brexit era.

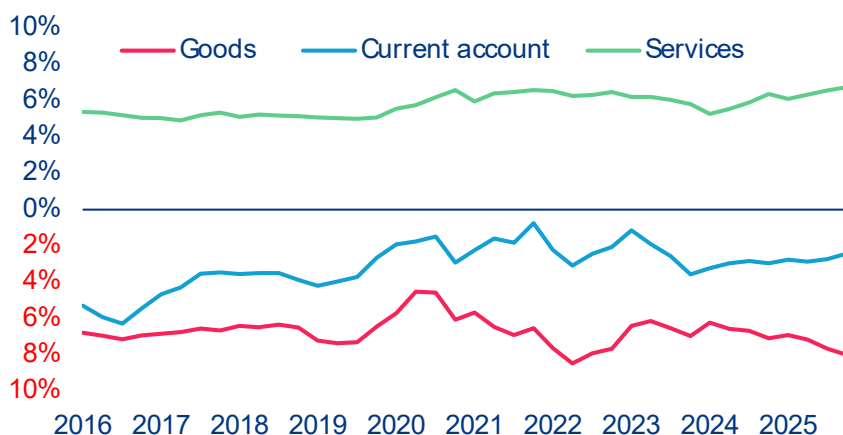


The UK's trade performance: goods imports substitution, services trade powering ahead

The UK's goods trade deficit has worsened since Brexit but rising services surplus have offset this rise. The UK's trade deficit in goods increased by +1.3pp of GDP (and by +1.5pp incl. gold) between 2016 and 2025, rising from 6.7% of GDP to 8.0% of GDP (Figure 10). The UK runs a goods trade deficit with most of its major trading partners. Around one-third of the total deficit is attributable to trade with the EU-27 (USD110bn), which represents a decline compared with 2016, when 70% of the deficit came from the EU-27). Commonwealth countries account for 8% of the deficit (22% incl. Gold), while China accounts for 28% and

the US 7%. On services, the UK's substantial surplus helps offset much this widening goods trade deficit. As a result, the current account balance has remained relatively stable over time and has even shown a modest improvement, reaching a deficit of USD97bn (2.4% GDP) by the end of 2025. This highlights the continuing importance of the UK's services sector – particularly financial, professional and business services – in mitigating the deterioration of its goods trade balance.

Figure 10: UK current account balance breakdown (four-quarter moving sum, % GDP)

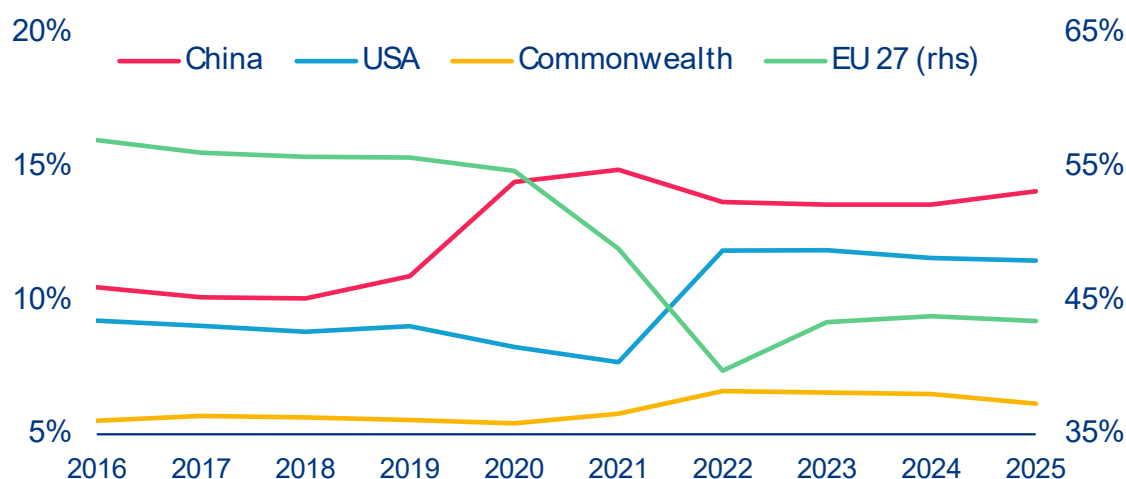


Sources: IMF BoP, Allianz research

Since the Brexit vote in 2016, the structure of UK goods exports toward foreign partners has not changed much. However, on the import side, major economic blocs have benefited from the reorientation of UK trade away from the EU. UK goods exports have decreased by 2.9pp of GDP since the Brexit vote, reaching 11.2% of GDP in 2025. The EU remains the UK's largest export market, accounting for 48% of total goods exports, a share that has remained stable from 50% in 2016. The shares of other major partners such as the US, the Commonwealth countries and China have also remained relatively stable. On the import side, however, changes have been much more significant. Total imports increased by 2pps of GDP between 2016 and 2025, reaching 18.7% of GDP. At the same time, the composition of imports has shifted substantially (Figure 11). The EU now accounts for 43% of the UK's total goods imports, compared with 57% a decade ago. Meanwhile, the importance of China and the US has grown, representing 14% (vs 10%) and 11% (vs 9%), of total imports, respectively. Two main factors explain these developments: First, imports from China surged since the pandemic, driven by strong demand for technology goods, and second, imports from the US increased significantly following the outbreak of the war in Ukraine, largely due to energy purchases.

The UK has diversified both its sources of supply and the types of goods it imports, notably through increased reliance on Chinese manufactured goods and US energy exports. Today, UK imports are dominated by machinery, chemicals, agrifood and automotive products. Compared with 2016, the composition of total imports across these products has become more balanced. Energy imports also gained importance, while chemicals, textiles, transport equipment and automotive products experienced a decline in their relative shares of total imports. A closer look at the changing composition of the UK's suppliers reveals substantial shifts between 2016 and 2025. China has made the most significant gains, particularly in electronics and electrical equipment (the UK is the top destination, with 35% total gross imports of this category), a trend similar to that observed in the EU's growing dependence on Chinese technology goods. China has also increased its presence in automotive products, machinery, chemicals and transport equipment. In short, it has expanded its market share across virtually all manufacturing sectors. Meanwhile, the US has strengthened its position in energy imports, increasing its share by 16pps of the UK's total energy imports. It has also gained ground in machinery and chemicals. Finally, Commonwealth countries have expanded their presence across nearly all sectors, particularly in textiles and chemicals, but the overall share is stable (+0.6pp).

Figure 11: Share of global gross goods imports of the UK by main partner (excluding gold)



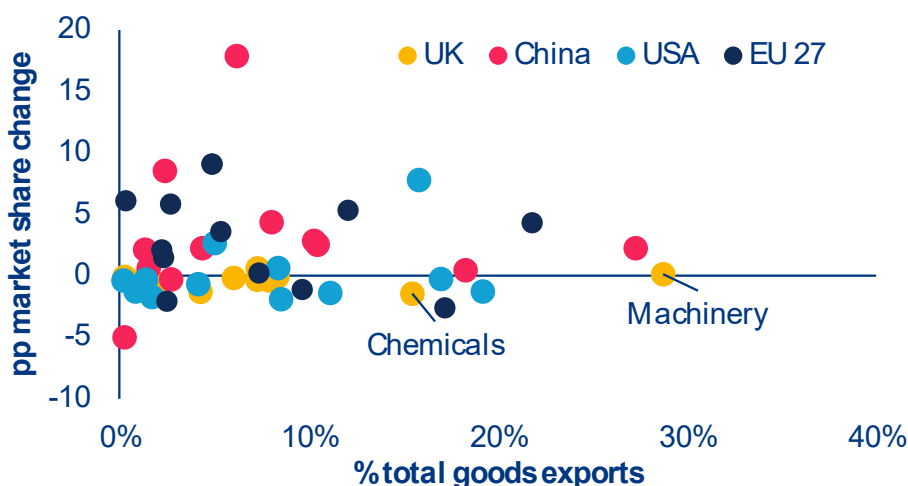
Sources: Trade Map, Allianz research

UK exports have reduced their market shares. Exports are more concentrated in machinery, with a loss of market shares in some traditional strengths (autos, chemicals). Machinery exports are primarily composed of turbojets, turbopropellers, other gas turbines used in the aerospace industry and specialized medical equipment. They now account for almost 30% of total gross exports. By contrast, the importance of the automotive and chemicals sectors has declined significantly. The share of chemicals in total goods exports fell from 20% to 15%, while the automotive sector's share decreased from 13% to 9%, a trend also reflected in broader transport equipment exports. These declines are visible not only in relative terms but also in absolute values, with automotive exports falling by 20% and chemical exports by 8% between 2016 and 2025. At the product level, the UK exports less to the EU across most products than it did in 2016 (as a share of total gross exports), with the notable exceptions of transport equipment and energy. However, these declines have not been offset by another economic bloc in relative share. Although the UK's export structure appears relatively stable in relative terms, with the same trading partners and some rebalancing across products, this masks significant shift towards machinery and energy exports, alongside a decline in sectors that have traditionally been among the

UK's strengths, namely automotive and chemicals. Figure 12 shows that overall, the UK has experienced a decline in its global market share across most of the products it exports compared to the other major economic blocks.

This analysis of UK trade excludes gold exports and imports, a category in which the UK has experienced strong growth in market share and which has had a significant impact on its trade structure as a result of the gold price rally in recent years. Including gold could distort the assessment of the underlying state of British manufacturing capabilities and the structural shifts currently taking place in global trade, such as the growing footprint of Asian economies — particularly China — across most goods categories, and the increasing role of the US in energy supply. When gold is included, the UK's degree of openness remains broadly unchanged at around 38% between 2016 and 2025. Excluding gold, however, the indicator points to a weakening integration of UK into global trades, with a share that decreased from 35% in 2016 to 30% in 2025.

Figure 12: Market share changes 2025 vs 2016 (vertical axis) and share of total gross exports of the country by product in 2025



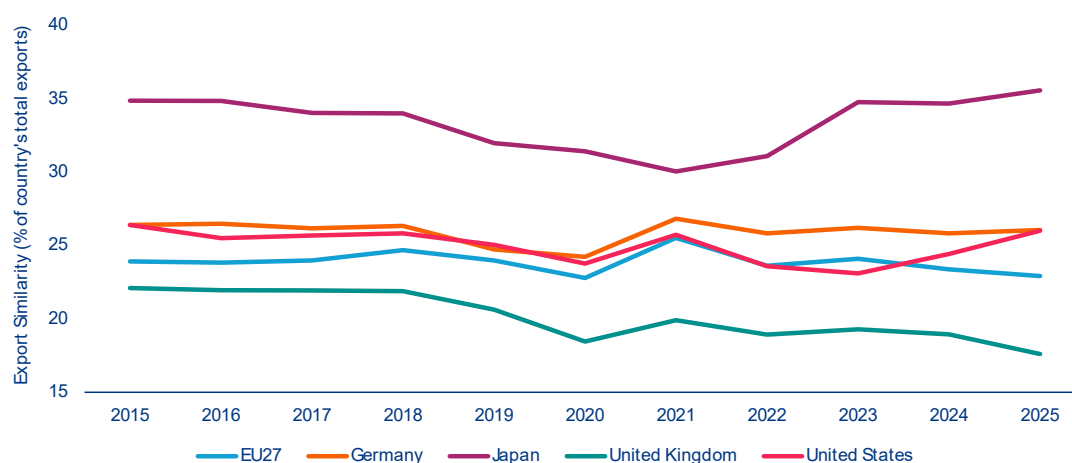
Sources: Trade Map (H2 classification), Allianz research

With regard to Chinese competition, the UK is less exposed. In 2025, about 15% of UK exports were in products among the top 25 of China's exports. In contrast, 25% of US exports and 26% of German exports compete directly with the Asian giant's top exports. As China has moved towards high-value manufacturing, its top exports now consist of telecommunication devices, integrated circuits and computers. The UK, historically a smaller exporter of electronics, is thus less exposed to Beijing's shift. Pharmaceuticals and aerospace, a large source of the UK's competitiveness advantage, are less in the crosshairs. Since the pandemic, a key area of competition is also the automobile industry. Cars represent the UK's second largest source of exports at over USD30bn per year. A falling similarity index could be a cause for concern. Indeed, a lack of competitiveness that erodes a sector's share in total exports will mechanically decrease the index. For the UK, car exports have increased by +1.1% in nominal value since 2022 while total exports (excl. gold) increased by +11.2%, highlighting a flailing automotive sector and thus decreasing the similarity index.

UK trade agreements have expanded market access, but they have not matched the scale of the economic relationship with the EU or fully offset Brexit-related trade losses. Although UK–EU tariffs remain relatively low, with an average applied tariff of 1.5% (reaching up to 19%

for certain products), the principal barriers to trade are non-tariff measures (NTMs). Since leaving the EU in 2020, the UK has introduced 58 trade-related measures affecting EU countries, including 25 export-related measures, 20 subsidies, seven tariff measures and five price controls. Over the same period, the EU has introduced 337 measures affecting the UK, predominantly subsidies (275), alongside export restrictions (19), tariff measures (15), licensing and quota measures (13), price controls (six), trade-protective measures (four) and one foreign direct investment (FDI) measure. These policy changes have contributed to a significant decline in bilateral trade integration. Structural gravity estimates suggest that Brexit reduced UK–EU trade by -21.3% for goods and -49.7% for services. This weaker bilateral performance reflects Brexit-related trade and supply-chain disruptions arising primarily from higher NTMs and, to a lesser extent, tariffs.

Figure 13: Similarity index to Chinese goods exports



Sources: ITC Trade Map, GTA, Allianz Research

Note: Using GTA methodology, similarity is calculated for each country as the sum of the export value of all their products appearing in China's top 25 HS4 codes, divided by their total exports. The HS code for cars is included for all years.

⁷ The UK's main recurrent residential property tax, Council Tax, is still based on property values assessed in England in 1991 (2003 in Wales). As a result: a £500,000 property and a £5 million property may face similar tax bills, effective tax rates fall as property values rise. The system is therefore regressive with respect to property wealth. Moreover, the UK taxes transactions heavily instead of ownership.

In parallel, the UK sought to offset some of the economic costs of Brexit through an independent trade policy. Since Brexit, the UK's free trade agreement (FTA) strategy has evolved from initially rolling over agreements previously negotiated through EU membership to pursuing new bilateral and plurilateral arrangements. Major milestones include agreements with Japan (2020), Australia (2021), New Zealand (2022), the EEA EFTA states (Norway, Iceland and Liechtenstein), digital trade agreements with Singapore and Ukraine, accession to the Comprehensive and Progressive Agreement for Trans-Pacific Partnership (CPTPP) in 2024, the India FTA signed in 2025, an upgraded agreement with South Korea and the Gulf Cooperation Council (GCC) agreement concluded in 2026 but not yet in force (Table 3). Collectively, these post-Brexit agreements account for approximately 21% of UK trade in 2025 (without the US, which is a prosperity deal to offset the impact of imposed tariffs rather than a FTA). The UK is also negotiating or upgrading agreements with Canada, Switzerland and Türkiye. Despite expanding the UK's global trade network, the scale of these agreements remains substantially smaller than the UK's economic relationship with the EU, which continues to account for the largest share of UK trade at 37% in 2025.

Deeper UK–EU trade ties could unlock billions in economic gains. Looking ahead, the UK and EU may find scope for greater alignment in goods trade, while the UK and US are more likely to deepen cooperation in services, technology, finance and regulation. However, only a

limited share of the trade frictions created by Brexit is likely to be reversed. Computable general equilibrium (CGE) simulations indicate that a deep UK–EU trade agreement could generate annual gains of USD25.2bn for the UK, comprising USD19.6bn in trade creation and USD5.6bn in trade diversion. This would increase the value of UK exports to the EU by +23.1%, with the largest proportional gains in sugars, food preparations and cereals, followed by vehicles, alcoholic beverages and textiles. EU countries could gain USD31.9bn annually (+17.4%), driven primarily by vehicles — reflecting the restoration of previously integrated value chains — as well as food preparations, dairy products and chocolate.

Table 3: UK Free Trade Agreements since Brexit

Partner	Agreement	Year	Key provisions	UK bilateral trade (share of total UK trade in 2025)
European Union	EU–UK Trade and Cooperation Agreement	2020 (effective 2021)	Necessary replacement for EU membership. Zero tariffs and quotas on qualifying goods, customs cooperation, services framework, fisheries, transport, energy, law-enforcement cooperation	37% with EU remaining the UK's largest trade partner
Japan	UK–Japan CEPA	2020	Goods market access, digital trade, financial services, data flows, investment protections.	1,50%
Australia	UK–Australia FTA	2021 (effective 2023)	Tariff elimination, services access, mobility provisions, digital trade, procurement.	1,30%
New Zealand	UK–New Zealand FTA	2022 (effective 2023)	Near-total tariff elimination, services liberalization, digital trade, investment access.	0,10%
Norway, Iceland, Liechtenstein	UK–EEA EFTA FTA	2021	Goods, services, investment, digital trade and mobility provisions.	2,30%
Singapore	Digital Trade Agreement	2022	Digital identities, e-commerce, fintech cooperation, cross-border data flows.	1,00%
Ukraine	Digital Trade Agreement	2023	Digital trade rules, e-signatures, cybersecurity cooperation, data flows.	0,20%
Comprehensive and Progressive Agreement for Trans-Pacific Partnership	UK accession to CPTPP	2023 (effective 2024)	Tariffs, services, investment, procurement, digital trade across 11 Pacific economies.	9,20%
India	UK–India FTA	2025 (not yet in force)	Tariff reductions, services access, professional mobility, customs simplification, investment provisions. Expected to increase bilateral trade substantially.	1,70%
South Korea	UK–South Korea FTA	2025 (updated from earlier EU agreement)	Replaces the continuity agreement inherited from EU membership. Modernizes rules of origin, services and digital trade, maintains tariff-free access on most goods, and strengthens access for automotive, financial services and technology sectors.	0,90%
United States*	Economic Prosperity Deal	2025 (partially implemented)	A non-binding framework intended to reduce the impact of US tariffs and improve market access. Includes preferential treatment for UK autos, aerospace, steel and aluminum, and a framework for future cooperation on pharmaceuticals and supply-chain security.	13,40%
Gulf Co-operation Council (Bahrain, Kuwait, Oman, Qatar, Saudi Arabia, UAE)	UK–GCC FTA	2026 (not yet in force)	The UK's first FTA with the GCC and the GCC's first FTA with a G7 country. Covers tariff reductions, customs facilitation, services, investment and market access.	2,60%

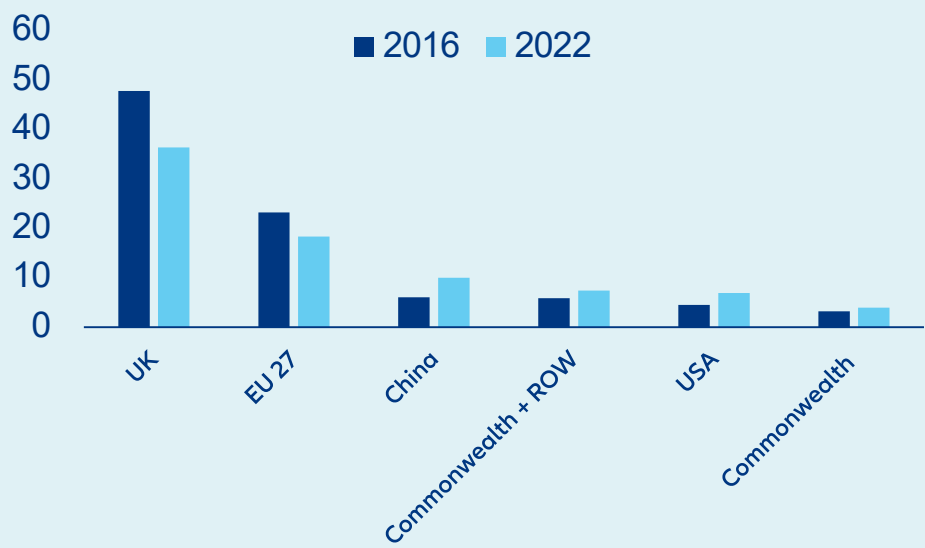
Sources: UNComtrade, UK House of Commons, Allianz Research. Notes: *The US Economic Prosperity Deal is not a FTA, but a deal to lower the impact of tariffs

The evolution of UK trade in global supply chains

From a supply-chain perspective, the UK's domestic demand has become increasingly dependent on industrial foreign suppliers' inputs, while the country has struggled to expand its export footprint in global supply chains. The OECD Trade in Value Added (TiVA) dataset has data only available up to 2022, yet it provides valuable insights into the changes to supply-chains. It looks at the foreign and domestic value added embodied in final demand. For the UK, the data indicates a gradual reorientation of domestic demand away from EU products. Although the EU remains the UK's largest trading partner in valued added, the older EU member states are no longer its preferred trading counterparts in supply chains as the relative importance of partners such as China, the US and Commonwealth countries has increased significantly over the past decade.

Indeed, between 2016 and 2022, the share of industrial foreign value added in UK final demand originating from the EU declined from 52% to 39% in the manufacturing sector. Over the same period, China's share increased by 8pps, making it the UK's largest single-country partner in 2022. The US gained 2pps and became the second-largest partner, while Germany's share fell by 8pps, causing it to drop from first to third place. Within the EU, only Ireland and Poland increased their shares, whereas most Western European countries experienced a decline. This shift is particularly pronounced in the computer, electronics, and optical products sector, where China's share has overtaken that of the EU. Similarly, in the motor vehicles industry, the US gained around 5pps, again at the expense of both EU and UK manufacturing value added. These trends suggest a gradual reconfiguration of the UK's supply chains away from its historical European partners and towards a more globally diversified network of suppliers. In terms of UK exports, between 2016 and 2022, the UK has lost a bit of market share in the trade in value added embodied in other countries' final demand for manufacturing (-0.4pp at 1.6% at the global level), visible for the main UK's industries (chemicals, electronics, machinery, automotive) .

Figure 14: Insurance penetration, selected countries (gross written premiums in percent of GDP)

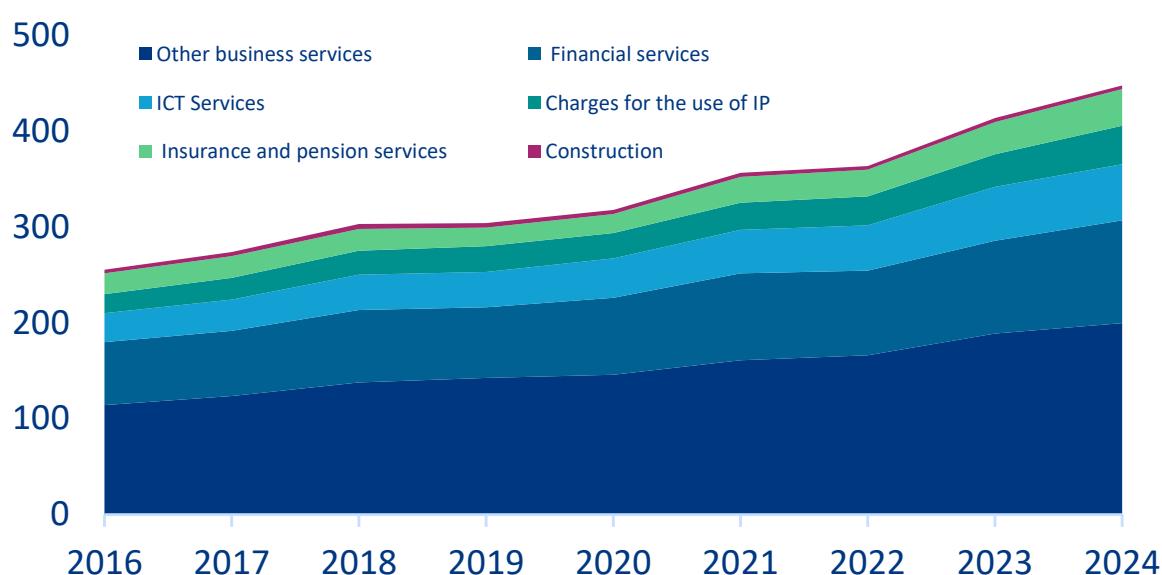


Source: OECD TiVA, Allianz research

While Brexit introduced additional compliance requirements and regulatory uncertainty, the services sector has largely avoided the trade frictions experienced in goods markets. UK services exports to the EU increased by USD96bn between 2026 and 2024, representing growth of +63%, only slightly below the pace of growth in UK services exports globally (+65%). Financial services have retained much of their international competitiveness despite the loss of preferential access to EU markets. While some trading activity, assets and jobs shifted to Amsterdam, Dublin and Paris, London remains Europe's leading center for foreign exchange, derivatives, insurance and international banking. Financial services exports to the EU grew by 58% between 2026 and 2024, compared with 63% globally. Moreover, while the UK's share of value added embodied in global final demand for financial services decreased by 0.3 percentage points between 2016 and 2022, its share in EU-27 final demand increased by 0.6 percentage points, suggesting a deeper integration of the UK financial sector into European value chains. The share of financial services in total UK service exports has remained broadly stable at around 17% in 2024⁹. The UK's trade surplus in financial services has increased over the period and the export destinations remain heavily concentrated in the US, which accounts for around 27% of UK financial services exports, followed by Ireland, Luxembourg, other EU

member states and a range of offshore financial centers. In the global financial services market, the UK's position has remained remarkably resilient. It continues to rank second worldwide, accounting for approximately 21% of global financial services exports, outperforming the EU and behind the US, which retain the first place. The main changes in the global ranking have occurred among the following positions, with Singapore notably strengthening its role and consolidating its position as the fourth-largest exporter of financial services. These trends suggest that while Brexit has not significantly enhanced the UK's dominance in financial services, neither has it undermined London's status as one of the world's leading financial centers. Actually, the UK is exporting mostly "other business services" more than financial services + insurance and pension services, including "R&D services", "consulting services" and "technical, trade-related services" (Figure 15). Finally, ICT services have done even better. Supported by digital trade provisions in the EU–UK Trade and Cooperation Agreement and continued EU data adequacy arrangements, UK ICT services exports to the EU grew by +94% since Brexit, outpacing growth to the rest of the world (+76%).

Figure 15: Distribution of UK's services exports (USD Bn)



Sources : WTO-OECD, Allianz research

⁹ From a value-added perspective, financial service activities accounted for 4.4% of total UK value added in 2023, down from 6.5% in 2016

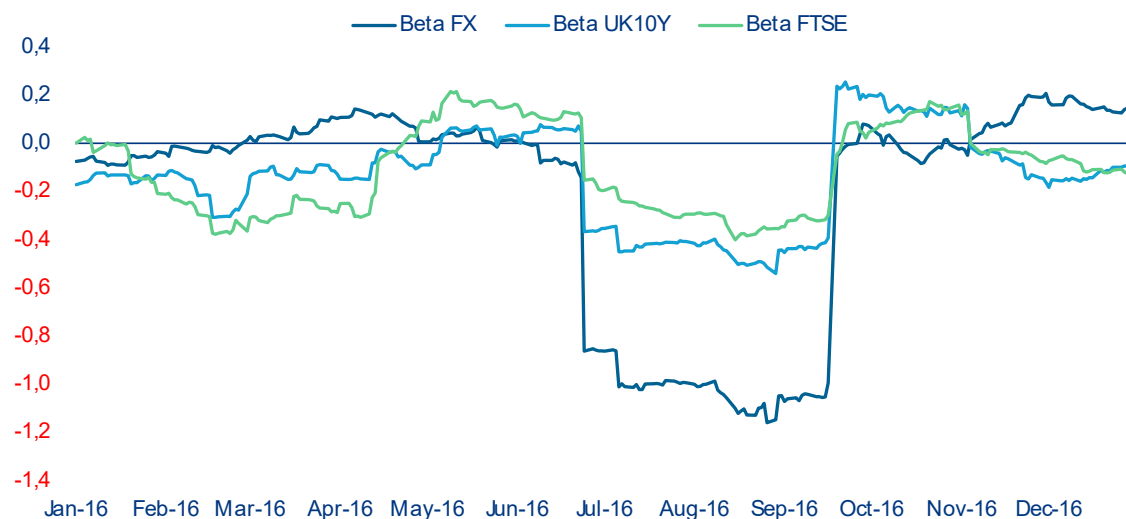


Financial markets: unofficial dollarization and eroding confidence but the City of London keeps (most of) its shine

From a capital market perspective, Brexit set in motion a sequence of structural changes — monetary, institutional and commercial — whose cumulative effects only became visible after the central bank's protective cover was removed.

The immediate market impact of the referendum was absorbed almost entirely by the exchange rate. From the moment Leave probabilities began rising in late 2015, sterling traded as the primary signal of UK political and growth uncertainty. The Gilt market was insulated: the Bank of England cut rates and relaunched asset purchases within weeks of the result, compressing term premia and anchoring the long end of the curve. The primary driver of Gilt yields throughout this period was the global rates cycle — Fed signaling above all — rather than any UK-specific risk premium. Quantitative easing masked the Brexit effect on rates; the FX market was the honest signal, the bond

market the managed one. On the day the referendum result was announced, 24 June 2016, the GBP/USD fell ~8.1% in a single day — a 14.8-sigma move relative to its normal daily volatility, by far the most extreme reaction of any asset. By contrast, the UK 10Y yield moved -28.8bp ($z = -5.3$) and the US 10Y just -16bp ($z = -3.1$), while the FTSE 100 fell -3.1% ($z = -3.3$). In standardized terms the currency shock was roughly three to five times larger than the rates and equity shocks. The Gilt had, in effect, become a UST proxy at a discount, its terminal rate tracking US conditions more closely than domestic fundamentals.

Figure 16: Rolling 61-day beta (z-score returns) vs policy index – Zoom 2016, Brexit 24/06

Sources : LSEG Datastream, Allianz Research

Beneath this surface stability, Brexit was accelerating a structural reorientation of the UK's external sector.

The share of UK exports invoiced in GBP fell from approximately 63% before 2015 to 38% by 2022, with USD invoicing rising from one-third to 45% over the same period. This transition was directly triggered by the currency-mismatch valuation shock of the 2016 depreciation and proved persistent. As the UK economy became progressively coupled with US financial conditions, the dependency on Fed policy transmission deepened. This dollarization of trade was the most consequential — and least visible — legacy of the referendum years. The risk premium currently visible in GILT was not observable right after Brexit.

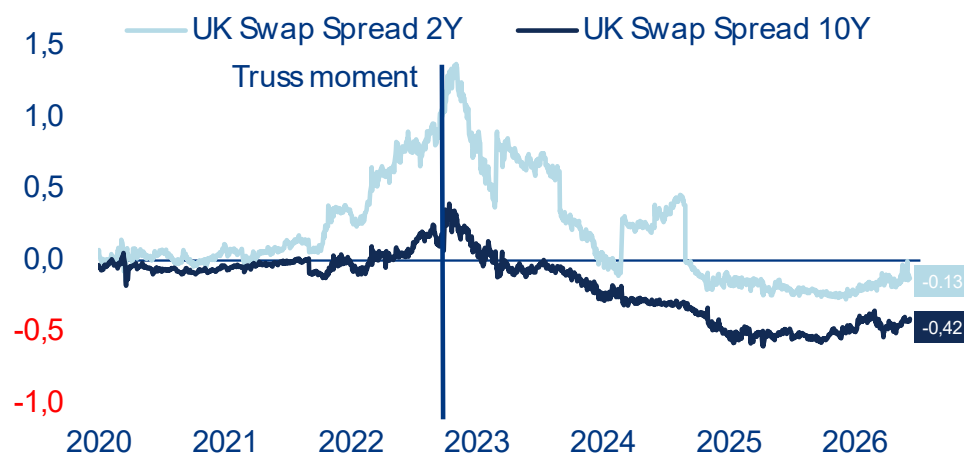
The City's resilience post-Brexit has proved more robust than pre-referendum predictions suggested, but headline stability conceals a meaningful structural shift at the margin. London's pre-eminence was built on two foundations: its historical role as the birthplace of the Eurodollar system and its subsequent emergence as the natural bridge between dollar and euro financial markets. It hosted the clearing infrastructure for euro-denominated derivatives on LCH, the primary market for rates and FX instruments across both currencies and the center of international banking for non-domestic entities operating in Europe.

The aggregate BIS data tells a story of remarkable persistence.

In April 2025, the UK accounted for 49.6% of global OTC interest rate derivatives turnover – up from 42.9% in 2022 – and handled 37.8% of global FX turnover, broadly stable versus the pre-Brexit baseline (BIS Triennial Survey, 2025). These figures point not only to an expansion in gross activity but also underline the UK's persistent global leadership. The depth of London's liquidity pool and the network externalities that make migration costly have proved more durable than political pressure alone could dismantle.

The euro clearing franchise is where nuance matters.

LCH retained approximately 81% of global euro IRS clearing as of end-2023, with Eurex holding the remaining 19% – a shift that occurred almost entirely in the 2019–2021 window driven by Brexit uncertainty and Eurex incentive programs, and which has not materially progressed since (ECB Financial Integration Report, June 2024). Around 95% of euro-denominated interest rate swaps were still cleared through LCH in 2024, and the European Commission's renewal of UK CCP equivalence until June 2028 is a tacit acknowledgement that the infrastructure cannot be relocated by political fiat. The exception is euro repo, the one segment where migration has been documented and sustained: London's banking ties to the Eurozone have loosened in part owing to the shift of euro repo clearing from London to Paris (BIS Bulletin No. 65, 2022; ICMA repo survey data, 2024).

Figure 17: Swap spread widening after the Truss moment.

Sources: ILSEG Datastream, Allianz Research

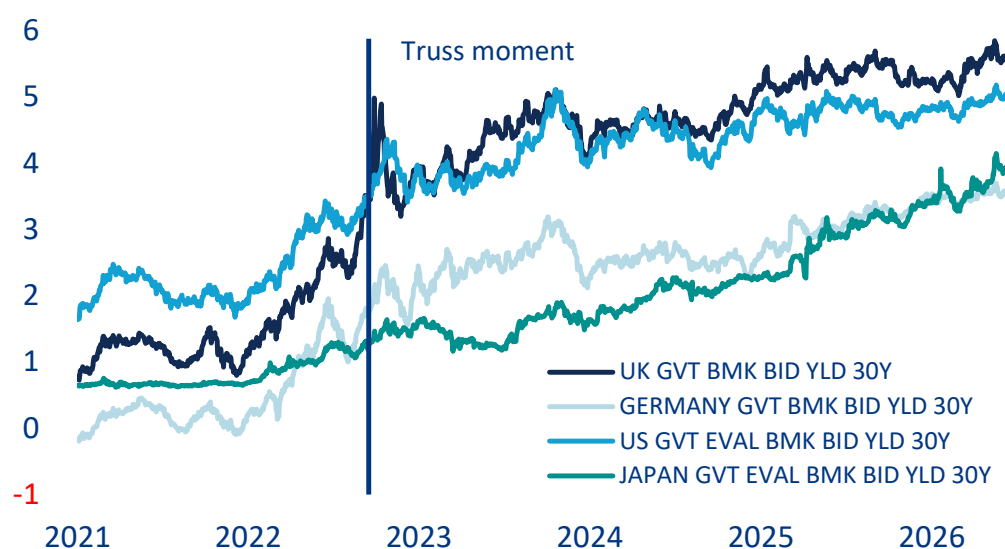
The honest assessment is that London's clearing and derivatives franchise stopped growing into its natural trajectory rather than declined outright.

Euro-denominated IRD trading surpassed US dollar-denominated instruments to account for the largest share of global IRD turnover in 2025 — growth that, absent Brexit, would have accrued disproportionately to London. The counterfactual loss is real even if absolute volumes remain strong. For the Gilt market, this matters through a second-order channel: a City that is less central to euro financial intermediation generates less natural structural demand for UK sovereign paper as collateral, reinforcing the shift toward a thinner, more price-sensitive foreign investor base.

The September 2022 fiscal episode is, in retrospect, the most consequential single event for the structural repricing of UK sovereign risk, more so than the referendum itself. It exposed the inherent fragility of a market dominated by concentrated, leveraged duration holders. The high-weighted average maturity of the Gilt stock (UK WAM is touching 13.5 year, against 7 year for Germany, or 6 year for US, at the end of 2025 according to the OCDE)) combined with the structural role of liability-driven investors and pension funds as dominant holders of long-dated paper, created a high vulnerability to duration shock and a self-reinforcing feedback loop that the Bank of England had to interrupt with emergency purchases of

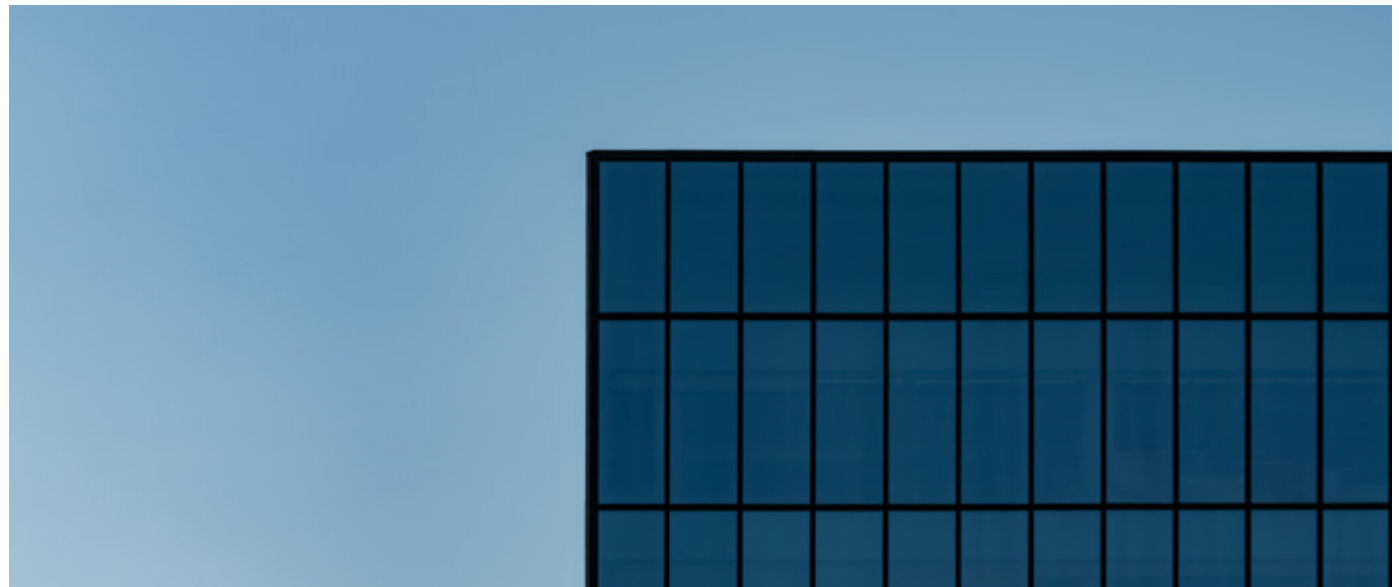
GBP65bn over 13 trading days. The episode established one structural fact that had not previously been priced: the central bank backstop for Gilts is conditional on fiscal credibility. Swap spreads, which had traded near zero prior to September 2022, moved to persistently negative territory, reflecting a banking sector constrained in its capacity to absorb newly issued sovereign paper, and a durable liquidity premium now embedded in the yield curve.

The transition from QE to quantitative tightening has compounded the vulnerabilities exposed in September 2022, but the causality is not purely mechanical. The Bank of England has pursued QT aggressively, in part as a deliberate signal to sovereign bond markets: a demonstration that institutional discipline had been restored after Truss. That signal has had a cost. The regime of scarce collateral and abundant reserves has given way to one of elevated Gilt supply and tighter bank funding conditions — GC rates now sit modestly above the BoE policy rate, consistent with a structural shift in money market. Banks depend more heavily on money market financing, reducing their capacity to absorb issuance and amplifying market sensitivity in stress episodes.

Figure 18: DM 30Y rates and GILT discount since the Truss moment

Sources: LSEG Datastream, Allianz Research

The BoE has credibly anchored the short end, what it has not recovered is control over the long end. Long-dated yields are increasingly set by supply dynamics, global uncertainty and term premia rather than by forward guidance or domestic macro anchors. US macro announcements — Fed decisions, Treasury supply, CPI prints — exert a more immediate and measurable effect on Gilt pricing than domestic budget decisions. This is the EMification dynamic in its precise formulation: not a loss of formal monetary sovereignty, but a structural compression of the market's capacity to price UK fundamentals independently of the global financial cycle. While this is a global phenomenon, the magnitude is strongly pronounced in the GILT market that exhibit a higher rate than the other DMs.



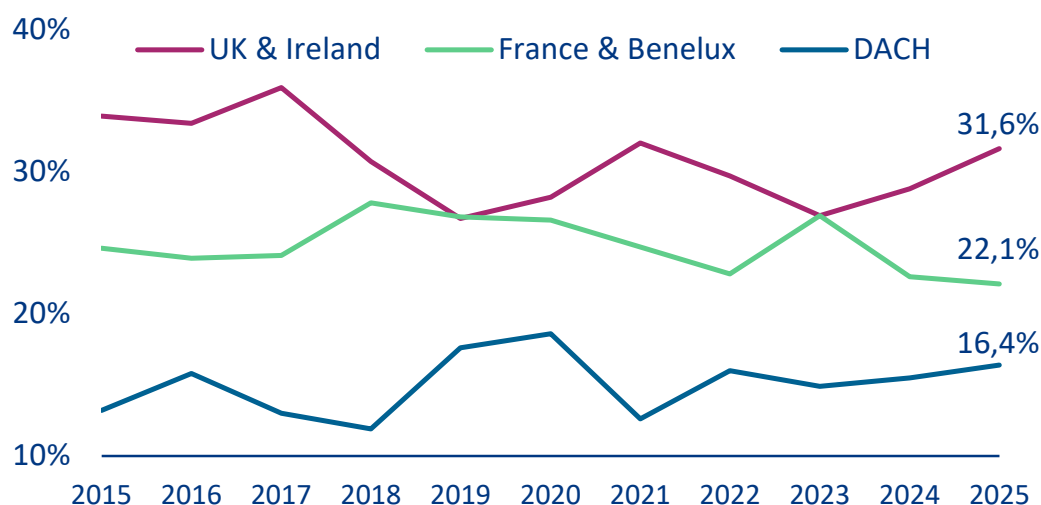
UK private equity market has lost its magic, low tech exposure prevails

A decade after the referendum, Brexit's imprint on UK private markets reads as a level shift that has now largely cleared the price. The UK is still Europe's dominant private capital hub, accounting for roughly one in five European deals and close to one in three fund closes, but it has stopped pulling away from the continent. The damage is real, concentrated and uneven: a one-off downshift in trend growth and relative attractiveness, since layered over by the pandemic, the energy shock and a higher-rate cycle that have blurred attribution. What remains is a market trading at a structural discount, increasingly financed from abroad, and now leaning on domestic policy to re-rate. Within that, the clearest bifurcation is in venture capital, the generalist fundraising base has hollowed out, yet the UK retains a relative structural edge in AI and deep-tech, where it continues to anchor the continent's largest rounds.

Brexit was a level shift, and it is now largely priced in. Estimates from Bloom et al. indicate that, by 2025, Brexit had reduced overall UK investment by around 12–18% relative to a no-Brexit counterfactual. Transmission ran through trade and labor frictions, the loss of passporting and equivalence, which pushed UK managers onto national private-placement regimes and EU-domiciled affiliates at higher operating cost, and through uncertainty itself, which depressed UK private equity volumes via a real-options "value of waiting" channel. Hundreds of financial firms have shifted staff, balance sheets and legal

entities into the EU, re-anchoring fund servicing across Dublin, Luxembourg, Paris and Frankfurt and embedding a more multipolar European capital map. The discrete adjustment is behind us; what persists is a structurally higher cost of capital for UK risk assets.

Private equity proved resilient, increasingly as a US-funded value trade. The UK retains roughly a third of European private equity deal value, a position held through the post-referendum period, yet its deal value growth over the decade since the vote has materially lagged Germany and France, and Germany has displaced France as the continent's second market. The structural mechanism behind the UK's resilience is partly a discount story: compressed equity valuations relative to US peers have turned listed UK companies into prime take-private targets, drawing a sustained and disproportionately large share of US capital into the market, a pattern consistent across the post-Brexit cycle and the highest rate of US participation of any major European PE market. Buyout activity has increasingly defined the deal mix, reflecting both the take-private dynamic and the relative underrepresentation of growth equity in the UK versus continental peers.

Figure 19: The UK retains roughly a third of European private equity deal value.

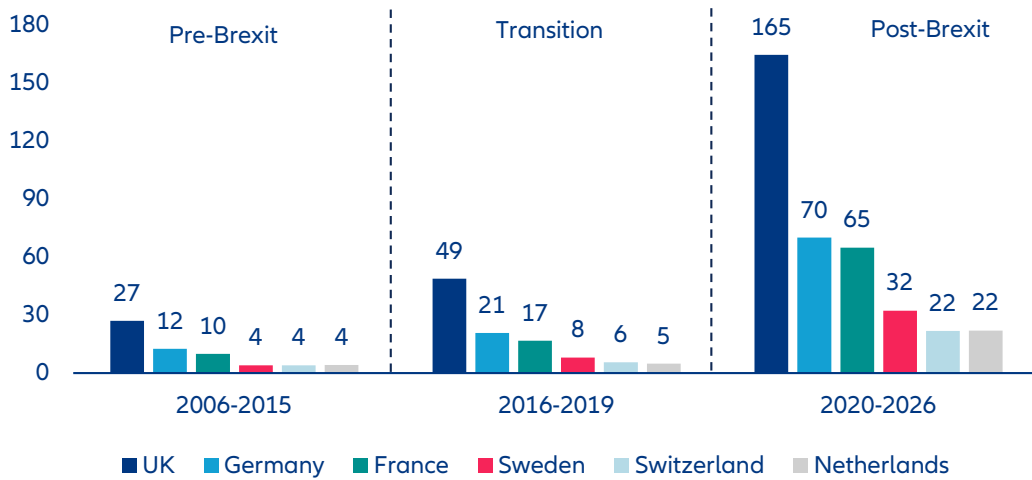
Sources : Pitchbook, Allianz Research

Notes: European PE deal value as a share of annual total by region.

The exit channel is the binding constraint, and policy has to be what clears it. Exit values have contracted sharply from the 2021-2022 peak and have been slow to recover, with disposals lagging deal formation and the ratio of PE-backed companies to exits widening to historically elevated levels, a backlog that accumulated across the rate-shock years and has yet to clear. The public listings market has been a structural drag, LSE has shed net listings across the post-Brexit decade, PE-backed IPO activity has been episodic rather than sustained and the equity re-rating visible in the FTSE 100 has yet to translate into a reliable exit route for private sponsors. Tax and domicile changes have added a directional headwind to manager incentives, reinforcing the pull of competing jurisdictions. Against that, the policy response is the most substantive in a decade, the Mansion House Accord targets GBP50bn of pension capital into private markets by 2030, PISCES provides an intermittent secondary venue short of a full listing and FCA deregulation is progressively lowering the post-Brexit compliance burden. Tentative green shoots are already visible in the exit data, PE exit value reached GBP20.9bn across 131 transactions in the first four months of 2026, pacing toward a record year by count, but the recovery is being led by buyouts rather than public listings, so the IPO channel most central to London's re-rating has yet to reassert itself.

The UK remains the most dynamic start-up ecosystem in Europe but the gap has narrowed since 2020. As Figure 21 shows, it still towers over Europe, drawing some USD 164bn of venture capital over 2020–2026, about two-and-a-half times Germany (68) or France (65) and roughly 44% of the six-country pool. Yet the UK's relative margin has quietly thinned since the referendum: its share of that pool eased from a 2016–2019 peak near 46% back to about 44%, and its lead over France narrowed from roughly 2.7x to 2.5x, as faster-growing rivals erupted around it. France and Germany remain the two heavyweight challengers, but the more telling shift is the rise of a second tier: Sweden, Switzerland and the Netherlands each pulled in between five and eight times more capital after 2020 than across the entire pre-Brexit decade (2006–2015) — Sweden, up eightfold to EUR32bn, has now overtaken both Switzerland and the Netherlands (22 each) to become Europe's third-largest hub. In short, Brexit left British innovation magnetic but no longer singular: the Continent is building its own pools of attraction, and the UK's edge, while still commanding, is slowly compressing.

Figure 20: Venture capital investment in UK and top European peers between 2006 and 2026



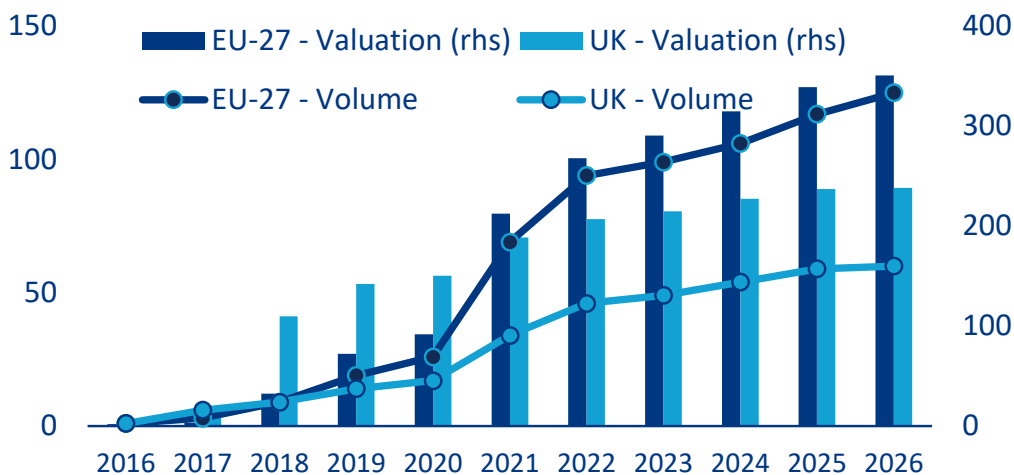
Sources : Dealrooms, Allianz Research

UK market dynamism hides a double concentration bias.

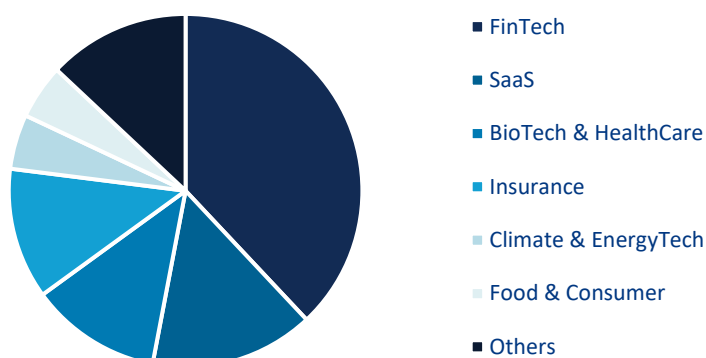
With 60 active unicorns, the UK is the fourth largest market worldwide behind the US, China and India. However that fleet leans on a handful of sectors. Fintech alone makes up roughly 40% of the pack, with software, health and insurtech filling most of the rest; the deep-tech and frontier layers that build lasting advantage barely register. And value clusters even tighter than companies do – three names carry around 25% of capital raised over the past decade and initially 40% of the entire pool valuation. That

fragility showed when the post-2021 sell-off hit. Having bet so hard on fintech and software, the very sectors that re-rated most brutally, the UK took the heaviest blow in Europe, unicorn fleet valuation being estimated in Q1 2026 at just 0.77x its last-round value, the most stressed major market, against a sturdy 0.97x in Sweden and 0.87x in Germany.

Figure 21: Venture capital investment in UK and top European peers between 2006 and 2026



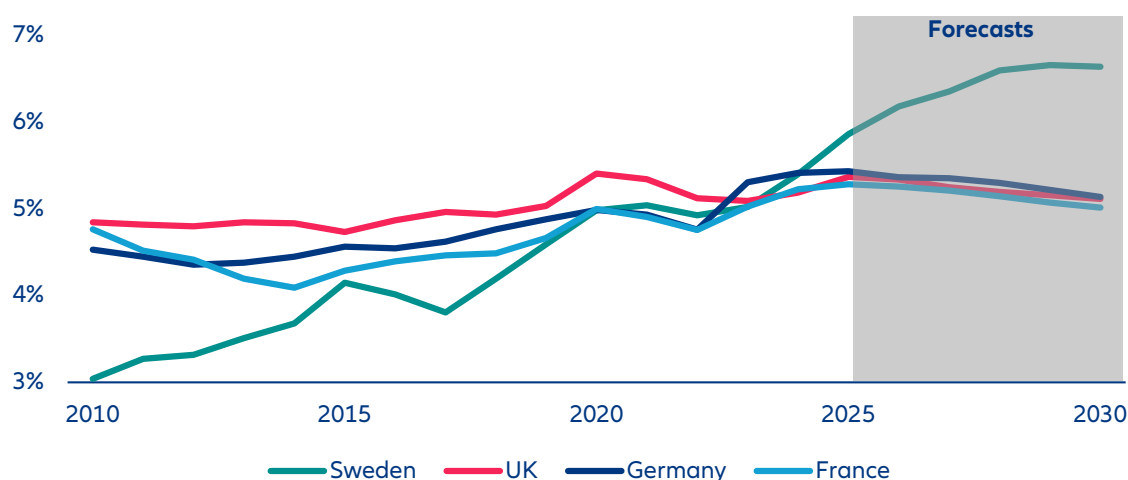
Sources : CBInsights Allianz Research

Figure 22: Sector breakdown of current unicorn ecosystem in UK (per volume)

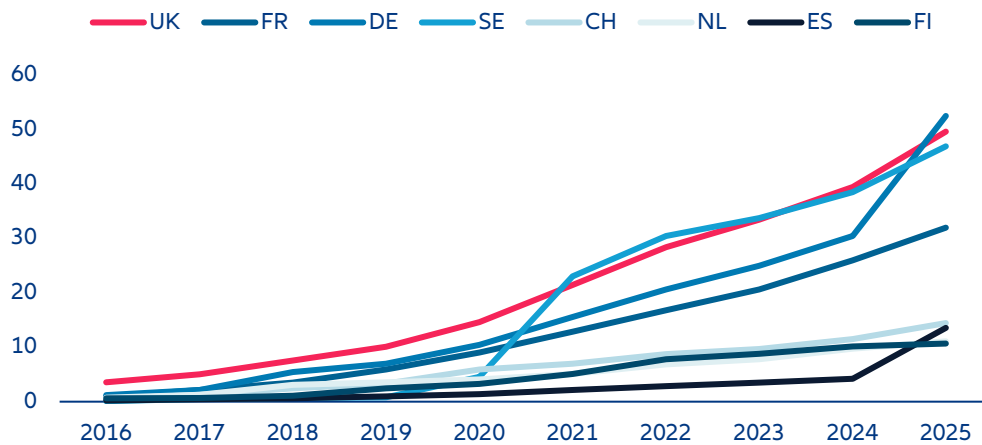
Sources : CBInsights Allianz Research

Diverging pattern: technology footprint is edging down in UK. ICT has grown into the UK's output more slowly than across the Channel — roughly 5.6% a year against 6% in France and 7% in Germany and Sweden — and the gap is widening, with Sweden breaking clear toward 6.5% of output by 2030 as Britain flattens near 5%. The causes compound. At home, finance, insurance and healthcare soak up the capital, starving the hardware and deep-tech that actually lift ICT's weight; the 2022 energy-and-inflation shock bit harder than on the Continent; and, cut off from EU structural funds, the UK now leans almost entirely on a private VC market that bankrolls software, not silicon. The deeper drag is structural: Europe's engineering schools feed a denser pipeline into

mechanical and frontier-AI work, while the UK's own innovation watchdog warns its science-and-engineering skills shortage is now structural rather than That is the crux, as the Cambridge Industrial Innovation Policy's UK Innovation Report concludes, closing the country's scale gap demands complementary capabilities like skills, supply-chain depth and manufacturing capacity, not simply more capital. And with no homegrown ICT champion — its highest-ranked tech firm sits just 389th in global R&D spending — there is no locomotive to pull a cluster along behind it.

Figure 23: Evolution of the weight of ICT sector into some European economies (based on gross output)

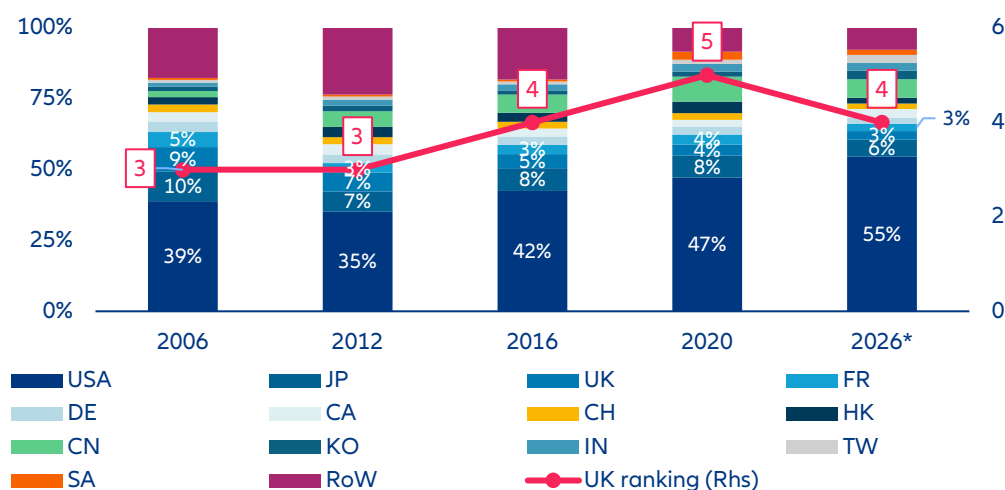
Sources : Oxford economics, Allianz Research

Figure 24: Cumulative private investment into AI firms in European countries since 2016

Sources :Center for Security and Emerging Technology (2026), Allianz Research

And the proof lands in AI itself. Narrow the lens to AI-specific capital and Britain's lead evaporates: Across 2020–2025 it drew around USD39bn, behind both Germany (USD45bn) and Sweden (USD46bn), with inflows growing just +21% a year lately against +24% in France and +37% in Germany – and a fast-rising Spain at +70%. The gap exposes where the UK sits in the value chain: strong in the materials layer – compound-semiconductor substrates, specialty chemicals – but absent from the parts that compound as comparatively to its European peers, the UK does not produce lithography machines or provide sovereign frontier-model and cloud services. London hosts among Europe's largest data-center fleets, but renting out the racks is not the same as owning the chips, models and clouds that fill them. That is the real verdict on Brexit-as-liberation: Far from a break-free moment to slip Europe's regulatory harness and chase the frontier, leaving the EU turned the UK inward — doubling down on the finance, insurance and software it already knew rather than the new domains where France, the Netherlands and Sweden are planting flags. The pandemic and the energy crisis weighed on the effort, but they cannot be the whole story, because the same storms crossed the Continent and its rivals emerged building, not retreating. The magnet still works – it is simply drawing capital into a narrower, more familiar field, while the frontier is mapped elsewhere.

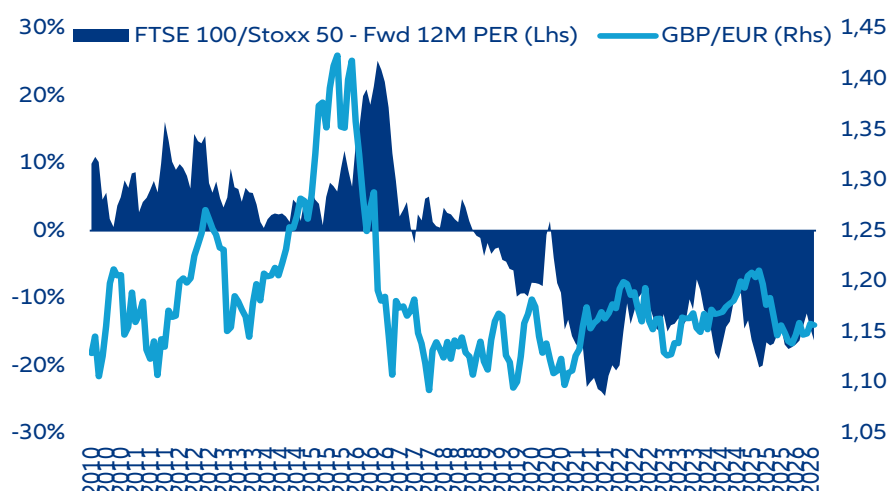
The UK stock market has also lost its shine. The same loss of altitude shows up in public markets. London remains the world's fourth-largest equity venue by market capitalization — behind the US, China and Japan — but that perch is now crowded, with South Korea, Canada and Taiwan each circling at roughly 3% share. Trading itself hasn't dried up: daily turnover has actually risen since the 2016 vote, to about GBP4.8bn from GBP3.8bn, though that merely tracks a global pick-up in equity activity that ran hotter still in the US and parts of Europe. What has unmistakably shrunk is relative weight. The UK's share of global market cap nearly halved from around 8.5% in 2006 to 5% by 2016, then slid again to roughly 3% by 2026 — even as the US vaulted from 42.5% to 54.5% in the past decade alone. And British equities now trade at a 15–20% discount to their Eurozone peers, a gap that has barely budged in years.

Figure 25: Market share of largest stock exchange place per market capitalization

Sources : LSEG Datastream, Allianz Research

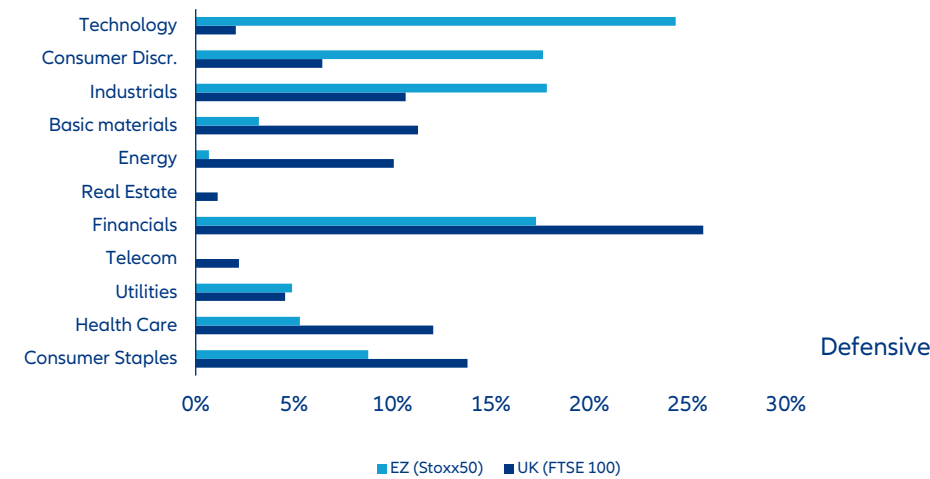
Persisting post-Brexit risk premia and low-technology profile burden stock attractiveness. The roots run deeper than Brexit — the slide was underway well before 2016 — but the divorce plainly didn't help. That lingering discount sits alongside a pound that never fully recovered against the euro or dollar, and a standing political-risk premium: global investors repriced UK assets downward after the referendum and never repriced them back. The bigger drag, though, is composition. The London index is structurally defensive — heavy in financials, energy, healthcare and staples, light on cyclicals — at the very moment markets have been relentlessly risk-on, powered by the AI trade. With technology barely 2% of the index, the UK simply lacks the exposure to ride that rally: its

make-up cushions the blows in volatile, risk-off spells but caps the upside precisely when, as now, the world is reaching for growth. The equity market, in short, mirrors the innovation economy above it – safe, familiar and increasingly on the wrong side of where the frontier is moving.

Figure 26: Relative valuation of UK equity toward Eurozone peers

Sources : Dealrooms, Allianz Research

Figure 27: Sector classification differences

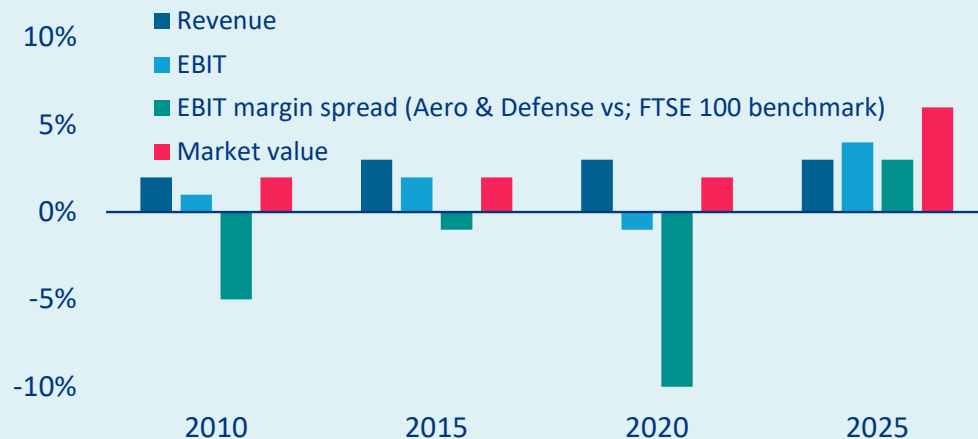


Sources :Center for Security and Emerging Technology (2026), Allianz Research

Box: Defense – Act or Lag

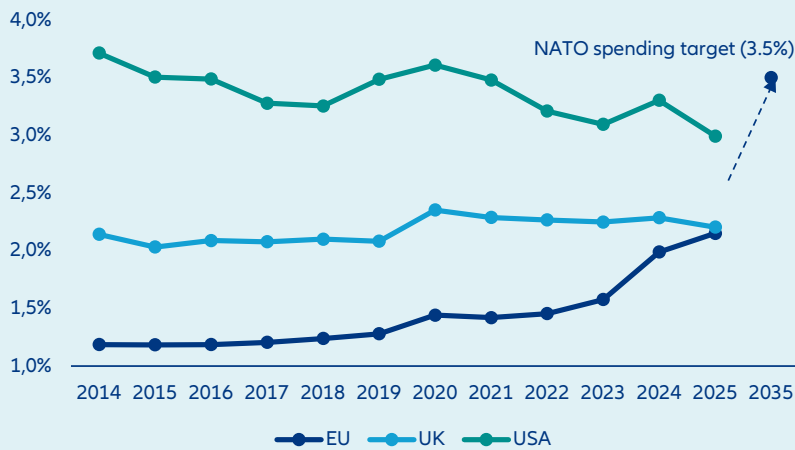
The Russia–Ukraine war handed European equity one of its scarcer assets — a genuine European growth story in which UK looks a lackluster against its peers. Continental names had a blockbuster 2025: the STOXX Europe Total Market Aerospace & Defence index returned roughly 68% over the twelve months into early 2026, taking the cumulative rally since Russia's February 2022 invasion to about 260%, with Rheinmetall — the rearmament proxy — up roughly 150% in 2025 alone and more than 1,000% above pre-war levels. Momentum has since cooled: the STOXX Europe Aerospace & Defence index is down ~1% year-to-date against ~5% for the STOXX 600 (late May 2026), and Rheinmetall sits about 38% below its January peak – a consolidation, not a reversal. The UK captures only a thin slice of that externality: core defense names in UK have roughly tripled their FTSE 100 market-cap weight since 2020, well ahead of the benchmark, but from such a low base that A&D stays a near-negligible index line. The US case is different again, and easily overstated as a Ukraine dividend: The S&P Aerospace & Defense Select index returned ~54% over the trailing year, comfortably ahead of the S&P 500 – but on an equal-weight basis that flattered smaller space and drone names) as much as giant domestic aerospace & aircraft. The marquee primes lagged: Lockheed fell ~13% over the trailing year and underperformed the S&P 500, leaving US primes at a discount to richly-rated European peers – a domestic budget-cycle trade (a ~USD1trn FY26 outlay) more than a war premium.

Figure 28: Weight of aerospace & defense core companies into FTSE 100 index benchmark (labeled in term of end fiscal 12-month trailing revenue, EBIT and market value)



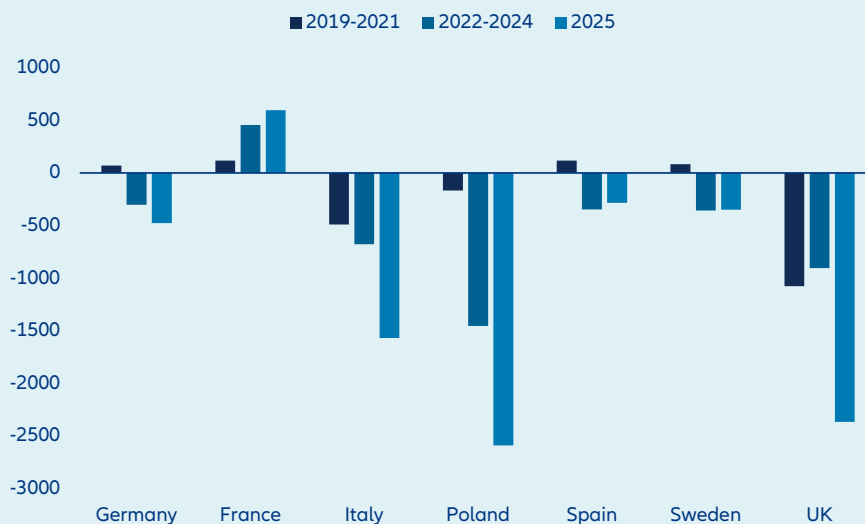
Source: LSEG Datatsream, Allianz Research

Figure 29: Evolution of military spending as % of GDP in EU, UK &



Source: NATO, Allianz Research

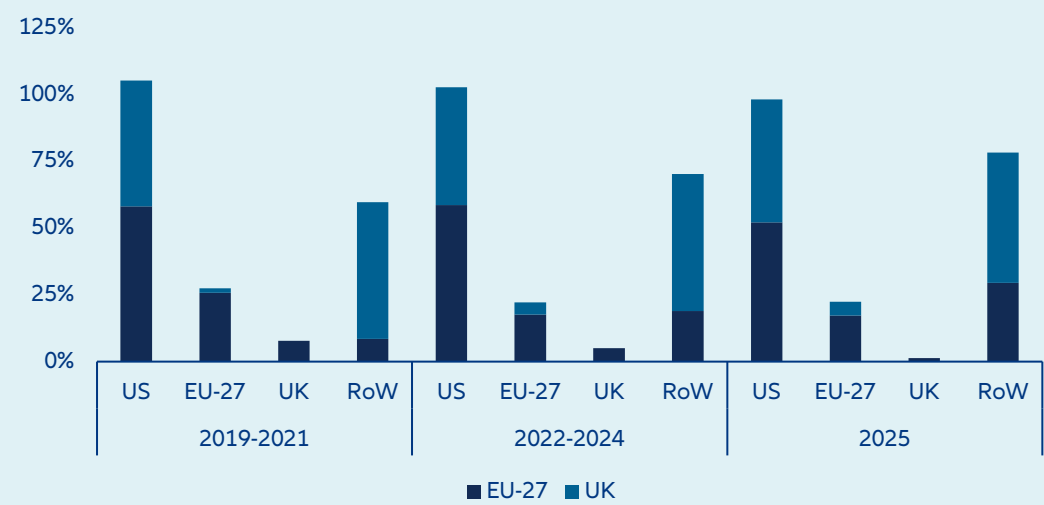
Figure 30: European countries trade balance with US in term of weapons and military equipment transfer (in mn of



Source: Sipri, Allianz Research

Neither war nor an American retreat drew Britain into Europe's defense fold. The chart tells it plainly: Across every window, including the 2022–24 war years, the UK's sourcing from EU-27 suppliers stays thin and by 2025 thins further toward the floor, while Britain remains a captive customer of US equipment, combat aircraft and munitions above all, for which it is Washington's only national "Level 1" industrial partner on the F-35, with roughly 15% of each airframe built in the UK. The corollary is a squandered windfall. Having opened negotiations to access the EU's EUR150bn SAFE rearmament facility after the May 2025 security pact, London walked away in late November over the entry fee and procurement terms — Brussels' demand reportedly ran to ~EUR6.75bn, even as Canada joined for around EUR10m — relegating UK firms to "third-country" status capped at 35% of any funded contract's value. And with the EUR150bn already parceled out to member states (Poland alone ~EUR44bn), that first-round ship has sailed. Global conflict handed Britain an open door to European defense-industrial integration and a multi-year demand pipeline to underpin its primes' earnings; it stayed welded to the dollar instead.

Figure 31: Regional distribution of arms & weapons imports by supplier breakdown (data labeled into millions of SIPRI trend-indicator values (TIVs))



Source: Sipri, Allianz Research

Stalling risks: funding constraints to hurt ambitious roadmap. On paper the trajectory is bold – 2.6% of GDP by 2027 en route to NATO's new 3.5% core target by 2035, 5% all-in — and the UK is embedded in a thick layer of partnership commitments, from AUKUS to the SAFE instrument and the trilateral GCAP fighter with Japan and Italy. In practice, a roughly GBP28bn funding gap and a Defence Investment Plan delayed several times have left those ambitions hostage to a Treasury unwilling to absorb more borrowing and a government wary that the tax rises needed to close the gap would render the agenda politically toxic. GCAP is the live illustration: lacking a long-term contract, the program has been running on a ~GBP686m bridge deal funding work only through end-June 2026, and frustration is rising in Italy and Japan that press UK government to offer funding guarantee. For the domestic base the signal is corrosive: immobilism today undercuts the case for capacity investment tomorrow and erodes the very argument primes need to attract capital – just as peers accelerate. The risk is not a reversal of intent but a creeping credibility gap between what Britain pledges and what it actually funds.

A photograph showing a group of diverse hands stacked on top of each other, resting on a tree trunk. The hands are of various skin tones and some have accessories like a black wristband and a silver ring. The background is a lush green forest. The text 'Our team' is overlaid on the image, with 'Our' in white and 'team' in yellow.

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Allianz Research | 18 June 2026

Public vs. private equity: The widest equity return gap in two decades

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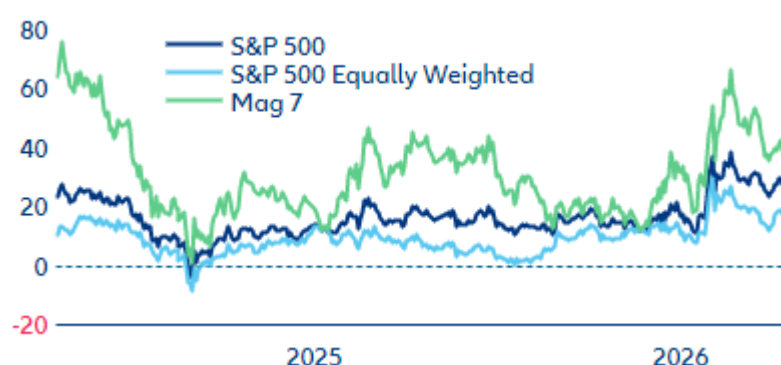
In Summary

- **Public equities have repriced the future while private equity is still monetizing the past.** Since the October 2022 trough, the S&P 500 has compounded at more than 20% a year for three straight years, while Private Equity buyout portfolios assembled at the 2020-2021 peak, and now carrying more expensive debt into a thin exit market, have lagged on every benchmark and horizon. At this point in time, the gap between liquid and illiquid equity returns seems to be the widest in two decades.
- **The rally that opened the gap is narrow and earnings-led, not a liquidity melt up.** The Magnificent Seven alone delivered more than half the S&P 500's three-year total return, strip them out and the index is an ordinary performer, and the equal weighted version has compounded at about half the pace, a gap unseen since the late 1990s. But the move rests on delivered earnings and a real AI capex super cycle rather than multiple expansion, so the premium looks more durable than the dot-com peak.
- **Yet private equity's long-run premium is real, this cycle has just interrupted it.** On a like for like basis, matched to public indices for geography, size, sector and leverage and net of fees and carry, buyout has beaten public markets over three decades, with MSCI estimating pooled direct alpha of roughly 400bps a year since 1994. The exception is the 2021-2023 vintages, which currently show a negative direct alpha of about ~800bps against the MSCI ACWI, the first consecutive run to trail public markets in the series. Consequently, the data points to rough PE performance ahead even allowing for the fact that these vintages are still young and J-curve distorted.
- **Private equity can no longer count on rising valuations and must now drive returns by growing the underlying business.** For over a decade, buyout returns came mostly from rising valuations, selling companies for a higher multiple than was paid. That is mostly gone as borrowing costs have roughly doubled and exit valuations have stopped climbing, so returns now come from growing the business. The contrast on the financing side is also large, with the largest listed companies still holding more cash than debt, so benefiting from higher rates, while buyout owned companies sit on floating rate debt due to be refinanced in 2026-2028, where higher rates only add to their interest bill.
- **As capital drains from private equity, private debt is catching the outflows.** The same jump in interest rates that made buyout deals more expensive made lending the more attractive trade. Big long term investors continue to like senior private credit because it pays a steady, contractual cash yield, its value barely moves and losses have been low, none of which buyout has delivered lately. So new money is rotating out of private equity and into private debt.
- **2026 will decide whether the public-private gap was a passing extreme or a lasting regime change.** Our central case assumes an economic mix that keeps returns positive but well below the post-Covid peaks with public equity settling in the low teens, while buyout follows in the mid-teens through 2027-2028. All in all, the way forward for private equity seems to be less about how much to own and more about how to own it. This means favoring managers and vintages with a demonstrable edge over generic exposure, treating liquidity and valuation discipline as features to be tested rather than assumed and resisting the temptation to buy the label without the underlying skill.

Public markets have sprinted ahead

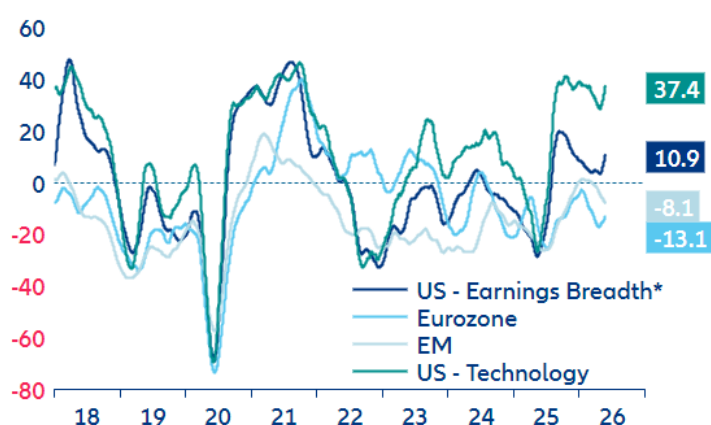
Since the October 2022 trough, public equities have staged one of the sharpest re-ratings in modern memory. The S&P 500 has compounded at more than 20% a year for three straight years, with other large-cap benchmarks close on its heels. But the headline flatters the market. The rally has been extraordinarily narrow: The Magnificent Seven (*Mag 7* - Apple, Microsoft, Alphabet (Google), Amazon, Nvidia, Meta (Facebook) and Tesla) alone delivered more than half of the S&P 500's three-year total return. Stripped of those seven names, the index is a thoroughly ordinary performer. The dispersion shows up in valuation, too. The top ten names now trade at a roughly 50% forward price/earnings premium to the rest of the index, wide by any historical yardstick. But that premium rests on an earnings path well above the dot-com peak of the 2000s, a sign that the fundamentals are more resilient now than they were then (Figure 1). Earnings breadth is mending as well (Figure 2), if grudgingly: Barely a third of S&P 500 members beat the index in 2025, up from a record low the year before. At the same time, small caps have scarcely shown up at all. The equal weighted cut of the same indices tells a more sober story still, with the S&P 500 equal weight compounding at about half the pace of its capitalization weighted parent, a gap unseen since the late 1990s. All in all, what reads at the index level as a broad bull market is, mechanically, the re-rating of a very small number of balance sheets.

Figure 1: Equity performance y/y%



Sources: LSEG Datastream, Allianz Research

Figure 2: Earnings breadth (3mma)



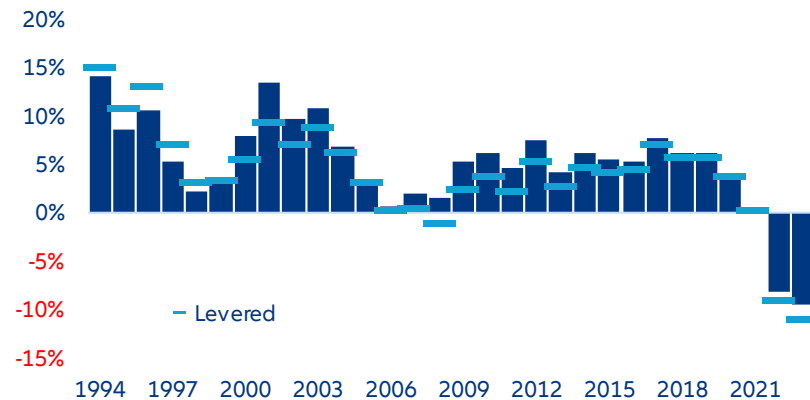
Sources: IBES, LSEG Datastream, Allianz Research

Note: Earnings breadth measures the share of index constituents receiving upward earnings estimate revisions.

The scale of the recent gap is easier to appreciate against private equity's long-run record. On a like-for-like basis, comparing each buyout fund with a public index matched for geography, size, sector and leverage, and measuring excess return net of fees and carry, private equity has, in fact, beaten public markets over the past three decades. MSCI estimates pooled buyout direct alpha of roughly 4% a year since 1994. That long-run premium is precisely what the current cycle has interrupted. Broken out by vintage, buyout funds delivered positive alpha against global equities in almost every cohort since the mid-1990s. The exception is the 2021-2023 vintages, which currently show

direct alpha of roughly -5% to -12% against the MSCI ACWI, the first run of consecutive vintages to trail public markets in the series. Of course, those readings come with an important caveat since the funds are young, and early fee drag (the J-curve) mechanically depresses returns that typically recover as portfolios mature, so those returns are provisional rather than final. But even allowing for that, the message is consistent: The vintages bought into the cheap-money peak are, so far, the worst relative performers in a generation and may still lead to some struggling years ahead (Figure 3).

Figure 3: Buyout direct alpha vs. public equity by vintage (in %)

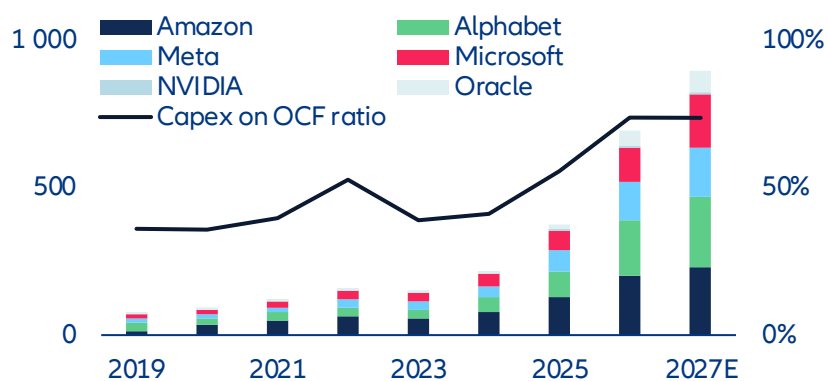


Sources: MSCI Private Capital Universe; Lester, O'Shea & Warren (2025), Allianz Research

Note: Direct alpha is the annualized excess return of buyout funds over a leverage- and sector-matched public index, net of fees and carry. 2021-2023 vintages are shaded to indicate they remain J-curve-distorted and are not yet representative of since-inception performance; 2024-2025 are omitted as too young to have generated representative exit data.

The character of the move matters as much as the magnitude. This has not been a liquidity-driven melt up of the 2020-2021 variety. It has been an earnings- and margin-led re-rating built on top of a genuine capex super cycle (Figure 4). Along these lines, forward earnings for the S&P 500 are up by roughly a third since the 2022 trough, and the bulk of the index's price return is explained by delivered earnings rather than multiple expansion. That distinction matters for how we read the gap to private markets performance (Figure 5). Public equity has not simply re-priced risk, it has also re-priced a specific, identifiable growth thesis around AI infrastructure, hyperscale compute and the operating leverage of asset-light platforms. Private equity, holding a portfolio assembled for a different regime, has had no comparable thesis to ride, and software, one of its biggest post-Covid allocations, has suffered from major potential disruptions. The second feature worth flagging is what the rally has done to the investable opportunity set itself. Free float in the largest names has shrunk as buybacks have accelerated and insider holdings have stayed concentrated, even allowing for the supply that the latest mega-IPOs have brought back. The effective supply of mega-cap equity available to incremental buyers is therefore tighter than headline market capitalization suggests (Figure 6). At the same time, the US now has roughly half as many listed companies as it did at its late 1990s peak, and the firms that do go public are older and larger than they used to be. So when a private equity sponsor comes to sell a company today, the public market it can exit into is smaller, more concentrated in a few large names and pickier about what it will absorb than the market sponsors were counting on when they bought in over 2018–2021. Consequently, the gap between public and private valuations in that light does not look like a temporary mispricing that will simply close, but a lasting result of a deeper shift in where growth capital is raised and how it is priced.

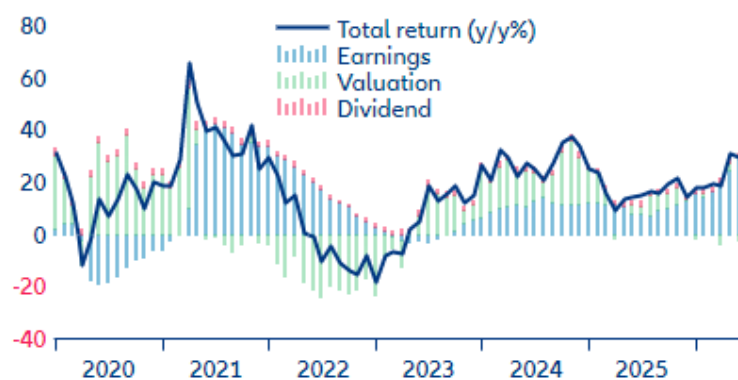
Figure 4: AI capital expenditure



Sources: Bloomberg, Allianz Research

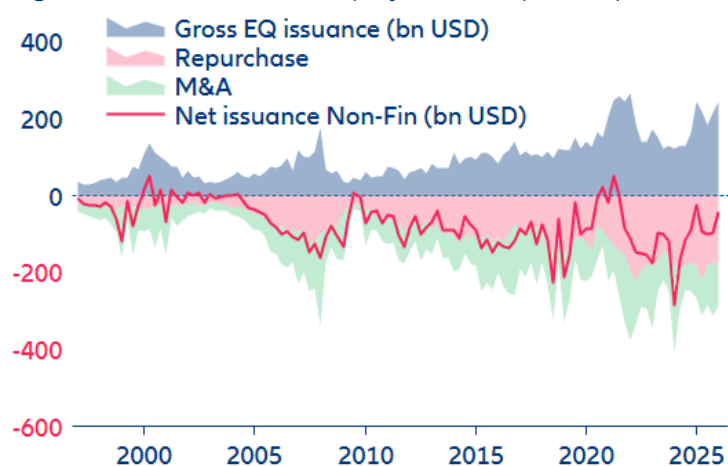
Note: OCF – Operating Cash Flow

Figure 5: S&P 500 return breakdown



Sources: LSEG Datastream, Allianz Research

Figure 6: US non-financial equity issuance (bn USD)

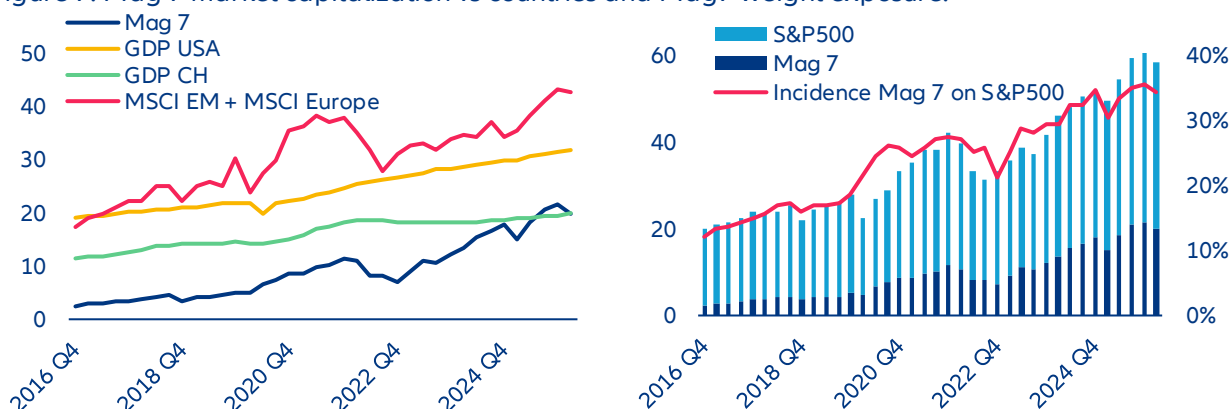


Sources: LSEG Datastream, Allianz Research

Public markets have absorbed private equity's growth engine

The narrowness of the public equity rally deserves closer inspection, because it is both the source of the divergence with private equity and a risk factor in its own right. The Mag 7 now represent roughly a third of S&P 500 market capitalization, up from around an eighth a decade ago, and their combined market value exceeds the GDP of every country except the US and China (Figure 7). Of course, the concentration is also self-reinforcing, because as these companies grow, passive and benchmark tracking flows mechanically direct more capital toward them, deepening their index weight independently of any fresh view on fundamentals.

Figure 7: Mag 7 market capitalization vs countries and Mag7 weight exposure.



Sources: Bloomberg, Allianz Research

Earnings concentration is also at extremes, and it now runs in parallel with the concentration in price. The same handful of mega-caps that dominate index weight also generate a disproportionate and rising share of S&P 500 earnings, so the benchmark's profits, not just its market value, increasingly depend on a single cohort. On valuation, the index trades well above its long-run average forward multiple, but the premium is almost entirely a mega-cap phenomenon; strip out the largest names and the rest of the market sits much closer to its historical norms, leaving a two-tier index in which the most expensive part is also the largest part (Figure 8). Because so much of the index now sits in long duration growth names, the multiple is also unusually sensitive to moves in rates and growth expectations; a small shift in either can swing the level materially. As an anchor for that sensitivity, a reversion of the cluster's valuation premium to its long-run average alone, before any change to underlying earnings, would cost the S&P 500 close to 10-20% of its value.

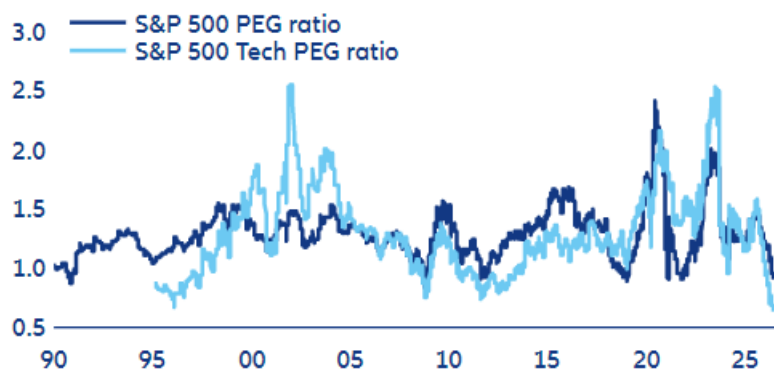
Figure 8: S&P 500 12m forward PE ratio



Sources: IBES, LSEG Datastream, Allianz Research

The same concentration invites two sharply opposing readings, and the contest between them is itself a market risk. The bullish case argues that the Mag 7 are nothing like the 2000 dot-com leaders as they generate hundreds of billions of dollars of free cash flow a year between them, sit on net cash balance sheets and dominate the decade's most powerful secular trend: AI compute, cloud infrastructure and advertising-data networks. Consensus forward earnings growth comfortably exceeds the broader market's, and on a PEG (Price/Earnings-to-Growth ratio) basis the cluster trades at parity or a discount. Nonetheless, the bearish case argues that the AI thesis rests on hyperscale capex sustaining at current levels, that enterprise AI revenue has yet to scale into the returns that capex implies and that the concentration itself is a source of fragility. An earnings disappointment from one or two names could de-rate the whole index in days. Both views can be right at once but which one prevails comes down to whichever is dominating the narrative at the time (Figure 9).

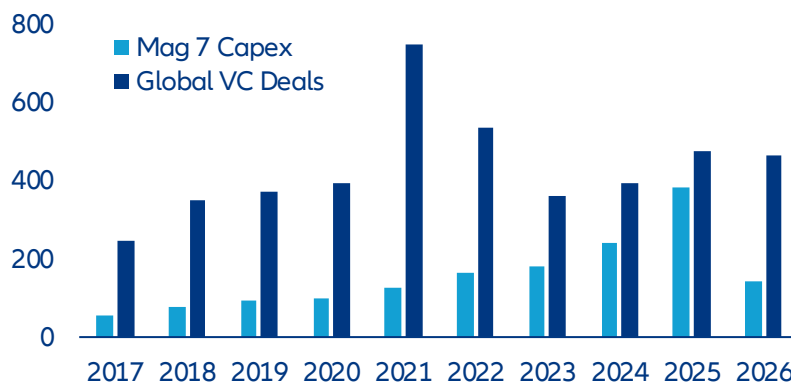
Figure 9: S&P 500 PE to growth ratio



Sources: IBES, LSEG Datastream, Allianz Research

Why the concentration matters for the public-private comparison is subtler than it first appears. The Mag 7 are themselves “long duration” growth assets, capex heavy and R&D intensive, with revenue streams that compound over a decade rather than monetize in a five-year LBO window. In effect, US public markets have absorbed the kind of growth equity exposure that used to migrate to private markets. Combined hyperscale capex now approaches the entire global venture capital (VC) deployment. Private investors who, in earlier cycles, would have funded the next generation of compute or AI infrastructure through late-stage venture rounds can now access that exposure through publicly listed mega caps, at any size and at low friction (Figure 10).

Figure 10: Mag 7 Capex vs VC deployment (USD Bn)



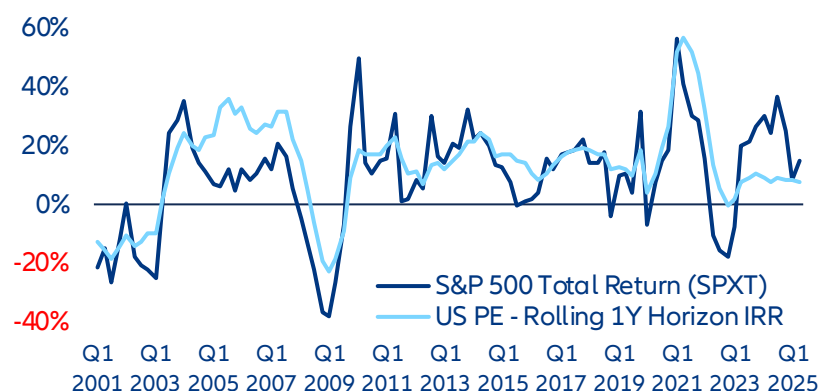
Sources: Bloomberg, PitchBook, Allianz Research

The corollary is uncomfortable for private markets. The universe of high-growth, scaled, profitable technology companies available exclusively to private investors has shrunk. At the time of writing, some of the private mega caps remain private holdings (though not for long, as SpaceX, Anthropic and OpenAI have IPO'd or filed to do so),

but the depth of comparable opportunity in private markets has narrowed precisely as fundraising and deployment has slowed.

Private equity, by contrast, is still digesting the previous cycle. The asset class entered the 2022 rate shock with peak fundraising, peak entry multiples, peak leverage and aggressive growth assumptions. The 2020-2021 vintages, in particular, were priced for a world of cheap money and continued multiple expansion, a world that ended within 18 months. Even top-quartile buyout funds returned single-digit IRRs in 2025, well behind both the S&P 500 and MSCI World, and the average vintage of the past decade barely cleared mid-single-digits (Figure 11).

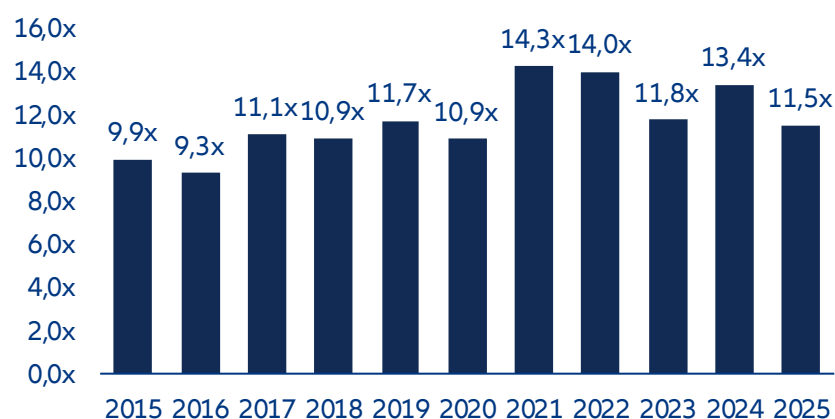
Figure 11: US Private Equity 1y IRR vs S&P 500



Sources: Bloomberg, PitchBook, Allianz Research

The lag has many causes, but the most mechanical is a reset in the cost of capital. A typical deal a decade ago cleared at around 10 times EBITDA, with half its capital in single-digit yield debt; today's clears closer to 13 times, with materially less leverage at far higher financing costs. Sponsors paid those record prices expecting to exit into further multiple expansion. That expansion has not arrived, and because thinner leverage cushions the entry, the drag on equity returns runs deeper than the change in multiples alone would suggest (Figure 12).

Figure 12: North America buyout entry multiples

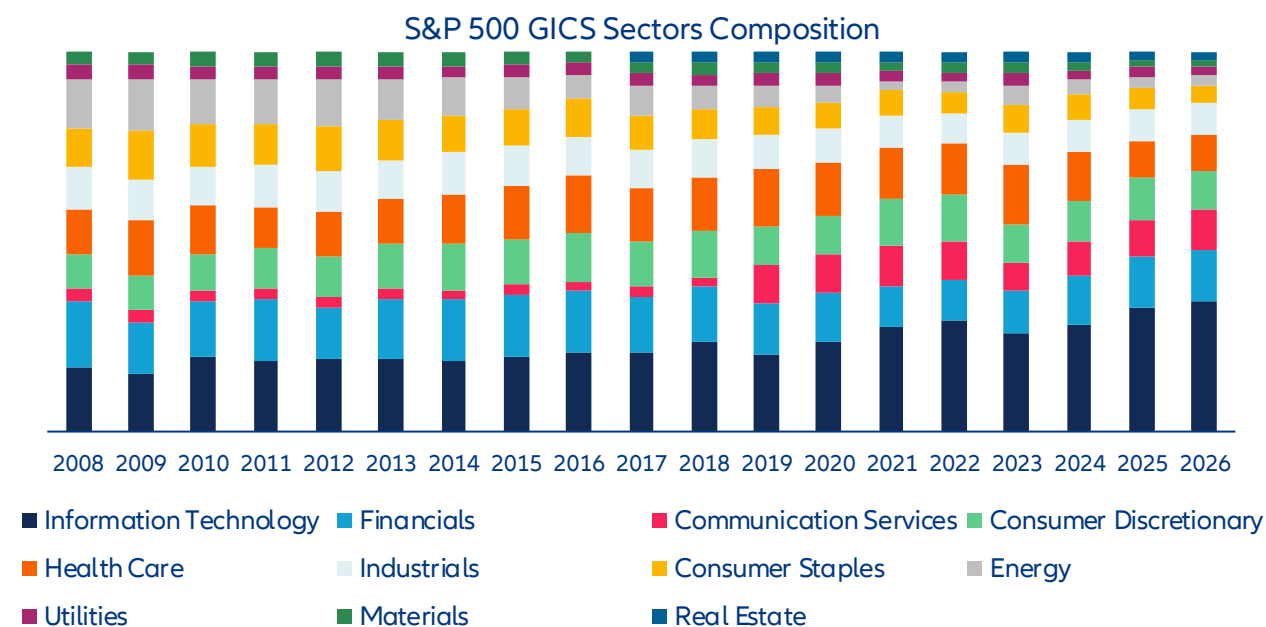


Sources: Pitchbook, Allianz Research

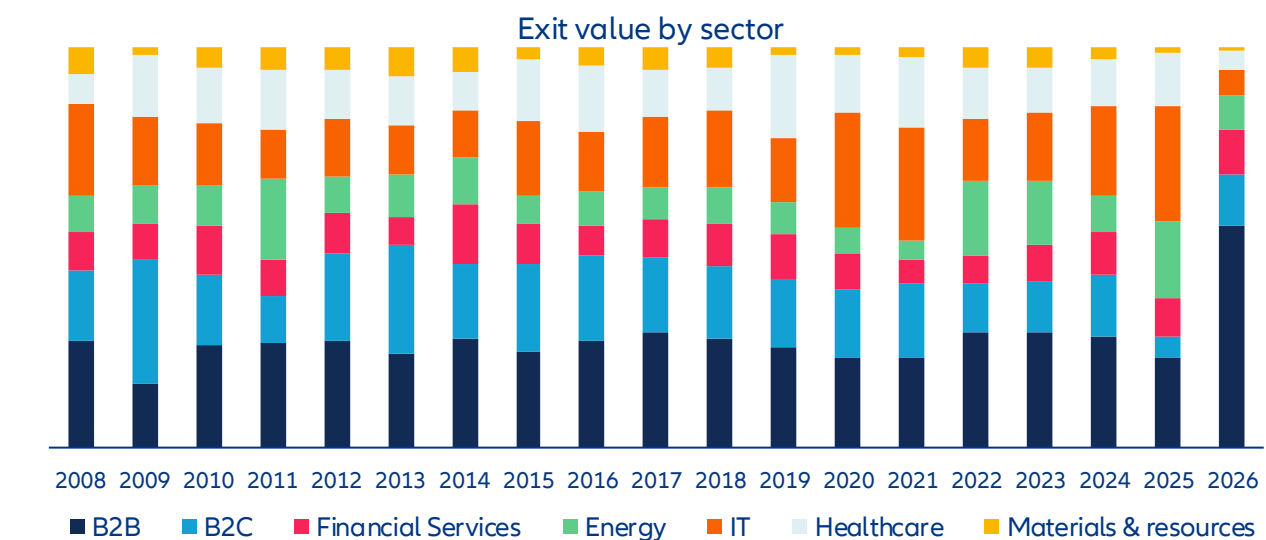
The mismatch is as much about sector mix as about timing. Public benchmarks are now dominated by mega-cap technology, semiconductors and communication services, the very cohort the AI capex cycle has re-rated, while financials, healthcare and industrials make up the balance. Private equity's deployment skews the other way, with software and technology being the single largest buyout sector for most of the past decade, peaking at roughly a third of deal value in the 2020-2021 vintages, with healthcare, business services and industrials filling out most of the remainder.

The catch is that the software the public market rewards and the software private equity owns are not the same exposure. Listed technology leadership is concentrated in hyperscale compute, AI infrastructure and platform businesses with net-cash balance sheets; the typical Private Equity-owned software asset is a mature, leverage-financed vertical or horizontal SaaS business bought on a high revenue multiple, precisely the cohort most exposed to slowing seat growth, AI-driven displacement risk and a higher discount rate. So even where the two universes appear to overlap on the word "software," they sit on opposite sides of the AI trade: public markets long the enabler, private portfolios long the potentially disrupted incumbent (Figure 13-15).

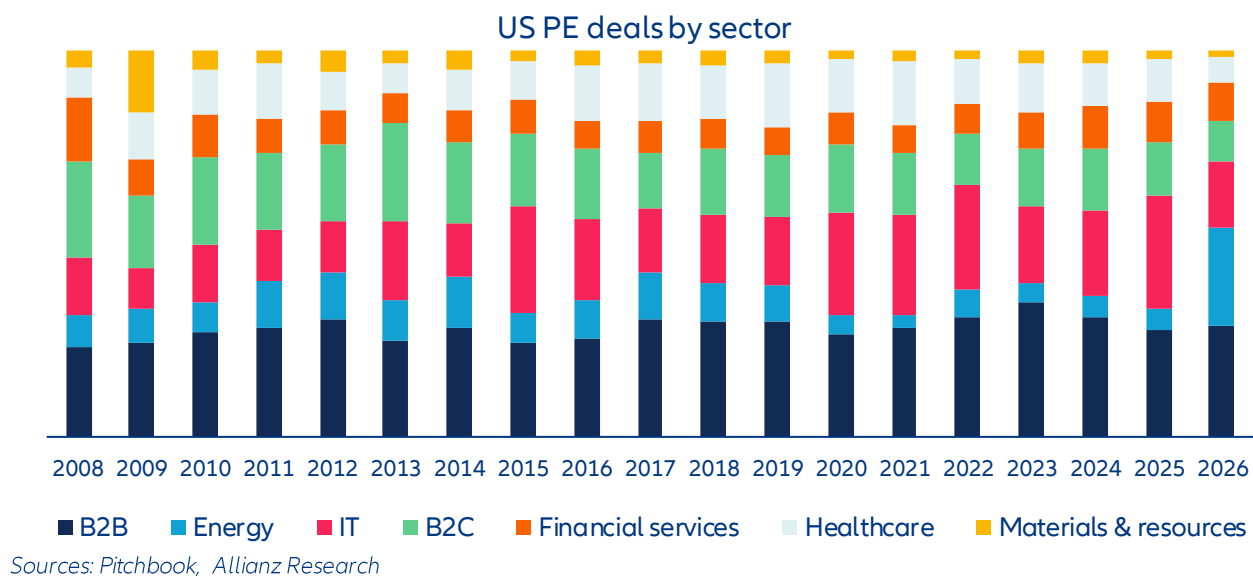
Figure 13-15: US Public vs Private EQ sector allocation (in %)



Sources: Bloomberg, Allianz Research



Sources: Pitchbook, Allianz Research



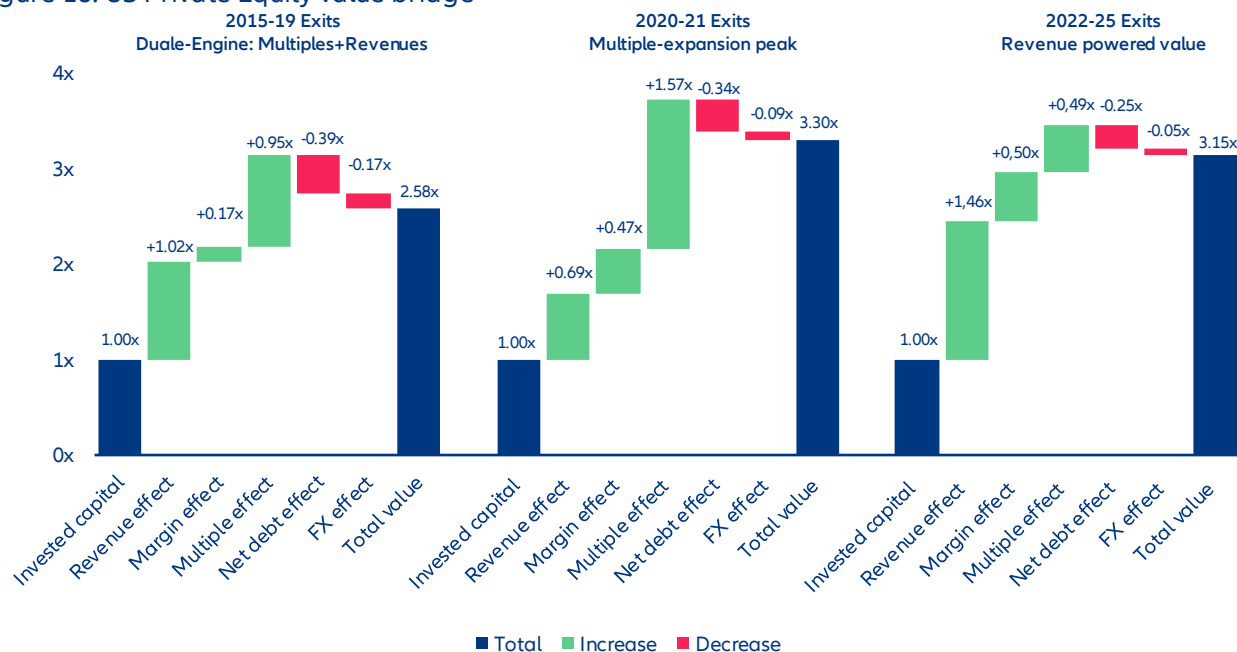
The Private Equity engine has changed, not just stalled

What ultimately separates public and private equity today is not how much they have returned, but what is driving those returns. Strip a buyout down to its parts and the value created comes from three places: the business growing its earnings, the multiple on those earnings rising between purchase and sale and the debt used to finance the deal. For most of the past decade, one of those did the heavy lifting. It was not, as is often assumed, leverage; across successive cohorts of exited deals, debt has been close to neutral and frequently a quiet drag once its rising cost is counted. The real engine was the multiple. Sponsors bought well, watched valuations climb and sold into a market that reliably paid more than they had, with growth in the underlying business playing only a supporting role.

That engine has stalled. Purchase multiples sit at or near record highs, so there is little room left for further re-rating, and exit multiples are no longer drifting up to reward patient owners. With cheap leverage gone and the multiple tapped out, only one source of value is left standing: the business itself. Revenue growth, once the junior partner, is now the dominant contributor to private equity returns, and the deals that work are the ones where the company genuinely grows rather than the ones where the multiple obligingly expands.

This is exactly where the two universes part ways. Public market returns are still, overwhelmingly, a re-rating story, but a narrow one, concentrated in a handful of net-cash mega caps that the AI trade has repriced upward. Private markets have been denied that same rerating, and the businesses sponsors own carry debt rather than cash, so their returns now have to be earned the hard way, one point of revenue growth at a time. Put simply, public equity is still being lifted by the market; private equity increasingly has to build its returns from the ground up. The bar has risen accordingly: only operationally well-versed managers in genuinely growing sectors can clear it (Figure 16).

Figure 16: US Private Equity value bridge



Data as of Q2 2025. Pooled value bridge across a subset of exited buyout holdings, weighted by invested capital.
Source: MSCI Private Capital Universe, MSCI Private Asset and Deal Metrics data

The exit market is where the Private Equity cycle's hangover shows most clearly. Buyout funds hold tens of thousands of unsold portfolio companies worth trillions of dollars, and median holding periods have stretched markedly. The starkest symptom is liquidity, the cash investors actually get back. Unlike a public equity, which can be sold on any trading day, a private holding returns capital only when the fund exits a deal, and that channel has narrowed sharply. Distributions to LPs (limited partners, the investors who fund the buyout) stayed below the 20% mark for a fourth straight year, the longest such stretch on record, and a depth last seen during the 2008-09 crisis. After three years of investors paying in more than they took out, net cash flow edged only modestly above breakeven in 2025, a change in direction rather than a true recovery.

The other exit routes tell the same uneven story. Deal and exit values rose sharply from a low base in 2025, but the rebound was narrow, driven by a handful of mega-deals while the mid-market stayed constrained. The IPO window has cracked open, but aftermarket pricing suggests public investors are still cautious on sponsor owned businesses, with Private Equity-backed listings tending to trade less buoyantly than venture backed ones. The secondary market has absorbed part of the gap, transacting a record volume in 2025 at a blended 87% of NAV (net asset value, the carrying value of the fund's holdings), with continuation vehicles taking a meaningful share, but it clears only a small slice of the unsold inventory each year. The plumbing is working again; the backlog behind it will probably take years to drain (Figure 17).

Figure 17: TTM US Private Equity buyout fund distributions as a share of beginning NAV



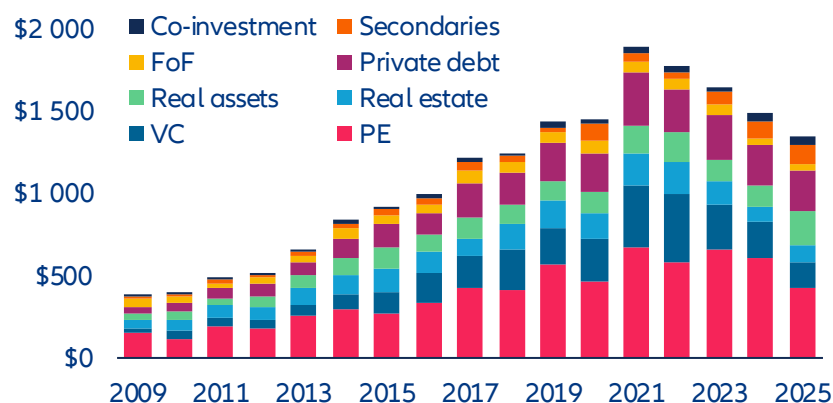
Source: Pitchbook, Allianz Research (Data as of Q3 2025)

Note: TTM = trailing twelve months. The line shows cash distributions paid to investors over the prior twelve months as a share of the fund's beginning net asset value (NAV); lower bars mean less cash is being returned.

Private credit has absorbed what private equity gave up

Private credit deserves its own section because it has been the structural beneficiary of the very forces that have pressured private equity. The same rate reset that raised LBO yields and made buyout math harder turned lending into the more attractive side of the trade, and as banks retreated from leveraged lending under regulatory pressure and sponsors still needed acquisition finance, direct lending stepped into the gap at unprecedented scale. Global direct-lending AUM has more than quadrupled since 2018 and is on track to keep growing through 2028. All-in yields on senior unitranche loans to mid-market, sponsor-backed borrowers remain in the low double digits, equity-like returns earned from a senior, secured position in the capital structure rather than the bottom of it. That combination is exactly what private equity has failed to deliver over the past three years: For long-duration insurers and pension funds, private credit has provided contractual cash yield, low mark-to-market volatility and a credit-loss experience that has stayed below long-run averages despite higher rates (Figure 18).

Figure 18: Private capital raised (USD bn) by strategy

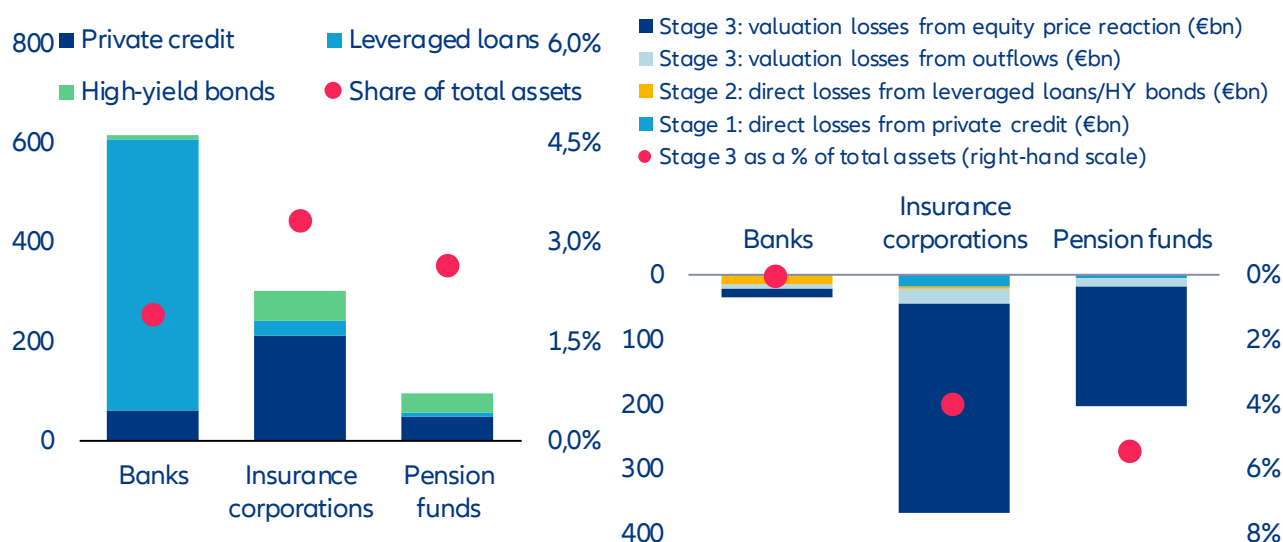


Sources: Pitchbook, Allianz Research

Two cautions are worth registering. First, private credit's default experience has been benign, with realized losses well below long-run averages, but the asset class has not yet been tested through a full credit cycle at current scale. The borrower base is also more levered than at any prior measurement point, and payment-in-kind (PIK) structures, in which coupons accrue rather than pay in cash, have grown sharply, suggesting some borrower-level stress that is being deferred rather than recognized. Second, private credit is increasingly intertwined with the insurance

balance sheets that own much of the asset class. The IMF and the ECB have flagged this as a systemic concern; when liquidity pressures emerge, insurance regulators may impose forced de-risking, creating sudden price-discovery events in a market built on stable, marked-to-model valuations. Neither concern is a near-term catalyst. Both argue for portfolio construction that favors senior secured exposure, granular underwriting and clear pricing discipline over chasing the highest-yielding mezzanine and second-lien strategies. (Figure 19-20)

Figure 19-20: Direct exposures of Eurozone entities to private credit and leveraged debt markets (left); Simulated losses for Eurozone financial entities from broad defaults in private credit and a severe shock in the software sector (right)



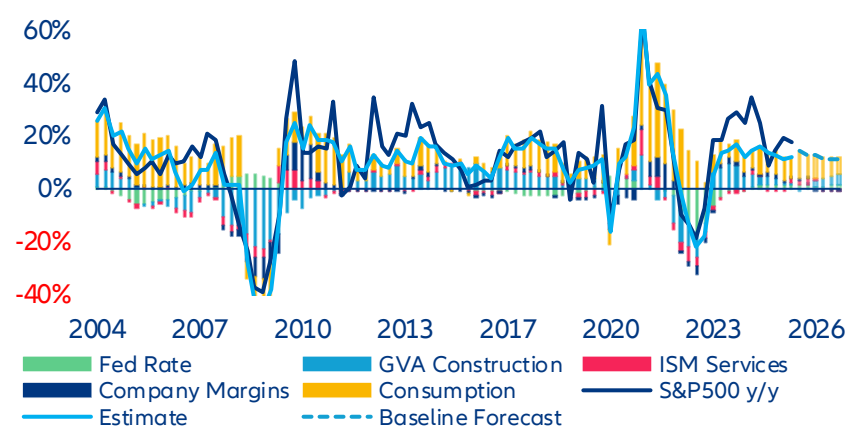
Sources: ECB, Allianz Research

The broader implication for private credit is that the asset class should be treated as a core, structural holding rather than a tactical reach for yield. The case does not depend on rates staying high. Even as financing costs normalize, the bank retreat from leveraged lending looks structural, the borrower base is largely captive and the asset's contractual cash yield and low mark to market volatility answer precisely the liquidity problem that has made private equity so hard to hold. For insurers and pension funds matching long-dated liabilities, that makes private credit a natural anchor allocation rather than a trade. But "core" tends to mean senior, not adventurous. The allocation belongs at the top of the capital structure, in senior secured exposure run by managers who can underwrite through a full cycle. Mezzanine, second-lien and opportunistic strategies still earn their place as return enhancers, but they are best treated more tactically, sized around that senior core rather than forming it. The suggested posture, in short, is a larger and more permanent allocation held more conservatively.

Outlook 2026: three drivers, three scenarios

The next 12 months will decide whether the 2022-2025 divergence between public and private market returns was a cyclical extreme or the start of a longer regime change. The central case assumes a prolonged but contained Middle East conflict, the Federal Reserve maintaining high rates and the AI capex cycle sustaining without a major guide down, a configuration that keeps returns positive across asset classes but well below the peaks of the post-Covid recovery. Under those conditions, public equity returns would be moderate into the low teens. Consumption and construction sentiment, the two biggest supports for the S&P 500 since 2010, hold up, while higher for longer rates shift from neutral to a modest drag. Private equity follows with a lag, with buyout returns settling in the mid-teens through 2027-2028 as financing conditions stabilize and the income flows that drive most of the variation in buyout returns recover from their 2023-2024 trough (Figure 21).

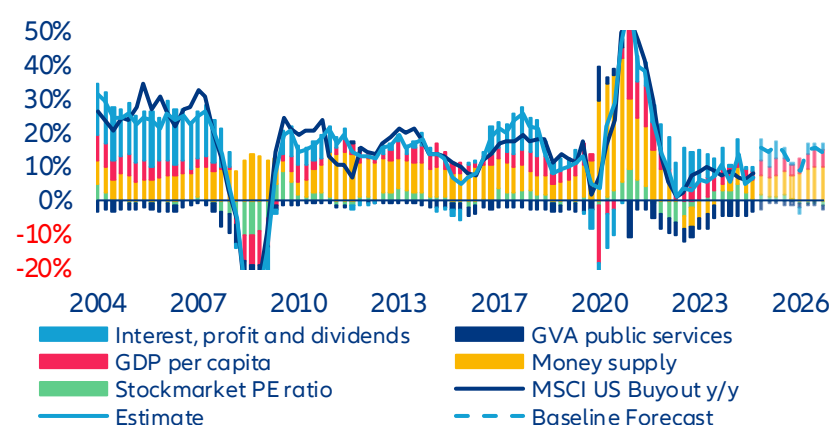
Figure 21: S&P 500 return decomposition and forecast (y/y%)



Sources: LSEG Datastream, Oxford Economics, Allianz Research

Note: Please note that these scenarios are produced using the Oxford Economics model and scenarios for comparability purposes. For official forecasts please refer to our quarterly scenario.

Figure 22: MSCI US Private Equity buyout return decomposition and forecast (y/y%)

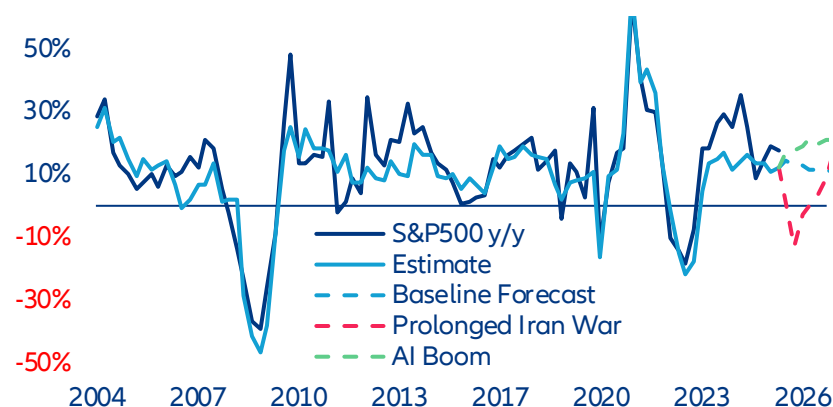


Sources: LSEG Datastream, Oxford Economics, MSCI, Allianz Research

Three drivers will decide which scenario plays out, each operating on a different horizon and through a distinct channel. In the near term, the resilience of the Middle East deal is the dominant variable. If oil and gas trade continue to normalize through the negotiated settlement, the path for both asset classes would look very different than if disruption to the Strait of Hormuz reignites beyond September. In that second case, a forced Fed tightening would turn rates from a modest drag into an active headwind for public equity. In this regard, consumption which is itself sensitive to rates through mortgage costs and credit availability, and the US consumer would enter any oil driven tightening from an already fragile position. The shock would also hit the real economy directly, with construction activity stalling as input cost visibility would collapse and commodity pass through would erode real

incomes. Under that path, public equity would turn negative in 2026, with buyouts following one to two quarters later as valuations reset with their characteristic lag (Figure 23).

Figure 23: S&P 500 scenario analysis (y/y%)

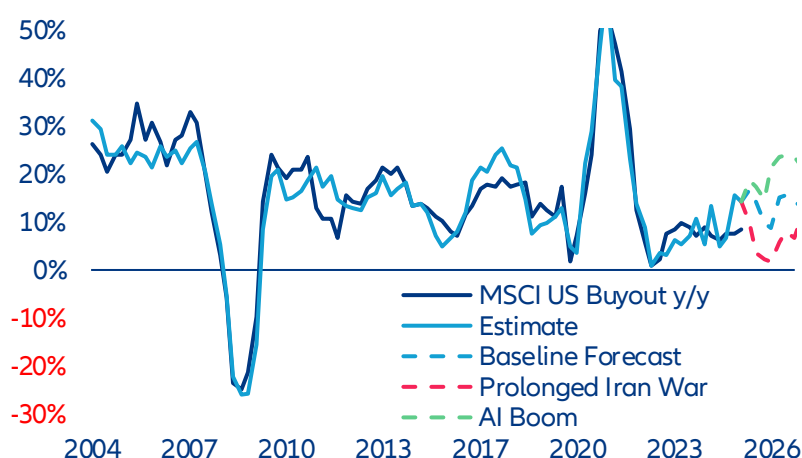


Sources: LSEG Datastream, Oxford Economics, Allianz Research

Note: Please note that these scenarios are produced using the Oxford Economics model and scenarios for comparability purposes. For official forecasts please refer to our quarterly scenario.

Over the medium term, the path of rates and the sustainability of AI capex are likely to become the binding constraints, and their interaction will determine whether the baseline recovery holds or extends. A higher for longer rate environment would delay the recovery of buyout profits, holding the downside path well below the cost of illiquidity in 2026. If AI capex were to guide down, whether through enterprise revenue disappointment, regulatory friction or balance sheet discipline, large cap tech margins would be hit first, and operating leverage means the damage would reach returns faster than the analyst revision cycle would suggest. From there it would propagate into buyout exit valuations. At that point a 10 to 15% public market de-rating is the single factor most capable of turning a baseline buyout year into the 4% downside outcome (Figure 24).

Figure 24: US Private Equity buyout scenario analysis (y/y%)



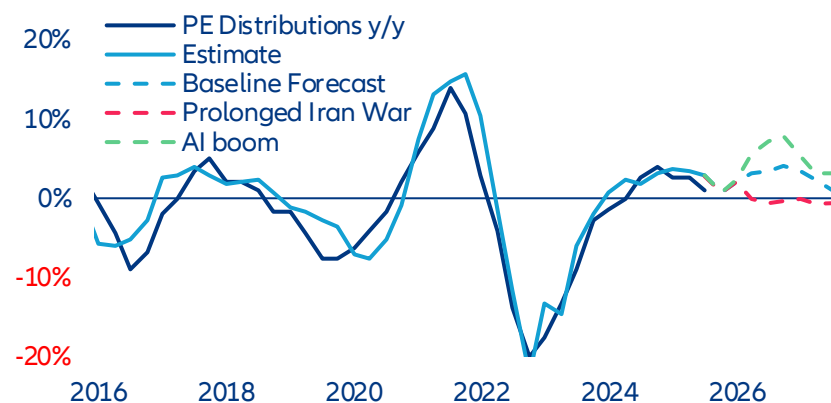
Sources: LSEG Datastream, MSCI, Oxford Economics, Allianz Research

The upside case requires both the near- and medium-term constraints to ease together, and when they do, the dynamics are mutually reinforcing in a way that produces outsized returns. In this scenario, with the Middle East conflict de-escalating after the June 2026 agreement, the Fed gains leeway and AI investment accelerates toward the 1990s capex cycle, with digital technology spending and business investment rising sharply. In that configuration the outcomes are assertive. Public equity peaks above 20% in 2027 as the consumption, margin and Fed rate drivers align at once, historically the setup that produces the sharpest single year public equity returns.

Buyout surpasses 20% shortly after as multiple expansion feeds through into exit valuations and money supply conditions ease, with the equity market PE ratio and the interest and profit rate drivers all contributing at the same time. Europe participates, but structurally less so. With the ECB's more gradual easing delivering less impulse than the Fed, Eurozone growth stays subdued and the AI preparedness gap, lower adoption capacity and smaller hyperscaler exposure mean the productivity gains accrue disproportionately to the US, compounding the transatlantic divergence of the post-2020 period.

The central message of the downside is not the magnitude of the return shock but its asymmetric transmission across asset classes. In the downside scenario, public equity would reprice immediately and sharply as margins and consumption deteriorate together under the stagflationary mix of input cost pressure and slowing growth, then would recover gradually as base effects and eventual policy normalization rebuild the factor stack. Private equity NAVs would adjust with a two to three quarter lag, technically positive in 2026 but close to the cost of illiquidity. The losses would be more contained in return space yet materially worse in liquidity terms, and nowhere is that gap more visible than in distributions. Historically, borrowing spreads and the CAPE ratio drive distribution activity, with rates the most persistent drag since 2022 and equity issuance, a proxy for IPO activity, the cyclical amplifier that has been conspicuously absent through the drought. The mega-IPO pipeline is where that factor becomes concrete. Successful mega-IPOs at premium multiples would lift the equity issuance driver, validate private AI marks across the buyout book and reopen the exit channel for sponsors holding inventory, converting years of paper gains into LP cash on a scale that dwarfs any prior exit event. Nonetheless, a collapse of the June agreement and renewed Iran escalation would break the sequence entirely, with spreads widening, the CAPE ratio deteriorating, rates staying higher and the IPO window closing before the queue clears (Figure 25).

Figure 25: US Private Equity distributions y/y abs change



Sources: LSEG Datastream, MSCI, Oxford Economics, Allianz Research

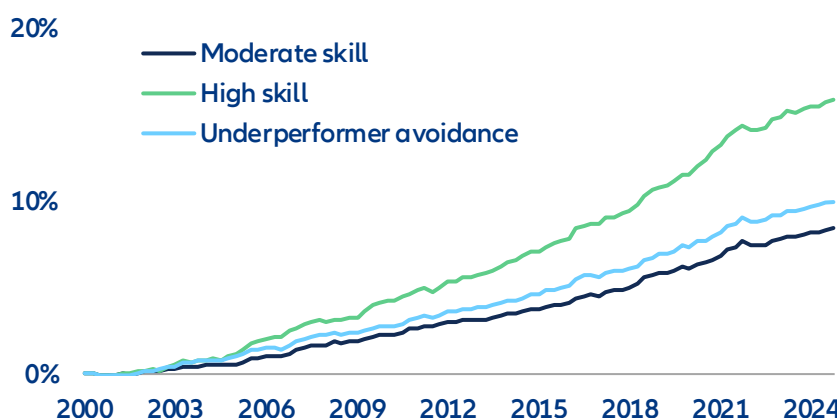
Note: Please note that these scenarios are produced using the Oxford Economics model and scenarios for comparability purposes. For official forecasts please refer to our quarterly scenario.

From beta to selection on both sides of the divide

All in all, the story is, at heart, a story about the fading power of beta. For most of the last two decades, simply being in the room was enough, public markets rode a long bull run and private equity compounded on cheap leverage and steadily expanding multiples. That era has closed on both sides of the divide. Public benchmarks now carry their returns on a narrow cohort of AI-exposed mega caps, while private equity is still working through vintages bought at the top of the cycle. What used to be a rising-tide question, how much to allocate to private markets, has become a question of which managers, which vintages, and which exposures. The center of gravity has shifted from beta to selection.

That shift is not a matter of conviction but of arithmetic. The dispersion between the best and worst private equity managers has widened to the point where the median fund tells you very little, and the returns that justify the asset class increasingly sit in its upper reaches. Skilled selection, on the evidence, is now worth more than the allocation decision itself, but it is a genuine skill, not a default setting, and an allocator without a durable edge is as likely to give that premium back as to capture it. Beta has become a commodity, the scarce input is judgement (Figure 26).

Figure 26: Average Private Equity buyout cumulative total portfolio alpha



Sources: Pitchbook, Allianz Research

For investors looking forward, that reframes the central question. The debate worth having is no longer "public or private," but "differentiated or undifferentiated" within each, because passive index beta and generic, broad-exposure buyout have quietly become the same crowded trade, priced for a tailwind that is no longer blowing. The democratization of private markets, through evergreen and semi-liquid wrappers, only sharpens the point as it widens access to an asset class precisely as access stops being the thing that pays, and the funds that actually generate the premium remain the hardest to reach. The way forward is less about how much to own and more about how to own it. Investors should favor managers and vintages with a demonstrable edge over generic exposure, treating liquidity and valuation discipline as features to be tested rather than assumed, and resisting the temptation to buy the label without the underlying skill. The next market correction will be the first real test of which private market structures were built to endure and which were built to sell, and the distinction between robust and fragile will become visible quickly. The alpha, on both sides of the liquidity divide, now belongs to those with the conviction and the capability to select it.

These assessments are, as always, subject to the disclaimer provided below.

FORWARD-LOOKING STATEMENTS

The statements contained herein may include prospects, statements of future expectations and other forward-looking statements that are based on management's current views and assumptions and involve known and unknown risks and uncertainties. Actual results, performance or events may differ materially from those expressed or implied in such forward-looking statements.

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